

# AK High-Dimensional Projection Structural Theory

## Version 21.0.0: Finite Globalization, Unified Bridge Calculus, and Certificate Complexity

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July 2026

### 1 Chapter 1: Scope, Migration, and the Dual-Core Constitution

**Scope Box (v21 conservative evolution; dual-Core constitution, finite globalization, unified Bridge calculus, and certificate complexity).**

All statements in this chapter, and every later statement labeled *Core-P*, *Core-DG*, *Core Interface*, or *Core governance*, are interpreted under the following v21.0 guard-rails.

- **Release purpose.** AK-HPDST v21.0 is the *Finite Globalization, Unified Bridge Calculus, and Certificate Complexity Release*. Its principal target is the typed chain

persistent certificate  $\longrightarrow$  finite diagram  $\longrightarrow$  global object  $\longrightarrow$  external Return.

Each arrow requires its own theorem, source and target type, equivalence strength, artifacts, and failure route.

- **Immutable lower baseline.** The complete frozen v20.0.0 source-controlled statement set is inherited as an immutable lower baseline. No v21 statement retroactively changes a v20 statement, hypothesis, comparison mode, proof artifact, verdict, non-claim, or permitted strength. A v21 strengthening receives a new v21 identity and does not overwrite the inherited item.
- **Conservative evolution.** “Core is immutable” means that an inherited theorem is not silently rewritten. It does not prohibit a later release from adding a separately typed Core, theorem, Interface, or strengthening. Such an addition is admissible only after its complete hypotheses, proof, dependencies, migration status, prohibited stronger readings, and artifact identity are displayed.
- **Dual-Core architecture.** The v21 mathematical baseline is organized as

$\text{Core}_P + \text{Core}_{DG}$ .

The symbol  $+$  denotes coordinated coexistence with typed Interfaces; it does not assert a product, co-product, equivalence, localization, or automatic mutual reconstruction.

- **Persistence Certificate Core.**  $\text{Core}_P$  retains the v20 repaired one-parameter persistence regime: window-first evaluation, hard threshold deletion, sharp two-sided threshold-gap transport, finite

detector certification, certificate complexity, higher Ext edges, actual-morphism Type IV diagnostics, transient-defect separation, certificate atlases, Restart, and finite-to-infinite reduction.

- **Finite Diagram and Globalization Core.**  $\text{Core}_{\text{DG}}$  is restricted by default to finite small categories or finite posets, finite-dimensional bounded complexes over a declared field, finite projective-generator probes, actual diagram morphisms, cones, finite Morita-type reconstruction, and explicitly effective finite descent. General constructible sheaves, stacks, hypersheaves,  $(\infty, 1)$ -categories, microlocal sheaves, wild multiparameter persistence, and unbounded derived categories are outside the stable Core unless separately promoted.
- **No automatic Core-P–Core-DG identification.** A barcode, persistence profile, or finite detector output does not itself determine a finite diagram object. A finite diagram probe does not itself determine a persistence object. Every passage between the two Cores carries a multidimensional Interface contract separating functoriality, exactness, reflection, information preservation, and permitted use.
- **Record globalization versus object globalization.** Chapter 6 may glue certificate records and establish a global record-level predicate. It does not construct a global mathematical object by itself. Object-level globalization requires Chapter 7 descent data, coherence, effectivity, and the declared target equivalence strength.
- **Selective HT–UB internalization.** Domain-independent finite reconstruction, cone detection, finite generator, finite-state or Noetherian reduction, explicit finite descent, noncompensating DAG composition, and model-relative certificate-complexity results may be promoted after proof-level migration. Iwasawa-, Serrin-, Tachikawa-, Leopoldt-, regulator-, or  $p$ -adic- $L$ -specific Returns remain Problem Interface, Validation, or source-specific Bridge material unless separately promoted. Editorial merger or citation does not change theorem status.
- **Typed Return strength.** A Return is valid only in the declared target category and at the declared relation: predicate, invariant, object class, isomorphism, exact, quasi-isomorphism, derived equivalence, pseudo-isomorphism, determinant, ideal, or another explicitly defined strength. No stronger Return is inferred from a weaker one without a proved promotion map.
- **Proof-DAG sovereignty.** Finite ordinary dependencies form an acyclic proof graph. A countable or infinite dependency scope additionally requires a well-founded rank or a separately certified finite-to-infinite reduction. A cycle is admissible only when a separately proved induction, coinduction, recursion, fixed-point, or mutually recursive theorem supplies the semantics. Validation results and benchmark truth values are downstream leaves and may not be imported as premises of the Core or Bridge route being validated.
- **Claim, evidence, verifier, workflow, and migration separation.** Document role, evidence status, invocation role, conformance status, verifier status, workflow status, and migration status are distinct axes. No favorable value on one axis promotes another axis.
- **Finite observation is not finite proof.** The inherited v20 distinction remains controlling. A finite list of values, ranks, maps, endpoints, local objects, overlap maps, generator probes, or numerical computations becomes a finite proof certificate only through a soundness/completeness or reconstruction theorem at the declared information, query, encoding, and verifier model.

- **Actual morphism and coherence discipline.** A matching, equal rank table, objectwise isomorphism class, or local equality is not automatically an actual morphism, natural transformation, cocycle, descent datum, or global isomorphism.
- **External-theorem separation.** A Core-P or Core-DG certificate is not an external-domain theorem. External Return requires source realization, every internal Core and HT/UB transfer, finite-to-infinite reduction when needed, coherence, descent, rigidity, source binding, and a final Return theorem at the requested strength.
- **Governance and manifest boundary.** Registration, complete serialization, hash agreement, successful compilation, replay, formal wiring, AI agreement, or human approval is not mathematical proof. These may support identity, auditability, or verification, but cannot compensate for a false, missing, ill-typed, circular, or unsupported mathematical obligation.

Accordingly, Chapter 1 fixes the constitutional language of v21, preserves the v20 lower baseline, defines the two Cores and their Interfaces, controls migration and theorem identity, types Return strength, fixes proof-DAG dependency semantics, and states the release gates and explicit non-claims.

## 1.1 Claim taxonomy and document discipline

**Definition 1.1** (v21 statement-role taxonomy). Every mathematically or operationally meaningful statement receives exactly one primary role record

$$\text{Role}(c) = (\text{Layer}(c), \text{Subtype}(c)).$$

The primary layer is one of the layers in Definition 1.10. The exact role subtype is one of:

- (i) **definition**;
- (ii) **lemma**;
- (iii) **theorem**;
- (iv) **Core micro-theorem**;
- (v) **metric-repair theorem**;
- (vi) **operational policy**;
- (vii) **audit axiom**;
- (viii) **Interface definition**;
- (ix) **Interface theorem**;
- (x) **Bridge candidate**;
- (xi) **Bridge program**;
- (xii) **Bridge theorem**;
- (xiii) **Return theorem**;

- (xiv) **toy bridge theorem**;
- (xv) **Validation result**;
- (xvi) **Search or Platform artifact**;
- (xvii) **Companion template**;
- (xviii) **Spec**;
- (xix) **successor-release candidate**;
- (xx) **non-claim**;
- (xxi) **deprecated or rejected claim**.

The layer and subtype are both mandatory. Thus, for example, a theorem in  $\text{Core}_P$ , a theorem in  $\text{Core}_{DG}$ , a theorem in a Problem Interface, and a Validation theorem share the subtype *theorem* but do not share the same layer or authority. Conversely, a layer name alone does not specify whether the item is a definition, theorem, policy, candidate, or non-claim.

**Definition 1.2** (Seven-axis statement typing). Every registered v21 statement is typed on seven independent axes:

- (i) **document role**, from Definition 1.1;
- (ii) **evidence status**, chosen from
 

proved, conditional, operational, spec, deprecated, rejected, unknown;
- (iii) **invocation role**, chosen from
 

theorem\_evidence, operational\_rule,  
specification\_target, validation\_target,  
provenance\_only;
- (iv) **conformance status**, recording whether the statement conforms to the declared v21 schema and dependency contract;
- (v) **verifier status**, chosen from
 

pass, reject, undefined, not\_invoked;
- (vi) **workflow status**, such as proposed, drafted, reviewed, frozen, superseded, or released;
- (vii) **migration status**, from Definition 1.11.

The axes are not interchangeable. In particular, workflow completion is not theorem proof; registration is not proof; conformance is not mathematical truth; validation success does not change the source theorem's role; and a migration status does not strengthen a statement.

**Policy 1** (Weaker-safe interpretation). *If source text, proof artifact, Claim Register, migration record, or consuming document disagree about role, hypotheses, scope, comparison mode, Return strength, or evidence status, the weakest source-supported reading governs until a new audited revision repairs the inconsistency.*

**Remark 1.3** (Registration and migration are not proof). A statement does not become proved by receiving a ClaimID, MigrationID, theorem number, source locator, hash, or release label. The proof or source-backed policy evidence remains controlling.

## 1.2 The v21 conservative-evolution constitution

**Definition 1.4** (Immutable v20 lower baseline). The immutable v20 lower baseline is the frozen set of registered v20 statements, proofs, source bindings, artifacts, permitted uses, non-claims, and verdicts. For every inherited v20 item  $c$ , v21 preserves:

- (i) its exact statement and hypotheses;
- (ii) its mathematical domain and codomain;
- (iii) its metric, coefficient, variance, endpoint, and evaluation conventions;
- (iv) its document role, evidence status, and permitted invocation roles;
- (v) its proof and artifact identity;
- (vi) its prohibited stronger readings;
- (vii) its historical verdict and repair ancestry.

**Definition 1.5** (Conservative v21 evolution). The v21 theory is a conservative evolution of the v20 baseline when:

- (i) every inherited v20 item remains available at its original strength;
- (ii) every strengthened, repaired, generalized, or differently typed statement receives a new v21 identity;
- (iii) no v21 proof relies on a stronger reading of an inherited source than the source supports;
- (iv) every new dependency is represented in the v21 proof DAG;
- (v) every migration records the old and new source locators, roles, strengths, artifacts, and prohibited readings;
- (vi) a user may still invoke the inherited v20 route through an exact binding to the frozen v20 source artifact, without invoking the v21 strengthening, provided its original scope is satisfied.

**Theorem 1.6** (No retroactive theorem mutation). *Under Definition 1.5, a new v21 theorem, Interface, migration, or editorial consolidation does not change the truth conditions, proof obligations, evidence status, or historical verdict of any inherited v20 item.*

*Proof.* The v20 item and the v21 item have distinct immutable identities. The migration relation records dependency, inheritance, supersession for future use, or strengthening, but it is not an equality that overwrites the source record. Therefore the inherited statement remains interpreted by its own source, proof, hypotheses, and artifact, while the v21 statement is interpreted by its new record.  $\square$

**Definition 1.7** (Dual-Core architecture). The v21 stable mathematical architecture consists of:

$\text{Core}_P :=$  the inherited and conservatively strengthened Persistence Certificate Core,  
and

$\text{Core}_{DG} :=$  the Finite Diagram and Globalization Core.

The dual-Core expression

$$\text{Core}_P + \text{Core}_{DG}$$

means that the two Cores have separate standing hypotheses, theorem namespaces, admissible objects, morphisms, certificates, and failure routes, together with explicitly declared Core Interfaces. It does not assert that either Core is reducible to the other.

**Definition 1.8** (Core-P standing scope). The default  $\text{Core}_P$  scope is the v20 constructible one-parameter persistence regime over a field, together with the declared finite-type filtered representatives, actual tower morphisms, detector families, certificate atlases, and finite-to-infinite reductions permitted by the inherited theorems and their v21 conservative strengthenings.

**Definition 1.9** (Core-DG standing scope). Fix a field  $k$ , a small category  $P$  with finite object and finite morphism sets, and the abelian category

$$\text{Diag}_P(k) := \text{Fun}(P, \text{Vect}_k).$$

The default  $\text{Core}_{DG}$  scope consists of

$$\text{Ch}^b(\text{Diag}_P(k)) \quad \text{and} \quad D^b(\text{Diag}_P(k)),$$

together with actual diagram morphisms, finite projective generators, generator probes, cones, finite Morita-type realization, finite Alexandrov models with declared variance, and explicitly effective finite descent. Every extension beyond this scope requires a separately proved Interface or Technical Extension theorem.

**Policy 2** (No silent Core transfer). *No statement proved in  $\text{Core}_P$  is consumed as a  $\text{Core}_{DG}$  theorem, and no statement proved in  $\text{Core}_{DG}$  is consumed as a  $\text{Core}_P$  theorem, unless a Core Interface theorem supplies the exact source object, target object, morphism action, variance, preserved or reflected predicate, equivalence strength, artifacts, and failure boundary.*

### 1.3 Layer taxonomy, migration calculus, and theorem identity

**Definition 1.10** (v21 layer taxonomy). Every v21 artifact and theorem-bearing statement belongs primarily to one of the following layers:

- (1)  $\text{Core}_P$ ;
- (2)  $\text{Core}_{DG}$ ;
- (3) Core Interface;
- (4) Technical Extension;
- (5) Problem Interface;
- (6) Bridge and Return;
- (7) Validation;
- (8) Search and Platform;
- (9) Governance.

A statement may have dependencies in several layers, but it has one primary role and does not acquire the authority of another layer by citation.

**Definition 1.11** (v21 migration-status record). The migration-status axis is a record with one *semantic status* and zero or more *disposition modifiers*. This separation is required because relocation, merger, deprecation, and source-only retention are not mutually exclusive with semantic inheritance or repair.

The semantic status is exactly one of:

- (i) *inherited*: retained at exactly the predecessor strength;
- (ii) *strengthened*: a distinct successor theorem has strictly greater declared claim strength under a registered comparison, for example by weaker hypotheses, a broader domain, a stronger conclusion, or a stronger equivalence/Return level;
- (iii) *repaired*: the predecessor is unavailable for v21 theorem use at the disputed reading and a corrected successor is supplied, while predecessor history remains immutable;
- (iv) *rejected*: unavailable for theorem use;
- (v) *successor\_only*: intentionally excluded from v21 and reserved for a later release.

The disposition modifiers are any compatible subset of:

- (a) *merged*: editorially or architecturally consolidated without theorem-strength promotion;
- (b) *relocated*: moved to another layer or document;
- (c) *deprecated*: preserved but discouraged for future use;
- (d) *source\_only*: retained only as source-specific, domain-specific, Validation, or provenance material.

No modifier changes the semantic status or theorem strength. In particular, *merged* does not mean proved, and *relocated* does not mean strengthened.

**Definition 1.12** (Migration record). A migration record contains at least:

- (i) MigrationID;
- (ii) predecessor theory version, artifact, UID, and exact locator;
- (iii) successor artifact, UID, and exact locator when one exists;
- (iv) semantic migration status and every disposition modifier;
- (v) old and new primary layers;
- (vi) old and new document roles and evidence statuses;
- (vii) an exact statement comparison;
- (viii) hypothesis and conclusion deltas;
- (ix) Return-strength delta, if any;
- (x) proof and artifact bindings;

- (xi) prohibited stronger readings;
- (xii) dependency changes;
- (xiii) whether the predecessor remains directly invocable;
- (xiv) reviewer and release-closure status.

**Policy 3** (Selective HT–UB promotion). *An HT or UB statement may enter the stable v21 Core only if:*

- (i) the promoted statement is domain-independent at the declared scope;*
- (ii) its complete proof and hypotheses are available in an immutable source artifact;*
- (iii) all source-specific assumptions and Return clauses are removed from the Core statement or remain explicitly typed as Interface hypotheses;*
- (iv) a conservative-extension analysis shows no conflict with v20;*
- (v) the theorem receives a v21 Core UID and proof locator;*
- (vi) every counterexample, failure route, and prohibited converse is preserved;*
- (vii) the promotion is recorded in the Migration Register and final Claim Register.*

*Editorial integration of HC, PE, SS, RL, IW/Return, HT, or UB content does not satisfy these conditions by itself.*

**Definition 1.13** (Theorem identity and namespace rule). The canonical identity of a v21 statement consists of theory version, primary layer, document or chapter namespace, stable local identifier, exact source locator, and immutable artifact identity.

The namespace rules are:

- (i) existing v20 ClaimIDs and registered UIDs are preserved;
- (ii) a v21 strengthening or repair receives a new v21 UID;
- (iii) the same mathematical statement is not independently registered in Main and Appendix under two theorem UIDs;
- (iv) Main carries the canonical statement when the result is a Main theorem, while the Foundation Appendix may carry the expanded proof under the same canonical identity and a distinct proof locator;
- (v) a Technical Extension, Problem Interface, Validation, or Companion statement is not promoted by copying its text into Main;
- (vi) theorem renaming requires an explicit alias or supersession record.

**Policy 4** (No duplicate mathematical authority). *If two artifacts reproduce the same theorem, exactly one source is canonical for theorem identity. Other copies are proof expansions, extracts, mirrors, or provenance copies. If the copies disagree, Policy 1 applies.*



## 1.4 Dependency constitution and proof-DAG sovereignty

**Definition 1.14** (Ordinary proof dependency graph). A v21 ordinary proof dependency graph is a directed graph

$$\mathfrak{D} = (V, E)$$

whose vertices are typed statements, artifacts, obligations, Interface records, reduction records, coherence cells, descent records, rigidity records, and Return records. An edge  $u \rightarrow v$  means that  $v$  consumes  $u$  at a declared invocation role and strength. For a finite target dependency scope, the ordinary graph is required to be acyclic. For a countable or infinite target scope, acyclicity alone is insufficient: the graph must also carry an independently checkable well-founded dependency rank, or the route must factor through a separately certified finite-to-infinite reduction.

**Definition 1.15** (Admissible cyclic semantics). A syntactic or conceptual cycle is admissible only when a separate theorem supplies one of the following semantics:

- (i) induction on a well-founded order;
- (ii) coinduction on a declared coalgebraic structure;
- (iii) structural recursion with a termination proof;
- (iv) a contraction or fixed-point theorem;
- (v) simultaneous construction with a separately proved existence and uniqueness theorem;
- (vi) another explicitly registered cycle-resolution theorem.

Without such a theorem, the cycle is a dependency failure rather than proof.

**Policy 5** (Layer direction and downstream non-interference). *The normal dependency direction is*

$$\text{Core foundations} \longrightarrow \text{Core Interfaces} \longrightarrow \text{verification} \longrightarrow \text{Bridge/Return} \longrightarrow \text{Validation}.$$

*A downstream Validation, Search, Platform, Companion, or release artifact may audit, instantiate, or falsify an upstream statement, but it may not silently change the statement or serve as a circular premise in the route it validates.*

**Policy 6** (Benchmark quarantine). *In Known-Theorem Recovery, the benchmark conclusion and every conclusion-derived witness are unavailable to the realization, finite certificate, finite-to-infinite reduction, and Return proof being validated, except as an explicitly quarantined final comparison target. Agreement with a benchmark does not repair circularity.*

**Remark 1.16** (Chapter dependency order). The intended Main dependency order is:

$$\begin{aligned} &\text{Ch.1} \longrightarrow \text{Ch.2} \longrightarrow \text{Ch.3} \longrightarrow \text{Ch.4}, \\ &(\text{Ch.2}, \text{Ch.3}, \text{Ch.4}) \longrightarrow \text{Ch.5}, \\ &(\text{Ch.2}, \text{Ch.3}, \text{Ch.5}) \longrightarrow \text{Ch.6}, \\ &(\text{Ch.4}, \text{Ch.5}, \text{Ch.6}) \longrightarrow \text{Ch.7}, \\ &\text{Ch.1-7} \longrightarrow \text{Ch.8} \longrightarrow \text{Ch.9} \longrightarrow \text{Ch.10}. \end{aligned}$$

Chapter 10 is a leaf layer and is not an input to Chapters 1–9.

## 1.5 Return-strength typing

**Definition 1.17** (Return-strength record). A Return-strength record is a tuple

$$\Sigma_{\text{Ret}} = (\mathcal{T}, \sim_{\mathcal{T}}, \mathcal{P}, \mathcal{W}),$$

where:

- (i)  $\mathcal{T}$  is the target category, class, or semantic domain;
- (ii)  $\sim_{\mathcal{T}}$  is the exact relation returned;
- (iii)  $\mathcal{P}$  is the returned predicate, invariant, object, morphism, or structure;
- (iv)  $\mathcal{W}$  is the list of weaker Returns that follow through separately registered weakening maps.

**Definition 1.18** (Canonical v21 Return-strength families). The standard v21 vocabulary includes:

- (i) predicate Return;
- (ii) scalar-invariant Return;
- (iii) graded or filtered-invariant Return;
- (iv) object-class Return;
- (v) object-isomorphism Return;
- (vi) morphism Return;
- (vii) exact-category Return;
- (viii) quasi-isomorphism Return;
- (ix) bounded-derived-equivalence Return;
- (x) pseudo-isomorphism Return;
- (xi) determinant Return;
- (xii) ideal or characteristic-ideal Return;
- (xiii) another source-defined strength with an explicit target relation.

**Policy 7** (Return strengths are partially ordered). *Return strengths are not forced into a single total order. A weakening edge*

$$\Sigma_1 \longrightarrow \Sigma_0$$

*may be registered only when a theorem proves that every Return at  $\Sigma_1$  induces a Return at  $\Sigma_0$  in the declared target regime. Incomparable strengths remain incomparable.*

**Policy 8** (Route-admissible Return set and conditional ceiling). *Let  $\mathfrak{S}$  be the partially ordered set of declared Return strengths. Every route component  $e$  records a downward-closed supported set  $\mathfrak{S}_e \subseteq \mathfrak{S}$ . The composite route supports exactly*

$$\mathfrak{S}_{\text{route}} := \bigcap_{e \in E_{\text{required}}} \mathfrak{S}_e.$$

*A requested strength  $\Sigma$  is admissible only if  $\Sigma \in \mathfrak{S}_{\text{route}}$ .*

*Because  $\mathfrak{S}$  is only partially ordered,  $\mathfrak{S}_{\text{route}}$  need not possess a unique greatest element. The phrase route ceiling may be used only when a greatest supported strength exists and is proved unique. Otherwise the supported lower set, not an invented “weakest link”, is the controlling object.*

## 1.6 Core-P–Core-DG Interface constitution

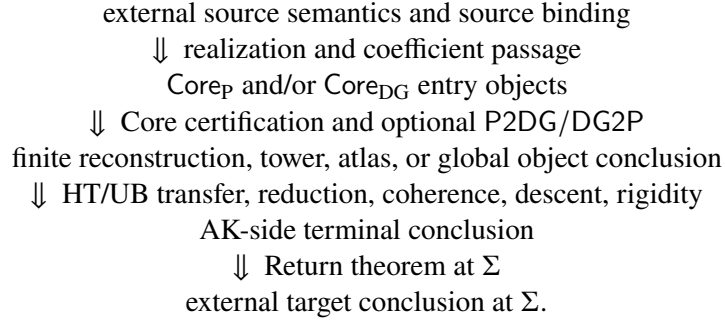
**Definition 1.19** (P-to-DG Interface record). A P2DG record contains:

- (i) a declared  $\text{Core}_P$  source object and profile stage;
- (ii) a finite critical set, finite grid, or other finite indexing category;
- (iii) a declared  $\text{Core}_{DG}$  target diagram;
- (iv) endpoint, variance, coefficient, and terminal-sentinel conventions;
- (v) an action mode: `functorial`, `partially_functorial`, or `objectwise_only`;
- (vi) an exactness profile: `exact`, `left_exact`, `right_exact`, `theorem_controlled`, or `not_asserted`;
- (vii) a reflection profile: `conservative`, `zero_reflecting`, `predicate_reflecting`, or `not_asserted`;
- (viii) an information profile: `lossless`, `feature_faithful`, `observation_only`, or `lossy`;
- (ix) the exact object, morphism, predicate, and structure data preserved, reflected, or discarded;
- (x) proof and immutable artifact support;
- (xi) prohibited stronger readings.

These fields are independent. Exactness does not imply conservativity, zero-reflection does not imply fullness or faithfulness, and objectwise losslessness does not manufacture a functor on morphisms.

**Definition 1.20** (DG-to-P Interface record). A DG2P record declares a finite diagram, a chosen chain or parameter direction, the extracted persistence profile, and the same independent action, exactness, reflection, and information profiles required by Definition 1.19. It additionally records every diagram direction, branching relation, extension class, or coherence datum forgotten by the extraction. It is not faithful, conservative, zero-reflecting, or reconstruction-complete unless a separate theorem proves that property on the declared class.

**Definition 1.21** (Unified v21 proof route). A unified v21 proof route is a typed sub-DAG of the form



Only invoked branches are required in an individual route. Thus a valid v21 route may use  $\text{Core}_P$  alone or  $\text{Core}_{DG}$  alone. The specific release witness required by G21-7 must consume both Cores and a proved Interface between them. No displayed arrow is supplied by adjacency in the diagram; each is an independently typed theorem or record.

**Definition 1.22** (Record-level globalization). Record-level globalization is the construction of one coherent global certificate record from local certificate records, overlap comparisons, coverage, and higher-simplex evidence. Its conclusion is a typed global record predicate. It does not by itself assert the existence of a global object in  $\text{Core}_{DG}$ .

**Definition 1.23** (Object-level globalization). Object-level globalization is a proved construction of a global object or morphism from local objects or morphisms, actual overlap data, cocycle or higher-coherence data, and an effective descent theorem in the declared target category. Its Return strength is bounded by the descent theorem used.

**Theorem 1.24** (No globalization by record gluing alone). *A record-level globalization theorem does not imply object-level globalization unless a separately proved Core-DG descent theorem supplies local object semantics, actual overlap maps, coherence, effectivity, and the target equivalence strength.*

*Proof.* A certificate record can encode agreement of predicates or observations without encoding objects, actual morphisms, cocycles, or an effective descent functor. Those data are logically independent of record coherence. Therefore the object-level conclusion requires the additional Core-DG theorem.  $\square$

**Definition 1.25** (Core-DG constitutional target theorems). Chapter 7 is required to prove, at explicitly displayed hypotheses, the following theorem families:

- (i) finite projective-generator zero detection;
- (ii) cone detection of isomorphisms in the declared derived category;
- (iii) enriched finite-diagram reconstruction;
- (iv) finite Morita-type realization;
- (v) finite Alexandrov realization with fixed variance;
- (vi) effective finite object and morphism descent;
- (vii) global zero and isomorphism certification;
- (viii) model-relative global certificate-complexity bounds.

This definition fixes release targets; it is not itself a proof of those theorems.

## 1.7 Constitutional gate architecture

**Definition 1.26** (Core-P gate). The v21 Core-P gate is the inherited B-Gate<sup>+</sup>, re-issued without weakening and conservatively extended only by separately proved Chapter 2–6 clauses. It certifies a declared persistence-certificate scope and does not certify Core-DG reconstruction or object-level descent.

**Definition 1.27** (Core-DG gate). A Core-DG gate passes only when every invoked required clause has verifier status `pass` and no required clause has status `undefined`, including as applicable:

- (i) finite category or poset scope;
- (ii) bounded finite-dimensional diagram or complex;
- (iii) actual source objects and morphisms;
- (iv) complete enriched generator probes;
- (v) generator zero-reflection or reconstruction theorem;
- (vi) cone data when an isomorphism is inferred;
- (vii) variance and coefficient conventions;
- (viii) local objects, overlap morphisms, cocycle or higher coherence, and effectivity when descent is invoked;
- (ix) the exact internal target category and conclusion/equivalence strength, together with Return strength only when a Return is invoked;
- (x) immutable proof, source, and artifact records.

A scalar rank table, objectwise zero list, pairwise local agreement, or manifest-complete record cannot discharge this gate without the corresponding Chapter 7 theorem.

**Definition 1.28** (Unified Bridge/Return gate). The unified Bridge/Return gate passes at requested strength  $\Sigma$  only when:

- (i) source semantics and source binding pass;
- (ii) realization and coefficient passage pass;
- (iii) every invoked Core-P and Core-DG gate passes;
- (iv) every HT and UB transfer component passes;
- (v) every finite-to-infinite, inverse-limit, coherence, descent, and rigidity obligation passes;
- (vi) the final Return theorem passes;
- (vii) every assumption context used by a required node or edge is preserved, discharged by a named theorem, or explicitly carried into the target statement;
- (viii) the requested strength belongs to the route-admissible Return set of Policy 8;
- (ix) the Chapter 8 proof-DAG and artifact closure pass.

**Definition 1.29** (v21 aggregate gate). For a declared run scope, the aggregate v21 gate is the typed conjunction of the invoked Core-P, Core-DG, verification, Bridge, Return, and release clauses. A clause outside scope is `not_invoked` only through an immutable scope-exclusion record. No pass cancels a reject or a required undefined clause.

## 1.8 v21 release gates and completion contract

**Policy 9** (G21-1: Migration completeness). *Every v20, HT, and UB theorem-bearing or governance-bearing statement in the declared migration inventory has a source locator, primary layer, semantic migration status, disposition modifiers, v21 destination or exclusion, and review status.*

**Policy 10** (G21-2: Core-P regression). *All mandatory v20 Core-P theorems, counterexamples, and gates pass the declared internal v21 regression suite without strengthened interpretation. Separately, every registered exact-strength v20 Known-Theorem Recovery track remains reproducible in Chapter 10 at its original external strength. That Validation success is not reclassified as Core-P theorem evidence.*

**Policy 11** (G21-3: Core-DG closure). *The Chapter 7 generator, cone, reconstruction, finite realization, effective descent, global zero/isomorphism, and boundary theorems are proved and artifact-bound at their exact strengths.*

**Policy 12** (G21-4: Core-P-to-Core-DG closure). *At least one nontrivial finite persistence profile is transported through a proved P2DG Interface into a finite diagram, with endpoint, morphism, variance, and information-loss obligations closed.*

**Policy 13** (G21-5: Object-level globalization). *At least one finite local-object system is globalized by an actual effective descent theorem, and the result is distinguished from record-level atlas gluing.*

**Policy 14** (G21-6: Certificate complexity). *For one common problem family and one fixed information/query/encoding/verifier model, v21 proves at least one nontrivial certificate upper bound and one nontrivial lower bound or sharpness result.*

**Policy 15** (G21-7: Unified external Return). *At least one complete external route consumes both  $\text{Core}_P$  and  $\text{Core}_{DG}$ , together with every required HT/UB, reduction, coherence, descent, rigidity, and Return component, and Returns at a precisely declared external strength.*

**Policy 16** (G21-8: Non-duplication and namespace closure). *No theorem has conflicting canonical identities across Main, Foundation, HT, UB, CM, TB, Technical Extension, Problem Interface, or Validation artifacts. Aliases, proof expansions, and migrated copies are explicit.*

**Policy 17** (G21-9: Release and artifact closure). *The final Claim Register, Migration Register, source bindings, manifests, dependency DAG, failure routes, proof and computation artifacts, replay records, non-claims, and publication package are internally consistent and independently reconstructible.*

**Policy 18** (v21 release acceptance). *The designation AK-HPDST v21.0 complete release is reserved for a document set in which G21-1 through G21-9 pass at their displayed strengths. A failed mandatory gate blocks the complete-release designation. A gate with missing required evidence is undefined and also blocks the designation.*

**Definition 1.30** (Chapter 1 completion condition). Chapter 1 is structurally complete when every v20, HT, and UB statement can be assigned:

- (i) a predecessor identity and exact locator;
- (ii) a primary v21 layer;
- (iii) a semantic migration status and every disposition modifier;
- (iv) a successor identity or explicit exclusion;

- (v) an invocation and Return-strength boundary;
- (vi) a dependency-DAG position;
- (vii) an artifact and review obligation.

This condition asserts schema completeness, not completion of the full Migration Register.

## 1.9 Immutable Core-P lower baseline and inherited finite-certificate contract

**Definition 1.31** (Conservative Core-P re-issuance). The v21 Core-P contract re-issues the complete v20 Core-P lower baseline. The v20 baseline was itself a conservative extension of the repaired v19.0.0 Core. The re-issuance is conservative when all of the following hold:

- (i) every v20 theorem retained by v21 is used with the same statement, hypotheses, metric convention, evaluation order, and prohibited stronger readings;
- (ii) each new detector, compression, higher-edge, defect, reduction, or Return theorem has an explicit source and target type;
- (iii) no v21 theorem retroactively changes a v20 verdict or artifact;
- (iv) no v21 theorem converts a previously non-claim, specification, candidate, toy theorem, or operational rule into theorem evidence without a new proof and registration;
- (v) every new dependency is recorded in the claim graph and immutable manifest.

**Definition 1.32** (Inherited v20 four-pillar Core-P contract). The inherited v20 Core-P program preserves four distinct pillars:

- (A) **Finite zero-reflecting detector calculus:** finite evidence for the exact zero/nonzero predicate of a repaired persistence or defect profile;
- (B) **Higher obstruction edge calculus:** conditional degree- $n$  transport from repaired persistence vanishing to derived  $\text{Ext}^n$ -vanishing;
- (C) **Finite proof compression:** family-relative certificate size, completeness bounds, lower bounds, sharpness, and no-go results;
- (D) **Problem-Bridge demonstration:** a problem-specific chain containing realization, Core certification, any finite-to-infinite reduction, and a Return Theorem.

The four pillars are logically distinct and non-compensating. No implication between them is asserted without a separate theorem, and progress in one pillar does not silently discharge another.

**Definition 1.33** (Finite observation and finite proof certificate). A *finite observation record* is any finite collection of sampled values, ranks, maps, endpoints, critical values, numerical approximations, or computed artifacts.

A *finite proof certificate* is a finite observation record together with all theorem, typing, completeness, provenance, and artifact data required to derive the declared mathematical predicate.

For a detector-based zero verdict, the required data include, as applicable:

- (i) the exact source profile and detector profile stage;

- (ii) monitored degree, window, threshold, and endpoint convention;
- (iii) a soundness theorem;
- (iv) a completeness, characterization, or coverage theorem;
- (v) a complete birth/germ set, coverage witness, or family-relative mesh certificate;
- (vi) the source and proof of any uniform threshold-excess lower bound;
- (vii) exact, metric-gap-certified, or theorem-controlled nonmetric comparison records used in transport;
- (viii) the certificate model  $\mathfrak{M} = (\mathcal{I}, \mathcal{Q}, \mathcal{E}, \mathcal{V})$ , including query transcript and acceptance semantics;
- (ix) the model-relative cost vector  $(q, b, t, a)$  and immutable evidence for each coordinate;
- (x) artifact identity, theorem identity, and claim-use preflight.

If a required item is absent, the detector-based zero conclusion is undefined.

**Definition 1.34** (Detector profile stage). Every detector record declares exactly one source stage:

`windowed_prethreshold,      repaired_postthreshold,`  
`kernel_defect,              cokernel_defect,`  
`diagram_probe,            descent_coherence.`

The stage determines the detector's source object and permitted conclusion. A theorem proved for one stage is not transferred to another stage without an explicit stage-transport theorem. The last two stages require actual Chapter 4 or Chapter 7 objects and the corresponding probe or coherence theorems; no barcode detector theorem is inherited by token substitution.

**Definition 1.35** (Finite detector certificate record). A finite detector certificate record contains at least:

- (i) a detector identifier and theorem identifier;
- (ii) the source profile identifier and stage;
- (iii)  $(n, W, \tau)$ , endpoint convention, and window-cofinal policy;
- (iv) detector mode, chosen from a registered set such as
 

`birth,    germ,    certified_stencil,    family_mesh;`
- (v) the finite sample or critical-value set;
- (vi) soundness evidence;
- (vii) completeness evidence and its non-circular source;
- (viii) exact, metric-gap-certified, or theorem-controlled nonmetric transport evidence when the source object is not the object whose predicate is consumed;
- (ix) the complete certificate model, query transcript, response precision, stopping rule, and verifier semantics;
- (x) the cost vector  $(q, b, t, a)$  and, when invoked, the model-relative Pareto frontier  $\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta)$ ;



(xi) immutable source, computation, proof, dependency, and replay artifacts.

The record proves only the predicate stated by its governing theorem.

**Definition 1.36** (Two-axis detector status). Detector certification and detector mathematical value are recorded on separate axes.

The certificate-status axis is:

`detector_certified,      detector_failed,      detector_pending.`

It maps to the canonical atomic status calculus by

`detector_certified ↦ pass,`  
`detector_failed ↦ reject,`  
`detector_pending ↦ undefined.`

where `detector_failed` means that a well-typed required detector certificate condition is evidenced to fail.

The value axis is:

`zero,      nonzero,      undefined.`

A value is mathematically consumable only when the certificate status is `detector_certified`. A certified nonzero value rejects a required zero clause; a certified zero value discharges that zero clause only at the strength of the governing completeness theorem. An uncertified zero-looking output remains undefined. When detector semantics are outside the claim scope, the canonical atomic status is `not_invoked`; no specialized certificate or value token is imputed.

**Policy 19** (Detector admission and non-compensation). *A detector may discharge a Core zero clause only when:*

- (i) *the detector is applicable to the exact declared profile stage;*
- (ii) *the certificate status is `detector_certified`;*
- (iii) *its zero-reflection theorem covers the declared object or family;*
- (iv) *every endpoint, birth/germ, coverage, mesh, cofinal, and family-membership obligation passes;*
- (v) *every exact or metric transport obligation passes;*
- (vi) *all referenced artifacts are immutable and reconstructible.*

*No favorable numerical reserve, detector score, compression ratio, or independent passing clause compensates for a failed or undefined detector obligation.*

**Definition 1.37** (Metric-safe detector transport chain). A metric-safe transfer of a detector-based zero predicate from a windowed pre-threshold profile  $B$  to a compared profile  $C$  has the typed chain

$$\begin{aligned}
 d_B(B, C) \leq \varepsilon & \quad + \quad \text{two-sided threshold-gap certificate} \\
 & \quad \Downarrow \\
 d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon & \quad + \quad 2\varepsilon < \tau \\
 & \quad \Downarrow \\
 T_\tau^{\text{bar}} B = 0 & \iff T_\tau^{\text{bar}} C = 0.
 \end{aligned}$$

If the zero statement on either side is established by a detector, the detector must be complete on that side. This definition records the required theorem chain; Chapter 2 and the Foundation metric appendix supply its mathematical components.

**Definition 1.38** (Higher obstruction contract). Fix an invoked degree  $n \geq 1$ . A standard  $n$ -eligible higher obstruction record consists of:

(i) an admissible representative  $C_{\tau, W}F$ ;

(ii) a declared realized object

$$A_n(F) := R_n(C_{\tau, W}F) \in D^b(k\text{-mod});$$

(iii) the declared standard amplitude regime, normally  $A_n(F) \in D^{[-n, 0]}(k\text{-mod})$ , or a separately proved admissible replacement;

(iv) a fixed edge extractor  $E_n$  on the declared repaired-profile class;

(v) an artifact-backed edge isomorphism

$$H^{-n}(A_n(F)) \xrightarrow{\sim} E_n(\text{Eval}_{n, W, \tau}(F));$$

(vi) zero compatibility

$$P = 0 \implies E_n(P) = 0.$$

Under these data, Chapter 4 may prove the one-way discharge

$$\text{Eval}_{n, W, \tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0.$$

**Definition 1.39** (Natural zero-reflection upgrade). A higher obstruction record admits a zero-reflection upgrade on a declared profile class  $\mathcal{P}_n$  only when:

(i)  $E_n$  is fixed independently of the desired verdict;

(ii)  $E_n$  is natural on the declared morphism class;

(iii) the edge-realization isomorphism is natural on the declared scope;

(iv) a theorem proves

$$E_n(P) = 0 \implies P = 0 \quad (P \in \mathcal{P}_n);$$

(v) representative-change and realization-change compatibility are separately discharged.

Only under these hypotheses may the vanishing equivalence

$$\text{Eval}_{n, W, \tau}(F) = 0 \iff \text{Ext}^n(A_n(F), k) = 0$$

be asserted on the declared scope. No object-level isomorphism  $\text{PH}_n \cong \text{Ext}^n$  follows from this vanishing equivalence.

**Remark 1.40** (Synthetic detector duality). For a vector space  $V$ , the shifted object  $V[n]$  satisfies a tautological derived duality with  $\text{Ext}^n$ . Applying this construction to a detector output can be mathematically valid, but it is synthetic unless a separately defined natural realization identifies that shifted object with an independently meaningful  $A_n(F)$ . Synthetic duality may be used as a toy theorem, calibration, or lower-level algebraic component; it is not promoted to the principal higher Bridge.

**Definition 1.41** (Family-relative proof compression). Fix a declared family-relative decision problem  $(\mathcal{F}, \Theta)$  and a certificate model

$$\mathfrak{M} = (I, Q, \mathcal{E}, \mathcal{V}).$$

A family-relative certificate scheme  $\Pi$  produces an object-specific record  $C_X$  for each  $X \in \mathcal{F}$  under the declared information and query rules and is verified by one sound-and-complete verifier rule. Its model-relative family cost is

$$\text{cost}_{\mathcal{F}}^{\mathfrak{M}}(\Pi) = (q_{\mathcal{F}}, b_{\mathcal{F}}, t_{\mathcal{F}}, a_{\mathcal{F}}) \in \overline{\mathbb{N}}^4,$$

using the aggregation fixed by the model; the default deterministic exact aggregation is coordinatewise worst case over  $\mathcal{F}$ . Let  $C_{\mathcal{F}}^{\mathfrak{M}, \text{fin}}(\Theta) \subseteq \mathbb{N}^4$  be the finite achievable scheme-cost set and define

$$\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta) := \text{ParetoMin}\left(C_{\mathcal{F}}^{\mathfrak{M}, \text{fin}}(\Theta)\right).$$

Thus the canonical v21 minimal-certificate object is a Pareto frontier of family-level scheme costs, not a collection of per-object costs and not a single scalar. The legacy scalar  $\text{MinCert}(\Theta)$  denotes only the worst-case query projection in the exact nonadaptive stencil model.

A family-relative proof-compression theorem must distinguish:

- (i) the information, query, encoding, and verifier models;
- (ii) an upper bound produced by births, germs, mesh, finite probes, or external structure;
- (iii) an information-model-relative lower bound or sharpness counterexample;
- (iv) adaptive/nonadaptive, deterministic/randomized, and exact/approximate semantics;
- (v) the producer/acquisition rule, family-cost aggregation, and whether object-dependent advice is charged or supplied in the information model;
- (vi) query, bit, verification, and artifact coordinates, including a finite encoding rule for every datum claimed to have finite bit cost;
- (vii) endpoint, boundary, family-promise, and unbounded-window conditions;
- (viii) the source of every family-wide excess, generator, or coherence bound.

A single already-classified zero object does not define a nontrivial family-relative minimal certificate merely because an empty transcript suffices retrospectively. Costs from different models are incomparable without a model-translation theorem.

**Definition 1.42** (Inherited Core-P Problem-Bridge proof chain). The inherited Core-P Problem-Bridge branch is a composable sequence of separately typed statements:

$$\begin{array}{c} \text{external source statement} \\ \Downarrow \text{realization/soundness-in} \\ \text{AK-readable filtered or persistence data} \\ \Downarrow \text{finite detector and Core certification} \\ \text{AK-side vanishing or defect conclusion} \\ \Downarrow \text{higher obstruction and/or finite-to-infinite reduction} \\ \text{AK-side theorem at the required infinite or derived scope} \\ \Downarrow \text{Return/soundness-out theorem} \\ \text{external mathematical conclusion.} \end{array}$$

Each arrow has its own role, hypotheses, artifacts, permitted use, and failure route. This branch remains valid when no Core-DG step is invoked; it is not the universal v21 route, which is Definition 1.21.

**Policy 20** (Return-Theorem and finite-to-infinite discipline). *No external-domain conclusion is inferred from a detector or Core pass alone. When the external target is infinite, asymptotic, analytic, tower-level, or global, the proof record must include:*

- (i) *a proved external-to-AK realization theorem;*
- (ii) *a certified finite detector or a separately proved non-finite Core route;*
- (iii) *a finite-to-infinite reduction theorem whenever finite data are used to establish an infinite statement;*
- (iv) *every invoked higher-edge and Type IV theorem;*
- (v) *a proved AK-to-external Return or soundness theorem.*

*Known-theorem recovery is valid only at the external strength actually returned by the final theorem.*

**Remark 1.43** (Chapter allocation). Chapter 2 retains the repaired metric calculus and zero transport. Chapter 3 supplies finite detector characterization, completeness, certificate-complexity bounds, and no-go results. Chapter 4 supplies persistence-to-finite-diagram reconstruction, higher Ext edges, and zero-reflection boundaries. Chapter 5 supplies towers, terminal/transient Type IV separation, and finite-to-infinite reduction. Chapter 6 supplies detector-aware record-level atlases, overlap, and Restart. Chapter 7 supplies Core<sub>DG</sub>, finite generator/cone reconstruction, and object-level finite globalization. Chapter 8 supplies canonical verifier, proof-DAG, manifest, and governance semantics. Chapter 9 supplies the unified HT–UB Bridge and Return calculus. Chapter 10 supplies downstream regression, validated Returns, counterexamples, and successor problems and is not an input to Chapters 1–9.

## 1.10 Inherited Core-P metric repair and the v21 Unified Collapse Contract

**Theorem 1.44** (Threshold-gap-safe stability under the standard bottleneck metric). *Let  $B$  and  $C$  be finite barcodes, possibly including infinite bars, under the standard bottleneck metric and the inherited v19 infinite-bar convention. Let  $\tau > 0$ ,  $\varepsilon \geq 0$ , and assume*

$$d_B(B, C) \leq \varepsilon.$$

*Assume that no finite bar of either barcode has length in*

$$[\tau - \varepsilon, \tau + \varepsilon].$$

*Then*

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

*This inherited v19 metric-repair theorem remains the mandatory transport component for every v21 metric inference through hard deletion.*

**Remark 1.45** (Proof location for the metric-repair theorem). The proof is supplied in Chapter 2, Theorem 2.46, and independently in Appendix A, Theorem 1.33 (the metric-repair theorem). Chapter 1 fixes the exact statement later chapters may consume.

**Definition 1.46** (Core-P Unified Collapse Contract: inherited and conservatively re-issued data). A v21 UCC mathematical contract contains the inherited repaired metric clauses and the following conservative extensions.

(i) **Hard deletion.**

$$T_{\tau}^{\text{bar}}(B) = \{I \in B \mid \ell(I) > \tau \text{ or } \ell(I) = +\infty\}.$$

(ii) **Barcode algebra.** The operation is endpoint-independent and idempotent, and

$$T_{\sigma}^{\text{bar}} T_{\tau}^{\text{bar}}(B) = T_{\max\{\sigma, \tau\}}^{\text{bar}}(B).$$

(iii) **Infinite-bar invariance.** If  $I = [a, \infty)$ , then

$$T_{\tau}^{\text{bar}}(I) = I.$$

(iv) **Rejected global non-expansiveness.** No inequality

$$d_B(T_{\tau}^{\text{bar}} B, T_{\tau}^{\text{bar}} C) \leq d_B(B, C)$$

is asserted for arbitrary barcodes.

(v) **Windowed repaired evaluation.**

$$\text{Eval}_{i, W, \tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

(vi) **Metric-use discipline.** Threshold-gap evidence is mandatory whenever a metric discrepancy is transported through hard deletion. Exact comparison uses a separately proved equality or isomorphism and is not automatically a positive-radius robustness statement.

(vii) **No unconditional crop–collapse commutation.** Such commutation may be used only under the exact criterion of Appendix A or a separately declared exact-admissible theorem.

(viii) **Representative compatibility.** If  $C_{\tau, W}F$  is used, then in every monitored degree

$$\mathbf{P}_i(C_{\tau, W}F) \cong \text{Eval}_{i, W, \tau}(F),$$

at the explicitly declared equivalence strength.

(ix) **Non-scalar obligations.** Representative existence, actual morphisms, eligibility, terminal tameness, higher coherence, claim status, and artifact identity remain independent proof obligations.

(x) **Exact-admissible strengthening.** Strict functoriality, exactness, or canonical tower comparison may be used only under a separately proved and registered strengthening.

(xi) **Finite-detector strengthening.** A finite detector may discharge a repaired-profile zero clause only through Definition 1.35, Policy 19, and the applicable Chapter 3 characterization or completeness theorem.

(xii) **Higher-obstruction strengthening.** An invoked degree- $n$  categorical clause uses Definition 1.38. The one-way discharge is the default strength. A vanishing equivalence requires Definition 1.39.

(xiii) **External-theorem separation.** A Core pass has no external-domain force without the complete Problem-Bridge chain of Definition 1.42 and Policy 20.

**Policy 21** (Core-P Unified Collapse Contract: audit policy). *A certified v21 run declares at least:*

- (i) *the barcode metric and infinite-bar convention;*
- (ii) *the windows, thresholds, monitored degrees, and deletion convention;*
- (iii) *the collapse policy*

`barcode_level, tau_split, exact_admissible;`

- (iv) *the evaluation order;*
- (v) *exactly one comparison mode for every consumed comparison:*

`exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared;`

- (vi) *in mode `exact`, the actual equality or persistence-profile isomorphism, its source and target, scope, equivalence strength, and immutable proof artifact;*
- (vii) *in mode `metric_gap_certified`, every required crop, window-edge, discrepancy, scalarization, and two-sided threshold-gap record;*
- (viii) *in mode `theorem_controlled_nonmetric`, the exact theorem identity, source and target types, transported predicate or relation, hypotheses, direction, naturality scope, artifacts, and prohibited stronger readings;*
- (ix) *mode `not_compared` only when the governing claim consumes no comparison inference; it cannot discharge an invoked overlap, transport, coherence, representative-comparison, or tower-comparison obligation;*
- (x) *representative and eligibility policies when invoked;*
- (xi) *the non-scalar obligation registry;*
- (xii) *for each metric-mode comparison, the metric quantale, bottleneck scalarization, finite protected profile family  $\mathfrak{B}$ , actual discrepancy upper-bound evidence, and strict safety inequality*

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B});$$

- (xiii) *the Type IV scope and, when invoked, the comparison artifact, terminal-tameness evidence, fixed  $(W, \tau, I_{\text{mon}})$ , and defect record;*
- (xiv) *Claim, CM, and TB references together with evidence status and invocation role;*
- (xv) *repair ancestry and immutable artifact references;*
- (xvi) *for every invoked detector, the detector ID, profile stage, mode, endpoint convention, sample set, completeness source, two-axis status, certificate size, and artifact identity;*
- (xvii) *for every invoked higher Ext clause, n-eligibility, the realized object, amplitude policy, edge extractor, edge-realization artifact, naturality scope, and zero-reflection status;*
- (xviii) *for every invoked transient-defect clause, the full kernel/cokernel profile artifacts and the complete defect-detector record;*

(xix) *for every external conclusion, realization, finite-to-infinite, higher-obstruction, and Return-Theorem records with their exact external source and target statements.*

*No metric ledger, positive threshold reserve, or gap certificate is fabricated for a purely exact comparison. Conversely, an exact comparison does not by itself certify positive-radius robustness. Non-scalar obligations cannot be discharged by numerical slack.*

*Proof and status.* Theorem 1.44 is the inherited metric theorem in the opening contract block; no detector rank-stability theorem is asserted. Definition 1.46 records mathematical data and admissibility clauses. Policy 21 records operational audit discipline.

### 1.11 UCC records, threshold-gap records, detectors, and repaired evaluation

**Definition 1.47** (v21 Core-P UCC-Contract). A v21 UCC-Contract consists of:

- (1) a metric-ledger quantale and, when required, a bottleneck scalarization;
- (2) a barcode metric and infinite-bar convention;
- (3) right-open windows and separate partition or overlap data;
- (4) a threshold policy or threshold sweep;
- (5) the hard deletion and evaluation rules;
- (6) a comparison-mode policy;
- (7) threshold-gap, crop, and window-edge policies;
- (8) representative and eligibility policies when invoked;
- (9) metric and non-scalar obligation policies;
- (10) gate and tower-comparison policies;
- (11) Claim, CM, TB, repair, and artifact-identity policies;
- (12) detector applicability, completeness, endpoint, coverage, and two-axis status policies;
- (13) higher-degree eligibility, edge, naturality, and zero-reflection policies;
- (14) finite-to-infinite reduction and Return-Theorem policies when an external conclusion is invoked.

**Definition 1.48** (Right-open windows, partition covers, and overlap covers). A window is a right-open interval

$$W = [u, u'), \quad u' \in \mathbb{R} \cup \{\infty\}.$$

A right-open partition cover is pairwise disjoint and ordered when sequential continuation is invoked. An overlap-admitting cover records every nonempty finite intersection used by the certificate logic and carries the required nerve, simplex, and face-compatibility data. The two cover structures are not silently identified.

**Definition 1.49** (Window restriction). Define

$$W_W : \text{Pers}_k^{\text{cons}} \longrightarrow \text{Pers}_k^{\text{cons}}$$

by restriction to  $W$ , with extension by zero outside  $W$  when a globally indexed object is required. At barcode level, interval summands are clipped to  $W$  and empty intersections are discarded. If  $u' < \infty$ , the infinite bar  $[a, \infty)$  becomes  $[\max\{a, u\}, u')$  whenever the intersection is nonempty.

**Definition 1.50** (Windowed pre-threshold profile). For  $F \in \text{FiltCh}(k)$ , define

$$B_{i,W}(F) := W_W \mathbf{P}_i(F).$$

All threshold clearance used for certification is measured on this profile before hard deletion.

**Definition 1.51** (Windowed repaired profile). Define

$$\mathbf{P}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

This is a repaired evaluation record. It is metric-gap-certified only after the relevant two-sided record is supplied.

**Definition 1.52** (Canonical comparison mode). Every comparison consumed by a v21 claim records exactly one mode:

$$\begin{array}{ll} \text{exact}, & \text{metric\_gap\_certified}, \\ \text{theorem\_controlled\_nonmetric}, & \text{not\_compared}. \end{array}$$

The meanings are:

- (i) **exact**: an actual equality or persistence-profile isomorphism between the declared source and target is proved and artifact-backed;
- (ii) **metric\_gap\_certified**: a real pre-threshold bottleneck upper bound and a two-sided threshold-gap certificate are supplied at the same protected budget, together with zero separation whenever the consumed conclusion transports exact vanishing;
- (iii) **theorem\_controlled\_nonmetric**: a named theorem transports the declared predicate or relation without asserting an exact isomorphism or a bottleneck estimate; its source, target, direction, hypotheses, naturality scope, artifacts, and prohibited stronger readings are explicit;
- (iv) **not\_compared**: the governing claim consumes no comparison conclusion between the declared objects.

No mode implies another merely by its status. An exact comparison is not automatically a positive-radius robustness certificate; a theorem-controlled nonmetric comparison is not automatically exact or metric; and **not\_compared** cannot discharge an invoked comparison obligation.

**Definition 1.53** (Exact comparison record). An exact comparison record contains:

- (i) immutable identifiers for the source and target profiles;
- (ii) the monitored degree, window, threshold, crop/restriction convention, and declared scope;
- (iii) an actual equality or persistence-profile isomorphism at the declared equivalence strength;
- (iv) the proof object and immutable artifact identifiers needed to reconstruct the exact comparison;



(v) status pass, reject, or undefined under the Chapter 8 atomic status calculus.

No scalarized metric budget or positive threshold reserve is inferred merely from this record.

**Definition 1.54** (Threshold clearance). For a barcode  $B$ ,

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|.$$

If  $B_{\text{fin}} = \emptyset$ , set  $\text{clear}_\tau(B) = +\infty$ . If a finite bar has length exactly  $\tau$ , then  $\text{clear}_\tau(B) = 0$ .

**Definition 1.55** (Threshold safety margin). For a finite family  $B_1, \dots, B_m$ , define

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_j \text{clear}_\tau(B_j).$$

For one barcode,  $\text{gap}_\tau(B) = \text{clear}_\tau(B)$ . The notation is a ledger-facing alias and introduces no second invariant.

**Definition 1.56** (Window-edge clearance and crop budget). A window-edge clearance record, denoted  $\text{clear}_{\partial W}(B)$ , measures the separation of relevant endpoints from active window boundaries. A crop budget  $\delta_{\text{crop}}(W)$  bounds discrepancy introduced by restriction, clipping, or moving boundaries before hard deletion. A source-level perturbation bound cannot be used as a windowed pre-threshold bound without a theorem or artifact controlling this crop step.

**Definition 1.57** (Metric threshold-gap record). A threshold-gap record is used only in metric-gap-certified mode. Its canonical mode label is

`metric_gap_certified.`

The record contains at least:

- (i) immutable identifiers for the compared pre-threshold profiles;
- (ii) the two profiles, window, threshold, and the fixed mode

`metric_gap_certified;`

- (iii) a declared  $\varepsilon$  and evidence for  $d_B(B, C) \leq \varepsilon$ ;
- (iv) crop and window-edge data when required;
- (v)  $\text{clear}_\tau(B)$ ,  $\text{clear}_\tau(C)$ , and  $\text{gap}_\tau(B, C)$ ;
- (vi) confirmation that neither finite-bar family meets  $[\tau - \varepsilon, \tau + \varepsilon]$ ;
- (vii) the scalarization and its discrepancy upper-bound evidence when the budget is ledger-derived;
- (viii) reconstructible artifact references;
- (ix) one specialized status:

`gap_certified, gap_failed, gap_pending.`

A one-sided record is not a v21 threshold-gap certificate; v21 inherits the same two-sided requirement. This record is not required for a purely exact comparison.

**Remark 1.58** (Specialized status mapping). The specialized gap statuses map to the canonical atomic statuses by

$$\begin{aligned}\text{gap\_certified} &\mapsto \text{pass}, \\ \text{gap\_failed} &\mapsto \text{reject}, \\ \text{gap\_pending} &\mapsto \text{undefined}.\end{aligned}$$

They do not create an independent audit-status algebra.

**Remark 1.59** (Exact comparison versus robustness certification). An exact equality or proved profile isomorphism may establish equality of specific repaired outputs without a positive metric budget. It does not prove robustness under positive perturbation. A profile with a finite bar of length exactly  $\tau$  may be exactly compared with itself, but it is not threshold-gap certified at  $\tau$ .

**Remark 1.60** (Bottleneck and interleaving metrics in the constructible one-parameter regime). In the constructible one-parameter persistence regime over a field, the isometry theorem identifies

$$d_{\text{int}}(P, Q) = d_{\text{B}}(\mathcal{B}(P), \mathcal{B}(Q)).$$

The identification is read under the inherited v19 infinite-bar convention. Interleaving-distance records may be used only when their barcode profiles are precisely those entering the threshold-gap record.

**Definition 1.61** (Typed pipeline ledger and scalarized budget). For one declared metric-gap-certified comparison, let  $\delta_j \in \mathbb{V}$  be its metric contributions. Define

$$\mathfrak{d}_{\text{met}} := \bigoplus_j \delta_j.$$

A certified scalarization satisfies

$$\text{val}_{\text{B}}(0_{\mathbb{V}}) = 0, \quad \text{val}_{\text{B}}(q \oplus q') \leq \text{val}_{\text{B}}(q) + \text{val}_{\text{B}}(q').$$

Set

$$\varepsilon_{\text{B}}(\mathfrak{d}_{\text{met}}) := \text{val}_{\text{B}}(\mathfrak{d}_{\text{met}}).$$

The record must also establish

$$d_{\text{B}}(B, C) \leq \varepsilon_{\text{B}}(\mathfrak{d}_{\text{met}})$$

for each comparison to which the budget is applied. Non-scalar obligations do not belong to  $\mathfrak{d}_{\text{met}}$ .

**Remark 1.62** (Sharpness of the  $\varepsilon$ -band). The sharp closed forbidden band for the two-sided theorem is

$$[\tau - \varepsilon, \tau + \varepsilon].$$

A half-width  $2\varepsilon$  policy may be used only as an explicitly non-sharp conservative policy and may not be described as the sharp Core constant.

**Remark 1.63** (Core sovereignty over Search). The Core accepts or rejects through declared mathematical conditions, typed obligations, diagnostics, budget checks, and artifact-supported records. No Search-side quantity upgrades a reject or undefined result.

## 1.12 Overlap Gate and B-Gate<sup>+</sup>

**Definition 1.64** (Overlap Gate). Let

$$W_{\alpha\beta} := W_\alpha \cap W_\beta.$$

For every monitored degree, both local objects are evaluated directly on  $W_{\alpha\beta}$  at one declared common threshold, with an explicit threshold-reconciliation record when their local thresholds differ. Exactly one comparison mode is then recorded.

An invoked pairwise Overlap Gate passes only under one of the following two branches.

- (i) **Exact branch.** Mode `exact` is declared and an exact overlap record supplies the directly evaluated common-window profiles and an actual equality or persistence-profile isomorphism

$$\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha) \xrightarrow{\sim} \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta),$$

together with immutable source, target, proof, crop/restriction, and threshold-reconciliation artifacts.

- (ii) **Metric branch.** Mode `metric_gap_certified` is declared; the directly evaluated pre-threshold profiles

$$B_{i, W_{\alpha\beta}}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad B_{i, W_{\alpha\beta}}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta)$$

satisfy an artifact-supported estimate

$$d_B(B_{i, W_{\alpha\beta}}^\alpha, B_{i, W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta, i};$$

a two-sided metric threshold-gap record protects both profiles; crop, window-edge, comparison, and threshold-reconciliation contributions are declared; and

$$\varepsilon_{\alpha\beta, i} < \text{gap}_\tau(B_{i, W_{\alpha\beta}}^\alpha, B_{i, W_{\alpha\beta}}^\beta).$$

Theorem 1.44 then yields

$$d_{\text{int}}(\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha), \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta)) \leq \varepsilon_{\alpha\beta, i}$$

in the declared constructible one-parameter regime.

Mode `not_compared` records that no overlap inference is consumed; it does not constitute a passing Overlap Gate and cannot discharge pairwise or higher Čech coherence. In either passing branch, every invoked representative, realization, and Type IV clause has compatible provenance; Type IV is either explicitly `not_invoked` by scope or is `certified_zero`; and every required artifact is present. If a detector-based zero statement is consumed on the overlap, the detector is applied to the directly evaluated common-window source profile and has a complete overlap-scope certificate. A zero detector on either unreduced local window does not substitute for this common-window record.

A passing pairwise gate does not by itself produce higher-simplex coherence, a global certificate atlas, a global filtered object, a finite-to-infinite reduction, or an external theorem.

**Remark 1.65** (Exact and metric layers of the Overlap Gate). The metric branch has two barcode levels: it consumes a discrepancy bound on the directly windowed pre-threshold profiles and transports that bound to the repaired outputs. A post-threshold distance estimate alone does not certify pre-threshold classification stability.

The exact branch instead proves the declared equality or profile isomorphism directly. It may bypass metric transport for that exact claim, but it supplies no positive-radius threshold robustness unless a separate metric certificate is also proved.

**Definition 1.66** (B-Gate<sup>+</sup> (inherited Core-P finite-certificate gate)). Fix a finite monitored degree set  $I_{\text{mon}} \subset \mathbb{Z}_{\geq 0}$  containing degree 1 whenever the baseline degree-one gate is invoked. B-Gate<sup>+</sup> passes on  $(F; W, \tau, I_{\text{mon}})$  if every required clause passes and no required clause is undefined.

- (1) **Topological clauses.** For every invoked  $n \in I_{\text{mon}}$ ,

$$\text{Eval}_{n,W,\tau}(F) = 0.$$

The zero statement is supported either by the full repaired-profile artifact or by a finite detector satisfying Definitions 1.35 and 1.36 together with Policy 19. A sampled zero without certified completeness is undefined. If an admissible compatible representative is supplied, the degree-one statement may also be recorded as

$$\text{PH}_1(C_{\tau,W}F) = 0.$$

- (2) **Comparison-mode clause.** Every consumed comparison declares exactly one mode from Definition 1.52. In mode `exact`, a valid exact comparison record is present. In mode `metric_gap_certified`, a valid two-sided metric threshold-gap record is present. In mode `theorem_controlled_nonmetric`, the governing theorem record directly proves the consumed predicate or relation at the declared strength and direction. Mode `not_compared` is admissible only when the governing scope consumes no comparison inference and cannot discharge an invoked comparison obligation.
- (3) **Detector clause.** Every detector used by the gate is applicable to the exact source stage and has certificate status `detector_certified`. Its value status is zero for a required zero clause. Birth/germ exhaustiveness, stencil coverage, family membership, excess-bound provenance, endpoint convention, cofinal policy, and artifact identity pass independently. A detector not used by the gate is explicitly `not_invoked` by scope.
- (4) **Categorical clauses.** For every invoked degree  $n$ , an admissible realization-compatible representative exists, the corresponding  $n$ -eligible regime holds, and

$$\text{Ext}^n(A_n(F), k) = 0.$$

The clause may be discharged by the Chapter 4 one-way theorem after exact repaired-profile vanishing has been established. A reverse inference requires Definition 1.39. Missing eligibility, edge realization, or naturality required by the consumed claim is undefined, not pass.

- (5) **Type IV and transient-defect clause.** Type IV is explicitly `not_invoked` or is `certified_zero` with a valid artifact and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

at one fixed  $(W, \tau, I_{\text{mon}})$ . If the governing claim additionally requires absence of transient or interior defects, the full kernel and cokernel profiles are supplied and the applicable complete defect detector is certified zero. Terminal-generic zero alone does not discharge that stronger clause.

- (6) **Metric-budget and zero-transport clause.** For every comparison whose mode is `metric_gap_certified`, the protected family  $\mathfrak{B}$  is finite, the scalarized value is proved to upper-bound the actual discrepancy, and

$$\text{val}_{\mathfrak{B}}(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B}).$$

If exact zero is transported across a positive bottleneck radius, the post-threshold separation condition, normally  $2\varepsilon < \tau$ , is separately discharged. No numerical budget or positive reserve is fabricated for a purely exact comparison.

- (7) **Non-scalar obligations.** Every required representative, morphism, eligibility, edge, naturality, detector applicability, endpoint, coverage, family, terminal-tameness, finite-to-infinite, coherence, claim-status, invocation-role, and artifact-identity obligation passes.
- (8) **Manifest and claim-use clause.** The record is complete and consistent under Chapter 8; every consumed claim has an admissible evidence-status–invocation-role combination; and no rejected, deprecated, unknown, unproved, operational, synthetic, toy, or spec item is promoted to theorem evidence.
- (9) **External conclusion clause.** This clause is outside the internal Core gate unless an external conclusion is actually invoked. In that case, the complete realization, finite-to-infinite, and Return chain of Policy 20 passes. An internal B-Gate<sup>+</sup> pass alone never supplies this clause.

**Remark 1.67** (Typed conjunction and no cancellation). B-Gate<sup>+</sup> is a typed conjunction. A passing clause cannot cancel a reject or required undefined clause, and numerical reserve cannot repair a missing non-scalar artifact.

**Remark 1.68** (Interpretation of B-Gate<sup>+</sup>). B-Gate<sup>+</sup> is a Core-P acceptance predicate for a declared object, window, threshold, monitored scope, and proof-object record. It is not a Search score, probabilistic judgment, external-domain oracle, or transport theorem.

**Remark 1.69** (No representative, no higher Ext-clause). A persistence zero clause may be evaluated from the full repaired profile or a certified complete detector. A degree- $n$  Ext clause may not be invoked unless a realization-compatible admissible representative, realized object, amplitude policy, edge extractor, edge-realization artifact, and every applicable eligibility condition are supplied.

### 1.13 Failure landscape and the Type IV canon

**Definition 1.70** (Failure landscape). The five failure types are preserved, with v21 detector and higher-degree specialization:

- **Type I: certified topological or persistence obstruction.** For at least one invoked monitored degree  $n$ ,

$$\text{Eval}_{n,W,\tau}(F) \neq 0.$$

A certified complete detector with value nonzero supplies a Type I witness for the corresponding required zero clause. An incomplete or inapplicable detector does not supply Type I; it leaves the clause undefined.

- **Type II: certified categorical obstruction in an eligible regime.** For at least one invoked  $n$ -eligible realized object,

$$\text{Ext}^n(A_n(F), k) \neq 0.$$

Missing eligibility, edge realization, or required naturality makes the clause undefined rather than categorically obstructed.

- **Type III: detector, operational, metric, comparison, or functorial failure.** This includes a well-typed failed detector applicability or completeness condition, failed birth/germ exhaustiveness, failed stencil coverage, invalid excess-bound source, failed exact or metric comparison, failed threshold-gap certification, invalid scalarization, insufficient discrepancy evidence, failed crop control, failed post-threshold zero-separation, missing representative or comparison morphism data, misuse of `not_compared`, failed threshold reconciliation, failed Restart transition control, or failed finite-to-infinite reduction. Missing evidence is normally undefined; an evidenced false required condition is reject/Type III.

- **Type IV: tower-level terminal-generic obstruction.** A valid comparison artifact exists and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0).$$

A transient interior defect detected with zero terminal diagnostics is recorded separately and does not become terminal-generic Type IV by renaming. Missing artifact data are undefined, not obstructed.

- **Type V: claim-status, manifest, provenance, or promotion failure.** This includes use of a rejected claim, promotion of an operational, synthetic, toy, Search, or spec item to theorem evidence, concealment of missing detector or Bridge evidence, identifying undefined with zero or pass, fabricating not\_invoked or not\_compared, treating a finite numerical observation as an infinite theorem, artifact-identity or hash mismatch, incompatible versions, or inconsistent repair ancestry. A Type III undefined clause may coexist with Type V when its absence is misreported or governance data are inconsistent.

**Definition 1.71** (Type IV scope status at the Chapter 1 contract level). Every run records exactly one Type IV status:

not\_invoked, undefined, certified\_zero, obstructed.

They mean, respectively: no terminal-generic kernel/cokernel comparison diagnostic is part of the claim; such a diagnostic is invoked but its required data are missing or ill-typed; a valid diagnostic exists and is zero; or a valid diagnostic exists and is nonzero. Direct or inverse tower semantics may remain active through a finite-to-infinite, Roos, stable-image, or apex-agreement lane while Type IV itself is not\_invoked.

**Definition 1.72** (Tower comparison artifact). A tower comparison artifact is actual declared morphism data

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

The source, target, morphism, kernel and cokernel profiles, and terminal-tameness evidence must be typed before Type IV diagnostics are computed. A bottleneck matching or expected limit correspondence is not automatically such a morphism.

**Definition 1.73** (Terminal-generic Type IV diagnostics at contract level). When the artifact is defined and terminal tame, set

$$\mu_{i,W,\tau} := \text{tgdim}_W(\ker \phi_{i,W,\tau}),$$

$$\nu_{i,W,\tau} := \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

Fix a declared window–threshold pair  $(W, \tau)$  and a declared finite set of monitored degrees  $I_{\text{mon}}$ . Define

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau},$$

$$\nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

An aggregate over infinitely many monitored degrees, or across multiple window–threshold pairs, requires a separately declared aggregation and summability theorem. Chapter 5 supplies the precise terminal-generic definitions.

**Remark 1.74** (Type IV canonical convention). The equality

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

means that no monitored terminal-generic kernel or cokernel defect is present. It does not imply full isomorphism, absence of finite interior defects, or object-level equivalence. Deprecated Latin-*u* Type IV cokernel notation has no normative use. The symbols  $\nu_{i,W,\tau}$  and  $\nu_{\text{Collapse}}$  are the sole canonical cokernel diagnostics in v21.0.

**Remark 1.75** (Normative  $\nu$ -migration rule). In theorem statements, hypotheses, gate clauses, diagnostic records, and manifest semantics, the cokernel diagnostic is written with  $\nu$ . The Latin symbol  $u$  may occur only in explicit historical deprecation or migration prose and carries no v21.0 diagnostic meaning.

**Definition 1.76** (Transient-defect detector contract). Assume a valid tower comparison artifact  $\phi_{i,W,\tau}$  and full constructible defect profiles

$$K_{i,W,\tau} := \ker \phi_{i,W,\tau}, \quad Q_{i,W,\tau} := \text{coker } \phi_{i,W,\tau}$$

are supplied. A transient-defect detector is a finite detector applied at source stage `kernel_defect` or `cokernel_defect` to certify whether

$$T_{\tau_{\text{def}}}^{\text{bar}}(K_{i,W,\tau}) \quad \text{or} \quad T_{\tau_{\text{def}}}^{\text{bar}}(Q_{i,W,\tau})$$

is zero, for a separately declared defect threshold  $\tau_{\text{def}} > 0$ .

The terminal-generic pair  $(\mu, \nu)$  and the transient-defect verdict answer different questions:

- (i)  $(\mu, \nu)$  detects right-cofinal terminal defects;
- (ii) the transient detector detects retained finite noncofinal defects above  $\tau_{\text{def}}$ ;
- (iii) zero results in both systems still imply full profile isomorphism only when every remaining shorter defect is excluded by the governing threshold or a separate no-residual-defect theorem.

**Remark 1.77** (Terminal zero and transient zero). The record distinguishes at least the following cases:

- (i) if  $(\mu, \nu) \neq (0, 0)$ , a terminal or cofinal defect is present, independently of the transient-detector value;
- (ii) if  $(\mu, \nu) = (0, 0)$  and the certified transient detector is `nonzero`, no monitored terminal defect is present but a retained interior defect is present;
- (iii) if  $(\mu, \nu) = (0, 0)$  and the certified transient detector is `zero`, no monitored terminal defect and no interior defect longer than  $\tau_{\text{def}}$  is present.

The final case is not a full-isomorphism theorem unless the shorter-defect regime is separately excluded.

**Definition 1.78** (Monitored Windowed Collapse Zone). Fix a declared finite degree set  $I_{\text{mon}}$ . Define

$$\mathfrak{C}_{I_{\text{mon}}, W, \tau} := \left\{ F \mid \begin{array}{l} \text{Eval}_{n, W, \tau}(F) = 0 \text{ for every invoked } n \in I_{\text{mon}}, \\ \text{Ext}^n(A_n(F), k) = 0 \text{ for every invoked degree-}n \text{ Ext clause} \end{array} \right\}.$$

The degree-one notation  $\mathfrak{C}_{W, \tau}$  abbreviates  $\mathfrak{C}_{\{1\}, W, \tau}$ . Every Ext condition is meaningful only in its eligible realization regime. No converse from Ext-vanishing to repaired-profile vanishing is asserted without a registered natural zero-reflection theorem.

**Remark 1.79** (Collapse Zone is not a complete gate). Membership in  $\mathfrak{C}_{I_{\text{mon}}, W, \tau}$  does not by itself imply detector completeness, threshold-gap certification, metric-ledger pass, Type IV or transient-defect cleanliness, manifest completeness, B-Gate<sup>+</sup> pass, finite-to-infinite validity, or an external-domain conclusion.

### 1.14 Minimal Manifest Axiom and proof-object discipline

**Axiom 1.1** (Minimal Manifest Axiom). *A proof-object candidate submitted for Core certification contains at least:*

- (i) *immutable RunID, theory version, schema version, Claim Register version, claim scope, repair ancestry, and artifact hashes;*
- (ii) *source and realized objects, windows, monitored degrees, finite  $I_{\text{mon}}$ , and overlap structure;*
- (iii) *metric, infinite-bar, endpoint, threshold, deletion, collapse, and evaluation policies;*
- (iv) *exactly one comparison mode for every consumed comparison;*
- (v) *in mode `exact`, the actual equality or profile-isomorphism proof, equivalence strength, and crop/restriction artifacts;*
- (vi) *in mode `metric_gap_certified`, every required threshold-gap, crop, window-edge, quantale, scalarization, actual discrepancy upper-bound, protected-family, strict-reserve, and post-threshold zero-separation record;*
- (vii) *in mode `theorem_controlled_nonmetric`, the theorem identity, exact source and target, transported predicate or relation, complete hypotheses, direction, naturality scope, proof artifacts, and prohibited stronger readings;*
- (viii) *for mode `not_compared`, the explicit scope exclusion proving that no comparison conclusion is consumed;*
- (ix) *for every invoked detector: detector ID, source profile ID, profile stage, detector mode,  $(n, W, \tau)$ , endpoint convention, sample or critical set, cofinal policy, soundness theorem, completeness source, family-membership evidence, excess-bound provenance when used, metric mode, certificate size, two-axis detector status, and immutable artifacts;*
- (x) *for every family-relative compression claim: information class  $\Theta$ , admissible detector class, upper and lower bounds, boundary convention, no-go applicability, and MinCert evidence;*
- (xi) *representative existence, compatibility, equivalence strength, actual morphism data, detector-preservation data when invoked, and realization compatibility;*
- (xii) *for every invoked degree- $n$  Ext clause: the realized object  $A_n(F)$ , amplitude policy, edge extractor, edge-realization isomorphism, zero compatibility, eligibility data, and the status of any natural zero-reflection upgrade;*
- (xiii) *a non-scalar obligation registry with invocation state, dependency set, atomic status, evidence, and failure route;*
- (xiv) *Type IV scope and, when invoked, comparison, source semantics, provenance authority, terminal-tameness, fixed  $(W, \tau, I_{\text{mon}})$ , degree-wise and total  $\mu, \nu$  diagnostics, and interior-defect status;*
- (xv) *when transient defects are invoked: the full kernel/cokernel profiles, defect threshold, complete detector records, and the exact relationship to terminal-generic diagnostics;*
- (xvi) *when an infinite, global, asymptotic, or external conclusion is invoked: realization, finite-to-infinite reduction, higher-obstruction, Return-Theorem, source-domain, target-domain, and external claim records;*



(xvii) *atomic clause statuses*

pass, reject, undefined, not\_invoked;

(xviii) *manifest state*

complete, incomplete, inconsistent;

(xix) *an aggregate typed-conjunction verdict and complete failure-routing record;*

(xx) *every Claim, CM, TB, synthetic, toy, candidate-only, and external-Bridge reference with its document role, evidence status, invocation role, exact source location, permitted strength, and prohibited stronger reading;*

(xxi) *a complete finite or countable clause registry, every justified scope exclusion, and immutable transitive artifact references sufficient for independent reconstruction and replay.*

**Remark 1.80** (Why the manifest axiom belongs to the Core). Concrete storage may live in a Companion or implementation layer, but the requirement that certification be supported by typed, reconstructible, artifact-backed records belongs to the Core audit discipline.

**Remark 1.81** (Canonical v21 manifest integration). Axiom 1.1 gives the foundational minimum. Chapter 3 extends repaired-evaluation and representative records. Chapter 8 supplies the canonical manifest, immutable identity, typed dependency closure, verdict reconstruction, replayability, and repair-as-new-record semantics.

**Remark 1.82** (Appendix F and Appendix G boundary). Appendix F elaborates obligation types, invocation semantics, atomic status algebra, dependency closure, non-compensation, failure routing, and certificate readiness. Appendix G elaborates the manifest field schema, immutable identifiers, reference slots, artifact links, and serialization-neutral proof-object structure. Neither appendix establishes a mathematical claim merely by recording it.

**Remark 1.83** (Claim, micro-theorem, and repair traceability). Every claim, micro-theorem, toy bridge, and repaired rule must be identified by its exact registered item and strength. This prevents silent reuse of rejected claims and silent promotion of toy or Search material.

**Definition 1.84** (Manifest-complete proof-object candidate). A manifest-complete proof-object candidate satisfies the required field, typing, identity, and consistency conditions of Axiom 1.1 and Chapter 8. Manifest completeness means that the required record is present and internally consistent; it does not establish the truth of the recorded mathematical claims.

**Definition 1.85** (Certified proof object). A certified proof object is a manifest-complete candidate such that every required mathematical claim has valid proof support, every computational claim has auditable artifact support, every required non-scalar obligation passes, every required gate clause passes, no required clause is reject or undefined, every registered claim is used within its permitted strength, and the aggregate verdict is pass.

**Remark 1.86** (Manifest completeness is not proof). A complete manifest may faithfully record a failed proof, an obstructed tower, or a rejected claim. Completeness supports auditability; it does not replace mathematical validity.

### 1.15 Forward contract for countable certificate atlases

**Policy 22** (Countable atlas forward contract). *Chapter 1 does not assert that countability, definability, finite Čech depth, and ordinary summability alone produce a global certificate. A countable local-to-global claim is admissible only through the typed record-level theorem of Chapter 6, in particular Theorem 6.87, or through a separately proved stronger theorem.*

*A countable atlas record declares at least:*

- (i) *an audit-locally finite right-open overlap cover;*
- (ii) *a finite Čech-depth bound;*
- (iii) *complete local records;*
- (iv) *when local detector certificates are used, each local record has a certified profile stage, endpoint convention, completeness theorem, sample set, and coverage artifact;*
- (v) *overlap-compatible birth/germ conventions and direct common-window evaluation of every detector source consumed on overlaps;*
- (vi) *every nonempty nerve simplex required by the declared cover and claim;*
- (vii) *face compatibility on every consumed restriction;*
- (viii) *a common-threshold or threshold-reconciliation policy;*
- (ix) *a local-to-global detector-coverage theorem proving that every globally retained bar is detected by at least one certified local or overlap stencil; local zero verdicts alone do not imply global zero;*
- (x) *every consumed face or simplex comparison uses one of*

`exact,      metric_gap_certified.`

*The modes `theorem_controlled_nonmetric` and `not_compared` cannot discharge coherence unless a separate theorem first constructs one of the two admitted simplex-comparison records;*

- (xi) *exact proof artifacts for exact-mode simplices;*
- (xii) *scalarization, actual discrepancy upper-bound evidence, and positive sharp threshold reserve for metric-mode simplices only;*
- (xiii) *summable scalarized exposure over exactly the metric-mode simplices when a countable exposure policy is invoked;*
- (xiv) *correctly typed Type IV scope;*
- (xv) *all required non-scalar obligations;*
- (xvi) *a complete countable clause registry and audit-readable artifact closure.*

*The conclusion is a Čech-coherent certificate atlas at record level. When detector certificates are invoked, the conclusion includes only the global zero predicate expressly proved by the local-to-global coverage theorem. No global filtered object, object-level descent, finite-to-infinite theorem, or external theorem follows without an additional theorem.*

**Remark 1.87** (Why the former countable-cover theorem is not retained). The earlier statement that a definable countable cover alone implies bounded global overlap error omitted typed scalarization, audit-local finiteness, higher-simplex coherence, face compatibility, Type IV scope discipline, and non-scalar obligations. The inherited repair delegates the valid mathematical theorem to Chapter 6.

## 1.16 Explicit non-claims of Chapter 1

- No finite set of scalar ranks is a complete finite-diagram reconstruction certificate by default.
- No projective-generator statement is invoked without the declared finite abelian or derived category and proof that the chosen object is a generator at the required strength.
- No equality

$$\mathbf{R}\mathrm{Hom}(G, E) \simeq 0 \implies E \simeq 0$$

is used outside a regime in which  $G$  is proved to generate the target category.

- No morphism is declared an isomorphism merely because source and target probes have equal dimensions. The cone or another complete morphism-detection theorem is required.
- No cone-zero statement is asserted from an object-level barcode comparison or rank table that does not construct the actual morphism and cone.
- No finite Morita, Yoneda, or generator probe is treated as an equivalence beyond its proved object, morphism, exact, Ext, or derived strength.
- No finite-poset diagram is identified with a sheaf or cosheaf until the Alexandrov topology, specialization order, variance, and restriction direction are fixed.
- No general constructible-sheaf, exit-path, microlocal, stack, hypersheaf, or  $(\infty, 1)$ -categorical theorem is part of the stable v21 Core-DG by terminology alone.
- No pairwise overlap isomorphisms imply a cocycle condition.
- No cocycle condition implies effective descent in an undeclared target category.
- No local object zero statement implies global object zero without a complete cover, compatible local-to-global theorem, and effectivity at the declared strength.
- No object-level descent is inferred from certificate-record gluing.
- No perfectness-locality theorem is identified with effective descent of an arbitrary derived object.
- No global certificate-complexity inequality is claimed independently of the information, query, encoding, and verifier models, nor without accounting for shared evidence and duplicate probes.
- No P2DG or DG2P route is called faithful, conservative, exact, zero-reflecting, or lossless without a theorem proving that property on the declared class.
- No HT or UB theorem becomes Core-DG merely because its notation or proof idea is reused.
- No domain-specific validated case is promoted to a universal Core-DG theorem by abstraction without a new proof.

- No v21 Main integration is represented as a literature-first discovery of standard Morita, Yoneda, cone, descent, inverse-limit, or sheaf-theoretic mathematics.

The following exclusions are part of the meaning of the v21 Core.

- No unconditional global equivalence

$$\text{PH}_n \iff \text{Ext}^n$$

is claimed for arbitrary  $n$ , objects, realizations, coefficients, or profile classes.

- No reverse implication from  $\text{Ext}^n$ -vanishing to repaired-profile vanishing is available without a separately proved natural zero-reflection or faithfulness theorem.
- No vanishing equivalence is read as an object-level isomorphism  $\text{PH}_n \cong \text{Ext}^n$ .
- No detector obtained merely by shifting its output into a derived degree is promoted from synthetic or toy duality to a natural higher-obstruction Bridge.
- No adaptive choice of edge extractor, realization, detector, window, threshold, or sample set is made after inspecting the desired verdict unless the governing theorem explicitly permits and audits that adaptivity.
- No finite observation is a finite proof certificate without the applicable soundness, completeness, endpoint, coverage, family, metric, and artifact obligations.
- No detector zero verdict is asserted from missing or incomplete birth, germ, coverage, mesh, family-membership, excess-bound, endpoint, or profile-stage data.
- No sound detector is assumed complete merely because every observed sample is zero.
- No complete detector for one profile stage is used on another stage without a stage-transport theorem.
- No birth-point detector is endpoint-decoration invariant by default. Decorated endpoints require the germ theorem or another proved endpoint-aware formulation.
- No bounded-window cofinal bar is ignored merely because it arose by clipping an unbounded bar.
- No family-relative excess lower bound is certified by fully reading the same target barcode when the theorem claims non-circular proof compression, unless that full read is explicitly included in the certificate cost.
- No single-instance retrospective empty stencil is promoted to a meaningful family-relative MinCert theorem.
- No uniform complete finite detector exists on a right-unbounded scope with unbounded possible births and no additional finite-birth, stabilization, global-bound, cofinal-reduction, or finite-to-infinite theorem.
- No detector rank, pointwise image dimension, sample count, or certificate size is globally bottleneck-stable by default.
- No metric detector verdict follows from a bottleneck bound alone. The inherited two-sided threshold-gap transport, post-threshold zero-separation, detector completeness, and all crop/window-edge data remain required.

- No use of  $2\varepsilon < \tau$  replaces the two-sided threshold-gap hypothesis; the two conditions solve different transport obligations.
- No positive-threshold Serre subcategory, exact reflector, abelian localization, or universal quotient is claimed in general.
- No global non-expansiveness of hard deletion under the standard bottleneck metric is claimed.
- No one-sided threshold-gap record is a v21 certificate.
- No finite bar of length exactly  $\tau$  is gap-safe.
- No exact comparison is automatically a positive robustness certificate.
- No metric ledger, positive gap, reserve, detector score, or compression score is fabricated merely to encode a purely exact comparison.
- No theorem-controlled nonmetric comparison is read as an exact isomorphism, bottleneck estimate, threshold-gap certificate, robustness radius, actual morphism, or inverse implication unless the cited theorem explicitly proves that target datum.
- No record whose comparison mode is `not_compared` discharges an invoked overlap, transport, coherence, representative comparison, detector transport, or tower-comparison obligation.
- No exact overlap certificate is automatically a positive-radius overlap-robustness certificate.
- No unconditional crop-collapse commutation is claimed.
- No crop-collapse criterion may ignore an infinite bar that becomes finite after bounded-window clipping.
- No global strict filtered lift  $C_\tau$  is asserted by default.
- No identity  $C_{\tau,W}F = (C_\tau F)|_W$  is asserted without a proved compatibility theorem.
- No persistence-profile equivalence automatically preserves detector sample maps, birth/germ sets, realization, or higher-edge data.
- No bottleneck matching is automatically a persistence morphism.
- No tower comparison morphism is automatic outside a proved exact-admissible regime.
- No missing Type IV artifact is interpreted as zero.
- No no-tower claim is represented by a fabricated

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

- No zero terminal-generic diagnostic implies absence of finite interior or transient defects.
- No zero terminal-generic diagnostic together with a thresholded transient zero verdict implies full isomorphism unless the shorter residual-defect regime is separately excluded.
- No aggregate Type IV or transient diagnostic is defined over an undeclared or non-summable monitored scope.

- No legacy Type IV  $u$ -notation has normative force outside explicit historical deprecation or migration prose.
- No unspecialized quantale element is compared directly with real-valued clearance.
- No specialized value is used without evidence that it upper-bounds the actual discrepancy consumed by the proof.
- No numerical budget repairs a missing non-scalar obligation.
- No local detector atlas proves global detector completeness without a local-to-global coverage theorem.
- No certificate-record gluing implies object-level descent.
- No finite detector certificate proves an infinite tower, continuum, asymptotic, global PDE, or  $p$ -adic analytic statement without a finite-to-infinite reduction theorem.
- No AK-side vanishing or B-Gate<sup>+</sup> pass is an arithmetic, PDE, geometric, categorical, physical, or cryptographic theorem without a separately proved Return or soundness theorem.
- No external-to-AK realization theorem is inferred from similarity, analogy, dimensional matching, shared notation, or numerical correlation.
- No Return Theorem is inferred from the realization theorem; the two directions are logically distinct.
- No Known-Theorem Recovery result proves more than the exact external theorem and hypotheses returned by its final Bridge.
- No Search, AI, advisory, spectral, ranking, or candidate-stencil artifact is a Core proof object or gate verdict.
- No scalar potential or agent consensus replaces the typed verifier.
- No manifest completeness is identified with mathematical truth.
- No Claim Register entry is proof merely because it is registered.
- No Claim Register entry may promote a statement beyond its underlying proof.
- No operational rule, specification target, synthetic construction, toy theorem, or companion template is used as theorem evidence merely because it is registered or procedurally satisfied.
- No Core micro-theorem is an external-domain theorem.
- No toy bridge is an arithmetic, PDE, geometric, categorical, or cryptographic theorem.
- No Bridge candidate or Bridge Program is a Bridge theorem.
- No successor-release candidate is a v21.0 theorem claim.
- No reference to Appendix CR-v21, Appendix CM-v21, Appendix TB-v21, Appendix F, Appendix G, a Problem Interface appendix, or a Companion enlarges the Core by citation alone.
- No concrete YAML, JSON, database, or proof-store serialization format is fixed by Chapter 1.

## 1.17 Terminology and notation

**Dual-Core notation.** The v21 stable kernels are denoted

$$\text{Core}_P \quad \text{and} \quad \text{Core}_{\text{DG}}.$$

For a finite small category  $P$ ,

$$\text{Diag}_P(k) = \text{Fun}(P, \text{Vect}_k).$$

The symbols

$$\text{P2DG} \quad \text{and} \quad \text{DG2P}$$

denote typed Interface records, not automatic functors or equivalences.

For a declared diagram morphism  $f : E \rightarrow E'$ ,  $\text{Cone}(f)$  denotes its cone in the declared complex or derived category. The notation  $\mathbf{RHom}(G, E)$  has theorem-bearing force only after the source and target categories, generator  $G$ , boundedness, and coefficient hypotheses are fixed.

A Return-strength record is denoted  $\Sigma_{\text{Ret}}$ . A migration record is identified by  $\text{MigrationID}$ .

Fix a base field  $k$ . We write  $\text{Vect}_k$  for finite-dimensional vector spaces and  $\text{FiltCh}(k)$  for finite-type filtered chain complexes. The degreewise persistence functor is  $\mathbf{P}_i$ , window restriction is  $W_W$ , the windowed pre-threshold profile is

$$B_{i,W}(F) := W_W \mathbf{P}_i(F),$$

and repaired evaluation is

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

For a finite stencil  $T \subseteq W$ , the canonical finite-map detector namespace is

$$D_{T,\tau}^{\text{stc}}(B) := \bigoplus_{t \in T} \text{im}(B(t) \rightarrow B(t + \tau)),$$

when every displayed map is defined. Chapter 3 fixes the exact birth, germ, stencil, mesh, and boundary conventions.

The detector profile-stage vocabulary is

$$\begin{aligned} &\text{windowed\_prethreshold}, \quad \text{repaired\_postthreshold}, \\ &\quad \text{kernel\_defect}, \quad \text{cokernel\_defect}, \\ &\quad \text{diagram\_probe}, \quad \text{descent\_coherence}. \end{aligned}$$

The detector certificate-status vocabulary is

$$\begin{aligned} &\text{detector\_certified}, \quad \text{detector\_failed}, \\ &\quad \text{detector\_pending}. \end{aligned}$$

and the detector value vocabulary is

$$\text{zero}, \quad \text{nonzero}, \quad \text{undefined}.$$

The two axes are not identified.

For a family decision problem  $(\mathcal{F}, \Theta)$  and certificate model  $\mathfrak{M}$ ,  $\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta)$  denotes the Pareto frontier of finite family-level scheme costs in  $(q, b, t, a)$ . The legacy notation  $\text{MinCert}(\Theta)$  denotes only the worst-case query projection in the exact nonadaptive stencil model and is  $+\infty$  when no finite-query complete scheme

exists in that model. For a windowed barcode  $B$ ,  $\text{exc}_{\tau, W}(B)$  denotes the window-relative threshold excess in the exact form fixed by Chapter 3.

The standard bottleneck distance is  $d_B$ , and the interleaving distance is  $d_{\text{int}}$ . Threshold clearance and safety margin are

$$\begin{aligned}\text{clear}_{\tau}(B) &= \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|, \\ \text{gap}_{\tau}(B_1, \dots, B_m) &= \min_j \text{clear}_{\tau}(B_j).\end{aligned}$$

Metric ledger values are compared with clearance through  $\text{val}_B$ . The canonical comparison modes are

$$\begin{aligned}\text{exact}, \quad & \text{metric\_gap\_certified}, \\ \text{theorem\_controlled\_nonmetric}, \quad & \text{not\_compared}.\end{aligned}$$

For an invoked degree  $n$ ,  $A_n(F)$  denotes the declared realized derived object and  $E_n$  the declared higher grade extractor. Neither notation supplies eligibility, naturality, or zero reflection by itself.

The Type IV scope vocabulary is

$$\text{not\_invoked}, \quad \text{undefined}, \quad \text{certified\_zero}, \quad \text{obstructed}.$$

The local diagnostics are

$$\mu_{i, W, \tau} = \text{tgdim}_W(\ker \phi_{i, W, \tau}), \quad \nu_{i, W, \tau} = \text{tgdim}_W(\text{coker } \phi_{i, W, \tau}).$$

For a fixed declared  $(W, \tau)$  and finite monitored degree set  $I_{\text{mon}}$ , the aggregate pair is

$$\left( \sum_{i \in I_{\text{mon}}} \mu_{i, W, \tau}, \sum_{i \in I_{\text{mon}}} \nu_{i, W, \tau} \right) = (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

Deprecated Latin- $u$  Type IV cokernel notation has no normative use;  $\nu_{i, W, \tau}$  and  $\nu_{\text{Collapse}}$  are canonical.

For traceability, ClaimID, CMID, TBID, and RunID denote Claim, Core Micro-Theorem, Toy Bridge, and immutable run identifiers.

## 1.18 Provenance and status labels

Constructible one-parameter persistence, interval decomposition, the standard bottleneck/interleaving isometry, bounded derived categories, cones, projective generators, finite functor categories, Morita/Yoneda methods, and effective descent are standard mathematical ingredients only at their exact published or proved hypotheses.

The AK-HPDST contribution asserted by this chapter is the conservative dual-Core architecture, migration calculus, Return-strength typing, proof-DAG and noncompensation discipline, selective HT-UB integration, finite-certificate and finite-globalization interfaces, and the exact-strength release and audit contract. This architecture does not convert standard ingredients into new literature-first theorems and does not convert validated cases into universal external results.

**Status labels for this chapter.** *Core constitutional theorems:* Theorem 1.6, Theorem 1.24, and the inherited Core-P metric theorem Theorem 1.44.

*Core definitions and contracts:* Definitions 1.1, 1.2, 1.4, 1.5, 1.7, 1.8, 1.9, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.17, 1.18, 1.19, 1.20, 1.21, 1.22, 1.23, 1.25, 1.26, 1.27, 1.28, 1.29, 1.30, together with the inherited Core-P definitions re-issued in Sections 1.9 and following.

*Operational policies and release gates:* Policies 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, and the inherited Core-P operational policies and manifest axiom.

*Explicit non-claims:* Section 1.16.



## 2 Chapter 2: Repaired Persistence, Threshold Geometry, and Metric Zero Transport

### 2.1 Standing Core-P contract and v21 division of responsibility

This chapter is the persistence-metric foundation of the v21 Corep. It inherits the complete v20 Chapter 2 and Foundation Appendix A theorem package at exactly its frozen strength. In particular, v21 does not weaken, strengthen, or reinterpret the inherited hard-threshold, threshold-gap, repaired-zero, crop-collapse, representative, or actual-morphism guard-rails.

The mathematical responsibility of this chapter is restricted to

$$\text{barcode and persistence profiles} \longrightarrow \text{typed comparison} \longrightarrow \text{repaired zero transport}.$$

It does not construct a finite diagram object, prove finite Morita reconstruction, perform object-level descent, or establish an external Return. Those operations belong respectively to Chapters 4, 7, and 9.

The inherited metric chain is

$$d_B(B, C) \leq \varepsilon + \text{two-sided threshold-gap certification} \implies d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon,$$

followed, inside the  $\tau$ -reduced image, by

$$2\varepsilon \leq \tau \implies (T_\tau^{\text{bar}} B = 0 \iff T_\tau^{\text{bar}} C = 0).$$

The v21 operational form retains the strict reserves

$$\varepsilon < \text{gap}_\tau(B, C), \quad 2\varepsilon < \tau.$$

**Policy 23** (Immutable v20 metric baseline). *Every v20 Chapter 2 or Appendix A theorem consumed by v21 is used with the same hypotheses, endpoint convention, infinite-bar convention, evaluation order, metric constant, equality boundary, and prohibited stronger reading. A stronger result requires a new v21 theorem identity and does not overwrite the inherited result.*

**Remark 2.1** (Conservative extension relative to v19 and v20). The v19 repair rejected global non-expansiveness and established the sharp two-sided threshold-gap theorem. v20 added the repaired-zero separation and detector-zero transport consequences without changing that repair. v21 inherits both layers and adds no stronger metric constant in this chapter.

**Remark 2.2** (Core-P sovereignty). A barcode statement has only barcode- or persistence-profile strength unless a separately proved Interface transports it to a chain, finite-diagram, or derived object. The existence of a finite barcode or a finite critical grid does not itself supply such an Interface.

**Remark 2.3** (Division of responsibility). Chapter 2 proves metric and exact profile transport. Chapter 3 proves detector soundness, completeness, and certificate-complexity results. Chapter 4 constructs the canonical finite linear-quiver reconstruction when its hypotheses hold. Chapter 7 handles general finite diagrams and effective object descent. Chapter 8 verifies proof-DAG closure, and Chapter 9 governs external Return.

**Remark 2.4** (What this chapter does not prove). This chapter proves no detector completeness theorem, higher  $\text{Ext}^n$ -edge, finite-diagram reconstruction, chain-level stability, derived equivalence, object-level globalization, arithmetic theorem, PDE theorem, or external Return theorem.

## 2.2 Standing Core-P regime

Fix a field  $k$ . Let

$$\text{Pers}_k := [(\mathbb{R}, \leq), \text{Vect}_k]$$

be the category of one-parameter persistence modules over  $k$ , indexed by the ordered real line.

**Definition 2.5** (Constructible persistence module). A persistence module  $M \in \text{Pers}_k$  is called *constructible* if:

- (i)  $M(t)$  is finite-dimensional over  $k$  for every  $t \in \mathbb{R}$ ;
- (ii) on every bounded interval  $J \subset \mathbb{R}$ , there is a finite critical set outside which all structure maps are isomorphisms;
- (iii) there is no accumulation of critical parameters inside a bounded interval.

The full subcategory of constructible persistence modules is denoted by

$$\text{Pers}_k^{\text{cons}} \subset \text{Pers}_k.$$

**Remark 2.6** (Working regime). All Core statements in this chapter are made inside the pointwise finite-dimensional constructible one-parameter regime over  $k$ , or inside finite barcode profiles obtained from declared windowed Core evaluations. Wild, nonconstructible, infinite-rank, or uncontrolled multi-parameter persistence data are outside the scope of this chapter unless a separate interface reduces them to declared constructible one-parameter profiles.

**Definition 2.7** (Finite-type filtered chain complex). Let  $\text{Ch}^b(k)$  be the category of bounded chain complexes of finite-dimensional  $k$ -vector spaces. A *finite-type filtered chain complex* is a pair

$$F = (C_\bullet, \{F^t C_\bullet\}_{t \in \mathbb{R}})$$

such that:

- (i)  $F^t C_\bullet \subseteq F^{t'} C_\bullet$  whenever  $t \leq t'$ ;
- (ii) the filtration is exhaustive and left-bounded;
- (iii) for each homological degree  $i$ , the persistence module

$$\mathbf{P}_i(F) : \mathbb{R} \rightarrow \text{Vect}_k, \quad t \mapsto H_i(F^t C_\bullet)$$

belongs to  $\text{Pers}_k^{\text{cons}}$ .

The category of such objects, with filtration-preserving chain maps, is denoted by

$$\text{FiltCh}(k).$$

**Remark 2.8** (Persistence layer first). The Core reads a filtered object  $F$  through monitored persistence profiles

$$\mathbf{P}_i(F).$$

Any filtered-level or derived-level use requires an additional representative policy. The barcode-level collapse operation by itself does not supply a global strict filtered lift.

### 2.3 Barcode normal form and endpoint discipline

**Definition 2.9** (Interval module). For an interval  $I \subseteq \mathbb{R}$ , the interval module  $k_I$  is the persistence module defined by

$$k_I(t) = \begin{cases} k, & t \in I, \\ 0, & t \notin I, \end{cases}$$

with structure maps equal to the identity whenever both source and target lie in  $I$ , and zero otherwise.

**Definition 2.10** (Bar and barcode). An interval summand  $k_I$  is called a *bar*. A barcode of a constructible persistence module  $M$  is an interval decomposition

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda},$$

with locally finite barcode support on bounded intervals. The corresponding multiset of bars is denoted by

$$\mathcal{B}(M) = \{I_\lambda\}_{\lambda \in \Lambda}.$$

**Theorem 2.11** (Constructible interval decomposition). *Every object  $M \in \text{Pers}_k^{\text{cons}}$  admits a locally finite interval decomposition*

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda}.$$

*The barcode is unique up to permutation of interval summands.*

*Proof sketch.* This is the standard interval-decomposition theorem in the constructible one-parameter regime over a field. On each bounded interval, the module is a finite representation of a linearly ordered finite poset and therefore splits into interval representations. Local finiteness assembles these bounded pieces into the global barcode. Krull–Schmidt uniqueness gives uniqueness up to permutation.  $\square$

**Definition 2.12** (Length of a bar). Let  $I \subset \mathbb{R}$  be a finite bar with endpoint values  $a < b$ . Its length is

$$\ell(I) := b - a.$$

If  $I = [a, \infty)$ , or differs from this only by endpoint bookkeeping at  $a$ , it is called an infinite bar and is assigned length

$$\ell(I) := +\infty$$

for thresholding purposes.

**Remark 2.13** (Endpoint independence). The length  $\ell(I)$  depends only on endpoint values, not on whether finite endpoints are open, closed, or half-open. All threshold rules in this chapter are therefore independent of harmless endpoint bookkeeping conventions.

**Definition 2.14** (Endpoint-discipline record). An endpoint-discipline record for a barcode or persistence-profile claim contains:

- (i) the interval-decoration convention used by the source profile;
- (ii) the rule by which finite endpoint decorations are forgotten for bar length and bottleneck cost;
- (iii) the birth-point or right-germ convention consumed by any downstream detector;

- (iv) the policy for a bar meeting the right endpoint of a bounded window;
- (v) the finite critical set or locally finite critical-data source used on the declared audit range.

Length-based thresholding is decoration-independent, but detector completeness, finite-grid reconstruction, and boundary coverage need not be.

**Definition 2.15** (Window-cofinal bar). Let  $W = [u, u')$  be bounded. A nonempty clipped bar is *window-cofinal* when its right endpoint is  $u'$  in the declared right-open convention. It is finite for threshold-length purposes, even when it is the crop of a globally infinite bar.

**Remark 2.16** (Critical values do not by themselves form a certificate). A finite critical set supplies a possible finite presentation of a constructible profile. It does not by itself prove endpoint completeness, zero reflection, reconstruction faithfulness, or a finite detector theorem. Those properties require the corresponding Chapter 3 or Chapter 4 theorem.

### 2.3.1 Standard bottleneck metric and infinite-bar convention

**Definition 2.17** (Standard bottleneck distance with infinite bars). Let  $B$  and  $C$  be finite barcode profiles, or locally finite constructible barcode profiles restricted to a declared bounded audit range. The bottleneck distance  $d_B(B, C)$  is computed by matchings between bars and, for finite bars, the diagonal.

The convention for infinite bars is fixed as follows.

- (i) A finite bar may be matched to the diagonal with cost  $\ell(I)/2$ .
- (ii) An infinite bar is never matched to the diagonal at finite cost.
- (iii) A finite bar and an infinite bar are never matched at finite cost.
- (iv) Two infinite bars  $[a, \infty)$  and  $[a', \infty)$  may be matched with cost

$$\text{cost}([a, \infty), [a', \infty)) := |a - a'|.$$

Endpoint bookkeeping at finite endpoints is ignored according to the endpoint-independence convention.

**Lemma 2.18** (Infinite-bar threshold invariance). *For every finite threshold  $\tau > 0$ , infinite bars are invariant under hard finite-threshold deletion:*

$$T_\tau^{\text{bar}}([a, \infty)) = [a, \infty).$$

*Moreover, infinite bars are never contained in a finite threshold band*

$$[\tau - \varepsilon, \tau + \varepsilon].$$

*Proof.* An infinite bar has length  $+\infty$ . Since  $T_\tau^{\text{bar}}$  deletes only finite bars of length at most  $\tau$ , an infinite bar is retained. The finite interval  $[\tau - \varepsilon, \tau + \varepsilon]$  contains only finite real values, so  $+\infty$  does not lie in it.  $\square$

**Remark 2.19** (Bottleneck–interleaving isometry). For q-tame or constructible one-parameter persistence modules over  $k$ , the usual isometry theorem identifies the interleaving distance and the bottleneck distance on barcodes:

$$d_{\text{int}}(M, N) = d_B(\mathcal{B}(M), \mathcal{B}(N)).$$

When infinite bars are present, this statement is read under the infinite-bar convention of Definition 2.17; namely infinite bars match infinite bars by birth-time displacement  $|a - a'|$ , and finite-to-infinite or infinite-to-diagonal matchings have infinite cost. Under this convention, a  $d_B$ -formulated stability statement may be transferred to a  $d_{\text{int}}$ -formulated comparison only when the persistence objects and barcode profiles being compared are exactly those appearing in the certified record.

## 2.4 Window-first evaluation

**Definition 2.20** (Right-open window). A domain window is a right-open interval

$$W = [u, u'),$$

where  $u \in \mathbb{R}$ ,  $u' \in \mathbb{R} \cup \{+\infty\}$ , and  $u < u'$ . If  $u' < +\infty$ ,  $W$  is called a bounded audit window. If  $u' = +\infty$ ,  $W$  is called a terminal audit window.

**Definition 2.21** (Window restriction). For  $W = [u, u')$ , define the window restriction

$$W_W(M)$$

by restricting  $M$  to  $W$ , and, when a globally indexed persistence module is required, extending by zero outside the declared window.

At the barcode level,  $W_W$  clips each bar  $I$  to

$$I \cap W,$$

discarding empty intersections.

**Remark 2.22** (Infinite bars under window restriction). The infinite-bar convention of Definition 2.17 applies before windowing and on terminal windows  $W = [u, \infty)$ . On a bounded audit window  $W = [u, u')$  with  $u' < \infty$ , an infinite bar  $[a, \infty)$  is clipped to

$$[\max\{a, u\}, u')$$

whenever the intersection is nonempty. Thus it becomes a finite bar inside the bounded window and is then thresholded according to its clipped finite length. On a terminal audit window  $W = [u, \infty)$ , it remains infinite and is never deleted by  $T_\tau^{\text{bar}}$ .

### 2.4.1 Core windowed evaluation

**Definition 2.23** (Windowed collapsed persistence evaluation). Let  $F \in \text{FiltCh}(k)$ , let  $i$  be a monitored degree, let  $W$  be a declared right-open window, and let  $\tau > 0$ . The Core windowed collapsed evaluation is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

**Remark 2.24** (Fixed order). The expression

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F))$$

means the following ordered pipeline:

$$F \mapsto \mathbf{P}_i(F) \mapsto B_{i,W}(F) = W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(B_{i,W}(F)).$$

The Core first restricts to the declared window and only then applies hard threshold deletion.

**Definition 2.25** (Repaired evaluation record). A repaired evaluation record for  $(F; i, W, \tau)$  contains:

- (i) immutable source-object and profile identifiers;
- (ii) the monitored degree  $i$ , declared window  $W$ , and threshold  $\tau$ ;

(iii) the windowed pre-threshold profile  $B_{i,W}(F)$ ;

(iv) the repaired profile

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F));$$

(v) the comparison mode, chosen from

`exact`, `metric_gap_certified`, `theorem_controlled_nonmetric`, `not_compared`;

(vi) reconstructible artifact references.

**Definition 2.26** (Threshold-gap-certified repaired evaluation comparison). A metric comparison between two repaired evaluation records is *threshold-gap-certified* only when the compared windowed pre-threshold profiles are exactly the profiles appearing in a valid two-sided record of Definition 2.44. The record must establish both

$$d_B(B, C) \leq \varepsilon_B$$

and

$$\varepsilon_B < \text{gap}_\tau(B, C).$$

The associated comparison mode is `metric_gap_certified`. A separately proved equality or profile isomorphism is recorded instead with comparison mode `exact`; it does not require a fabricated positive-radius gap certificate and does not by itself assert robustness under perturbation.

**Remark 2.27** (Evaluation is definable without certification). The profile  $\text{Eval}_{i,W,\tau}(F)$  is definitionally available once the windowed barcode profile is given. Whenever a metric discrepancy is transported through hard deletion—including metric overlap, Restart, representative, or tower comparisons—the stronger two-sided threshold-gap comparison record is required. A separately proved exact equality or profile isomorphism may instead be used in comparison mode `exact`; such an exact record is not automatically a positive-radius robustness certificate.

## 2.5 Hard threshold deletion

**Definition 2.28** (Hard barcode threshold deletion). Let  $B$  be a barcode profile. For  $\tau > 0$ , define hard barcode threshold deletion by

$$T_\tau^{\text{bar}}(B) := \{ I \in B \mid \ell(I) > \tau \text{ or } \ell(I) = +\infty \}.$$

Equivalently,  $T_\tau^{\text{bar}}$  deletes precisely the finite bars of length at most  $\tau$ . Thus:

$$\begin{aligned} \ell(I) < \tau &\implies I \text{ is deleted,} \\ \ell(I) = \tau &\implies I \text{ is deleted,} \\ \ell(I) > \tau &\implies I \text{ is retained,} \\ \ell(I) = +\infty &\implies I \text{ is retained.} \end{aligned}$$

For a constructible persistence module  $M$ , the notation  $T_\tau^{\text{bar}}(M)$  means threshold deletion applied to  $\mathcal{B}(M)$ , read at the barcode-profile level.

**Remark 2.29** (Status of  $T_\tau^{\text{bar}}$ ). In the general Core regime,  $T_\tau^{\text{bar}}$  is an operation on barcode normal forms and hence on isomorphism classes of constructible persistence modules. It is not asserted, without additional hypotheses, to be a strict exact endofunctor on all morphisms of  $\text{Pers}_k^{\text{cons}}$ .

**Proposition 2.30** (Basic properties of hard threshold deletion). *For every  $\tau > 0$ , the operation  $T_\tau^{\text{bar}}$  satisfies:*

(i) *endpoint independence*;

(ii) *idempotence*:

$$T_\tau^{\text{bar}} T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}}(B);$$

(iii) *infinite-bar preservation*;

(iv) *threshold monotonicity*: if  $0 < \tau \leq \sigma$ , then every bar retained by  $T_\sigma^{\text{bar}}(B)$  is retained by  $T_\tau^{\text{bar}}(B)$ .

*Proof.* All assertions follow directly from the definition. Deleting all finite bars of length at most  $\tau$  twice has the same effect as deleting them once. Infinite bars have length  $+\infty$ , hence are never deleted by a finite threshold. Endpoint choices do not affect length. Finally, increasing the threshold can only delete additional finite bars.  $\square$

**Proposition 2.31** (Threshold composition). *For  $\tau, \sigma > 0$ ,*

$$T_\sigma^{\text{bar}} T_\tau^{\text{bar}}(B) = T_{\max\{\tau, \sigma\}}^{\text{bar}}(B)$$

*at the barcode-profile level.*

*Proof.* Applying  $T_\tau^{\text{bar}}$  deletes bars of finite length at most  $\tau$ . Applying  $T_\sigma^{\text{bar}}$  afterwards deletes bars of finite length at most  $\sigma$  among those remaining. The total effect is deletion of all finite bars of length at most  $\max\{\tau, \sigma\}$ .  $\square$

## 2.6 Negative categorical and metric results

The following observation remains one of the categorical guard-rails of the repaired Core.

**Proposition 2.32** (Positive-threshold short-bar objects are not Serre in general). *Let  $\tau > 0$ . The class of constructible persistence modules whose barcode summands all have finite length at most  $\tau$  is not extension-closed in general. Consequently, it is not a Serre subcategory of  $\text{Pers}_k^{\text{cons}}$  in general.*

*Proof.* It suffices to give a counterexample. Take  $\tau = 1$  and consider the short exact sequence of interval modules

$$0 \longrightarrow k_{[1,2)} \longrightarrow k_{[0,2)} \longrightarrow k_{[0,1)} \longrightarrow 0.$$

The subobject  $k_{[1,2)}$  has length 1, and the quotient  $k_{[0,1)}$  also has length 1. However, the middle object  $k_{[0,2)}$  has length 2. Thus an extension of two short-bar objects can contain a longer bar. Therefore the class of bars of length at most  $\tau$  is not extension-closed in general.  $\square$

**Corollary 2.33** (No global exact reflector claim). *For  $\tau > 0$ , hard threshold deletion is not asserted in the general Core to arise from an exact Serre localization, an abelian exact reflector, or a global exact endofunctor on all of  $\text{Pers}_k^{\text{cons}}$ .*

**Remark 2.34** (Why this guard-rail matters). If positive-threshold deletion were treated as an exact Serre reflector without additional hypotheses, then exactness, functoriality, tower comparison maps, and filtered lifts would inherit unsupported categorical strength. The repaired formulation keeps the barcode-level operation while requiring stronger claims to be explicitly justified by  $\tau$ -split or exact-admissible hypotheses.

### 2.6.1 Counterexample to global non-expansiveness

The next example is the metric repair point inherited through v20.

**Example 2.35** (Failure of global non-expansiveness). Fix  $\tau > 0$ , and let  $0 < \eta < \tau/4$ . Consider the one-bar barcodes

$$B = \{[0, \tau + \eta)\}, \quad C = \{[0, \tau - \eta)\}.$$

Before threshold deletion, matching the two bars to each other has cost  $2\eta$ . Matching both bars to the diagonal has cost

$$\max \left\{ \frac{\tau + \eta}{2}, \frac{\tau - \eta}{2} \right\} = \frac{\tau + \eta}{2} > 2\eta,$$

where the strict inequality follows from  $\eta < \tau/4$ . Therefore

$$d_B(B, C) = 2\eta.$$

After hard threshold deletion,

$$T_\tau^{\text{bar}}(B) = \{[0, \tau + \eta)\}, \quad T_\tau^{\text{bar}}(C) = \emptyset.$$

Therefore the remaining bar of  $T_\tau^{\text{bar}}(B)$  must be matched to the diagonal, giving

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) = \frac{\tau + \eta}{2}.$$

Moreover,

$$\frac{\tau + \eta}{2} > 2\eta = d_B(B, C),$$

again because  $\eta < \tau/4$ . Consequently,

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq d_B(B, C)$$

does not hold in general.

**Remark 2.36** (Meaning of the counterexample). The failure occurs exactly at the threshold boundary. A small endpoint perturbation can move a finite bar from length  $> \tau$  to length  $\leq \tau$ , thereby changing whether it is retained or deleted. Thus the correct stability statement must exclude bars near the threshold.

**Corollary 2.37** (Deprecated v18 metric claim). *The statement that hard positive-threshold deletion  $T_\tau^{\text{bar}}$  is globally non-expansive under the standard bottleneck metric is rejected in v19. Any occurrence of that claim in earlier formulations is deprecated and may not be used in v21 proof construction.*

### 2.7 Threshold clearance

**Definition 2.38** (Threshold clearance). Let  $B$  be a barcode profile. Define the finite-bar threshold clearance by

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|,$$

where  $B_{\text{fin}}$  denotes the finite bars of  $B$ . If  $B$  has no finite bars, set

$$\text{clear}_\tau(B) := +\infty.$$

If  $B$  contains a finite bar of length exactly  $\tau$ , then

$$\text{clear}_\tau(B) = 0.$$



**Definition 2.39** (Threshold gap as ledger alias). For a barcode profile  $B$ , define the ledger safety margin

$$\text{gap}_\tau(B) := \text{clear}_\tau(B).$$

For a finite family of barcode profiles  $B_1, \dots, B_m$ , define

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_{1 \leq j \leq m} \text{clear}_\tau(B_j).$$

Thus  $\text{gap}_\tau$  is not a second invariant distinct from threshold clearance; it is the ledger-facing alias for the minimum finite-bar clearance at the deletion threshold.

**Definition 2.40** (Bottleneck scalarization and typed metric budget). Let  $(V, \oplus, 0_V, \preceq)$  be the declared ledger quantale. A *bottleneck scalarization* is a monotone, subadditive map

$$\text{val}_B : V \longrightarrow [0, +\infty], \quad \text{val}_B(0_V) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

For a metric ledger element  $\mathfrak{d}_{\text{met}} \in V$ , define the real-valued bottleneck budget

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

Whenever the established shorthand

$$\Sigma\delta < \text{gap}_\tau$$

is used in this chapter, it means the typed real inequality

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau$$

for the declared protected comparison family, together with theorem-level, verified-computation, or artifact evidence that the scalarized value upper-bounds the actual bottleneck discrepancy consumed by the proof. Only metric defects belong to  $\mathfrak{d}_{\text{met}}$ . Representative existence, comparison morphisms, eligibility, terminal tameness, claim status, and manifest completeness are non-scalar proof obligations and cannot be repaired by numerical slack.

**Remark 2.41** (Sharp threshold-safety margin versus scalarization tightness). Under an  $\varepsilon$ -bottleneck matching, the length of a finite bar may change by at most  $2\varepsilon$ . For a *two-sided* threshold-gap certificate, however, the sharp forbidden band is

$$[\tau - \varepsilon, \tau + \varepsilon],$$

not  $[\tau - 2\varepsilon, \tau + 2\varepsilon]$ . Consequently, for a scalarized budget

$$\varepsilon = \varepsilon_B(\mathfrak{d}_{\text{met}}),$$

the condition

$$\varepsilon < \text{gap}_\tau(B) = \text{clear}_\tau(B)$$

means that all finite bars of the relevant pre-threshold profile avoid

$$[\tau - \varepsilon, \tau + \varepsilon].$$

The former factor  $1/2$  is not part of the v21.0 two-sided Core certificate; v21 inherits this correction through the frozen v20 baseline. The word *sharp* refers to the optimal half-width of the threshold band for a given true bottleneck bound; it does not assert that a chosen scalarization  $\text{val}_B$  is numerically tight.

**Definition 2.42** (Windowed pre-threshold profile). For  $F \in \text{FiltCh}(k)$ , monitored degree  $i$ , and declared window  $W$ , define the *windowed pre-threshold profile*

$$B_{i,W}(F) := W_W \mathbf{P}_i(F).$$

This is the barcode profile on which threshold clearance and threshold gap are measured. The repaired evaluation is then

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

**Remark 2.43** (Clearance is measured before deletion, after windowing). The quantity  $\text{clear}_\tau$  is never measured on the already deleted profile  $\text{Eval}_{i,W,\tau}(F)$ . It is measured on the windowed pre-threshold profile

$$B_{i,W}(F) = W_W \mathbf{P}_i(F).$$

This order matters: the threshold gap certifies that hard deletion will not change classification under the declared perturbation budget. Once a bar has already been deleted, its distance from the threshold can no longer be audited.

**Definition 2.44** (Threshold-gap comparison record). A threshold-gap comparison record for two windowed pre-threshold profiles

$$B := B_{i,W}(F), \quad C := B_{i,W}(G)$$

at one declared monitored degree  $i$ , common window  $W$ , and threshold  $\tau$  contains:

- (i) immutable identifiers for  $B$  and  $C$ ;
- (ii) the two pre-threshold profiles, the monitored degree, the common window, and the threshold;
- (iii) the finite-bar length data or certified summaries for both profiles;
- (iv)  $\text{clear}_\tau(B)$ ,  $\text{clear}_\tau(C)$ , and

$$\text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\};$$

- (v) the metric ledger element  $\mathfrak{d}_{\text{met}}$ , the declared scalarization  $\text{val}_B$ , and

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}});$$

- (vi) theorem-level, verified-computation, or artifact evidence that

$$d_B(B, C) \leq \varepsilon_B;$$

- (vii) confirmation that neither finite-bar family meets the closed band

$$[\tau - \varepsilon_B, \tau + \varepsilon_B],$$

equivalently the strict two-sided condition

$$\varepsilon_B < \text{gap}_\tau(B, C);$$

- (viii) every crop, window-edge, endpoint, or threshold-reconciliation component used to obtain the windowed comparison bound;
- (ix) reconstructible artifact references;

(x) one specialized status

`gap_certified, gap_failed, gap_pending,`

mapping respectively to pass, reject, and undefined.

The record is two-sided by definition. Here `gap_failed` means that a well-typed required gap condition is proved or artifact-evidenced to fail. Missing, incomplete, stale, or pending evidence is recorded as `gap_pending`, never as `gap_failed`. A single-profile clearance record may be retained as a readiness record, but it is not a metric comparison certificate and does not establish transport through hard deletion.

**Remark 2.45** (Length exactly  $\tau$  is not gap-safe). A finite bar of length exactly  $\tau$  gives  $\text{clear}_\tau = 0$ . Such a profile may still be evaluated by  $T_\tau^{\text{bar}}$ , and the bar is deleted because deletion uses the rule  $\ell(I) \leq \tau$ . However, the profile is not positively gap-ready at threshold  $\tau$ . It cannot support positive-radius metric transport through hard deletion at that threshold. A separately proved exact identity may still be recorded in exact comparison mode, but it is not a robustness certificate.

## 2.8 Sharp threshold-gap stability

**Theorem 2.46** (Metric-repair theorem: sharp threshold-gap-safe stability). *Let  $B$  and  $C$  be barcodes with finitely many bars, possibly including infinite bars, interpreted under the bottleneck convention of Definition 2.17. Let  $\tau > 0$  and  $\varepsilon \geq 0$ . Assume*

$$d_B(B, C) \leq \varepsilon.$$

*Assume also that no finite bar of either  $B$  or  $C$  has length in the closed threshold band*

$$[\tau - \varepsilon, \tau + \varepsilon].$$

*Then*

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon.$$

*Proof.* First consider the degenerate case  $\varepsilon = 0$ . Then  $d_B(B, C) = 0$ , so  $B$  and  $C$  are identical as barcode profiles under the stated barcode convention. The closed threshold band is the singleton  $\{\tau\}$ , and the hypothesis says that no finite bar has length exactly  $\tau$ . Applying the same deterministic deletion rule to identical barcodes gives

$$T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}}(C),$$

hence the conclusion follows.

Now assume  $\varepsilon > 0$ . Choose a bottleneck matching between  $B$  and  $C$  of cost at most  $\varepsilon$ . Such a matching exists for the finite barcode profiles under consideration. We construct a matching between the retained barcodes after applying  $T_\tau^{\text{bar}}$  by modifying the chosen matching only by deleting matched pairs that are removed on both sides.

Let  $I \in B$  be a bar retained by  $T_\tau^{\text{bar}}$ . If  $I$  is infinite, then by Definition 2.17 it cannot be matched at finite cost to the diagonal or to a finite bar. Hence it is matched to an infinite bar of  $C$ , which is also retained.

Suppose next that  $I$  is finite. Since  $I$  is retained,  $\ell(I) > \tau$ . Because no finite bar of  $B$  has length in  $[\tau - \varepsilon, \tau + \varepsilon]$ , this implies

$$\ell(I) > \tau + \varepsilon.$$

There are two cases.

If  $I$  is matched to the diagonal with cost at most  $\varepsilon$ , then the same diagonal match may be kept after threshold deletion. It contributes cost at most  $\varepsilon$  to the retained-barcode matching. This case is possible when

$\ell(I)/2 \leq \varepsilon$ , and it is the reason the proof must not assume that every retained finite bar has a finite matched partner.

If  $I$  is matched to a finite bar  $J \in C$ , then the endpoint displacement bound gives

$$|\ell(I) - \ell(J)| \leq 2\varepsilon.$$

If  $J$  were deleted, then  $\ell(J) \leq \tau$ . Since  $C$  also avoids  $[\tau - \varepsilon, \tau + \varepsilon]$ , this would force

$$\ell(J) < \tau - \varepsilon.$$

Together with  $\ell(I) > \tau + \varepsilon$ , this gives

$$\ell(I) - \ell(J) > 2\varepsilon,$$

contradicting  $|\ell(I) - \ell(J)| \leq 2\varepsilon$ . Therefore  $J$  is retained. Finite-to-infinite matching is impossible at finite cost by Definition 2.17.

The same argument, with  $B$  and  $C$  interchanged, shows that every retained finite bar of  $C$  that is not kept on the diagonal is matched to a retained finite bar of  $B$ , while every retained infinite bar is matched to a retained infinite bar. Hence the chosen matching restricts, with the permitted retained-to-diagonal matches, to a bottleneck matching between  $T_\tau^{\text{bar}}(B)$  and  $T_\tau^{\text{bar}}(C)$  of cost at most  $\varepsilon$ . Thus

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon.$$

□

**Example 2.47** (Sharpness of the  $\varepsilon$ -band). The constant  $\varepsilon$  in the two-sided closed threshold band is sharp. Let  $0 < c < \varepsilon$ , choose  $0 < \eta < \varepsilon - c$ , and take  $\tau$  large enough that the diagonal costs below exceed  $\varepsilon$ . Set

$$L_+ := \tau + c + \eta, \quad L_- := \tau - c - \eta,$$

and consider the centered finite bars

$$I = \left[-\frac{L_+}{2}, \frac{L_+}{2}\right), \quad J = \left[-\frac{L_-}{2}, \frac{L_-}{2}\right).$$

Then both lengths avoid the smaller closed band  $[\tau - c, \tau + c]$ , while

$$d_B(\{I\}, \{J\}) = c + \eta < \varepsilon.$$

However  $I$  is retained and  $J$  is deleted by  $T_\tau^{\text{bar}}$ . Hence

$$T_\tau^{\text{bar}}(\{I\}) = \{I\}, \quad T_\tau^{\text{bar}}(\{J\}) = \emptyset.$$

For  $\tau$  sufficiently large,  $L_+/2 > \varepsilon$ , so

$$d_B(T_\tau^{\text{bar}}(\{I\}), T_\tau^{\text{bar}}(\{J\})) = \frac{L_+}{2} > \varepsilon.$$

Thus no two-sided closed band of half-width  $c < \varepsilon$  suffices in general. The sharp two-sided half-width is exactly  $\varepsilon$ .

**Remark 2.48** (Two-sided certificate). The threshold-gap certificate is two-sided. It is not enough for  $B$  alone to avoid the threshold band. If  $C$  contains a finite bar in  $[\tau - \varepsilon, \tau + \varepsilon]$ , then a matched pair may cross the deletion threshold and change retained/deleted status. For one-sided certificates, a  $2\varepsilon$ -type margin is a natural conservative constant; it is not the sharp constant for the two-sided v19 Core theorem.

**Corollary 2.49** (Scalarized-ledger-safe threshold classification). *Let  $B$  and  $C$  be finite barcode profiles, possibly with infinite bars. Let  $\mathfrak{d}_{\text{met}} \in \mathcal{V}$  be a declared metric ledger and set*

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

*Suppose*

$$d_B(B, C) \leq \varepsilon_B$$

*and*

$$\varepsilon_B < \text{gap}_\tau(B, C).$$

*Then*

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon_B.$$

*Proof.* The strict inequality

$$\varepsilon_B < \text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\}$$

implies that no finite bar of either barcode has length in

$$[\tau - \varepsilon_B, \tau + \varepsilon_B].$$

Apply Theorem 2.46 with  $\varepsilon = \varepsilon_B$ . □

## 2.9 Reduced image and distance to zero

**Definition 2.50** ( $\tau$ -reduced barcode profile). A barcode profile  $P$  is  $\tau$ -reduced if

$$T_\tau^{\text{bar}}(P) = P.$$

Equivalently, every finite bar of  $P$  has length strictly greater than  $\tau$ , while every infinite bar is retained. Every repaired profile  $T_\tau^{\text{bar}}(B)$  is  $\tau$ -reduced.

**Definition 2.51** (Zero radius of a finite barcode profile). For a finite barcode profile  $P$ , define

$$\text{zrad}(P) := d_B(P, 0),$$

where  $0$  denotes the empty barcode. Under the standard convention,

$$\text{zrad}(P) = \begin{cases} 0, & P = 0, \\ \frac{1}{2} \max_{I \in P} \ell(I), & P \neq 0 \text{ and every bar is finite,} \\ +\infty, & P \text{ contains an infinite bar.} \end{cases}$$

**Lemma 2.52** (Distance-to-zero formula). *Definition 2.51 gives the exact bottleneck distance from  $P$  to the zero profile.*

*Proof.* When  $P$  has only finite bars, every bar must be matched to the diagonal, and the matching cost is the maximum of the diagonal costs  $\ell(I)/2$ . If an infinite bar is present, it cannot be matched to the diagonal at finite cost. The empty profile has distance zero from itself. □

**Theorem 2.53** (Sharp zero separation for  $\tau$ -reduced profiles). *Let  $P$  and  $Q$  be finite  $\tau$ -reduced barcode profiles, possibly including infinite bars. Let  $\varepsilon \geq 0$ , and suppose*

$$d_B(P, Q) \leq \varepsilon \quad \text{and} \quad 2\varepsilon \leq \tau.$$

*Then*

$$P = 0 \iff Q = 0.$$

*The constant is sharp: if  $2\varepsilon > \tau$ , the conclusion can fail.*

*Proof.* Assume  $P = 0$ . If  $Q$  contained an infinite bar, then  $d_B(P, Q) = +\infty$ , contrary to the hypothesis. If  $Q \neq 0$  and all its bars were finite,  $\tau$ -reducedness would give

$$\ell(I) > \tau \geq 2\varepsilon$$

for every bar  $I \in Q$ . Lemma 2.52 would then imply

$$d_B(0, Q) = \frac{1}{2} \max_{I \in Q} \ell(I) > \varepsilon,$$

again a contradiction. Hence  $Q = 0$ . The converse is symmetric.

For sharpness, suppose  $2\varepsilon > \tau$ . Choose  $L$  with  $\tau < L \leq 2\varepsilon$ , let  $P = 0$ , and let  $Q$  consist of one centered finite bar of length  $L$ . Then  $Q$  is  $\tau$ -reduced,  $d_B(P, Q) = L/2 \leq \varepsilon$ , but  $P = 0$  and  $Q \neq 0$ .  $\square$

**Remark 2.54** (Strict operational reserve). The theorem is valid at the closed boundary  $2\varepsilon = \tau$ , because repaired finite bars have length *strictly* greater than  $\tau$ . The Chapter 1 audit contract uses  $2\varepsilon < \tau$  to retain a positive reserve and to avoid equality-sensitive scalarization or rounding policies. The strict policy is conservative, not mathematically sharp.

**Proposition 2.55** (Exact zero transport). *If  $P \cong Q$  as persistence profiles, then*

$$P = 0 \iff Q = 0.$$

*No metric ledger or threshold-gap certificate is required for this exact claim. The actual isomorphism and its immutable proof artifact are required.*

*Proof.* The zero object is invariant under isomorphism.  $\square$

## 2.10 Metric repaired-zero transport

**Theorem 2.56** (Metric-safe repaired-zero transport). *Let  $B$  and  $C$  be finite barcode profiles, possibly including infinite bars. Let  $\tau > 0$ ,  $\varepsilon \geq 0$ , and assume*

$$d_B(B, C) \leq \varepsilon.$$

*Assume that no finite bar of either profile has length in*

$$[\tau - \varepsilon, \tau + \varepsilon],$$

*and assume*

$$2\varepsilon \leq \tau.$$

*Then*

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0.$$

*Proof.* Theorem 2.46 gives

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon.$$

Both repaired profiles are  $\tau$ -reduced by Definition 2.50. Apply Theorem 2.53. □

**Corollary 2.57** (Scalarized metric-safe repaired-zero transport). *Let  $\mathfrak{d}_{\text{met}} \in \mathbb{V}$ , set*

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}}),$$

*and suppose the record proves*

$$d_B(B, C) \leq \varepsilon_B, \quad \varepsilon_B < \text{gap}_\tau(B, C), \quad 2\varepsilon_B < \tau.$$

*Then*

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0.$$

*This is the strict operational form consumed by the Chapter 1 metric-safe detector chain.*

**Definition 2.58** (Post-threshold zero-separation record). A post-threshold zero-separation record contains:

- (i) immutable identifiers for the repaired profiles  $P = T_\tau^{\text{bar}}(B)$  and  $Q = T_\tau^{\text{bar}}(C)$ ;
- (ii) the common threshold  $\tau$ , the proved bound  $d_B(P, Q) \leq \varepsilon$ , and its derivation;
- (iii) evidence that  $P$  and  $Q$  are  $\tau$ -reduced;
- (iv) the zero-separation inequality  $2\varepsilon \leq \tau$ , or the stricter operational inequality  $2\varepsilon < \tau$ ;
- (v) the theorem reference Theorem 2.53;
- (vi) exact source, scalarization, computation, proof, and artifact identities.

Missing evidence yields `undefined`. An evidenced false required inequality yields `reject`. No new independent status algebra is created.

## 2.11 Crop-collapse criterion

**Proposition 2.59** (No unconditional crop-collapse commutation). *In general,*

$$W_W T_\tau^{\text{bar}}(B) \neq T_\tau^{\text{bar}} W_W(B).$$

*Proof.* Let  $B = \{[0, 10]\}$ , let  $W = [0, 1]$ , and take  $\tau = 2$ . Since the global bar has length  $10 > \tau$ , it survives  $T_\tau^{\text{bar}}$ . Therefore

$$W_W T_\tau^{\text{bar}}(B) = \{[0, 1]\}.$$

On the other hand,

$$W_W(B) = \{[0, 1]\},$$

whose length is  $1 \leq \tau$ . Thus

$$T_\tau^{\text{bar}} W_W(B) = \emptyset.$$

Hence the two operations do not commute in general. □

**Definition 2.60** (Window-edge clearance). A window-edge clearance record for a source-level perturbation problem records a declared margin between the bar endpoints or critical parameters relevant to  $\mathbf{P}_i(F)$  and the window boundary points  $u, u'$ . Its purpose is to certify that passage from source-level perturbation to windowed pre-threshold profile comparison does not silently gain or lose bars at the crop boundary.

**Definition 2.61** (Crop budget). A crop budget is a declared ledger component

$$\delta_{\text{crop}}(i, W)$$

bounding the discrepancy introduced when moving from a source-level comparison to a comparison of windowed pre-threshold profiles:

$$B_{i,W}(F) = W_W \mathbf{P}_i(F).$$

Whenever a proof uses a source-level perturbation statement to infer

$$d_B(B_{i,W}(F), B_{i,W}(G)) \leq \varepsilon,$$

the contribution of window restriction, boundary clipping, endpoint uncertainty, and crop instability must either be proved zero by a window-edge clearance record or included in the ledger as  $\delta_{\text{crop}}(i, W)$ .

**Remark 2.62** (Gap-safe stability consumes windowed profile distance). Theorem 2.46 does not assert that a raw perturbation of filtered objects automatically gives a bottleneck bound after window restriction. It consumes the hypothesis

$$d_B(B, C) \leq \varepsilon$$

for the barcode profiles to which threshold deletion is applied. In AK applications, those profiles are the windowed pre-threshold profiles

$$B_{i,W}(F) = W_W \mathbf{P}_i(F).$$

Therefore window-edge stability belongs to the ledger and is separate from threshold-gap stability.

**Example 2.63** (Infinite-bar obstruction to the former finite-only condition). Let

$$B = \{[0, +\infty)\}, \quad W = [0, 1), \quad \tau = 2.$$

The infinite bar is retained by global threshold deletion, so

$$W_W T_\tau^{\text{bar}}(B) = \{[0, 1)\}.$$

After windowing first, however, the bar has finite clipped length  $1 \leq \tau$ , and hence

$$T_\tau^{\text{bar}} W_W(B) = \emptyset.$$

Thus any commutation criterion that checks only globally retained finite bars is insufficient: globally retained infinite bars must also be tested after clipping to a bounded window.

**Definition 2.64** ( $\tau$ -window-adapted condition). A pair  $(B, W)$  is called  $\tau$ -window-adapted if every bar  $I \in B$  retained by  $T_\tau^{\text{bar}}$ , including every infinite bar, satisfies one of the following:

- (i)  $I \cap W = \emptyset$ ;
- (ii) the clipped bar  $I \cap W$  is retained by threshold deletion on the window, equivalently

$$\ell(I \cap W) > \tau \quad \text{or} \quad \ell(I \cap W) = +\infty.$$



For a bounded window, clause (ii) requires every nonempty crop of a globally retained infinite bar to have finite clipped length strictly greater than  $\tau$ . For a terminal window, a nonempty crop of an infinite bar remains infinite and automatically satisfies clause (ii).

**Proposition 2.65** (Exact crop–collapse commutation criterion). *For a barcode profile  $B$ , a right-open window  $W$ , and  $\tau > 0$ , the following are equivalent:*

(i)  $(B, W)$  is  $\tau$ -window-adapted;

(ii)

$$W_W T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}} W_W(B)$$

as barcode multisets, including multiplicities.

*Proof.* Assume first that  $(B, W)$  is  $\tau$ -window-adapted. If a finite bar  $I$  is deleted globally, then  $\ell(I) \leq \tau$ ; clipping cannot increase length, so every nonempty crop  $I \cap W$  is also deleted after windowing. If a bar  $I$  is retained globally, the adaptedness condition says that it is either disjoint from  $W$  or its nonempty crop is retained after windowing. Thus every bar contributes with the same multiplicity in both orders.

Conversely, suppose the two barcode multisets are equal and let  $I$  be retained globally with  $I \cap W \neq \emptyset$ . If  $I \cap W$  were a finite bar of length at most  $\tau$ , then the occurrence of  $I \cap W$  contributed by  $I$  would appear in  $W_W T_\tau^{\text{bar}}(B)$  but would be deleted from  $T_\tau^{\text{bar}} W_W(B)$ . The multiplicities would differ, contradicting equality. Hence the crop is retained, and  $(B, W)$  is  $\tau$ -window-adapted.  $\square$

**Remark 2.66** (Default policy and audit use). The default Core policy is not to assume  $\tau$ -window-adaptedness. Gate-relevant quantities are evaluated by the fixed order

$$\mathbf{P}_i(F) \mapsto W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Conditional commutation may be used only when Definition 2.64 is verified for all globally retained bars, including infinite bars, and the verification is recorded in the manifest.

## 2.12 Comparison-mode calculus

**Definition 2.67** (Canonical v21 comparison modes). Every comparison inference consumed by this chapter records exactly one mode:

$$\begin{array}{ll} \text{exact,} & \text{metric\_gap\_certified,} \\ \text{theorem\_controlled\_nonmetric,} & \text{not\_compared.} \end{array}$$

Their meanings are:

- (i) **exact**: an actual equality or persistence-profile isomorphism is proved for the displayed source and target;
- (ii) **metric\_gap\_certified**: an actual pre-threshold bottleneck bound, the sharp two-sided threshold-gap hypotheses, and every zero-separation hypothesis needed by the conclusion are supplied;
- (iii) **theorem\_controlled\_nonmetric**: a named theorem directly transports the declared predicate or relation without asserting a bottleneck estimate or an exact isomorphism;
- (iv) **not\_compared**: no comparison conclusion is consumed.

The mode is part of the claim type. It is not selected retrospectively from whichever interpretation gives the desired verdict.

**Definition 2.68** (Theorem-controlled nonmetric comparison record). A theorem-controlled nonmetric comparison record contains the exact source and target profiles, the transported predicate or relation, the theorem identity, all hypotheses, direction, naturality scope, artifacts, and prohibited stronger readings. It does not create a bottleneck bound, threshold-gap certificate, robustness radius, actual morphism, or inverse implication unless the cited theorem explicitly proves that item.

**Theorem 2.69** (Comparison-mode separation). *No canonical comparison mode implies another mode merely by its status. In particular:*

- (i) *an exact comparison need not have positive threshold clearance;*
- (ii) *a metric-gap-certified comparison need not provide an actual profile isomorphism;*
- (iii) *a theorem-controlled nonmetric comparison need not supply either a metric bound or an exact comparison;*
- (iv) *not\_compared discharges no invoked comparison obligation.*

*A conversion between modes is permitted only through a separate theorem whose conclusion is exactly the target mode's required data.*

*Proof.* The four modes record logically different evidence types. Exactness is an objectwise relation; metric-gap certification is a quantitative perturbation certificate; theorem-controlled transport is a named implication at its own strength; and not-compared is a scope exclusion.

For clause (i), exact self-comparison of a barcode containing a finite bar of length exactly  $\tau$  is valid, while every positive threshold-gap record at that threshold fails because the clearance is zero. For clause (ii), choose a finite bar  $I = [0, L)$  with  $L \neq \tau$ , and let  $J = [\delta, L + \delta)$ , where

$$0 < \delta < \min\{|L - \tau|, L/2\}.$$

The one-bar profiles are metric-gap-certified and have bottleneck distance  $\delta$ , but they are not equal as barcode profiles and their interval modules are not isomorphic as  $\mathbb{R}$ -indexed persistence modules. Clauses (iii) and (iv) follow because a theorem-controlled record and a scope-exclusion record do not contain the defining fields of an exact or metric record. Thus no mode conversion is available without an additional theorem producing the required target data.  $\square$

### 2.13 Profile discrepancy versus higher-level discrepancy

**Definition 2.70** (Typed discrepancy level). A discrepancy record declares exactly one mathematical level among

`barcode, persistence_module, filtered_chain, finite_diagram, derived.`

It also declares the source and target objects, the relation or metric, the coefficient and indexing conventions, the comparison mode, and the theorem that permits any change of level.

**Definition 2.71** (Barcode discrepancy). For barcode profiles in the standing regime, barcode discrepancy is measured by the standard bottleneck distance  $d_B$  under the declared infinite-bar and endpoint conventions.

**Definition 2.72** (Persistence-module discrepancy). For constructible one-parameter persistence modules, persistence-module discrepancy may be measured by interleaving distance  $d_{\text{int}}$ . The bottleneck–interleaving isometry identifies this value with barcode discrepancy only for the exact modules and barcode conventions covered by the isometry theorem.

**Definition 2.73** (Higher-level discrepancy placeholder). A filtered-chain, finite-diagram, or derived discrepancy is not assigned a canonical scalar by this chapter. Such a record is meaningful only after a later chapter declares the object category, equivalence relation or metric, functorial comparison data, and the theorem relating that level to the persistence profile.

**Theorem 2.74** (No automatic upward discrepancy transport). *Let a barcode or persistence-module comparison be supplied.*

- (i) *A positive-radius bound, or any bound not accompanied by an exact identification theorem, does not imply chain isomorphism, chain-homotopy equivalence, filtered quasi-isomorphism, equality of finite diagrams, quasi-isomorphism, derived equivalence, or stability of kernels, cokernels, cones, or descent data.*
- (ii) *Even an exact barcode or persistence-module isomorphism does not by itself identify filtered-chain representatives, branching finite diagrams, derived enhancements, cones of independently chosen morphisms, or descent data.*
- (iii) *An exact persistence-module isomorphism does induce a natural isomorphism after restriction to any fixed finite linearly ordered subcategory, but only by applying that declared restriction functor; it does not supply a canonical artifact-level representative or any stronger reconstruction statement.*

*Every other change of level requires a separately proved Interface with its own functoriality, exactness, reflection, and information profiles.*

*Proof.* A positive bottleneck or interleaving radius allows non-isomorphic persistence modules, so it cannot imply any exact relation at a higher level. Moreover, the monitored persistence module forgets chain-level extension, representative, and enhancement data, and a barcode matching is not a morphism of those objects. This proves clauses (i) and (ii). Clause (iii) is ordinary functoriality of restriction: a natural isomorphism remains a natural isomorphism after precomposition with the inclusion of the finite ordered subcategory. The qualification is essential because restriction neither chooses the original representatives nor proves endpoint completeness, full-profile reconstruction, grid independence, branching reconstruction, or descent.  $\square$

**Remark 2.75** (Downward transport is also typed). A chain-, diagram-, or derived-level comparison may imply a persistence or barcode comparison only through the declared persistence functor and its stability theorem. The absence of automatic upward transport does not license an automatic downward transport.

## 2.14 Core-DG input record and handoff boundary

**Definition 2.76** (Finite critical-grid datum). For a constructible profile  $P$  on a declared window  $W$ , a finite critical-grid datum is a finite ordered set

$$\Gamma = \{t_0 < \cdots < t_m\} \subset W$$

together with endpoint sentinels or germ data sufficient for the declared finite presentation task. The record states whether  $\Gamma$  is complete, merely sampled, or supplied by an external characterization theorem.

**Definition 2.77** (Core-P to Core-DG input record). A P2DG input record is the tuple

$$\text{P2DGInput} = (P, \Gamma, W, \tau, \text{endpt}, \text{variance}, \text{comparison}, \text{loss}, \text{artifacts}),$$

where:

- (i) the source kind is declared as `actual_module` or `barcode_profile`, and  $P = \mathbf{P}_i(F)$ ,  $B_{i,W}(F)$ , or a separately declared repaired profile is identified exactly;
- (ii)  $\Gamma$  is a finite critical-grid datum;
- (iii)  $(W, \tau)$  and the window-first evaluation stage are fixed;
- (iv) `endpt` records endpoint and germ conventions;
- (v) `variance` records the direction of all proposed diagram arrows;
- (vi) `comparison` records one mode from Definition 2.67;
- (vii) `loss` lists every feature not certified to survive the proposed passage;
- (viii) `artifacts` binds the source profile, critical data, computation, and proof records.

**Definition 2.78** (P2DG readiness status). A P2DG input has one readiness status:

`ready, incomplete, inconsistent, not_invoked.`

Ready means that the input fields required by the chosen Chapter 4 or Chapter 7 Interface are present and internally consistent. It does not mean that the target diagram exists, that reconstruction is faithful, or that the Interface theorem has passed. These are workflow-readiness values, not the Chapter 8 atomic verifier statuses `pass`, `reject`, `undefined`, and `not_invoked`.

**Theorem 2.79** (P2DG handoff boundary). *Let a ready P2DG input record be given.*

- (i) *If the source kind is `actual_module`, restriction to the finite ordered grid  $\Gamma$  canonically gives the preliminary linear diagram  $P|_{\Gamma}$ .*
- (ii) *If the source kind is `barcode_profile`, the barcode determines the isomorphism class of the one-parameter persistence module in the standing interval-decomposable regime, but it does not canonically choose an actual representative, structure-map artifact, or immutable object identity.*
- (iii) *In either case, readiness does not prove that the finite diagram is endpoint-complete, reconstructs the full windowed profile, preserves a detector, reflects zero, is independent of the chosen grid, or extends to a branching  $\text{Core}_{\text{DG}}$  object.*

*The canonical finite linear-order reconstruction and its loss/reflection claims are supplied, when their hypotheses hold, by Chapter 4. Direct passage to Chapter 7 requires a separately proved Interface whose source is this record.*

*Proof.* For an actual persistence module, evaluation on the objects of a finite linearly ordered subcategory and use of its structure maps define the restriction diagram. Barcode data determine the persistence-module isomorphism class in the standing interval-decomposable regime, but do not fix a chosen representative or artifact-level maps. Neither construction alone contains the endpoint-completeness, reconstruction, reflection, branching, or descent theorems listed in clause (iii).  $\square$

**Remark 2.80** (What a barcode does and does not determine). In the standing one-parameter regime, a barcode determines the isomorphism class of the persistence module. It does not canonically choose an actual representative with artifact identity, a filtered chain model, a branching finite-category representation, a natural transformation to another object, or a derived enhancement.

## 2.15 Downstream Core-P handoff records

The following inherited records are retained because later chapters consume them. They do not enlarge the mathematical responsibility of Chapter 2 beyond persistence-profile typing and comparison.

### 2.15.1 Detector-zero transport and rank-instability boundary

**Definition 2.81** (Metric-safe detector-zero transport record). A metric-safe detector-zero transport record from  $B$  to  $C$  contains:

- (i) the complete Chapter 3 detector certificate establishing  $T_\tau^{\text{bar}}(B) = 0$ , including its exact profile stage;
- (ii) the two-sided threshold-gap comparison record for  $B$  and  $C$ ;
- (iii) the repaired-profile distance bound produced by Theorem 2.46;
- (iv) the post-threshold zero-separation record of Definition 2.58;
- (v) the conclusion  $T_\tau^{\text{bar}}(C) = 0$ ;
- (vi) immutable detector, comparison, theorem, and replay artifacts.

The record transports only the exact zero predicate. It does not transport the detector sample values, ranks, birth set, germ set, sample count, or certificate size.

**Corollary 2.82** (Detector-certified metric zero transport). *Assume a complete detector certificate proves*

$$T_\tau^{\text{bar}}(B) = 0.$$

*If the hypotheses of Corollary 2.57 hold for  $(B, C)$ , then*

$$T_\tau^{\text{bar}}(C) = 0.$$

*The converse holds after exchanging  $B$  and  $C$ . Detector completeness is required only for the profile whose zero statement is obtained from detector data; the metric theorem transports the already certified zero predicate.*

*Proof.* The detector certificate supplies the source zero statement. Apply Corollary 2.57. □

**Definition 2.83** (Pointwise span-rank observation). For a persistence module  $M$ , a point  $t$ , and a span  $s > 0$ , define the pointwise span-rank observation

$$r_{t,s}(M) := \dim_k \text{im}(M(t) \longrightarrow M(t+s)),$$

when the displayed map is defined. This is a finite observation, not by itself a complete detector certificate.

**Example 2.84** (Arbitrarily small perturbation can change a pointwise detector rank). Fix  $s > 0$  and  $L > 0$ . For any  $0 < \delta < \min\{L, (s+L)/2\}$ , let

$$B_\delta = \{[0, s+L)\}, \quad C_\delta = \{[\delta, s+L+\delta)\}.$$

Matching the two bars gives

$$d_B(B_\delta, C_\delta) = \delta.$$

Both bars have the same length  $s + L$ , so they have the same clearance from any threshold depending only on that length. Nevertheless, at  $t = 0$ ,

$$r_{0,s}(B_\delta) = 1, \quad r_{0,s}(C_\delta) = 0.$$

Thus pointwise span-rank observations may change under arbitrarily small bottleneck perturbations, even without any bar crossing a length threshold.

**Proposition 2.85** (No bottleneck stability theorem for raw pointwise detector ranks). *There is no general implication of the form*

$$d_B(B, C) \leq \varepsilon \implies r_{t,s}(B) = r_{t,s}(C)$$

*for a fixed sample point  $t$  and span  $s$ , even for one-bar finite barcodes and arbitrarily small positive  $\varepsilon$ .*

*Proof.* Apply Example 2.84 with  $0 < \delta \leq \varepsilon$ . □

**Remark 2.86** (What metric transport preserves). The v21 metric chain preserves the repaired zero/nonzero predicate under the sharp separation hypotheses. It does not preserve the raw output of a fixed-point detector. Birth, germ, coverage, mesh, endpoint, and stage conditions remain Chapter 3 non-scalar obligations and cannot be replaced by metric slack.

### 2.15.2 Overlap comparison handoff

The Overlap Gate has two different metric layers. This distinction is essential in v19.

**Definition 2.87** (Overlap pre-threshold profiles). For two local inputs  $F_\alpha, F_\beta$  and an overlap window

$$W_{\alpha\beta} := W_\alpha \cap W_\beta,$$

define the overlap pre-threshold profiles by

$$B_{i,W_{\alpha\beta}}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad B_{i,W_{\alpha\beta}}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

The overlap repaired profiles are

$$\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) = T_\tau^{\text{bar}}(B_{i,W_{\alpha\beta}}^\alpha), \quad \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) = T_\tau^{\text{bar}}(B_{i,W_{\alpha\beta}}^\beta).$$

**Remark 2.88** (Two-level structure of the Overlap Gate). The metric comparison clause of the Overlap Gate compares the *post-threshold* repaired profiles:

$$d_{\text{int}}(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)).$$

By contrast, the threshold-gap clause is computed on the *pre-threshold* profiles:

$$B_{i,W_{\alpha\beta}}^\alpha, \quad B_{i,W_{\alpha\beta}}^\beta.$$

Thus the gate deliberately uses two barcode levels:

gap is measured before deletion,      agreement is checked after deletion.

**Proposition 2.89** (Overlap transfer under a scalarized threshold-gap certificate). *Let  $\mathfrak{d}_{\alpha\beta,i}$  be the declared overlap metric ledger and set*

$$\varepsilon_{\alpha\beta,i} := \text{val}_B(\mathfrak{d}_{\alpha\beta,i}).$$

*Suppose*

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i}.$$

*If the two-sided threshold-gap record satisfies*

$$\varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta),$$

*then*

$$d_B(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}.$$

*Equivalently, by the bottleneck–interleaving isometry in the declared constructible regime, the same bound may be read using  $d_{\text{int}}$ .*

*Proof.* This is Corollary 2.49 applied to the two overlap pre-threshold profiles and the overlap metric ledger  $\mathfrak{d}_{\alpha\beta,i}$ . The final sentence follows from Remark 2.19.  $\square$

**Corollary 2.90** (Overlap repaired-zero equivalence). *Under the hypotheses of Proposition 2.89, assume in addition*

$$2\varepsilon_{\alpha\beta,i} \leq \tau.$$

*Then*

$$\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) = 0 \iff \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) = 0.$$

*For audit consumption, the strict reserve  $2\varepsilon_{\alpha\beta,i} < \tau$  may be required.*

*Proof.* Proposition 2.89 supplies the repaired-profile bottleneck bound. Both repaired profiles are  $\tau$ -reduced. Apply Theorem 2.53.  $\square$

### 2.15.3 Exact-admissible optional regimes

The general Core does not treat  $T_\tau^{\text{bar}}$  as a global exact functor. Nevertheless, stronger exact calculus may be available under separately declared hypotheses.

**Definition 2.91** ( $\tau$ -split object). An object  $M \in \text{Pers}_k^{\text{cons}}$  is called  $\tau$ -split if it is equipped with a declared decomposition

$$M \cong M_{\leq \tau} \oplus M_{> \tau}$$

such that:

- (i) every finite bar of  $M_{\leq \tau}$  has length at most  $\tau$ ;
- (ii) every finite bar of  $M_{> \tau}$  has length greater than  $\tau$ ;
- (iii) all infinite bars, if any, lie in  $M_{> \tau}$ ;
- (iv) the decomposition is part of the declared data, not merely inferred informally.

**Definition 2.92** ( $\tau$ -split morphism). A morphism

$$f : M \rightarrow N$$

between  $\tau$ -split objects is called  $\tau$ -split if it respects the declared decompositions:

$$f(M_{\leq \tau}) \subseteq N_{\leq \tau}, \quad f(M_{> \tau}) \subseteq N_{> \tau}.$$

**Definition 2.93** ( $\tau$ -split regime). A  $\tau$ -split regime is a declared working subcategory or workflow domain in which all objects and all morphisms under consideration are  $\tau$ -split in the above sense.

**Proposition 2.94** (Functorial deletion in a  $\tau$ -split regime). *Inside a declared  $\tau$ -split regime, the assignment*

$$M \mapsto M_{>\tau}$$

*is functorial and agrees with  $T_\tau^{\text{bar}}(M)$  on barcode profiles.*

*Proof.* Since every morphism respects the declared short/long decomposition, it restricts to a morphism

$$M_{>\tau} \rightarrow N_{>\tau}.$$

Therefore  $M \mapsto M_{>\tau}$  is functorial in the  $\tau$ -split regime. By construction,  $M_{>\tau}$  consists exactly of the bars retained by  $T_\tau^{\text{bar}}$ .  $\square$

**Definition 2.95** (Exact-admissible collapse regime). An exact-admissible collapse regime is a declared  $\tau$ -split regime in which the retained-long-bar assignment

$$M \mapsto M_{>\tau}$$

is additionally verified to preserve the exact sequences used in the relevant Core argument.

**Remark 2.96** (Exactness is conditional). Exactness of positive-threshold collapse is not a global Core theorem. It is a conditional property available only in an exact-admissible regime. Whenever such a regime is used, the proof object must declare:

- (i) the  $\tau$ -split decomposition policy;
- (ii) the morphism compatibility policy;
- (iii) the exact sequences being tested;
- (iv) the artifact or proof establishing exact preservation.

#### 2.15.4 Filtered representative handoff

Because  $T_\tau^{\text{bar}}$  is treated in the general Core as a barcode-level operation, filtered-level collapse must be phrased in terms of representatives.

**Definition 2.97** (Windowed collapsed persistence profile). For  $F \in \text{FiltCh}(k)$ , monitored degree  $i$ , window  $W$ , and threshold  $\tau$ , define the monitored windowed collapsed profile by

$$\mathcal{P}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

**Definition 2.98** (Admissible filtered representative). Let  $F \in \text{FiltCh}(k)$ , let  $W$  be a declared window, and let  $\tau > 0$ . A filtered object

$$C_{\tau,W}F$$

is called an admissible filtered representative of the windowed collapse of  $F$  if, for every monitored degree  $i$ ,

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathcal{P}_{i,W,\tau}(F)$$

at the persistence barcode level, up to the declared representative equivalence policy.



**Remark 2.99** (No global strict lift by default). The general Core does not assert a strict global endofunctor

$$C_\tau : \text{FiltCh}(k) \rightarrow \text{FiltCh}(k)$$

realizing positive-threshold collapse in every degree and on every morphism. The default object is the windowed representative  $C_{\tau,W}F$ , whose existence and properties must be declared in the proof object or implementation record.

**Definition 2.100** (Representative equivalence). Two admissible representatives  $C_{\tau,W}F$  and  $C'_{\tau,W}F$  are considered equivalent for Core gate purposes if their monitored persistence profiles agree:

$$\mathbf{P}_i(C_{\tau,W}F) \cong \mathbf{P}_i(C'_{\tau,W}F)$$

for all monitored degrees  $i$ , under the declared barcode equivalence policy.

**Remark 2.101** (Filtered quasi-isomorphism). When filtered quasi-isomorphism data are available, representative equivalence may be strengthened to filtered quasi-isomorphism. However, filtered quasi-isomorphism is not required merely to define the windowed collapsed barcode profile. It becomes necessary only when the argument uses filtered-level or derived-level constructions.

### 2.15.5 B-Gate input handoff

The inherited repaired calculus affects how the Core gate is read.

**Definition 2.102** (Windowed  $B$ -side input). The  $B$ -side input for a monitored degree  $i$ , window  $W$ , and threshold  $\tau$  is the collapsed profile

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)),$$

or an admissible filtered representative  $C_{\tau,W}F$  realizing this profile.

**Definition 2.103** (Windowed topological clause). The topological clause of  $B\text{-Gate}^+$  on  $(F; W, \tau)$  is read as

$$\text{Eval}_{1,W,\tau}(F) = 0$$

at the monitored persistence-profile level. Equivalently, if an admissible representative  $C_{\tau,W}F$  is supplied and realizes the declared profile, the clause may be read as

$$\text{PH}_1(C_{\tau,W}F) = 0.$$

**Definition 2.104** (Windowed categorical clause). The categorical clause of  $B\text{-Gate}^+$  may be invoked only when an admissible filtered representative  $C_{\tau,W}F$  is supplied and the eligible realization condition holds:

$$R(C_{\tau,W}F) \in D^{[-1,0]}(k\text{-mod}).$$

Only then is the Ext-clause

$$\text{Ext}^1(R(C_{\tau,W}F), k) = 0$$

well-typed for the Core bridge.

**Definition 2.105** (Comparison-typed  $B\text{-Gate}^+$  input). A  $B\text{-Gate}^+$  input records:

- (i) the post-threshold repaired profile  $\text{Eval}_{1,W,\tau}(F)$ ;
- (ii) the pre-threshold profile  $B_{1,W}(F)$ ;

- (iii) the comparison mode for every comparison involving that profile;
- (iv) for comparison mode `metric_gap_certified`, the two compared pre-threshold profiles, their two-sided threshold-gap comparison record, and every ledger component needed to justify the actual bottleneck bound, including any crop budget;
- (v) for comparison mode `exact`, the equality or profile-isomorphism proof and its artifact reference.

A single-profile clearance value is readiness information only and is not substituted for a two-sided metric comparison record.

**Remark 2.106** (PH before Ext). The barcode-level profile  $\text{Eval}_{1,W,\tau}(F)$  is the primary persistence datum. The Ext-clause is not a replacement for it. Ext is read only through an eligible representative and the one-way bridge established later.

### 2.15.6 Tower and Type IV input handoff

Since  $T_\tau^{\text{bar}}$  is not a global strict exact functor in the general Core, tower comparison maps must be declared rather than assumed. The full terminal-generic diagnostic is defined in Chapter 5; the present section fixes only the well-typed input contract.

**Definition 2.107** (Windowed collapsed tower objects). Let

$$F_0 \rightarrow F_1 \rightarrow F_2 \rightarrow \cdots$$

be a directed tower in the declared filtered or persistence regime. For a monitored degree  $i$ , window  $W$ , and threshold  $\tau$ , write

$$M_{n;i,W,\tau} := \text{Eval}_{i,W,\tau}(F_n).$$

The sequence of objects  $M_{n;i,W,\tau}$  is not by itself a directed system: actual profile morphisms and their composition laws must be declared, unless supplied by a  $\tau$ -split or exact-admissible regime.

**Definition 2.108** (Tower comparison artifact). A tower comparison artifact for  $(i, W, \tau)$  is declared data specifying:

- (i) a repaired tower-profile system with actual persistence-profile morphisms;
- (ii) a source profile  $\text{ColimProfile}_{i,W,\tau}(F_\bullet)$  with declared provenance;
- (iii) an apex profile  $\text{ApexProfile}_{i,W,\tau}(F_\infty)$ ;
- (iv) an actual comparison morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty);$$

- (v) the typing, kernel/cokernel, terminal-tameness, and artifact data required by Chapter 5.

A sequence of barcode multisets or bottleneck matchings without actual profile morphisms is not a tower comparison artifact.

**Remark 2.109** (No automatic tower comparison). The comparison morphism  $\phi_{i,W,\tau}$  is not produced automatically by hard threshold deletion, by a barcode matching, or by the existence of a source tower and an apex object. It must be explicitly constructed and typed, or supplied canonically by a declared exact-admissible regime.

**Definition 2.110** (Typed Type IV scope status). Every Chapter 2 run records exactly one Type IV scope status:

`not_invoked`,      `undefined`,      `certified_zero`,      `obstructed`.

They mean respectively:

- (i) no terminal-generic kernel/cokernel comparison diagnostic belongs to the claim; a direct or inverse tower may still be used by a finite-to-infinite, Roos, stable-image, or apex-agreement lane;
- (ii) a terminal-generic Type IV diagnostic is invoked, but the artifact or a required terminal-generic input is missing or ill-typed;
- (iii) a valid artifact exists and the Chapter 5 terminal-generic kernel/cokernel diagnostic is zero;
- (iv) a valid artifact exists and that diagnostic is nonzero.

A missing required Type IV artifact is `undefined`, never a numerical zero and never `not_invoked`. The latter status is available only when the scope policy excludes the terminal-generic comparison diagnostic itself.

**Definition 2.111** (Collapse diagnostics notation contract). For a valid tower comparison artifact whose kernel and cokernel defect profiles are terminal-tame on  $W$ , Chapter 5 defines

$$\mu_{i,W,\tau} := \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} := \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

On a right-unbounded window,  $\text{tgdim}_W$  is the eventual generic dimension and counts defect bars cofinal to  $+\infty$ . On a bounded window, it is the left limit of the dimension at the right endpoint and counts defect bars cofinal to that endpoint; it must not be replaced by the dimension after zero extension to  $+\infty$ . For a fixed declared pair  $(W, \tau)$  and a declared finite monitored degree set  $I_{\text{mon}}$ , define

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad \nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

An infinite monitored scope requires a separately declared aggregation or summability theorem. Deprecated Latin- $u$  diagnostic notation is confined to migration metadata. The symbols  $\nu_{i,W,\tau}$  and  $\nu_{\text{Collapse}}$  remain the sole canonical cokernel diagnostics in v21.0.

**Remark 2.112** (Type IV remains Core, but zero is typed). Type IV remains part of the Core gate discipline, but its comparison map is not silently inherited from threshold deletion. The shorthand

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

may be used only under status `certified_zero`. Under `obstructed`, a valid diagnostic is nonzero. Under `undefined` or `not_invoked`, no numerical pair is assigned. In particular, missing or excluded tower data are never represented by a fabricated zero pair. The zero pair means absence of monitored terminal-generic kernel and cokernel defect relative to the declared artifact; it does not imply that the comparison morphism is an isomorphism on finite interior bars.

**Remark 2.113** (Object-distance control does not automatically control defect profiles). A bottleneck bound between the source and target objects of two comparison morphisms does not, by itself, produce bottleneck bounds between their kernels or cokernels. Metric transport of a detector on `kernel_defect` or `cokernel_defect` therefore requires actual comparison morphisms between the defect profiles, or a separately proved kernel/cokernel stability theorem with the required commutative data. Object proximity cannot manufacture Type IV or transient-defect evidence.

### 2.15.7 Manifest handoff

A certified proof object using Chapter 2 must record the following data.

**Definition 2.114** (Collapse policy manifest entry). A collapse policy manifest entry contains:

- (i) the threshold  $\tau$ , monitored degrees  $i$ , and window  $W$ ;
- (ii) the window type: bounded audit window or terminal audit window;
- (iii) the evaluation order
 
$$\mathbf{P}_i \rightarrow W_W \mathbf{P}_i \rightarrow T_\tau^{\text{bar}}(W_W \mathbf{P}_i);$$
- (iv) the collapse status
 
$$\text{barcode\_level}, \quad \text{tau\_split}, \quad \text{exact\_admissible};$$
- (v) the repaired evaluation record and comparison mode for every comparison;
- (vi) for each metric discrepancy transported through hard deletion, the two-sided threshold-gap comparison record; for each exact comparison, the exact equality or profile-isomorphism proof;
- (vii) for every metric transport of a repaired zero predicate, the post-threshold zero-separation record, including the bound  $2\varepsilon \leq \tau$  or the stricter operational reserve;
- (viii) for every detector-derived zero statement, the complete Chapter 3 detector certificate and, when transported, the metric-safe detector-zero transport record;
- (ix) the metric ledger  $\mathfrak{d}_{\text{met}}$ , bottleneck scalarization  $\text{val}_B$ , and scalarized budget  $\varepsilon_B$  whenever metric mode is invoked;
- (x) window-edge clearance or crop-budget data when source-level perturbations are transported through window restriction;
- (xi) any  $\tau$ -window-adapted commutation certificate, including explicit verification of globally retained infinite bars on bounded windows;
- (xii) whether an admissible filtered representative  $C_{\tau, W} F$  is supplied;
- (xiii) the typed Type IV status and, if invoked, the tower comparison artifact, terminal-generic inputs, monitored degree set, and canonical  $\mu/\nu$  diagnostics;
- (xiv) the repair record documenting the replacement of global non-expansiveness by threshold-gap-safe stability, the repair of the former finite-only crop condition, and the migration from legacy  $u$  to canonical  $\nu$  notation.

**Definition 2.115** (Metric and crop repair record). A Chapter 2 repair record contains:

- (i) the deprecated claim being avoided: global bottleneck non-expansiveness of  $T_\tau^{\text{bar}}$ ;
- (ii) the replacement theorem: Theorem 2.46;
- (iii) the counterexample reference: Example 2.35;
- (iv) the relevant pre-threshold profiles and two-sided threshold-gap certificate;

(v) the metric ledger, scalarization, and typed inequality

$$\text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau;$$

(vi) any crop budget required to obtain the windowed pre-threshold profile comparison;

(vii) if crop–collapse commutation is used, the exact criterion of Proposition 2.65;

(viii) explicit confirmation that globally retained infinite bars were included in that criterion;

(ix) the Type IV scope status and artifact reference when tower semantics are invoked;

(x) every post-threshold zero-separation record consumed by the proof;

(xi) explicit confirmation that no raw pointwise detector rank or certificate-size stability was inferred from bottleneck proximity;

(xii) confirmation that  $\nu_{i,W,\tau}$  and  $\nu_{\text{Collapse}}$  are used canonically and that legacy  $u$ -notation is confined to deprecation metadata.

**Remark 2.116** (Audit consequence). A proof object that simply states “apply  $T_\tau$ ” is incomplete under v21. It must state whether  $T_\tau$  means barcode-level hard threshold deletion, a  $\tau$ -split functorial model, or an exact-admissible collapse regime. If a metric discrepancy is transported through hard deletion, it must cite a two-sided scalarized threshold-gap comparison record rather than global non-expansiveness. If the transported conclusion is a repaired zero predicate, it must additionally cite a post-threshold zero-separation record; threshold-gap stability alone bounds distance but does not identify zero and nonzero profiles. A separately proved exact comparison is recorded in comparison mode `exact` with its proof and is not misrepresented as a positive-radius gap certificate. If crop–collapse commutation is used, it must verify all globally retained bars, including infinite bars clipped by bounded windows. If Type IV is invoked, it must provide an actual comparison morphism and the typed terminal-generic diagnostic input of Chapter 5.

## 2.16 Operating summary

For each monitored object  $F$ , degree  $i$ , window  $W$ , and threshold  $\tau$ , the repaired Chapter 2 pipeline remains

$$F \longmapsto \mathbf{P}_i(F) \longmapsto B_{i,W}(F) = W_W \mathbf{P}_i(F) \longmapsto T_\tau^{\text{bar}}(B_{i,W}(F)) = \text{Eval}_{i,W,\tau}(F).$$

Threshold clearance is measured on  $B_{i,W}(F)$ , not on the repaired profile. The metric-safe repaired-zero chain is

$$\begin{aligned} \mathbf{d}_{\text{met}} &\xrightarrow{\text{val}_B} \varepsilon_B \geq d_B(B, C), & \varepsilon_B < \text{gap}_\tau(B, C) \\ &\Downarrow \\ &d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B \\ &\Downarrow \quad 2\varepsilon_B \leq \tau \\ &T_\tau^{\text{bar}} B = 0 \iff T_\tau^{\text{bar}} C = 0. \end{aligned}$$

The mathematically sharp zero-separation boundary is  $2\varepsilon_B \leq \tau$ ; the v21 operational contract uses the strict reserve  $2\varepsilon_B < \tau$ .

A complete detector may establish the source zero predicate. Chapter 2 then transports that predicate through the displayed metric chain. It does not transport raw detector ranks, sample values, birth/germ sets, coverage, certificate size, or completeness. Those remain Chapter 3 obligations.

The default evaluation order is window first and threshold second. Crop–collapse commutation is available only under Proposition 2.65, including the bounded-window treatment of globally infinite bars. Source-level perturbations require crop and window-edge control before they become bounds on the actual pre-threshold profiles.

Filtered or derived arguments require an admissible representative  $C_{\tau, W}F$ . Tower and transient-defect arguments require actual comparison morphisms. Metric proximity of tower objects does not automatically control kernel or cokernel defect profiles. Exactness or strict functoriality requires a declared  $\tau$ -split or exact-admissible regime.

## 2.17 Explicit non-claims

The following are not asserted in this chapter.

- (i) No P2DG readiness record is a finite-diagram realization theorem.
- (ii) No barcode or persistence discrepancy is silently retyped as a filtered-chain, finite-diagram, or derived discrepancy.
- (iii) No theorem-controlled nonmetric comparison is represented as exact or metric-gap-certified unless a separate theorem constructs the required target record.
- (iv) No finite critical grid proves completeness, faithfulness, or reconstruction by finiteness alone.
- (v) No object, morphism, cone, descent datum, or global object of  $\text{Core}_{\text{DG}}$  is constructed in this chapter.
- (vi) No global non-expansiveness of hard positive-threshold deletion under the standard bottleneck metric is asserted.
- (vii) No claim is made that positive-threshold short-bar objects form a Serre subcategory in general.
- (viii) No claim is made that  $T_{\tau}^{\text{bar}}$  is a global exact endofunctor on all of  $\text{Pers}_k^{\text{cons}}$ .
- (ix) No claim is made that  $T_{\tau}^{\text{bar}}$  is a global abelian reflector or Serre localization.
- (x) No unconditional commutation
$$W_W T_{\tau}^{\text{bar}} = T_{\tau}^{\text{bar}} W_W$$
is asserted.
- (xi) No finite-only window-adapted condition is sufficient: globally retained infinite bars must also be checked after clipping.
- (xii) No raw source-level perturbation implies a windowed pre-threshold profile bound without a crop-stability argument or crop budget.
- (xiii) No unscalarized quantale element is compared directly with the real-valued clearance  $\text{gap}_{\tau} = \text{clear}_{\tau}$ .
- (xiv) No numerical budget repairs a missing representative, comparison morphism, eligibility certificate, terminal-tameness certificate, claim status, or manifest record.
- (xv) No threshold-gap certificate is one-sided.
- (xvi) No single-profile clearance or gap-readiness record is treated as a two-sided metric comparison certificate.

- (xvii) No finite bar of length exactly  $\tau$  is treated as gap-safe at threshold  $\tau$ .
- (xviii) No  $2\varepsilon$ -band is claimed to be sharp for the two-sided theorem, and no factor convention  $\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$  is used.
- (xix) No global strict filtered lift  $C_\tau : \text{FiltCh}(k) \rightarrow \text{FiltCh}(k)$  is asserted by default.
- (xx) No tower comparison map after threshold deletion is automatic unless an actual artifact or exact-admissible regime is declared.
- (xxi) No barcode matching by itself defines the kernel or cokernel of a tower comparison.
- (xxii) No undefined Type IV diagnostic is interpreted as zero, and no no-tower claim fabricates a numerical zero instead of recording `not_invoked`.
- (xxiii) No equality  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$  implies full comparison-map isomorphism without a no-interior-defect theorem.
- (xxiv) No deprecated Latin-*u* cokernel notation has normative diagnostic meaning in v21.0.
- (xxv) No bounded-window cofinal bar is called a global infinite bar.
- (xxvi) No Ext-clause is well-typed without an admissible filtered representative and an eligible realization condition.
- (xxvii) No Search artifact, advisory invariant, spectral indicator, or AI-generated ranking replaces the windowed collapsed persistence evaluation or any typed proof obligation.
- (xxviii) No repaired-zero equivalence is inferred from threshold-gap stability alone; the zero-separation inequality is an additional theorem hypothesis.
- (xxix) No metric zero transport is asserted when  $2\varepsilon > \tau$ ; counterexamples exist even inside the  $\tau$ -reduced image.
- (xxx) No raw pointwise detector rank, birth set, germ set, sample count, coverage witness, or certificate size is bottleneck-stable by default.
- (xxxi) No detector soundness statement is promoted to completeness by metric proximity or by repeated zero observations.
- (xxxii) No metric-safe detector transport changes the detector profile stage; a stage-transport theorem is required.
- (xxxiii) No object-level bottleneck bound automatically gives a bound on kernel or cokernel defect profiles.
- (xxxiv) No finite detector certificate supplied by Chapter 3 is proved in this chapter merely because its zero predicate can be transported metrically.

## 2.18 Provenance, migration, and status

The constructible one-parameter persistence setting, interval decomposition, barcode uniqueness, bar-length convention, and bottleneck/interleaving isometry are standard persistence-theoretic inputs at their displayed hypotheses.

The complete v20 Chapter 2 and Foundation Appendix A metric/crop package is inherited by v21 without change of theorem strength. This includes the negative Serre and global-nonexpansiveness results, sharp two-sided threshold-gap theorem, exact all-retained-bars crop criterion, distance-to-zero formula,  $\tau$ -reduced zero separation, and repaired-zero transport.

The genuinely new v21 material in this chapter is architectural rather than a stronger persistence theorem and is synchronized with the Chapter 1 four-mode comparison constitution:

- (i) the four-mode comparison calculus, including `theorem_controlled_nonmetric`;
- (ii) the typed discrepancy-level separation theorem;
- (iii) the P2DG input and readiness contract;
- (iv) the explicit prohibition on constructing a `CoreDG` object inside Chapter 2.

These additions do not strengthen the inherited bottleneck, threshold-gap, zero-separation, or crop constants.

**Inherited Core-P mathematical theorems.** Theorem 2.11, Theorem 2.46, Theorem 2.53, Theorem 2.56, and their displayed corollaries and correction results.

**New v21 comparison and Interface guard-rails.** Theorem 2.69 is a comparison-evidence separation theorem; Theorem 2.74 is a level-change guard-rail; and Theorem 2.79 is a Core Interface theorem. None strengthens the inherited persistence-metric constants.

**Operational and Interface records.** Policy 23, Definition 2.67, Definition 2.68, Definition 2.70, Definition 2.77, and Definition 2.78.

**Explicit exclusions.** Subsection 2.17.



### 3 Chapter 3: Finite Detector Calculus and Certificate Complexity

**Chapter 3 v21 contract.** This chapter is the finite-certificate and certificate-complexity layer of Corep. It inherits, at unchanged mathematical strength, the v20 birth, right-germ, finite-stencil, family-mesh, metric-zero-composition, and right-unbounded no-go results. It separates a finite observation from a finite proof certificate by requiring a declared certificate model

$$\mathfrak{M} = (I, Q, \mathcal{E}, \mathcal{V}),$$

where  $I$  specifies available information,  $Q$  the permitted access pattern,  $\mathcal{E}$  the representation and cost accounting, and  $\mathcal{V}$  the verifier's computational power and acceptance semantics.

The standard persistence detector reads a declared windowed pre-threshold profile

$$B_{n,w}(F) = W_w \mathbf{P}_n(F)$$

through finitely many structure maps

$$B(t) \longrightarrow B(t + \tau).$$

A finite observation becomes a finite proof certificate only after its source stage, endpoint convention, soundness theorem, completeness or coverage theorem, family membership, query transcript, encoding, verifier, cost vector, and immutable artifacts are certified. Missing completeness evidence yields an undefined zero verdict, never a fabricated vanishing theorem.

The v21 detector-stage taxonomy is

$$\begin{array}{ll} \text{windowed\_prethreshold,} & \text{repaired\_postthreshold,} \\ \text{kernel\_defect,} & \text{cokernel\_defect,} \\ \text{diagram\_probe,} & \text{descent\_coherence.} \end{array}$$

The last two stages are Interface stages. They do not convert the barcode stencil theorems into diagram or descent theorems. A diagram probe requires a Chapter 4 or Chapter 7 object and a proved zero-reflecting probe family; a descent-coherence probe requires the actual local objects, overlap morphisms, and coherence cells of Chapter 7.

Certificate cost is not collapsed to one scalar. An object-specific certificate record has a cost vector

$$\text{cost}^{\mathfrak{M}}(C_X) = (q, b, t, a) \in \overline{\mathbb{N}}^4.$$

A family-relative certificate scheme  $\Pi$  supplies such a record for every admissible input and is charged according to the aggregation rule fixed by the model; the default deterministic exact rule is coordinatewise worst-case cost over the family. The family-relative quantity  $\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta)$  is the Pareto frontier of finite achievable scheme-cost vectors, not a silently chosen scalar.

This chapter proves nontrivial upper bounds from exhaustive births, augmented meshes, and certified finite probe families; lower bounds from information-model indistinguishability and omitted private witness cells; and finite composition inequalities under a declared compositional verifier. No unrestricted global inequality obtained by summing local  $\text{MinCert}$ -values is asserted here. Such a theorem requires the fixed encoding, coherence, descent, and reuse hypotheses of Chapter 7.

The chapter does not prove a higher categorical bridge

$$\text{PH}_n \Longrightarrow \text{Ext}^n,$$

does not by itself construct a  $\text{Core}_{\text{DG}}$  object and does not infer object-level globalization from certificate composition. Chapter 4 receives the typed representative handoff and owns endpoint-complete finite linear  $A_{[m]}$ -diagram reconstruction (the standard Dynkin quiver  $A_{m+1}$ ). Chapter 7 owns arbitrary finite categories, branching diagrams, general generators and cones, and effective descent; Chapter 8 owns canonical verification and manifest semantics.

### 3.1 Scope and standing discipline

All statements in this chapter are made in the v21 conservative-evolution regime fixed by Chapters 1–2 and Appendix A. The repaired v19 metric calculus and the complete v20 finite-detector theorem pack are inherited without weakening. The new v21 material is model-relative certificate complexity and the typed Interface extension to diagram and descent probes.

In particular:

- (i) persistence data live in constructible one-parameter persistence over a field  $k$ ;
- (ii) barcode-level threshold deletion  $T_{\tau}^{\text{bar}}$  deletes precisely finite bars of length  $\leq \tau$  and preserves infinite bars before bounded-window clipping;
- (iii) global non-expansiveness of  $T_{\tau}^{\text{bar}}$  under the standard bottleneck metric is not asserted;
- (iv) the replacement stability principle is sharp two-sided threshold-gap-safe stability with closed band
 
$$[\tau - \varepsilon, \tau + \varepsilon];$$
- (v) the ledger margin is
 
$$\text{gap}_{\tau} = \text{clear}_{\tau},$$
 with  $\text{gap}_{\tau}$  retained as the ledger-facing alias of  $\text{clear}_{\tau}$ ;
- (vi) right-open windows  $W = [u, u')$  are part of the certified data;
- (vii) no unconditional crop–collapse commutation is used;
- (viii) a filtered object  $C_{\tau, W}F$  is an admissible representative of a repaired windowed barcode profile, not a global collapse functor;
- (ix) the metric ledger lives in a declared commutative quantale

$$(\mathbf{V}, \oplus, 0_{\mathbf{V}}, \preceq),$$

and is compared with real-valued clearance only after application of a declared monotone subadditive bottleneck scalarization  $\text{val}_{\mathbf{B}}$ ;

- (x) no numerical budget repairs a missing representative, realization, comparison morphism, eligibility certificate, claim-status entry, manifest field, or artifact identity;
- (xi) if crop–collapse commutation is invoked, the Chapter 2 / Appendix A condition is checked on every globally retained bar, including infinite bars clipped by bounded windows;
- (xii) each comparison is explicitly typed as exactly one of

`exact,      metric_gap_certified,`  
`theorem_controlled_nonmetric,      not_compared.`

- (xiii) every detector declares exactly one profile stage and may be consumed only at the strength proved for that stage;
- (xiv) exact point detectors use a declared left-closed right-open barcode convention; decorated endpoints use the right-germ theorem or another separately proved decoration-safe theorem;
- (xv) detector soundness and detector completeness are separate mathematical obligations, while Chapter 1 governs their joint admission as a Core zero certificate;
- (xvi) family-mesh completeness uses an augmented mesh containing both boundary gaps of the effective detector domain;
- (xvii) a family-wide threshold-excess lower bound must come from an independently declared model, separation theorem, filtration rule, or external artifact rather than retrospective full reading of the target barcode when proof compression is claimed.

**Remark 3.1** (Why Chapter 3 moved before the Ext-edge). Earlier drafts placed eligibility and the Ext-edge in Chapter 3. The v19 metric repair made the record discipline logically prior. The v20 extension adds a second logically prior layer: one must know whether a finite detector is sound, complete, stage-correct, endpoint-correct, and artifact-backed before its vanishing predicate is passed to a higher edge. For this reason all eligible realization and Ext-edge statements remain deferred to Chapter 4.

### 3.2 Pre-threshold windowed profiles

**Definition 3.2** (Pre-threshold windowed profile). Let  $F \in \text{FiltCh}(k)$ , let  $i$  be a monitored degree, and let

$$W = [u, u')$$

be a declared right-open window. The *pre-threshold windowed profile* is the barcode profile

$$B_{i,W}(F) := W_W \mathbf{P}_i(F).$$

It is the object on which threshold clearance is measured and to which any metric discrepancy consumed by hard deletion must apply.

**Definition 3.3** (Repaired windowed evaluation). For a threshold  $\tau > 0$ , the repaired windowed evaluation is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

**Remark 3.4** (The two profiles are different). The pre-threshold profile  $B_{i,W}(F)$  and the repaired evaluation  $\text{Eval}_{i,W,\tau}(F)$  are not interchangeable. The former is used for threshold clearance and metric-gap records. The latter is used for post-collapse gate clauses and repaired-profile comparison. Thus a valid record keeps both:

$$\text{Pre}_{i,W}(F) := B_{i,W}(F), \quad \text{Post}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F).$$

**Remark 3.5** (Fixed order). The Core order is

$$F \mapsto \mathbf{P}_i(F) \mapsto W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

No identity

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is used without a separately declared window-adapted or exact-admissible hypothesis. The window-adapted condition is the necessary-and-sufficient condition of Chapter 2 and Appendix A: every globally retained bar, including an infinite bar, must be disjoint from the window or remain retained after nonempty clipping. A finite-only check is insufficient.

### 3.3 Threshold clearance and ledger gap

**Definition 3.6** (Threshold clearance on a windowed profile). Let  $B$  be a barcode profile with finitely many bars in the monitored record, possibly including infinite bars. For  $\tau > 0$ , define

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|,$$

where  $B_{\text{fin}}$  denotes the finite bars of  $B$ . If  $B_{\text{fin}} = \emptyset$ , set

$$\text{clear}_\tau(B) := +\infty.$$

For an AK evaluation record, write

$$\text{clear}_\tau(F; i, W) := \text{clear}_\tau(B_{i,W}(F)).$$

**Definition 3.7** (Ledger gap). The ledger gap is the clearance itself:

$$\text{gap}_\tau(B) := \text{clear}_\tau(B).$$

For a finite family of monitored pre-threshold profiles  $B_1, \dots, B_m$ , define

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_{1 \leq j \leq m} \text{clear}_\tau(B_j).$$

Thus  $\text{gap}_\tau$  is not a half-clearance in v19.0.0; it is a ledger-facing alias for the actual threshold clearance.

**Remark 3.8** (Why  $\text{gap}_\tau = \text{clear}_\tau$ ). The sharp two-sided threshold-gap-safe theorem uses the closed band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Let  $\mathfrak{d}_{\text{met}} \in \mathbb{V}$  be a gate-scoped metric ledger and let

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

The strict readiness condition

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(B) = \text{clear}_\tau(B)$$

ensures that no finite bar of  $B$  lies in

$$[\tau - \varepsilon_B(\mathfrak{d}_{\text{met}}), \tau + \varepsilon_B(\mathfrak{d}_{\text{met}})].$$

For metric transport, the same budget must also upper-bound the bottleneck discrepancy between two declared profiles, and the condition must hold on both. The shorthand  $\Sigma\delta < \text{gap}_\tau$  is permitted only when the run manifest identifies  $\Sigma\delta$  with this scalarized real budget and records the discrepancy theorem or artifact.

**Remark 3.9** (Bars of length exactly  $\tau$ ). If  $B$  contains a finite bar of length exactly  $\tau$ , then

$$\text{clear}_\tau(B) = \text{gap}_\tau(B) = 0.$$

Such a profile is not positively gap-ready. It may still be evaluated by  $T_\tau^{\text{bar}}$ , and the bar is deleted because the convention is  $\ell(I) \leq \tau$ . A separately proved exact identity may still be used in exact mode, but it is not a positive-radius robustness certificate.

### 3.4 Typed scalarized metric ledger

**Definition 3.10** (Bottleneck scalarization and metric ledger). Let

$$(V, \oplus, 0_V, \preceq)$$

be the declared commutative ledger quantale used by the run. A *bottleneck scalarization* is a monotone subadditive map

$$\text{val}_B : V \longrightarrow [0, +\infty]$$

satisfying

$$\text{val}_B(0_V) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

For a gate-scoped metric ledger  $\mathfrak{d}_{\text{met}} \in V$ , define its real bottleneck budget by

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

A scalarized ledger may be consumed only when a theorem, verified computation, or reconstructible artifact proves the relevant pre-threshold profile bound

$$d_B(B, C) \leq \varepsilon_B(\mathfrak{d}_{\text{met}}).$$

Monotonicity is understood relative to  $\preceq$  on  $V$  and the usual order on  $[0, +\infty]$ .

**Definition 3.11** (Non-scalar record obligations). The following are non-scalar proof obligations whenever they are in scope: representative existence, profile-isomorphism data, realization existence and compatibility, actual comparison morphisms, terminal-tameness certificates, eligibility, claim-status entries, manifest completeness and consistency, and artifact provenance and identity. They are not elements of  $V$  and cannot be discharged by increasing  $\varepsilon_B(\mathfrak{d}_{\text{met}})$ .

**Remark 3.12** (Sharp band versus scalarization tightness). The adjective *sharp* refers to the optimal half-width  $\varepsilon$  of the forbidden threshold band for a given true bottleneck bound. It does not assert that a chosen scalarization  $\text{val}_B$  is numerically tight. A conservative scalarization may enlarge the certified band, but it does not change the sharp constant in the metric theorem.

### 3.5 Repaired evaluation and comparison records

**Definition 3.13** (Repaired evaluation record). A *repaired evaluation record* for  $(F; i, W, \tau)$  contains at least:

- (i) an immutable identifier for the filtered object or persistence source  $F$ ;
- (ii) the monitored degree  $i$ ;
- (iii) the right-open window  $W = [u, u')$ , including whether  $u' = +\infty$ ;
- (iv) the threshold  $\tau > 0$ ;
- (v) the pre-threshold profile

$$B_{i,W}(F) = W_W \mathbf{P}_i(F);$$

- (vi) the repaired post-threshold profile

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F));$$

- (vii) the finite-bar deletion convention  $\ell(I) \leq \tau$ ;
- (viii) the infinite-bar and endpoint conventions used by the barcode metric;
- (ix) artifact references sufficient to reconstruct the data.

**Remark 3.14** (Definition, not certification). A repaired evaluation record is only a typed record. It does not by itself certify stability, transport, overlap compatibility, representative compatibility, or B-Gate<sup>+</sup> pass. Those uses require a two-sided metric comparison record, a separately proved exact comparison, and every applicable non-scalar obligation.

**Definition 3.15** (Single-profile threshold-gap readiness record). A *single-profile threshold-gap readiness record* for  $(F; i, W, \tau)$ , relative to a real test radius  $\varepsilon_B \geq 0$ , contains:

- (i) the immutable identifier of  $B_{i,W}(F)$ ;
- (ii) the clearance

$$\text{clear}_\tau(B_{i,W}(F));$$

- (iii) the equality

$$\text{gap}_\tau(B_{i,W}(F)) = \text{clear}_\tau(B_{i,W}(F));$$

- (iv) the strict check

$$\varepsilon_B < \text{gap}_\tau(B_{i,W}(F));$$

- (v) the resulting closed-band exclusion

$$\ell(I) \notin [\tau - \varepsilon_B, \tau + \varepsilon_B]$$

for every finite bar  $I \in B_{i,W}(F)$ ;

- (vi) reconstructible evidence for the finite-bar summary and clearance.

For  $\varepsilon_B = 0$ , the strict check means  $\text{clear}_\tau(B_{i,W}(F)) > 0$ .

This record certifies only that one profile avoids the threshold band at a declared radius. It is *not* a metric comparison certificate, does not supply a second profile, and does not establish transport through hard deletion. The historical label `def:ch3-gap-certified-record` is retained for reference stability, but the normative role is readiness rather than comparison certification.

**Definition 3.16** (Comparison record and two-sided metric certificate). Let  $F$  and  $G$  be two filtered objects or certified profile sources. A comparison record on  $(i, W, \tau)$  declares exactly one mode:

$$\begin{array}{ll} \text{exact,} & \text{metric\_gap\_certified,} \\ \text{theorem\_controlled\_nonmetric,} & \text{not\_compared.} \end{array}$$

In mode `exact`, the record contains a separately proved equality or profile isomorphism between the relevant pre-threshold or repaired profiles, its exact source and target, and reconstructible proof or artifact evidence. No positive-radius robustness follows merely from exactness.

In mode `theorem_controlled_nonmetric`, the record contains the exact theorem identity, source and target, transported predicate or relation, direction, hypotheses, naturality scope, artifacts, and prohibited stronger readings. This mode supplies neither a bottleneck radius nor an actual profile isomorphism unless the cited theorem explicitly proves it.

In mode `metric_gap_certified`, the record is a two-sided threshold-gap comparison certificate and contains:

(i) immutable identifiers for

$$B := B_{i,W}(F), \quad C := B_{i,W}(G);$$

(ii) one metric ledger  $\mathfrak{d}_{\text{met}}$ , a declared scalarization  $\text{val}_B$ , and

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}});$$

(iii) theorem-level, verified-computation, or artifact evidence that

$$d_B(B, C) \leq \varepsilon_B;$$

(iv)  $\text{clear}_\tau(B)$ ,  $\text{clear}_\tau(C)$ , and

$$\text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\};$$

(v) the strict two-sided reserve

$$\varepsilon_B < \text{gap}_\tau(B, C),$$

equivalently confirmation that neither finite-bar family meets  $[\tau - \varepsilon_B, \tau + \varepsilon_B]$ ;

(vi) every declared metric component used to establish the windowed comparison bound, including crop, endpoint, measurement, realization-metric, computation, and threshold-reconciliation terms;

(vii) reconstructible artifact references;

(viii) one specialized status

$$\text{gap\_certified}, \quad \text{gap\_failed}, \quad \text{gap\_pending},$$

mapping respectively to pass, reject, and undefined.

Here `gap_failed` means that a well-typed required gap condition is proved or artifact-evidenced to fail. Missing, incomplete, stale, or pending evidence is recorded as `gap_pending`, never as `gap_failed`.

The metric certificate is two-sided by definition. Two single-profile readiness records without the evidenced discrepancy bound do not constitute a comparison certificate. In mode `not_compared`, no comparison inference is permitted. No mode is upgraded to another without a separate theorem producing the target mode's required data.

**Proposition 3.17** (Gap-certified stability transfer). *A comparison record on  $(F, G; i, W, \tau)$  whose mode is `metric_gap_certified`, with*

$$\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}}),$$

*implies*

$$d_B(\text{Eval}_{i,W,\tau}(F), \text{Eval}_{i,W,\tau}(G)) \leq \varepsilon_B.$$

*Proof.* Apply the sharp threshold-gap-safe stability theorem of Chapter 2 and Appendix A to the pre-threshold profiles

$$B_{i,W}(F), \quad B_{i,W}(G).$$

The comparison record supplies the real bottleneck bound and the two-sided closed-band exclusion at that same budget. The theorem therefore yields the asserted post-threshold bound.  $\square$

**Remark 3.18** (Isometry convention). In the constructible one-parameter regime,  $d_B$  and  $d_{\text{int}}$  are identified by the barcode isometry theorem under the same infinite-bar cost convention fixed in Chapter 2 and Appendix A. Infinite bars are compared by birth-time displacement and are not finitely matched to finite bars or to the diagonal. A certificate may record the comparison using  $d_B$  or  $d_{\text{int}}$  only when the modules and barcodes are precisely those named in the comparison record and the metric convention is declared.

### 3.6 Crop budget and window-edge clearance

**Definition 3.19** (Window-edge clearance record). A *window-edge clearance record* for a comparison of  $F$  and  $G$  on  $(i, W)$  records the stability of passage from the source-level data to

$$B_{i,W}(F) = W_W \mathbf{P}_i(F), \quad B_{i,W}(G) = W_W \mathbf{P}_i(G)$$

near the window endpoints. It contains:

- (i) the window endpoints  $u, u'$  and the bounded/terminal policy;
- (ii) immutable identifiers for the source and resulting windowed profiles;
- (iii) the relevant bar endpoints or critical parameters near  $u$  and  $u'$ , when required;
- (iv) a declared metric crop component

$$\delta_{\text{crop}}(i, W) \in \mathcal{V};$$

- (v) the theorem, verified computation, or artifact supporting that component and its use in the windowed bottleneck bound.

**Remark 3.20** (Threshold gap does not replace crop control). Threshold clearance controls movement across the length threshold  $\tau$  after windowing. It does not prove that a perturbation of  $F$  and  $G$  produces a controlled perturbation of their windowed profiles. The latter is a crop-stability question and must be supported separately through a crop component or a declared window-edge theorem. This is the Chapter 3 record-level expansion of the crop-budget discipline fixed in Chapter 2 and Appendix A; it is not an independent replacement definition.

**Definition 3.21** (Profile-level metric-ledger decomposition). A profile comparison ledger may be decomposed in  $\mathcal{V}$  as

$$\mathbf{d}_{\text{met}} := \delta_{\text{source,met}} \oplus \delta_{\text{crop}} \oplus \delta_{\text{measurement}} \oplus \delta_{\text{realization,met}} \oplus \delta_{\text{computation}} \oplus \delta_{\text{other,met}}.$$

Every displayed component is a declared *metric discrepancy* component; in particular,  $\delta_{\text{realization,met}}$  measures only an evidenced metric distortion induced by a realization step. Realization existence, compatibility, and eligibility remain non-scalar obligations.

The real bottleneck budget is

$$\varepsilon_B := \text{val}_B(\mathbf{d}_{\text{met}}).$$

When all components are nonnegative real upper bounds and  $\oplus$  is ordinary addition, this specializes to the usual sum. The threshold-gap comparison record consumes  $\varepsilon_B$  only after the run establishes

$$d_B(B_{i,W}(F), B_{i,W}(G)) \leq \varepsilon_B.$$

Undeclared or non-metric quantities do not belong to this decomposition.

**Remark 3.22** (No shortcut from source perturbation). A source-level perturbation estimate is not automatically a threshold-gap comparison certificate. The certified chain is

$$\begin{aligned} \text{source comparison} &\implies (\mathbf{d}_{\text{met}}, \text{val}_B) \\ &\implies d_B(B_{i,W}(F), B_{i,W}(G)) \leq \varepsilon_B \\ &\implies d_B(\text{Eval}_{i,W,\tau}(F), \text{Eval}_{i,W,\tau}(G)) \leq \varepsilon_B, \end{aligned}$$

where  $\varepsilon_B = \text{val}_B(\mathbf{d}_{\text{met}})$ , and the last implication additionally consumes the two-sided gap reserve. Each arrow requires its own theorem or artifact.



### 3.7 Infinite bars and window policy

**Definition 3.23** (Infinite-bar policy for windowed evaluation). Infinite bars are treated as follows.

- (i) Before windowing, an infinite bar has length  $+\infty$  for thresholding.
- (ii) Under the standard infinite-bar bottleneck convention,  
 $[a, \infty)$  matches  $[a', \infty)$  with cost  $|a - a'|$ .
- (iii) A finite bar and an infinite bar are not matched at finite bottleneck cost.
- (iv) If  $W = [u, u']$  is bounded, window restriction may clip an infinite bar to a finite bar.
- (v) If  $W = [u, \infty)$ , an infinite bar intersecting  $W$  remains infinite after windowing.

**Remark 3.24** (Where infinite bars affect the gate). Hard finite-threshold deletion never deletes an unwindowed infinite bar. However, bounded windowing may turn it into a finite clipped bar before thresholding. Therefore the infinite-bar convention must be read together with the declared window policy. The Core never lets an unwindowed infinite-bar rule override the actual windowed profile  $B_{i,W}(F)$ . If crop-collapse commutation is claimed, every globally retained infinite bar must be checked after clipping; on a bounded window it may become a finite bar of length at most  $\tau$ , which destroys commutation.

### 3.8 Representative discipline

**Definition 3.25** (Admissible filtered representative). Let  $F \in \text{FiltCh}(k)$ , let  $W$  be a declared window, let  $\tau > 0$ , and let  $I_{\text{mon}}$  be the declared set of monitored degrees. A filtered object

$$C_{\tau,W}F \in \text{FiltCh}(k)$$

is an *admissible filtered representative* of the repaired windowed profile family if, for every  $i \in I_{\text{mon}}$ , it is equipped with a declared profile isomorphism

$$\rho_i : \mathbf{P}_i(C_{\tau,W}F) \xrightarrow{\cong} \text{Eval}_{i,W,\tau}(F).$$

The isomorphisms, their source and target identities, and their equivalence strength are part of the representative data. Profile admissibility alone does not imply chain isomorphism, filtered chain-homotopy equivalence, filtered quasi-isomorphism, or derived equivalence.

**Definition 3.26** (Representative record). A representative record for  $C_{\tau,W}F$  contains:

- (i) the repaired evaluation records it realizes;
- (ii) an immutable identifier for  $C_{\tau,W}F$ ;
- (iii) the monitored set  $I_{\text{mon}}$ ;
- (iv) the actual profile isomorphisms

$$\rho_i : \mathbf{P}_i(C_{\tau,W}F) \xrightarrow{\cong} \text{Eval}_{i,W,\tau}(F), \quad i \in I_{\text{mon}};$$

- (v) the declared equivalence strength;

- (vi) artifact references for the representative construction and the isomorphisms;
- (vii) whether filtered, derived, realization, or Ext-level use is requested;
- (viii) the representative-record status

not\_invoked,      undefined,      certified;

- (ix) the claim-status, dependency, manifest, and artifact-identity entries governing the assertion.

A missing in-scope representative or profile-isomorphism artifact yields `undefined`; it is never repaired by metric slack.

**Remark 3.27** (Representative is not a global collapse functor). The notation  $C_{\tau, W}F$  does not assert a global functor

$$C_{\tau} : \text{FiltCh}(k) \longrightarrow \text{FiltCh}(k).$$

It also does not assert

$$C_{\tau, W}F = (C_{\tau}F)|_W.$$

It means only that a filtered representative has been supplied for the repaired windowed profile family in the declared monitored degrees.

**Definition 3.28** (Realization-compatible representative). For a declared later use, a representative  $C_{\tau, W}F$  is called *realization-compatible* when it is supplied with:

- (i) a typed realization route  $\mathcal{R}(C_{\tau, W}F)$ ;
- (ii) source and target categories and functors;
- (iii) the comparison or compatibility morphisms required by that route;
- (iv) artifact evidence and the additional hypotheses required by the eligible realization regime of Chapter 4.

Realization compatibility is not automatic from persistence-profile admissibility.

**Remark 3.29** (Eligibility deferred). This chapter does not define or certify the full Ext-eligible regime. It only records whether a representative is intended for such use and whether the required realization data are present. The amplitude, edge-realization, and Ext-edge conditions are stated in Chapter 4.

### 3.9 Representative invariance at the profile level

**Proposition 3.30** (Profile-level representative invariance). *Let  $C_{\tau, W}F$  and  $C'_{\tau, W}F$  be admissible representatives of the same repaired windowed profile family, and let  $i$  be a common monitored degree. Then*

$$\mathbf{P}_i(C_{\tau, W}F) = 0 \iff \mathbf{P}_i(C'_{\tau, W}F) = 0.$$

Moreover, each condition is equivalent to

$$\text{Eval}_{i, W, \tau}(F) = 0.$$

Thus the monitored topological vanishing clause is independent of the choice of admissible representative at profile-isomorphism strength.

*Proof.* Both monitored persistence profiles are isomorphic to  $\text{Eval}_{i,W,\tau}(F)$ . The zero object is preserved and reflected by an isomorphism, so the three vanishing statements are equivalent.  $\square$

**Remark 3.31** (Derived invariance is stronger). Profile-level representative invariance does not imply chain-level, derived, or Ext-level representative invariance. If a certificate invokes such constructions, it must supply realization-compatible representative data and, when required, a filtered quasi-isomorphism, derived isomorphism, or explicit comparison theorem at the consumed strength.

### 3.10 Evaluation use in B-Gate<sup>+</sup>

For invoked higher degrees  $n \in I_{\text{mon}}$ , the corresponding topological and categorical clauses are governed by the Chapter 1 gate contract and Chapter 4. The definitions below preserve the baseline degree-one reading inherited from v19. The finite detector calculus developed later in this chapter is degree-generic.

**Definition 3.32** (Topological clause input). The topological input for B-Gate<sup>+</sup> on  $(F; W, \tau)$  is

$$\text{Eval}_{1,W,\tau}(F) = T_{\tau}^{\text{bar}}(B_{1,W}(F)).$$

The topological clause passes when

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

If an admissible representative  $C_{\tau,W}F$  and its degree-one profile isomorphism are supplied, the same clause may be read as

$$\text{PH}_1(C_{\tau,W}F) = 0.$$

**Definition 3.33** (Scalarized gap-ready topological input). The topological input is *gap-ready at radius  $\varepsilon_B \geq 0$*  if its pre-threshold profile satisfies

$$\varepsilon_B < \text{gap}_{\tau}(B_{1,W}(F)).$$

Equivalently, every finite bar of  $B_{1,W}(F)$  avoids

$$[\tau - \varepsilon_B, \tau + \varepsilon_B].$$

This is a single-profile readiness property. A stable perturbation claim also requires a second profile and a comparison record in mode `metric_gap_certified`, including evidence that the same  $\varepsilon_B$  upper-bounds their pre-threshold bottleneck discrepancy. The historical label is retained for reference compatibility.

**Remark 3.34** (Gate value versus gate safety). The equality

$$\text{Eval}_{1,W,\tau}(F) = 0$$

is a post-threshold value statement. Single-profile gap readiness is a pre-threshold band-separation statement. A robustness claim is stronger: it requires the full two-sided comparison record. None of these numerical or profile statements supplies missing representative, realization, claim-status, manifest, or artifact evidence.

**Remark 3.35** (Ext-clause not discharged here). This chapter does not discharge the Ext<sup>1</sup>-clause. It prepares the representative and evaluation records needed for the later one-way bridge. No implication

$$\text{Eval}_{1,W,\tau}(F) = 0 \implies \text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0$$

is asserted in Chapter 3.

### 3.11 Overlap comparison discipline

**Definition 3.36** (Overlap pre-threshold profile). Let  $W_\alpha$  and  $W_\beta$  be overlapping windows and write  $W_{\alpha\beta} := W_\alpha \cap W_\beta$ . The pre-threshold overlap profiles are

$$B_{i,W_{\alpha\beta}}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad B_{i,W_{\alpha\beta}}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

They are evaluated directly on the common overlap window, not obtained by silently reusing already-deleted local outputs.

**Definition 3.37** (Overlap repaired evaluation). At one declared common threshold  $\tau$ , the repaired overlap evaluations are

$$\begin{aligned} \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) &:= T_\tau^{\text{bar}}(B_{i,W_{\alpha\beta}}^\alpha), \\ \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) &:= T_\tau^{\text{bar}}(B_{i,W_{\alpha\beta}}^\beta). \end{aligned}$$

**Remark 3.38** (Two-layer structure of the Overlap Gate). In metric-gap mode, the Overlap Gate records one scalarized budget

$$\varepsilon_{\alpha\beta,i} := \text{val}_B(\mathfrak{d}_{\alpha\beta,i}),$$

together with the pre-threshold statements

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i},$$

$$\varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta).$$

The metric-repair theorem then yields the post-threshold bound

$$d_{\text{int}}(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}$$

in the declared constructible isometry regime. Thus the gap and consumed discrepancy are pre-threshold, while the resulting agreement is post-threshold. In exact mode, a separately proved equality or isomorphism is recorded instead and no positive-radius robustness is inferred.

Higher overlap coherence, simplex records, typed Type IV status, and countable-atlas conditions belong to Chapter 6 and are not created by a pairwise comparison record.

**Proposition 3.39** (Scalarized overlap threshold safety). Let  $\mathfrak{d}_{\alpha\beta,i}$  be a declared overlap metric ledger and set

$$\varepsilon_{\alpha\beta,i} := \text{val}_B(\mathfrak{d}_{\alpha\beta,i}).$$

Suppose

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i}$$

and

$$\varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta).$$

Then

$$d_B(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}.$$

*Proof.* The two assumptions supply a comparison record in mode `metric_gap_certified`. Apply Proposition 3.17 on the common overlap window.  $\square$

### 3.12 Detector source stages

**Definition 3.40** (Detector profile stage). Every detector record declares exactly one source stage:

windowed\_prethreshold,      repaired\_postthreshold,  
kernel\_defect,      cokernel\_defect,  
diagram\_probe,      descent\_coherence.

The standard finite stencil, birth, and right-germ theorems of this chapter are proved for `windowed_prethreshold`. A theorem proved for one stage is not reused at another stage without an explicit stage-transport theorem. The kernel and cokernel stages are consumed only after Chapter 5 supplies an actual comparison morphism and well-typed defect profiles. The `diagram_probe` stage is available only after Chapter 4 or Chapter 7 supplies an actual finite diagram or derived object. The `descent_coherence` stage is available only after Chapter 7 supplies actual local objects, overlap morphisms, and coherence cells. These Interface stages inherit no barcode completeness theorem merely from their names.

### 3.13 Information model

**Definition 3.41** (Certificate information model). An information model  $\mathcal{I}$  specifies the input data available before queries are issued. It contains the source class, exactness or precision, promise conditions, provenance, and permitted primitive views. Canonical primitive views include

`full_barcode`,   `critical_value_list`,   `point_evaluations`,   `rank_access`,  
`enriched_projective_probes`,   `local_object_restrictions`,   `overlap_morphisms`.

The model records whether each view is given explicitly, computed from another view, obtained from an oracle, or only promised to exist. A proof may not use information outside  $\mathcal{I}$ .

**Remark 3.42** (Information is not a query transcript). The information model describes what can in principle be accessed. It does not count which accesses were made, whether they were adaptive, or how their answers were encoded. Those belong respectively to  $\mathcal{Q}$  and  $\mathcal{E}$ .

### 3.14 Query model

**Definition 3.43** (Certificate query model). A query model  $\mathcal{Q}$  specifies:

- (i) the query alphabet and response type;
- (ii) whether the strategy is adaptive or nonadaptive;
- (iii) whether it is deterministic or randomized;
- (iv) whether responses are exact or approximate;
- (v) stopping rules, worst-case or expected query accounting, and, for a randomized model, the random seed and error parameters;
- (vi) whether producer-side and verifier-side source accesses are both charged, and how repeated or cached accesses are counted;
- (vii) for approximate queries, the tolerance, decision gap, and theorem converting approximate responses into the claimed exact or approximate verdict.

For an object  $X$ , the resulting ordered query–answer record is denoted  $\text{Tr}_Q(X)$ . Every object-dependent source access not already supplied as fixed input in  $\mathcal{I}$  must be charged according to this query policy.

**Policy 24** (Exact-zero use of randomized or approximate queries). *A randomized or approximate procedure is not consumed as an exact zero certificate unless its verifier semantics and a theorem establish the exact claim at the declared error strength. A Monte Carlo confidence value, finite precision zero, or empirical absence of a witness is not silently retyped as a deterministic exact vanishing theorem.*

### 3.15 Encoding model

**Definition 3.44** (Certificate encoding model). An encoding model  $\mathcal{E}$  fixes canonical representations for queries, answers, witnesses, theorem references, dependency references, verifier traces, and artifacts. It also fixes whether a content-addressed object may be referenced rather than duplicated, and how that reference is counted. The default cost vector of a certificate  $C$  is

$$\text{cost}^{\mathfrak{M}}(C) := (q(C), b(C), t(C), a(C)) \in \overline{\mathbb{N}}^4,$$

where:

- (i)  $q$  is the number of logical oracle or probe calls;
- (ii)  $b$  is total encoded bit length, including query descriptions, responses, witnesses, and non-free dependency references;
- (iii)  $t$  is verifier cost in the declared machine, field-operation, or proof-checking model;
- (iv)  $a$  is the persistent artifact footprint required for independent replay.

The four coordinates are not identified. A one-query full-barcode oracle can have small  $q$  and large  $b$ ; many low-bandwidth probes can have large  $q$  with different bit and verification costs.

**Remark 3.45** (Finite bit complexity requires a finite code). An exact endpoint, matrix entry, or oracle response contributes finite  $b$ -cost only when  $\mathcal{E}$  supplies a finite representation, for example a rational, algebraic, finite-field, symbolic, or content-addressed code. An arbitrary exact real value with no finite coding rule has  $b = +\infty$  for this accounting. Consequently, a finite query upper bound does not by itself imply a finite bit-complexity or artifact-size upper bound.

**Definition 3.46** (Coordinatewise cost order). Write  $\overline{\mathbb{N}} := \mathbb{N} \cup \{+\infty\}$ . On  $\overline{\mathbb{N}}^4$ , define

$$(q, b, t, a) \preceq_{\text{cost}} (q', b', t', a')$$

when all four coordinate inequalities hold. This is a partial order, not a total order. No scalar weighting of the four coordinates is canonical unless the model explicitly supplies one.

### 3.16 Verifier model

**Definition 3.47** (Certificate verifier model). A verifier model  $\mathcal{V}$  declares the algorithms, arithmetic, proof rules, trusted theorem registry, artifact access, resource bounds, and acceptance semantics available to the checker. It distinguishes:

- (i) producer computation from verifier computation;

- (ii) an oracle answer from a locally recomputed value;
- (iii) a trusted dependency reference from an unverified external claim;
- (iv) deterministic acceptance from probabilistic acceptance;
- (v) mathematical rejection from undefined verification caused by missing or stale evidence.

A certificate is valid only relative to its declared  $\mathcal{V}$ .

**Definition 3.48** (Certificate model). A certificate model is the typed quadruple

$$\mathfrak{M} = (\mathcal{I}, \mathcal{Q}, \mathcal{E}, \mathcal{V}).$$

Two certificate costs are compared only after confirming that they use the same model or after applying a proved model-translation theorem that accounts for all added information, queries, encoding, and verification work.

**Definition 3.49** (Family-relative certificate scheme and family cost). Fix a family  $\mathcal{F}$ , a decision problem  $\Theta$ , and a certificate model  $\mathfrak{M}$ . A *family-relative certificate scheme*  $\Pi$  consists of:

- (i) a producer or acquisition strategy that, for each  $X \in \mathcal{F}$ , uses only  $\mathcal{I}$  and the accesses permitted by  $\mathcal{Q}$  to construct an object-specific certificate record  $C_X$ ;
- (ii) the declared verifier in  $\mathcal{V}$ ;
- (iii) a proof that the accepted verdict is sound and complete for  $\Theta$  on every  $X \in \mathcal{F}$ ;
- (iv) a family-cost aggregation rule fixed by the model.

In the default deterministic exact regime, the family cost is the coordinatewise worst-case vector

$$\text{cost}_{\mathcal{F}}^{\mathfrak{M}}(\Pi) := \left( \sup_{X \in \mathcal{F}} q(C_X), \sup_{X \in \mathcal{F}} b(C_X), \sup_{X \in \mathcal{F}} t(C_X), \sup_{X \in \mathcal{F}} a(C_X) \right) \in \overline{\mathbb{N}}^4.$$

A randomized or average-case model may use another explicitly declared aggregation, but it then defines a different certificate model and is not compared directly with the default worst-case model.

**Definition 3.50** (Oracle-closed certificate model). An exact query model is *oracle-closed* for a family  $\mathcal{F}$  when all object-dependent information available to both producer and verifier is either fixed in  $\mathcal{I}$  or obtained through charged query responses. Every certificate field, adaptive query, and verdict must be a function of public data, declared randomness if any, and the charged transcript. No omniscient prover message, hidden full-object access, or uncharged object-dependent advice is permitted.

The indistinguishability and right-unbounded lower bounds below apply only to such oracle-closed models. They need not hold when, for example, a full barcode is supplied in  $\mathcal{I}$  or an independently verified witness is available without the counted query access.

### 3.17 Finite probe schemas beyond barcode stages

**Definition 3.51** (Typed finite probe family). For a declared family  $\mathcal{F}$  and predicate  $\Theta$ , a typed finite probe family is a finite list

$$\mathcal{P} = (p_1, \dots, p_m)$$

with a common source stage, response types, actual evaluation procedures, and a theorem candidate relating the response tuple  $(p_1(X), \dots, p_m(X))$  to  $\Theta(X)$ . In stage `diagram_probe`, examples include declared  $\text{Hom}(G_j, X)$  or  $\text{RHom}(G_j, X)$  probes. In stage `descent_coherence`, examples include prescribed cocycle squares, triple-overlap cells, or effectivity comparison maps. Merely listing such probes does not prove that they are complete.

**Theorem 3.52** (Certified finite-probe upper bound). *Let  $\mathcal{P} = (p_1, \dots, p_m)$  be a typed finite probe family for  $(\mathcal{F}, \Theta)$ . Assume a proved zero-reflection or decision theorem of the exact form*

$$\Theta(X) = 0 \iff p_j(X) = 0 \text{ for every } j = 1, \dots, m$$

*for all  $X \in \mathcal{F}$ , with all source-stage and family hypotheses verified. Then querying all probes defines a nonadaptive exact family-relative certificate scheme with worst-case query count at most  $m$ . For each input, its transcript is the object-specific certificate record. The  $(b, t, a)$ -coordinates are determined by  $\mathcal{E}$  and  $\mathcal{V}$ .*

*Proof.* For each input, query all  $m$  probes and apply the assumed decision theorem. The same producer and verifier rules work uniformly over  $\mathcal{F}$ , and completeness and soundness are exactly the two implications of that theorem. Only the worst-case query-count coordinate follows from finiteness of the list; the other coordinates remain model-dependent.  $\square$

**Remark 3.53** (No automatic generator or descent completeness). Theorem 3.52 is a conditional certificate schema. It does not assert that a proposed projective object is a generator, that  $\text{RHom}(G, -)$  reflects zero, that pairwise overlap tests imply a cocycle, or that cocycle verification implies effective descent. Those are Chapter 7 theorems and hypotheses.

### 3.18 Effective detector domain and exact stencil regime

**Definition 3.54** (Effective detector domain). Let  $W = [a, b)$ , where  $b \in \mathbb{R} \cup \{+\infty\}$ , and let  $\tau > 0$ . Define

$$W_\tau := \{t \in W \mid t + \tau \in W\}.$$

Thus

$$W_\tau = \begin{cases} [a, b - \tau), & b < +\infty \text{ and } b - a > \tau, \\ \emptyset, & b < +\infty \text{ and } b - a \leq \tau, \\ [a, +\infty), & b = +\infty. \end{cases}$$

Only sample points in  $W_\tau$  may be used by the exact  $\tau$ -stencil detector.

**Remark 3.55** (Window-relative length). All bars in the detector calculus are bars of the already windowed profile. A bar reaching the right endpoint of a bounded window is therefore a finite window-cofinal bar. Its length is its clipped window-relative length, not  $+\infty$ . A bar remains genuinely infinite only on a right-unbounded window.



### 3.19 Exact finite stencil and active-start regions

**Definition 3.56** (Exact finite stencil detector). Let  $B$  be a constructible persistence module on  $W$ , let  $T \subseteq W_\tau$  be finite, and fix the left-closed right-open interval registration convention. Define

$$D_{T,\tau}^{\text{stc}}(B) := \bigoplus_{t \in T} \text{im}(B(t) \longrightarrow B(t + \tau)).$$

Its detector value is zero precisely when every displayed image space is zero. The direct sum is finite and is counted with vector-space multiplicity.

**Definition 3.57** (Retained bars and active-start region). For a windowed barcode  $B$ , let

$$\text{Ret}_{\tau,W}(B) := \{I \in \mathcal{B}(B) \mid \ell_W(I) > \tau\},$$

with multiplicity, where a genuinely infinite bar has  $\ell_W(I) = +\infty$ . For an interval summand  $I$ , define its  $\tau$ -active-start region by

$$A_\tau(I) := \{t \in W_\tau \mid t, t + \tau \in I\}.$$

If  $I = [p, q)$  in the left-closed right-open convention, then

$$A_\tau(I) = \begin{cases} [p, q - \tau), & q < +\infty \text{ and } q - p > \tau, \\ [p, +\infty), & q = +\infty, \\ \emptyset, & q - p \leq \tau. \end{cases}$$

In particular,  $A_\tau(I) \neq \emptyset$  exactly when  $I$  is retained by  $T_\tau^{\text{bar}}$ .

**Proposition 3.58** (Interval decomposition of a stencil image). *Under the hypotheses of Definition 3.56,*

$$\dim_k \text{im}(B(t) \rightarrow B(t + \tau))$$

*equals the number, counted with multiplicity, of interval summands  $I$  of  $B$  for which  $t \in A_\tau(I)$ . Consequently,*

$$D_{T,\tau}^{\text{stc}}(B) \neq 0 \iff \exists I \in \text{Ret}_{\tau,W}(B) \text{ with } T \cap A_\tau(I) \neq \emptyset.$$

*Proof.* Decompose  $B$  into interval modules. On an interval summand  $k_I$ , the structure map from  $t$  to  $t + \tau$  is the identity exactly when both parameters lie in  $I$ , and is zero otherwise. Direct sums preserve images componentwise in the finite-dimensional constructible regime. Summing over  $t \in T$  gives the equivalence.  $\square$

**Theorem 3.59** (Finite-stencil soundness). *For every finite  $T \subseteq W_\tau$ ,*

$$D_{T,\tau}^{\text{stc}}(B) \neq 0 \implies T_\tau^{\text{bar}}(B) \neq 0.$$

*Equivalently,*

$$T_\tau^{\text{bar}}(B) = 0 \implies D_{T,\tau}^{\text{stc}}(B) = 0.$$

*No coverage or mesh hypothesis is required for this direction.*

*Proof.* A nonzero stencil contains a nonzero component at some  $t \in T$ . Proposition 3.58 then supplies an interval summand whose active-start region contains  $t$ . Such an interval has window-relative length  $> \tau$ , hence survives hard deletion.  $\square$

**Remark 3.60** (Soundness is not completeness). A zero stencil may miss a retained bar whose active-start region contains no sample. Thus the converse of Theorem 3.59 requires a birth, germ, coverage, or family-mesh completeness theorem. Operational admission of a Core zero verdict follows the Chapter 1 detector-certificate contract.

### 3.20 Birth detector characterization

**Definition 3.61** (Exhaustive birth set). For a finite left-closed right-open windowed barcode

$$\mathcal{B}(B) = \{[p_\lambda, q_\lambda)\}_{\lambda \in \Lambda},$$

define the set of distinct birth values and its effective  $\tau$ -stencil by

$$T_{\text{birth}}(B) := \{p_\lambda \mid \lambda \in \Lambda\}, \quad T_{\text{birth}, \tau}(B) := T_{\text{birth}}(B) \cap W_\tau.$$

A birth-set certificate is *exhaustive* when it proves that every interval summand of the declared windowed profile has its left endpoint in the recorded finite set. Only the effective subset is sampled. Every retained interval has its birth in  $W_\tau$ , whereas a short interval born too near the right boundary may be omitted because its  $\tau$ -map is not defined inside the window. Multiplicities belong to the profile artifact even though repeated birth values are sampled only once.

**Theorem 3.62** (Birth detector characterization). *Let  $B$  be a finite windowed barcode in the left-closed right-open convention, and let its exhaustive birth set be certified. Then*

$$D_{T_{\text{birth}, \tau}(B), \tau}^{\text{stc}}(B) \neq 0 \iff T_\tau^{\text{bar}}(B) \neq 0.$$

Equivalently,

$$D_{T_{\text{birth}, \tau}(B), \tau}^{\text{stc}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.$$

*Proof.* Soundness is Theorem 3.59. Conversely, let  $I = [p, q)$  be a retained interval. Then  $q - p > \tau$ , so  $p + \tau < q$ . The interval summand contributes nontrivially to

$$\text{im}(B(p) \rightarrow B(p + \tau)),$$

and  $p \in T_{\text{birth}, \tau}(B)$ . Hence the birth stencil is nonzero.  $\square$

**Remark 3.63** (Endpoint dependence of the point-birth theorem). The proof evaluates  $B(p)$ , so it consumes the left-closed registration convention. It is not decoration-invariant. Decorated intervals are handled canonically by the right-germ theorem below.

### 3.21 Right-germ detector characterization

**Definition 3.64** (Right germ and shifted germ map). For a constructible persistence module  $B$  and  $p \in \mathbb{R}$ , define the right germ

$$B(p^+) := \varinjlim_{\epsilon \downarrow 0} B(p + \epsilon).$$

For  $\tau > 0$ , the structure maps induce the shifted germ map

$$\gamma_{p, \tau} : B(p^+) \longrightarrow B((p + \tau)^+)$$

from the compatible family

$$B(p + \epsilon) \longrightarrow B(p + \tau + \epsilon).$$

This definition consumes a declared constructible right-germ convention.

**Definition 3.65** (Germ detector). Let  $T_{\text{gbirth}}(B)$  be the finite set of geometric lower endpoint values of the decorated interval summands of the declared windowed barcode, and put

$$T_{\text{gbirth},\tau}(B) := T_{\text{gbirth}}(B) \cap W_\tau.$$

Define

$$D_\tau^{\text{germ}}(B) := \bigoplus_{p \in T_{\text{gbirth},\tau}(B)} \text{im}(\gamma_{p,\tau}).$$

An exhaustive geometric-birth certificate is part of the detector record. Every retained interval has its geometric birth in the effective subset.

**Theorem 3.66** (Decoration-invariant germ characterization). *Let  $B$  be a finite decorated windowed barcode with an exhaustive geometric birth set. Then, independently of open or closed endpoint decoration,*

$$D_\tau^{\text{germ}}(B) \neq 0 \iff T_\tau^{\text{bar}}(B) \neq 0.$$

Equivalently,

$$\boxed{D_\tau^{\text{germ}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.}$$

*Proof.* Consider one decorated interval with geometric endpoints  $p < q$ . Its contribution to the shifted germ map is nonzero exactly when points strictly to the right of both  $p$  and  $p + \tau$  remain inside the interval for sufficiently small positive displacement. This is equivalent to  $p + \tau < q$ , hence to  $q - p > \tau$ . When  $q - p = \tau$ , the target right germ is taken strictly to the right of  $q$  and is zero even if the endpoint  $q$  itself is closed. Summing over the exhaustive geometric birth set gives the equivalence.  $\square$

### 3.22 Externally certified stencil completeness

**Definition 3.67** (Stencil coverage certificate). A stencil coverage certificate for  $(B, W, \tau, T)$  is an artifact-backed proof of

$$\forall I \in \text{Ret}_{\tau,W}(B), \quad T \cap A_\tau(I) \neq \emptyset.$$

The record declares the source of coverage evidence and includes bounded-window cofinal bars. Coverage is a non-scalar obligation.

**Theorem 3.68** (Certified stencil completeness). *Let  $T \subseteq W_\tau$  be finite and suppose a valid stencil coverage certificate is supplied. Then*

$$\boxed{D_{T,\tau}^{\text{stc}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.}$$

*Proof.* The implication from right to left is Theorem 3.59. If the repaired profile is nonzero, there is a retained interval  $I$ . Coverage supplies  $t \in T \cap A_\tau(I)$ , and Proposition 3.58 makes the stencil nonzero. Taking contrapositives gives the other implication.  $\square$

**Remark 3.69** (Logical validity versus proof compression). The theorem is mathematically valid for any genuine coverage proof. A claim of finite proof compression additionally records a non-circular source for coverage, such as a family generation rule, finite critical-data theorem, quantized endpoint lattice, separation theorem, combinatorial filtration constraint, or external-domain theorem. Coverage reconstructed only by first enumerating the complete target barcode may still certify the theorem, but it does not by itself establish a meaningful compression gain.

### 3.23 Family threshold excess and augmented mesh

**Definition 3.70** (Window-relative threshold excess). For a windowed barcode  $B$ , define

$$\text{exc}_{\tau, W}(B) := \inf_{I \in \text{Ret}_{\tau, W}(B)} (\ell_W(I) - \tau),$$

with  $\inf \emptyset := +\infty$ . A genuinely infinite retained interval contributes  $+\infty$ ; a bounded-window cofinal interval contributes its finite clipped excess.

For a nonempty barcode family  $\mathcal{F}$ , a *uniform excess lower bound certificate* is a declared number

$$\underline{e}_{\tau, W}(\mathcal{F}) > 0$$

with artifact-backed proof that

$$\underline{e}_{\tau, W}(\mathcal{F}) \leq \text{exc}_{\tau, W}(B) \quad \text{for every } B \in \mathcal{F}.$$

**Remark 3.71** (Independent source of excess). When family-mesh completeness is advertised as proof compression, the lower bound must be obtained independently of full retrospective reading of the target barcode. Permitted sources include a model-class theorem, endpoint lattice, Lipschitz or separation theorem, filtration constraint, external domain theorem, or certified family generator. An absent or unverifiable source makes detector completeness undefined.

**Definition 3.72** (Augmented mesh). Assume  $W = [a, b]$  is bounded and  $b - a > \tau$ . Put

$$c := b - \tau, \quad W_\tau = [a, c), \quad L_{\text{eff}} := c - a = b - a - \tau.$$

For

$$T = \{t_1 < \dots < t_m\} \subseteq [a, c),$$

define the augmented mesh

$$h_{W_\tau}(T) := \max\{t_1 - a, t_2 - t_1, \dots, t_m - t_{m-1}, c - t_m\}.$$

For  $T = \emptyset$ , set

$$h_{W_\tau}(\emptyset) := L_{\text{eff}}.$$

The endpoints  $a, c$  are virtual boundary points for gap measurement; they are not automatically detector samples.

**Lemma 3.73** (Augmented-mesh hitting lemma). Let  $J = [p, r) \subseteq [a, c)$ . If

$$r - p > h_{W_\tau}(T),$$

then

$$J \cap T \neq \emptyset.$$

*Proof.* If  $J \cap T = \emptyset$ , then  $J$  lies in one component between two successive augmented boundary or sample points. Its length is at most the corresponding augmented gap, hence at most  $h_{W_\tau}(T)$ , a contradiction.  $\square$

**Theorem 3.74** (Family-mesh completeness). *Let  $W = [a, b]$  be bounded, let  $\mathcal{F}$  be a nonempty family of left-closed right-open windowed barcodes, and suppose a valid uniform excess lower bound*

$$\underline{e}_{\tau, W}(\mathcal{F}) > 0$$

*is supplied. If a finite stencil  $T \subseteq W_\tau$  satisfies*

$$h_{W_\tau}(T) < \underline{e}_{\tau, W}(\mathcal{F}),$$

*then  $T$  is complete for every  $B \in \mathcal{F}$ :*

$$\boxed{\forall B \in \mathcal{F}, \quad D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.}$$

*If  $b - a \leq \tau$ , then  $W_\tau = \emptyset$  and every windowed finite bar has length  $\leq \tau$ , so the zero conclusion is trivial.*

*Proof.* Let  $I = [p, q]$  be any retained interval of any  $B \in \mathcal{F}$ . Its active-start region is

$$A_\tau(I) = [p, q - \tau)$$

and has length

$$q - p - \tau \geq \underline{e}_{\tau, W}(\mathcal{F}) > h_{W_\tau}(T).$$

The hitting lemma gives  $T \cap A_\tau(I) \neq \emptyset$ . Thus  $T$  satisfies the coverage hypothesis of Theorem 3.68.  $\square$

**Theorem 3.75** (Strict mesh sharpness and equality boundary). *Fix a bounded effective domain  $W_\tau = [a, c]$  and a finite stencil  $T$ .*

(i) *If  $h_{W_\tau}(T) > e > 0$ , then there exists a one-bar barcode  $B$  with*

$$\text{exc}_{\tau, W}(B) > e, \quad T_\tau^{\text{bar}}(B) \neq 0, \quad D_{T, \tau}^{\text{stc}}(B) = 0.$$

(ii) *The strict hypothesis in Theorem 3.74 cannot be replaced uniformly by  $h_{W_\tau}(T) \leq e$ . For example, if the left boundary gap satisfies  $t_1 - a = e$ , the retained interval*

$$I = [a, a + \tau + e)$$

*has active region  $[a, t_1)$ , which contains no detector sample.*

(iii) *At equality  $h_{W_\tau}(T) = e$ , completeness is boundary- and gap-location-sensitive. No universal pass or fail conclusion is asserted without additional endpoint or placement data.*

*Proof.* For (i), choose an augmented gap of length  $h_{W_\tau}(T)$ , choose  $e'$  with  $e < e' < h_{W_\tau}(T)$ , and place a half-open interval  $[p, p + e')$  strictly inside that gap, with the left-boundary case allowed to start at  $a$ . Set

$$I = [p, p + \tau + e').$$

Then  $I \subseteq W$ , its excess is  $e' > e$ , and its active-start region is the chosen sample-free interval. This gives the claimed barcode. Item (ii) is the displayed boundary construction. Item (iii) follows because an internal equality gap has different left-endpoint behavior from the left-boundary equality gap under the fixed half-open convention.  $\square$

### 3.24 Detector certificate and two-axis status

**Definition 3.76** (Finite detector certificate record). A finite detector certificate record contains at least:

- (i) detector, theorem, source-profile, and immutable run identifiers;
- (ii) the profile stage and the exact source object;
- (iii) the monitored degree  $n$ , window  $W$ , threshold  $\tau$ , and endpoint convention;
- (iv) detector mode
 

`birth, germ, certified_stencil, family_mesh;`
- (v) the finite sample, birth, or geometric-birth set;
- (vi) soundness evidence;
- (vii) completeness, characterization, or coverage evidence;
- (viii) family membership and the independently sourced excess lower bound when family mesh is used;
- (ix) exact or metric transport records when the consumed predicate belongs to a compared profile;
- (x) certificate size, any family-relative model-typed MinCert bound, and replay artifacts;
- (xi) specialized certificate and value statuses.

**Definition 3.77** (Two-axis detector status). Detector certificate status and detector mathematical value are independent fields.

The certificate axis is

`detector_certified, detector_failed, detector_pending,`

mapping respectively to

`pass, reject, undefined.`

Here `detector_failed` means that a well-typed required detector condition is evidenced false. Missing, incomplete, stale, or pending evidence is `detector_pending`, not failure.

The value axis is

`zero, nonzero, undefined.`

Under the Chapter 1 admission policy, a zero value discharges a zero clause only with certified completeness. A certified nonzero value rejects a required zero clause. A raw zero-looking computation with missing completeness remains undefined. The mathematical soundness theorem remains available at its exact strength, but an operational gate consumes only the registered status combination permitted by Chapter 1 and Chapter 8.

**Proposition 3.78** (Detector invariance under profile isomorphism). *Let*

$$\eta : B \xrightarrow{\sim} C$$

*be an isomorphism of persistence modules on the declared window. Then for every finite stencil  $T \subseteq W_\tau$ ,*

$$D_{T,\tau}^{\text{stc}}(B) \cong D_{T,\tau}^{\text{stc}}(C).$$

*The corresponding right-germ detectors are also isomorphic when the same geometric-birth convention is used. Hence detector zero/nonzero is invariant under an actual Chapter 3 profile isomorphism.*

*Proof.* Naturality of  $\eta$  gives a commutative square for each structure map  $B(t) \rightarrow B(t + \tau)$ . Vertical isomorphisms induce an isomorphism of image spaces, and finite direct sums preserve it. Passing to right-germ colimits gives the germ statement.  $\square$

### 3.25 Metric-safe detector verdict composition

**Theorem 3.79** (Metric zero-verdict composition). *Let  $B$  and  $C$  be windowed pre-threshold profiles. Suppose:*

(i) *valid complete detector certificates give*

$$D_{T_B, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0$$

and

$$D_{T_C, \tau}^{\text{stc}}(C) = 0 \iff T_{\tau}^{\text{bar}}(C) = 0;$$

(ii)  $d_B(B, C) \leq \varepsilon$ ;

(iii) *the Chapter 2 two-sided threshold-gap hypotheses hold at the same  $\varepsilon$ ;*

(iv)  $2\varepsilon \leq \tau$ .

Then

$$\boxed{D_{T_B, \tau}^{\text{stc}}(B) = 0 \iff D_{T_C, \tau}^{\text{stc}}(C) = 0.}$$

The operational v21 certificate uses the strict reserve  $2\varepsilon < \tau$ .

*Proof.* Chapter 2 gives

$$T_{\tau}^{\text{bar}}(B) = 0 \iff T_{\tau}^{\text{bar}}(C) = 0$$

under the stated metric, gap, and zero-separation hypotheses. Compose this equivalence with the two detector zero-reflection certificates.  $\square$

**Remark 3.80** (Zero predicate, not pointwise-rank stability). The theorem does not compare detector ranks at common sample points. It composes complete zero-reflection with repaired-zero metric transport. Chapter 2 supplies explicit counterexamples to pointwise detector-rank stability under arbitrarily small bottleneck perturbations.

**Definition 3.81** (Boundary-safe robust stencil). Let  $B$  and  $C$  be equipped with actual  $\varepsilon$ -interleaving morphisms, and define the boundary-safe sample domain

$$W_{\tau, \varepsilon}^{\text{rob}} := \{t \mid t - \varepsilon, t + \tau + \varepsilon \in W\}.$$

For finite  $T \subseteq W_{\tau, \varepsilon}^{\text{rob}}$ , define

$$D_{T, \tau, \varepsilon}^{\text{rob}}(B) := \bigoplus_{t \in T} \text{im}(B(t - \varepsilon) \longrightarrow B(t + \tau + \varepsilon)).$$

For  $W = [a, b)$ , the boundary-safe domain is  $[a + \varepsilon, b - \tau - \varepsilon)$  when nonempty.

**Proposition 3.82** (Robust-to-exact sandwich). *Let  $B, C$  carry specified  $\varepsilon$ -interleaving morphisms on the required boundary-safe domain. Then*

$$D_{T, \tau, \varepsilon}^{\text{rob}}(B) \neq 0 \implies D_{T, \tau}^{\text{stc}}(C) \neq 0,$$

and symmetrically with  $B$  and  $C$  interchanged.

*Proof.* For each  $t \in T$ , the structure map

$$B(t - \varepsilon) \longrightarrow B(t + \tau + \varepsilon)$$

factors, by the interleaving identities and naturality, as

$$B(t - \varepsilon) \longrightarrow C(t) \longrightarrow C(t + \tau) \longrightarrow B(t + \tau + \varepsilon).$$

If the composite is nonzero, the middle exact structure map of  $C$  is nonzero. Taking finite direct sums proves the implication.  $\square$

**Remark 3.83** (Supporting status of the sandwich). The sandwich is one-way and requires actual interleaving morphisms, not merely a scalar distance value. It is a supporting detector theorem, not the main zero-equivalence theorem and not a substitute for completeness.

### 3.26 Family-relative certificates and vector-valued cost

**Definition 3.84** (Detector information class). A detector decision problem is a tuple

$$\Theta = (\mathcal{F}, W, \tau, \text{stage}, \text{endpt}, \text{predicate}, \text{promises}),$$

where  $\mathcal{F}$  is a nonempty declared family and the remaining fields fix the source stage, endpoint convention, target predicate, and all promise conditions. A certificate is family-relative when one fixed verifier rule is sound and complete for every  $X \in \mathcal{F}$ , rather than being designed after full inspection of one target object.

**Definition 3.85** (Achievable scheme-cost set and family-relative minimal certificate). Fix  $(\mathcal{F}, \Theta, \mathfrak{M})$ . Let

$$C_{\mathcal{F}}^{\mathfrak{M}, \text{fin}}(\Theta) \subseteq \mathbb{N}^4$$

be the set of finite family-cost vectors  $\text{cost}_{\mathcal{F}}^{\mathfrak{M}}(\Pi)$  of all valid family-relative certificate schemes that are sound and complete for  $\Theta$  under  $\mathfrak{M}$ . Define

$$\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta) := \text{ParetoMin}\left(C_{\mathcal{F}}^{\mathfrak{M}, \text{fin}}(\Theta)\right),$$

where  $\text{ParetoMin}$  denotes the set of coordinatewise minimal elements. If no finite-cost family-relative scheme exists, the finite achievable set and the Pareto frontier are both empty.

For an individual coordinate  $x \in \{q, b, t, a\}$ , define the projection lower envelope over valid schemes by

$$\text{MinCert}_x^{\mathfrak{M}}(\Theta) := \inf\{x_{\mathcal{F}}(\Pi) : \Pi \text{ is valid for } (\mathcal{F}, \Theta, \mathfrak{M})\},$$

where  $x_{\mathcal{F}}(\Pi)$  is the corresponding coordinate of  $\text{cost}_{\mathcal{F}}^{\mathfrak{M}}(\Pi)$ . The value is  $+\infty$  when no finite value is achievable. The legacy scalar  $\text{MinCert}(\Theta)$  is retained only as the worst-case  $q$ -projection in the exact nonadaptive stencil model; it is not the v21 canonical certificate cost.

**Theorem 3.86** (Well-posedness of the finite Pareto frontier). *If  $C_{\mathcal{F}}^{\mathfrak{M}, \text{fin}}(\Theta) \neq \emptyset$ , then  $\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta)$  is a nonempty finite set. Moreover, every finite achievable cost vector dominates at least one element of this frontier.*

*Proof.* The coordinatewise order on  $\mathbb{N}^4$  is a well-quasi-order by Dickson's lemma. Hence every nonempty subset has finitely many minimal elements, and every element lies above a minimal element obtained by descending inside its finite coordinate box. These minimal elements are exactly the defined Pareto frontier.  $\square$

**Remark 3.87** (Why the frontier is set-valued). The cost order is partial. A certificate using fewer queries can require a larger response encoding, more verifier work, or a larger replay artifact. Without a declared scalar utility function, two incomparable finite scheme-cost vectors must both remain on the Pareto frontier. Theorem 3.86 ensures that this frontier is nevertheless finite whenever at least one finite-cost scheme exists.



### 3.27 Upper-bound calculus

**Proposition 3.88** (Birth-set upper bound). *Let  $\mathfrak{M}_{\text{birth}}$  be the exact nonadaptive model in which a query evaluates one birth stencil map and the exhaustive family birth set*

$$T_{\text{birth}, \tau}(\mathcal{F}) := \bigcup_{B \in \mathcal{F}} T_{\text{birth}, \tau}(B)$$

*is finite and certified independently of the target verdict. Then there is a valid family-relative certificate scheme with a finite family-cost vector whose query coordinate satisfies*

$$q \leq |T_{\text{birth}, \tau}(\mathcal{F})|.$$

Consequently,

$$\text{MinCert}_q^{\mathfrak{M}_{\text{birth}}}(\Theta) \leq |T_{\text{birth}, \tau}(\mathcal{F})|.$$

The  $(b, t, a)$ -coordinates are determined by the declared encoding and verifier models.

*Proof.* Query the finite union of birth locations. The birth characterization detects every retained bar of every family member. The argument fixes only the number of logical queries and does not identify the remaining coordinates.  $\square$

**Definition 3.89** (Minimal augmented-mesh stencil number). For effective length  $L_{\text{eff}} \geq 0$  and target augmented mesh  $\gamma > 0$ , define

$$m_{\text{mesh}}(L_{\text{eff}}, \gamma) := \max \left\{ 0, \left\lceil \frac{L_{\text{eff}}}{\gamma} \right\rceil - 1 \right\}.$$

**Theorem 3.90** (Exact query count for an augmented-mesh target). *Let  $W_\tau = [a, c)$  have length  $L_{\text{eff}} = c - a$ . In the exact nonadaptive point-stencil model:*

(i) *there exists a stencil  $T \subseteq [a, c)$  with*

$$|T| = m_{\text{mesh}}(L_{\text{eff}}, \gamma), \quad h_{W_\tau}(T) \leq \gamma;$$

(ii) *every stencil satisfying  $h_{W_\tau}(T) \leq \gamma$  has*

$$|T| \geq m_{\text{mesh}}(L_{\text{eff}}, \gamma).$$

*Thus the displayed number is the exact  $q$ -coordinate for the augmented-mesh design problem. It does not determine  $b, t, a$ .*

*Proof.* Let

$$N := \lceil L_{\text{eff}}/\gamma \rceil.$$

If  $N = 0$ , the domain is empty. If  $N \geq 1$ , divide the interval into  $N$  equal gaps and place  $N - 1$  internal sample points. Each gap has length  $L_{\text{eff}}/N \leq \gamma$ , proving existence. Conversely,  $m$  samples determine  $m + 1$  augmented gaps whose lengths sum to  $L_{\text{eff}}$ . If each is at most  $\gamma$ , then  $L_{\text{eff}} \leq (m + 1)\gamma$ , which gives the lower bound after taking ceilings.  $\square$

**Corollary 3.91** (Family mesh upper bound for vector-valued MinCert). *Under the hypotheses of Theorem 3.74, choose*

$$0 < \gamma < \underline{e}_{\tau, W}(\mathcal{F}).$$

*In the exact nonadaptive stencil model there exists a valid family-relative scheme with finite family-cost vector  $(q, b, t, a)$  satisfying*

$$q \leq m_{\text{mesh}}(L_{\text{eff}}, \gamma).$$

*Hence the same number is an upper bound for the  $q$ -projection of  $\text{MinCert}_{\mathcal{F}}^{\mathfrak{M}}(\Theta)$ . It need not be the exact Pareto frontier for a family with additional structure.*

### 3.28 Information-model-relative lower bounds

**Definition 3.92** (Private witness cells). Fix a deterministic exact oracle-closed certificate model with query universe  $\mathcal{U}$ , response function  $\text{Ans}_X(u)$ , a predicate  $\Theta$ , and a family containing a baseline  $X_0$  and witnesses  $X_1, \dots, X_m$ . Pairwise disjoint subsets  $U_1, \dots, U_m \subseteq \mathcal{U}$  are *private witness cells* when

$$\Theta(X_j) \neq \Theta(X_0)$$

and

$$\text{Ans}_{X_j}(u) = \text{Ans}_{X_0}(u) \quad \text{for every } u \notin U_j.$$

Thus witness  $X_j$  is indistinguishable from the baseline until a query in its private cell is issued.

**Theorem 3.93** (Indistinguishability lower bound). *Under Definition 3.92, every deterministic exact oracle-closed family-relative certificate scheme that is sound and complete on  $\{X_0, X_1, \dots, X_m\}$  must, on the baseline transcript, query at least one element of each  $U_j$ . Consequently its worst-case family query cost satisfies*

$$q_{\mathcal{T}}(\Pi) \geq m.$$

*The conclusion applies to both adaptive and nonadaptive deterministic schemes.*

*Proof.* Run the scheme on  $X_0$ . If it issues no query in some  $U_j$ , every charged answer is identical on  $X_0$  and  $X_j$ . Oracle-closedness implies, by induction over an adaptive transcript, that the producer emits the same certificate fields, asks the same subsequent queries, and supplies the same verifier-visible record on both objects. The verifier therefore returns the same verdict, contradicting soundness and completeness because the predicate values differ. Hence every private cell must be hit. Pairwise disjointness gives at least  $m$  queries.  $\square$

**Corollary 3.94** (Omitted-region, omitted-generator, and omitted-cell lower bounds). *Theorem 3.93 specializes as follows.*

- (i) *For point-stencil persistence queries, pairwise disjoint active-start regions that carry private one-bar witnesses give a birth-region lower bound.*
- (ii) *For `diagram_probe`, a family of private objects whose only distinguishing probes lie in disjoint generator-probe cells gives an omitted-generator lower bound, provided Chapter 7 supplies the actual probe semantics.*
- (iii) *For `descent_coherence`, private failures supported on disjoint coherence cells give an omitted-coherence-cell lower bound, provided the local objects and overlap maps are actual Chapter 7 data.*

*These are information-model-relative statements; a stronger information model may destroy the indistinguishability hypothesis.*

**Remark 3.95** (Randomized lower bounds require a distributional theorem). The deterministic adversary theorem does not by itself give an optimal randomized lower bound. A randomized claim must specify worst-case or average-case error and provide a distributional or minimax argument. No such argument is fabricated from deterministic indistinguishability alone.

**Remark 3.96** (Why oracle-closedness is indispensable). If the full object, a complete barcode, or an independently verified object-dependent witness is supplied outside the charged query transcript, the private-cell lower bound may fail: the producer can encode the verdict or witness without probing every cell. The theorem is therefore a black-box information-model lower bound, not an unconditional statement about all proof systems.

### 3.29 Right-unbounded-window no-go theorem

**Theorem 3.97** (Uniform finite-detector no-go on an unbounded birth family). *Let*

$$W = [a, +\infty), \quad \tau > 0, \quad e > 0,$$

*and consider the translation family*

$$\mathcal{U}_{a,\tau,e} := \{B_p = \{[p, p + \tau + e)\} \mid p \geq a\}.$$

*In every deterministic exact oracle-closed point-stencil model whose family-relative scheme halts after finitely many parameter queries on every input, adaptive or nonadaptive, there exists  $B_p \in \mathcal{U}_{a,\tau,e}$  with*

$$D_{T,\tau}^{\text{stc}}(B_p) = 0, \quad T_{\tau}^{\text{bar}}(B_p) \neq 0,$$

*where  $T$  is the finite set of parameters queried on the zero-profile transcript. Consequently no finite-query sound-and-complete family-relative scheme exists,*

$$\text{MinCert}_q^{\mathfrak{M}_{\text{stencil}}} \mathcal{U}(\Theta) = +\infty,$$

*and the finite achievable scheme-cost set, hence its Pareto frontier, is empty in that model. More generally, the same conclusion holds for any family that admits retained active-start regions arbitrarily far to the right of every finite transcript.*

*Proof.* On the zero-profile transcript a finitely halting scheme asks only finitely many parameters, hence a bounded set  $T$ . Choose  $p > \max T$ , with the empty-set case arbitrary. The one-bar witness has active-start region  $[p, p + e)$ , disjoint from every issued query, and therefore produces the same answers as the zero profile throughout that transcript. Oracle-closedness forces the same certificate and verdict. Yet its length is  $\tau + e > \tau$ , so it survives deletion. The scheme cannot correctly certify zero on both inputs. This is also the private-witness argument with a witness cell placed beyond the finite transcript.  $\square$

**Remark 3.98** (Model-relative escape conditions). The theorem does not exclude finite certificates on a right-unbounded window under a stronger model. Possible escape conditions include a finite known birth set, a global birth bound, eventual zero, eventual stabilization, a finite-to-infinite reduction theorem, a full-barcode input, or a separately certified countable cofinal detector. Finite observed stability alone does not establish any of these conditions.

### 3.30 Certificate composition calculus

**Definition 3.99** (Composable certificate schemes). Two family-relative schemes  $\Pi_1, \Pi_2$  are composable when they use one compatible certificate model and family, their source stages and predicates are typed, their dependencies form an acyclic verifier subgraph, and  $\mathcal{V}$  has a rule for deriving the requested conjunction or typed composite predicate from the two accepted verdicts. For each input  $X$ , shared queries, answers, theorem references, and artifacts in the object-specific records  $C_{1,X}, C_{2,X}$  are reusable only through identical content addresses and a declared reuse rule.

**Definition 3.100** (Composition overhead and reuse discipline). For a compositional verifier, let

$$h_{12} = (h_q, h_b, h_t, h_a) \in \overline{\mathbb{N}}^4$$

record the query, encoding, verification, and artifact overhead of the composition node. Pure logical conjunction may have  $h_q = 0$ . If the composite predicate requires an additional compatibility probe, dependent

source query, or cross-certificate reconciliation, that access is included in  $h_q$ . A probe derived from previous answers is not counted as a new oracle query only when its derivation is replayable by the verifier; otherwise it is counted. Duplicate artifacts are counted once only when one immutable object is actually referenced by both certificates. A bare textual assertion that evidence is shared creates no cost reduction.

**Theorem 3.101** (Finite certificate-scheme composition inequality). *Let  $\Pi_1, \Pi_2$  be composable finite-cost family-relative schemes with family costs  $c_i = (q_i, b_i, t_i, a_i)$ . Then there exists a family-relative scheme  $\Pi_1 \otimes \Pi_2$  for the verifier-approved composite predicate satisfying*

$$\text{cost}_{\mathcal{F}}^{\mathfrak{M}}(\Pi_1 \otimes \Pi_2) \preceq_{\text{cost}} c_1 + c_2 + h_{12}.$$

*For each input  $X$ , if  $Q_i(X)$  are the actual content-addressed query sets and  $H_{12}(X)$  is the set of additional composition queries, the object-specific composite record may use*

$$q_X(\Pi_1 \otimes \Pi_2) = |Q_1(X) \cup Q_2(X) \cup H_{12}(X)|.$$

*Consequently its worst-case family query coordinate is at most  $q_1 + q_2 + h_q$ . Exact reductions in  $b, t, a$  require the reuse rules of Definition 3.100 and are not inferred merely from overlapping subject matter.*

*Proof.* For each input, run or replay the two object-specific certificate records, store each distinct query once, and append the verifier's composition node. Concatenation gives the per-input coordinatewise upper bound, while content-addressed query deduplication, together with the declared additional composition-query set, gives the union formula. Taking the model's coordinatewise worst-case aggregation gives the displayed family bound. The verifier rule proves the composite predicate. No claim is made that this construction is Pareto-optimal.  $\square$

**Remark 3.102** (Local composition is not global certificate complexity). Theorem 3.101 is finite and model-relative. It does not prove

$$\text{MinCert}_{\text{global}} \leq \sum_i \text{MinCert}_i + \dots.$$

A global theorem must account for overlap coherence, dependency cycles, shared encodings, object descent, and reusable artifacts under one fixed model. That theorem belongs to Chapter 7.

### 3.31 Representative handoff to Chapter 4

**Definition 3.103** (Certificate-to-representative handoff). A Chapter 3 handoff record is

$$\begin{aligned} \text{Cert2ReplInput} := & (\text{source, stage, } \Theta, \text{verdict, } \mathfrak{M}, \\ & \text{certificate, representative\_candidate, strength, artifacts}). \end{aligned}$$

It records exactly which profile-stage predicate the object-specific certificate proves, the family and model relative to which the governing scheme is complete, the object-specific and family-level cost vectors, and any candidate representative named for Chapter 4. The representative field may be absent; when present, it is only a candidate identifier until the actual profile isomorphisms and realization-compatible data of Chapter 4 are proved.

**Theorem 3.104** (Representative handoff boundary). *A valid finite detector certificate can establish the declared predicate at its source stage and can populate Cert2ReplInput. It does not, without a separate construction theorem, produce a filtered representative, a finite diagram, a chain equivalence, a derived object, an Ext edge, or an external Return. Every such promotion requires the corresponding Chapter 4, Chapter 7, or Chapter 9 theorem.*

*Proof.* The certificate relation is defined only on the declared source object and predicate. The additional objects and equivalences are not data in that relation. Therefore they cannot be recovered without an extra theorem and its artifacts.  $\square$

### 3.32 Legacy whole-barcode $E_1$ detector

**Definition 3.105** (Residual degree-one bar mass). Let  $\mathcal{B}_{1,W,\tau}^{\text{res}}(F)$  denote the residual degree-one barcode of  $\text{Eval}_{1,W,\tau}(F)$ , counted with multiplicity. Define

$$E_1(F; W, \tau) := \sum_{I \in (\mathcal{B}_{1,W,\tau}^{\text{res}}(F))_{\text{fin}}} \ell(I).$$

If an infinite residual degree-one bar is present, set

$$E_1(F; W, \tau) := +\infty,$$

unless a separately declared truncated detector is being used. Empty or zero-length intervals are not barcode summands in the monitored constructible profile.

**Proposition 3.106** (Sound  $E_1$ -vanishing detector). *If*

$$E_1(F; W, \tau) = 0,$$

*then*

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

*Proof.* By definition, the presence of an infinite residual bar would give  $E_1 = +\infty$ . Every nonempty finite interval summand has strictly positive length and contributes positively, with multiplicity, to the displayed sum. Therefore  $E_1 = 0$  implies that no residual degree-one bar is present, so the repaired degree-one profile is zero.  $\square$

**Corollary 3.107** (Whole-barcode  $E_1$  characterization). *For a finite residual barcode under the stated convention,*

$$E_1(F; W, \tau) = 0 \iff \text{Eval}_{1,W,\tau}(F) = 0.$$

*Proof.* The forward implication is Proposition 3.106. The reverse implication is immediate because the zero residual barcode has empty length sum.  $\square$

**Remark 3.108** (Why  $E_1$  is not finite proof compression). The scalar  $E_1$  reads the already reconstructed residual barcode. It is a whole-profile vanishing functional, not a finite-map stencil certificate. It does not replace the exact profile statement, is not asserted to be bottleneck-Lipschitz, is not a threshold-gap budget, and does not by itself establish finite proof compression. A truncated variant carries its own policy and cannot silently inherit the characterization above.

### 3.33 Manifest consequences

**Definition 3.109** (Chapter 3 evaluation and representative manifest extension). The Chapter 3 manifest is an extension of the foundational Chapter 1 Minimal Manifest Axiom. A proof object invoking Chapter 3 data records:

- (i) repaired evaluation records and immutable profile identities;

- (ii) pre-threshold profiles  $B_{i,W}(F)$  and repaired evaluations  $\text{Eval}_{i,W,\tau}(F)$ ;
- (iii) clearance and gap values and any single-profile readiness records;
- (iv) the comparison mode for every comparison;
- (v) for metric-gap mode, every metric ledger, its aggregation law, scalarization, real budget, evidenced discrepancy bound, two-sided gap reserve, specialized status, and artifact reference;
- (vi) for exact mode, the exact equality or profile-isomorphism proof and its precise source and target;
- (vii) for theorem-controlled nonmetric mode, the theorem identity, source, target, transported predicate, direction, hypotheses, artifacts, and prohibited stronger readings;
- (viii) crop budgets or window-edge records when source-level comparisons are transported to windowed profiles;
- (ix) any claimed  $\tau$ -window-adapted commutation certificate, including checks for globally retained infinite bars on bounded windows;
- (x) infinite-bar and bounded/terminal-window policies;
- (xi) representative records, actual profile isomorphisms, monitored degrees, equivalence strengths, and typed statuses whenever  $C_{\tau,W}F$  is in scope;
- (xii) realization-compatibility data when filtered, derived, or Ext-level use is intended;
- (xiii) all in-scope non-scalar obligations, dependency states, atomic statuses, and artifact identities;
- (xiv) detector identifiers, all six source stages, monitored degrees, windows, thresholds, endpoint conventions, and detector modes;
- (xv) the complete certificate model  $\mathfrak{M} = (\mathcal{I}, \mathcal{Q}, \mathcal{E}, \mathcal{V})$ , including adaptive/nonadaptive, deterministic/randomized, and exact/approximate declarations;
- (xvi) the producer/acquisition rule, family-cost aggregation, and whether the model is oracle-closed, together with every object-dependent input supplied without charged queries;
- (xvii) query transcripts, response precision, stopping rules, error semantics, and any approximate-to-exact separation theorem;
- (xviii) object-specific and family-level certificate cost vectors  $(q, b, t, a)$ , the coordinatewise order, finite encoding policy, achievable scheme-cost evidence, and any Pareto-frontier claim;
- (xix) lower-bound adversary, private-cell, or indistinguishability artifacts;
- (xx) certificate-composition nodes, overheads, shared content addresses, dependency references, and reuse rules;
- (xxi) every Cert2ReplInput handoff and its prohibited stronger readings;
- (xxii) finite stencil, birth, or germ sets and their immutable computation artifacts;
- (xxiii) detector soundness and completeness theorem identities;

- (xxiv) birth/germ exhaustiveness, stencil coverage, family membership, augmented-mesh, and cofinal-window evidence;
- (xxv) every uniform excess lower bound and its independently declared source;
- (xxvi) detector certificate status and detector value status on separate axes;
- (xxvii) certificate size, family-relative model-typed MinCert bounds, and any unbounded-window no-go or escape-condition record;
- (xxviii) actual interleaving morphisms and boundary-safe domains when the robust-to-exact sandwich is invoked;
- (xxix) claim-status references, schema/version identifiers, repair ancestry, and artifact hashes sufficient for audit and replay.

**Remark 3.110** (Canonical manifest hierarchy). Chapter 1 supplies the foundational manifest axiom; this chapter expands it for repaired evaluation, comparison mode, metric-ledger, crop, window, and representative records; Chapter 4 adds eligibility records. Chapter 8 integrates these extensions into the canonical v21 audit manifest with immutable run identity, completeness/consistency status, typed clause verdicts, replayability, and repair-as-new-record semantics. For final serialization and verdict reconstruction, the Chapter 8 contract is canonical.

**Remark 3.111** (Record completeness and failure semantics). A record containing only  $\text{Eval}_{i,w,\tau}(F)$  can state a post-threshold value, but cannot support stability, overlap transfer, representative compatibility, detector completeness, or Ext eligibility. Missing required Chapter 3 data yield an `undefined` audit clause, not a passing clause and not a mathematical counterexample. A detector condition or inequality known by valid evidence to be false yields `reject`; missing evidence yields `undefined`. An internally inconsistent record is routed to Type V rejection under Chapter 8.

### 3.34 Explicit non-claims of Chapter 3

The following are not asserted in this chapter.

- (i) No global non-expansiveness of  $T_\tau^{\text{bar}}$  is asserted.
- (ii) No band of half-width  $2\varepsilon$  is claimed to be sharp for the two-sided theorem; the sharp half-width is  $\varepsilon$ .
- (iii) No half-clearance convention
$$\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$$
is used.
- (iv) No single-profile readiness record is a metric comparison certificate.
- (v) No exact equality or profile isomorphism is automatically a positive-radius robustness certificate.
- (vi) No unconditional crop–collapse commutation is asserted.
- (vii) No unscalarized quantale element is compared directly with the real-valued clearance  $\text{gap}_\tau = \text{clear}_\tau$ .
- (viii) No numerical budget repairs a missing representative, profile isomorphism, realization, comparison morphism, eligibility certificate, claim-status entry, manifest field, or artifact identity.
- (ix) No realization existence or compatibility obligation is hidden inside  $\delta_{\text{realization},\text{met}}$ .

- (x) No finite-only window-adapted certificate is sufficient; globally retained infinite bars must also be checked after clipping.
- (xi) No pairwise overlap record supplies higher Čech coherence or countable-atlas gluing without Chapter 6 hypotheses.
- (xii) No Chapter 3 manifest fragment supersedes the canonical Chapter 8 audit manifest.
- (xiii) No source-level perturbation is automatically converted into a windowed-profile perturbation.
- (xiv) No global strict filtered collapse functor  $C_\tau$  is asserted.
- (xv) No identity  $C_{\tau,W}F = (C_\tau F)|_W$  is asserted.
- (xvi) Persistence-profile isomorphism does not automatically imply chain isomorphism, homotopy equivalence, filtered quasi-isomorphism, or derived equivalence.
- (xvii) No derived or Ext-level conclusion follows from an admissible representative without the eligible realization regime of Chapter 4.
- (xviii) No reverse implication  $\text{Ext}^1 \Rightarrow \text{PH}_1$  is asserted.
- (xix) No undefined or not-invoked Type IV state is represented by a fabricated numerical pair:

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

- (xx) No problem-specific arithmetic, PDE, geometric, categorical, or cryptographic theorem is proved.
- (xxi) No Search artifact, advisory invariant, scalar potential, ranking, or AI confidence value affects a Core repaired-evaluation verdict.
- (xxii) No finite observation set is a finite proof certificate without a governing soundness and completeness theorem.
- (xxiii) No zero stencil implies repaired-profile zero without exhaustive birth/germ data, certified coverage, family mesh, or another proved zero-reflection theorem.
- (xxiv) No missing birth, germ, endpoint, coverage, family-membership, excess, profile-stage, or artifact evidence is interpreted as detector zero.
- (xxv) No point-birth detector is decoration-invariant; decorated endpoints require the right-germ theorem or another proved replacement.
- (xxvi) No theorem proved for `windowed_prethreshold` is silently transferred to repaired, kernel-defect, or cokernel-defect stages.
- (xxvii) No scalarized metric reserve proves stencil coverage or family completeness.
- (xxviii) No family excess bound obtained only by retrospectively reading the complete target barcode is advertised as independent proof compression.
- (xxix) No family-mesh completeness is asserted from an ordinary internal sample gap that omits boundary gaps.
- (xxx) No universal conclusion is asserted at the equality  $h_{W_\tau}(T) = \underline{e}_{\tau,W}$  without boundary-location data.



- (xxxi) No  $h > \underline{e}$  stencil is complete for the unrestricted one-bar family covered by the sharpness theorem.
- (xxxii) No object-relative empty stencil for an already known zero barcode is promoted to a nontrivial family-relative MinCert theorem.
- (xxxiii) No exact augmented-mesh stencil count is identified with the exact MinCert of an arbitrary structured family.
- (xxxiv) No uniformly complete finite stencil exists for the unbounded translation family of Theorem 3.97.
- (xxxv) No finite prefix or observed stabilization proves an unbounded-window escape condition or finite-to-infinite reduction theorem.
- (xxxvi) No bottleneck bound by itself provides actual interleaving morphisms for the robust-to-exact sandwich.
- (xxxvii) No metric zero-verdict theorem asserts equality of pointwise detector ranks or equality of sample maps.
- (xxxviii) No whole-barcode  $E_1$  functional is represented as a finite-map proof-compression certificate.
- (xxxix) No detector theorem is transferred to `diagram_probe` or `descent_coherence` merely by changing the source-stage token.
  - (xl) No  $\mathbf{RHom}(G, -)$  or enriched projective probe is zero-reflecting without a proved generator or conservativity theorem.
  - (xli) No pairwise or finite coherence-cell observation implies effective descent without the Chapter 7 effectivity theorem.
  - (xlii) No query count is identified with bit length, verifier time, or artifact footprint.
  - (xliii) No single scalar MinCert is canonical for v21; the default object is a Pareto frontier in  $\overline{\mathbb{N}}^4$ .
  - (xliv) No certificate costs are compared across different information, query, encoding, or verifier models without a model-translation theorem.
  - (xlv) No deterministic indistinguishability lower bound is silently promoted to an optimal randomized lower bound.
  - (xlvi) No approximate or randomized zero-looking transcript is an exact zero certificate without explicit error and separation semantics.
  - (xlvii) No shared evidence, dependent probe, or dependency reference is free unless the encoding and verifier models provide a replayable reuse rule; additional compatibility queries contribute to  $h_q$ .
  - (xlviii) No finite certificate-composition inequality implies a global local-to-global certificate bound or object-level descent.
  - (xlix) No per-object certificate cost is silently substituted for the family-level cost of a sound-and-complete certificate scheme.
    - (l) No private-witness or right-unbounded query lower bound is asserted in a model with uncharged object-dependent advice, hidden full-object access, or an omniscient prover message.
    - (li) No finite  $q$ -bound implies finite  $b$ ,  $t$ , or  $a$  when exact real-valued data lack a finite encoding.
    - (lii) No Cert2ReplInput handoff constructs a representative, finite diagram, derived object, Ext edge, or Return.

### 3.35 v21 completion criteria for Chapter 3

Chapter 3 is complete for v21 only if the release artifacts exhibit all of the following under at least one common declared certificate model:

- (i) one nontrivial family-relative upper bound for a valid certificate scheme with a finite family-cost vector;
- (ii) one nontrivial information-model-relative lower bound or sharpness theorem, with its oracle/advice boundary stated explicitly;
- (iii) an explicit adaptive/nonadaptive distinction and a declared deterministic/randomized policy;
- (iv) separate query-count and bit-complexity accounting;
- (v) compatibility with the right-unbounded no-go theorem;
- (vi) one replayable certificate-composition example with shared-evidence accounting;
- (vii) one typed Cert2ReplInput handoff whose scope does not exceed the certified source-stage predicate.

No completion item authorizes the unrestricted global certificate inequality reserved for Chapter 7.

### 3.36 Provenance and status

This chapter depends on:

- (i) Chapter 1 for v21 claim discipline, finite detector admission, family-relative proof compression, B-Gate<sup>+</sup>, ledger discipline, Type IV status semantics, and manifest discipline;
- (ii) Chapter 2 for repaired windowed evaluation, comparison modes, crop control, sharp threshold-gap-safe stability, repaired-zero separation, metric detector-zero transport, and pointwise-rank guard-rails;
- (iii) Appendix A for the detailed proof of the metric repair and the exact crop-collapse criterion;
- (iv) Appendix B for fuller representative discipline when filtered or derived constructions are used;
- (v) Chapter 4 for  $n$ -eligible realization, the higher one-way discharge, and any separately natural zero-reflecting Ext theorem;
- (vi) Chapter 6 for detector-aware record atlases and overlap discipline;
- (vii) Chapter 7 for finite diagrams, generator and cone probes, coherence, object-level descent, and model-fixed global certificate bounds;
- (viii) Chapter 8 for the canonical audit manifest and typed verdict semantics.

**Status labels for this chapter.** *Inherited v19 Core definitions and propositions.* Definitions [3.2](#), [3.3](#), [3.6](#), [3.7](#), [3.10](#), [3.11](#), [3.13](#), [3.15](#), [3.16](#), [3.23](#), [3.25](#), [3.26](#), [3.32](#), and [3.33](#); Propositions [3.17](#), [3.30](#), and [3.39](#).

*Inherited v20 finite-detector definitions.* Definitions [3.40](#), [3.54](#), [3.56](#), [3.57](#), [3.61](#), [3.64](#), [3.65](#), [3.67](#), [3.70](#), [3.72](#), [3.76](#), [3.77](#), [3.81](#), [3.84](#), [3.85](#), and [3.89](#).

*Inherited v20 detector theorems and supporting results.* Proposition [3.58](#), Theorem [3.59](#), Theorem [3.62](#), Theorem [3.66](#), Theorem [3.68](#), Lemma [3.73](#), Theorem [3.74](#), Theorem [3.75](#), Proposition [3.78](#), Theorem [3.79](#), Proposition [3.82](#), Proposition [3.88](#), Theorem [3.90](#), Corollary [3.91](#), and Theorem [3.97](#).

*New v21 certificate-model and complexity definitions.* Definitions [3.41](#), [3.43](#), [3.44](#), [3.46](#), [3.47](#), [3.48](#), [3.49](#), [3.50](#), [3.51](#), [3.92](#), [3.99](#), [3.100](#), and [3.103](#).

*New v21 certificate-complexity and Interface results.* Theorem [3.52](#), Theorem [3.86](#), Theorem [3.93](#), Corollary [3.94](#), Theorem [3.101](#), and Theorem [3.104](#).

*Inherited and strengthened whole-barcode detector.* Definition [3.105](#), Proposition [3.106](#), and Corollary [3.107](#).

*Operational policies and manifest records.* Definitions [3.19](#), [3.21](#), [3.28](#), [3.36](#), [3.37](#), and [3.109](#).

*Explicit exclusions.* Subsection [3.34](#).

## 4 Chapter 4: Derived Obstruction and Persistence-to-Diagram Reconstruction

**Chapter 4 contract (v21 finite linear-interface closure).** This chapter is the first  $\text{Core}_P$ -to- $\text{Core}_{DG}$  interface chapter. It consumes the repaired persistence profiles and certificate handoffs of Chapters 2–3, restricts an endpoint-complete finite one-parameter profile to a declared finite linear quiver  $A_{[m]}$ , reconstructs its barcode by finite rank/Möbius data, and then preserves the inherited v20 higher Ext-edge at exactly its proved strength.

The finite-grid route is

$$P \mapsto \text{P2DG}_\Gamma(P) = (V_0 \rightarrow V_1 \rightarrow \cdots \rightarrow V_m) \mapsto \{\text{rk}(V_i \rightarrow V_j)\}_{i \leq j} \mapsto \mathcal{B}(P).$$

It is lossless only on the declared endpoint-complete  $\Gamma$ -constructible class. On that class, enriched restriction is an exact equivalence with  $\text{Rep}_k(A_{[m]})$ . Outside that class, the same finite samples are only observations and may lose endpoint, tail, morphism, or barcode data.

This chapter also preserves the unconditional field-coefficient identification

$$\text{Ext}^n(A, k) := \text{Hom}_{D^b(k\text{-mod})}(A, k[n]) \xrightarrow{\sim} \text{Hom}_k(H^{-n}(A), k)$$

and the conditional one-way discharge

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n,F,W,\tau}, k) = 0$$

under a complete  $n$ -eligible realization record. The quiver Ext signature and this field-valued higher Ext-edge are different constructions and are never identified by notation alone.

Natural realization, purpose-faithful realization, synthetic encoding, and target-shaped construction are separately classified. A diagram produced by restriction from an actual persistence module is not the same thing as a target-shaped object manufactured to force a desired Ext verdict. A reverse vanishing implication requires a separately proved zero-reflection theorem.

General finite categories, branching posets, generator/cone reconstruction, and effective object descent belong to Chapter 7. This chapter is restricted to one-parameter finite linear order and the  $A_{[m]}$ -quiver interface. Its cone output is only a typed handoff record for Chapter 7.

Missing grid completeness, endpoint data, terminal policy, actual morphisms, eligibility, edge realization, detector completeness, or artifacts yields `undefined`; it is never interpreted as reconstruction, derived equivalence, or categorical vanishing.

### 4.1 Scope and placement in the v21 dual Core

All statements in this chapter are made in the v21 conservative-evolution regime fixed by Chapters 1–3 and Appendix A. The degree-one bridge and the higher field-coefficient edge are inherited from v20 without weakening. The new v21 mathematics is the endpoint-complete finite linear-quiver interface and its reconstruction calculus.

The standing scope is:

- (i) the source is a finite barcode profile or a constructible one-parameter persistence module over a field  $k$ , on a declared right-open window  $W$ ;
- (ii) the repaired profile is

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F));$$

(iii) every consumed comparison declares exactly one canonical mode

`exact, metric_gap_certified, theorem_controlled_nonmetric, not_compared;`

(iv) a Chapter 3 `Cert2ReplInput` record can name a candidate source, but does not itself construct a persistence module, a finite quiver object, a representative, or a derived realization;

(v) finite reconstruction is performed only after an endpoint-complete atom grid  $\Gamma$ , variance convention, and terminal policy have been fixed;

(vi) the finite diagram is linear:

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow m;$$

arbitrary finite categories, branching posets, and descent are excluded from this chapter and assigned to Chapter 7;

(vii) restriction of an actual persistence module and objectwise encoding of a barcode isomorphism class have different action modes;

(viii) a categorical  $\text{Ext}^n$ -clause is invoked only after the corresponding  $n$ -eligible realization record is complete and consistent;

(ix) realization and edge-extraction policies are fixed at declared scope level and are not selected after inspection of the desired verdict;

(x) audited use retains the independent metric, detector, Type IV, manifest, claim-status, and Return obligations of Chapters 1–3, 5, 8, and 9.

**Remark 4.1** (Relocation of the previous failure-taxonomy chapter). Earlier drafts placed failure types, tower comparison artifacts, and Type IV diagnostics in Chapter 4. After the v19 Chapter 3 repair, that material is logically downstream of the one-way bridge and should be read as the next failure-diagnostics chapter. The present Chapter 4 is reserved for finite linear persistence-to-diagram reconstruction, eligibility, the inherited  $\text{Ext}^1$ -edge, and its controlled higher-degree extension.

**Remark 4.2** (Metric repair is consumed, not reproved). This chapter does not reprove threshold-gap-safe stability. It assumes that the repaired evaluation records consumed here have already been formed according to Chapters 2–3, with any required threshold-gap certification supplied separately. The bridge below is a categorical discharge result, not a stability theorem for threshold deletion.

## 4.2 Realization classes and the synthetic boundary

**Definition 4.3** (Two-axis realization classification). Every construction from a persistence source to a diagrammatic or derived target carries two independent classification fields.

The *construction-provenance axis* records exactly one of

`natural_realization, synthetic_encoding, target_shaped.`

Here `natural_realization` means a scope-level construction fixed from source semantics, with its declared action on morphisms and comparison data; `synthetic_encoding` means a mathematically valid encoding of selected source data without an independent theorem identifying the target as the intended external or Core object; and `target_shaped` means that the target was chosen or modified so that the desired conclusion is built into its definition.

The *semantic-fidelity axis* records exactly one of

$$\begin{array}{ll} \text{purpose\_faithful}, & \text{purpose\_preserving\_only}, \\ \text{observation\_only}, & \text{not\_established}. \end{array}$$

The label `purpose_faithful` requires a theorem proving both preservation and reflection of the declared predicate or invariant on the stated source class. It does not by itself assert fullness, faithfulness, or essential surjectivity of the underlying functor.

A *purpose-faithful realization* therefore carries the natural-realization provenance label and the purpose-faithful semantic label. A natural realization need not be purpose-faithful, and a synthetic encoding may be faithful to an internal statistic without acquiring the external semantics of a natural realization. A target-shaped construction cannot support an independent Bridge or Return unless a separate theorem removes the verdict dependence and proves the required source semantics.

**Policy 25** (No verdict-conditioned realization). *The grid, representative, realization functor, edge extractor, and target category must be fixed from source semantics and declared scope before the target zero/nonzero verdict is evaluated. Any construction selected after reading the desired conclusion is classified as `target_shaped` and is not admissible as Core Interface evidence.*

**Remark 4.4** (Restriction is natural; barcode selection may be objectwise). Evaluation of an actual persistence module on a fixed finite grid is a genuine functor on actual persistence morphisms. By contrast, choosing an interval decomposition of a barcode isomorphism class and then writing down a direct sum of finite interval representations is generally only objectwise unless the decomposition and morphism action are supplied functorially. The two routes must not share one action-mode label.

### 4.3 Endpoint-complete finite atom grid

**Definition 4.5** (Finite endpoint set). Let  $P$  be a finite persistence profile on a right-open window  $W = [u, u')$ , where  $u' \in \mathbb{R} \cup \{+\infty\}$ . A finite set

$$K = \{c_0 < \cdots < c_r\} \subseteq W$$

is *endpoint-complete* for  $P$  when it contains  $u$ , every finite geometric birth and death value of every interval summand of  $P$  lying in  $W$ , and every additional critical value at which the declared module or morphism data change. For a family or a morphism, endpoint completeness is required simultaneously for every source, target, and comparison object in scope.

**Definition 4.6** (Atom decomposition and enriched grid). Given  $W$  and an endpoint-complete  $K$ , decompose  $W$  into its nonempty ordered atoms:

$$\{c_i\}, \quad (c_i, c_{i+1}), \quad [u, c_0), \quad (c_r, u')$$

with the obvious omissions when an atom is empty and with  $(c_r, +\infty)$  allowed on a terminal window. Enumerate the nonempty atoms in increasing order as

$$\mathcal{A}_\Gamma = (A_0 < \cdots < A_m).$$

Choose one probe  $\gamma_j \in A_j$  when  $A_j$  is not a singleton, and take  $\gamma_j = c_i$  for a singleton atom. The resulting ordered probe set

$$\Gamma = \{\gamma_0 < \cdots < \gamma_m\}$$

together with the atom labels is called the *enriched finite grid*.

The atom projection

$$q_\Gamma : W \longrightarrow [m] = \{0 < \dots < m\}$$

sends every point of  $A_j$  to  $j$ . It is monotone.

**Remark 4.7** (Why critical values alone are insufficient). Sampling only the geometric endpoint values can lose open/closed decoration. The enriched grid keeps singleton critical atoms and adjacent open-stratum atoms as distinct nodes. Consequently, left and right germ behavior at each critical value is retained. This is the endpoint-complete replacement for an undecorated list of critical values.

**Definition 4.8** (Terminal sentinel policy). On a bounded window, the final nonempty atom below  $u'$  is the terminal bounded atom; no value at the excluded endpoint  $u'$  is invented. On a terminal window  $W = [u, +\infty)$ , the final unbounded atom is represented by one terminal sentinel probe and the record declares `cofinal_tail`. The module is required to be constant on that tail. A nonzero terminal value then records bars cofinal to  $+\infty$ . A finite sentinel without the cofinal-tail declaration does not certify an infinite bar.

**Definition 4.9** ( $\Gamma$ -constructible profile). A persistence module  $P$  on  $W$  is  $\Gamma$ -constructible when  $P(x)$  is finite-dimensional for every  $x \in W$  and, for every atom  $A_j$  and every  $x \leq y$  in  $A_j$ , the structure map

$$P(x \leq y) : P(x) \longrightarrow P(y)$$

is an isomorphism. On a declared cofinal-tail atom this is the required eventual constancy condition. Thus all changes occur only when one passes between consecutive atoms. This is an intrinsic condition on  $P$ ; it does not assume in advance the reconstruction theorem proved below.

**Definition 4.10** (Grid completeness record). A grid completeness record contains:

- (i) source profile, window, coefficient field, and profile stage;
- (ii) the endpoint set  $K$ , atom list  $\mathcal{A}_\Gamma$ , and grid  $\Gamma$ ;
- (iii) proof that every finite endpoint and module/morphism critical value in scope belongs to  $K$ ;
- (iv) the endpoint-decoration convention and the singleton/open-atom distinction;
- (v) the bounded-terminal or cofinal-tail sentinel policy;
- (vi) proof that every source and target is  $\Gamma$ -constructible;
- (vii) the action mode on morphisms;
- (viii) immutable artifacts and prohibited stronger readings.

An incomplete endpoint list or unproved tail policy gives undefined, not a lossy-but-complete reconstruction.

**Remark 4.11** (Source of endpoint completeness and non-circular use). Endpoint completeness may be certified either from a previously available full barcode or from an independent critical-data theorem, filtration rule, endpoint lattice, or source artifact. In the first case the reconstruction theorem remains a valid serialization and consistency theorem, but it is not by itself a discovery or proof-compression result. Any claim of compression, finite discovery, or reduced information access must identify a source of the endpoint set that does not already contain the complete target barcode.

#### 4.4 Linear-quiver encoding

**Definition 4.12** (Finite linear-quiver category). Let  $A_{[m]}$  denote the linearly oriented quiver

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow m.$$

It has  $m + 1$  vertices and is the Dynkin quiver usually denoted  $A_{m+1}$ . The notation  $A_{[m]}$  is used here to keep the index set  $[m] = \{0, \dots, m\}$  visible and to eliminate the off-by-one ambiguity of the informal phrase “ $A_m$ -quiver.” Write

$$\text{Rep}_k(A_{[m]}) := \text{Fun}([m], \text{Vect}_k^{\text{fd}}).$$

An object is a finite diagram

$$V_0 \xrightarrow{v_0} V_1 \xrightarrow{v_1} \cdots \xrightarrow{v_{m-1}} V_m.$$

Kernels, cokernels, and exactness are computed pointwise.

**Definition 4.13** (Grid restriction and step realization). Define

$$\text{Res}_\Gamma(P) := (P(\gamma_0) \rightarrow \cdots \rightarrow P(\gamma_m)) \in \text{Rep}_k(A_{[m]}).$$

Conversely, define the step realization

$$\text{Step}_\Gamma(V) := q_\Gamma^* V.$$

Both constructions act on actual morphisms by evaluation and pullback.

**Lemma 4.14** (Canonical atom transport). *Let  $P$  be  $\Gamma$ -constructible and let  $x \in A_j$ . Define*

$$\theta_{P,x} : P(\gamma_j) \longrightarrow P(x)$$

*by the structure map  $P(\gamma_j \leq x)$  when  $\gamma_j \leq x$ , and by the inverse of  $P(x \leq \gamma_j)$  when  $x \leq \gamma_j$ . Then the family  $\theta_P = (\theta_{P,x})_{x \in W}$  is a natural isomorphism*

$$\theta_P : \text{Step}_\Gamma \text{Res}_\Gamma(P) \xrightarrow{\sim} P,$$

*natural in every actual persistence morphism between  $\Gamma$ -constructible modules.*

*Proof.* Every map used in the definition is invertible because its source and target lie in one atom. For  $x \leq y$ , functoriality of the persistence structure maps, with cancellation of the within-atom isomorphisms, gives

$$P(x \leq y) \theta_{P,x} = \theta_{P,y} (\text{Step}_\Gamma \text{Res}_\Gamma(P))(x \leq y).$$

This proves naturality in the parameter. The same commutative squares for an actual persistence morphism prove naturality in  $P$ .  $\square$

**Theorem 4.15** (Finite-grid encoding equivalence). *Let  $\text{Pers}_k^\Gamma(W)$  be the full subcategory of  $\Gamma$ -constructible persistence modules on  $W$ . Then*

$$\text{Res}_\Gamma : \text{Pers}_k^\Gamma(W) \longrightarrow \text{Rep}_k(A_{[m]})$$

*and*

$$\text{Step}_\Gamma : \text{Rep}_k(A_{[m]}) \longrightarrow \text{Pers}_k^\Gamma(W)$$

*are quasi-inverse exact equivalences. In particular, the restriction functor is full, faithful, conservative, and zero-reflecting on this declared subcategory.*



*Proof.* By construction,

$$\text{Res}_\Gamma \text{Step}_\Gamma(V) = V$$

strictly. Lemma 4.14 supplies a natural isomorphism

$$\text{Step}_\Gamma \text{Res}_\Gamma(P) \xrightarrow{\sim} P$$

for every intrinsically  $\Gamma$ -constructible  $P$ . Thus the two functors are quasi-inverse.

A natural transformation between two step modules is determined uniquely by its components at the probes, and any natural transformation of the finite quiver representations pulls back to one of the step modules. Hence restriction is full and faithful. Pointwise kernels and cokernels of a map between  $\Gamma$ -constructible modules again have isomorphism structure maps inside every atom, so  $\text{Pers}_k^\Gamma(W)$  is closed under the relevant pointwise kernels and cokernels. Evaluation and pullback preserve them. Therefore the equivalence is exact. Conservativity and zero-reflection follow from the equivalence.  $\square$

**Proposition 4.16** (Probe-choice invariance on a fixed atom decomposition). *Let  $\Gamma$  and  $\Gamma'$  choose possibly different probes in the same ordered atom decomposition  $\mathcal{A} = (A_0 < \cdots < A_m)$ . For every  $\mathcal{A}$ -constructible persistence module  $P$ , within-atom transport gives a natural isomorphism of finite quiver representations*

$$\alpha_P : \text{Res}_\Gamma(P) \xrightarrow{\sim} \text{Res}_{\Gamma'}(P).$$

*The family  $\alpha_P$  is natural in actual persistence morphisms. Consequently, the quiver isomorphism class, rank table, interval multiplicities, and decorated-barcode Return do not depend on the probe point chosen inside each fixed atom.*

*Proof.* At node  $j$ , define  $\alpha_{P,j} : P(\gamma_j) \rightarrow P(\gamma'_j)$  by the structure map in the appropriate order, using its inverse when  $\gamma'_j < \gamma_j$ . This map is an isomorphism because both probes lie in  $A_j$ . Functoriality of the persistence structure maps makes the squares with consecutive quiver arrows commute. The same naturality squares for a persistence morphism show that  $\alpha_P$  is natural in  $P$ . All listed invariants are preserved by quiver isomorphism.  $\square$

**Remark 4.17** (Different grid refinements are not literally the same diagram). If one endpoint-complete grid refines another by splitting an atom, the two restricted objects lie in representation categories with different indexing quivers. They are compared through their common step realization as the same persistence module, or through an explicitly declared subdivision/contraction functor whose inserted arrows are proved invertible. No literal equality of the two finite diagrams is asserted merely because both grids are complete. The persistence-module and decorated-barcode Returns remain invariant once that comparison is supplied.

**Remark 4.18** (Exactness does not belong to hard deletion). The exactness in Theorem 4.15 concerns restriction of actual  $\Gamma$ -constructible persistence modules. It does not make hard barcode deletion  $T_\tau^{\text{bar}}$  an exact functor. If the repaired source exists only as a barcode isomorphism class, the resulting finite diagram is objectwise until an actual module and morphism action are supplied.

**Remark 4.19** (Ambient equivalence versus the repaired-image restriction). Theorem 4.15 is an ambient equivalence for all  $\Gamma$ -constructible modules, including interval modules supported only on a singleton critical atom. A repaired source

$$P = T_\tau^{\text{bar}}(B), \quad \tau > 0,$$

satisfies the stronger support predicate that every finite geometric interval has length strictly greater than  $\tau$ . Its finite-quiver image therefore lies in the corresponding full essential image inside  $\text{Rep}_k(A_{[m]})$ , and this

image is generally proper. Thus the repaired Corep-to-diagram route is full, faithful, conservative, and zero-reflecting onto its essential image, but it is not asserted to be essentially surjective onto all of  $\text{Rep}_k(A_{[m]})$ . This distinction also prevents the ambient exact equivalence from being read as an exactness theorem for hard threshold deletion.

**Definition 4.20** (Objectwise barcode encoding). For a finite barcode  $B$  whose intervals are unions of consecutive atoms, define

$$\text{GridRep}_\Gamma(B) := \bigoplus_{I \in B} k_{J_\Gamma(I)}$$

where  $J_\Gamma(I) \subseteq [m]$  is the consecutive node interval formed by the atoms contained in  $I$ . This construction preserves multiplicities and endpoint decoration through the atom dictionary. Without a declared action on barcode morphisms, it has action mode `objectwise_only`.

## 4.5 Exactness and information-loss record

**Definition 4.21** (Grid encoding strength record). A grid encoding strength record independently declares:

- (i) object status: `reconstructive` or `observational`;
- (ii) morphism status:
 

`full_faithful, functorial,`  
`partially_functorial, objectwise_only;`
- (iii) exactness status: `exact`, `theorem_controlled`, or `not_asserted`;
- (iv) barcode status: `endpoint_complete`, `undecorated_only`, or `incomplete`;
- (v) terminal status: `bounded_terminal`, `cofinal_tail`, or `unresolved`;
- (vi) information profile: `lossless`, `feature_faithful`, `observation_only`, or `lossy`;
- (vii) the exact data retained, reflected, and discarded.

These fields are independent. For example, an objectwise barcode encoding may be lossless on objects while having no asserted morphism functor.

**Theorem 4.22** (Endpoint-complete reconstruction). *Let  $P$  be a finite  $\Gamma$ -constructible persistence profile and assume the grid completeness record includes all finite geometric endpoints and the terminal policy. Then:*

- (i)  $P$  is reconstructed up to persistence-module isomorphism from  $\text{Res}_\Gamma(P)$ ;
- (ii) each decorated barcode interval of  $P$  corresponds bijectively, with multiplicity, to an interval representation on a consecutive node set of  $A_{[m]}$ ;
- (iii) the atom dictionary reconstructs the geometric endpoints, endpoint decorations, and finite/cofinal terminal type.

*Proof.* The first item is Theorem 4.15. Every interval summand is constant on each atom and is supported on a consecutive sequence of atoms, so restriction produces an indecomposable interval representation of  $A_{[m]}$ . Conversely, the union of the atoms in a consecutive node interval is one decorated real interval, with singleton critical atoms determining inclusion of endpoints. The terminal atom and its declared sentinel policy distinguish a bounded cofinal interval from a genuinely infinite tail. Krull–Schmidt uniqueness on the finite type  $A_{[m]}$ -category preserves multiplicities.  $\square$

**Remark 4.23** (Loss modes). If the grid omits a critical value, merges a singleton endpoint atom with an adjacent open atom, or omits the terminal policy, distinct persistence profiles may restrict to the same finite observations. In that regime the encoding is not reconstructive, even when all displayed vector spaces and maps are computed exactly.

## 4.6 Rank invariant and Möbius barcode reconstruction

**Definition 4.24** (Finite rank table). For  $V \in \text{Rep}_k(A_{[m]})$  and  $0 \leq i \leq j \leq m$ , define

$$r_V(i, j) := \text{rk}(V_i \longrightarrow V_j),$$

where the map is the unique composite along the linear quiver. Set  $r_V(-1, j) = 0$ ,  $r_V(i, m+1) = 0$ , and  $r_V(-1, m+1) = 0$ .

**Theorem 4.25** (Finite Möbius multiplicity formula). *Let  $V \in \text{Rep}_k(A_{[m]})$ . The multiplicity  $\mu_V(i, j)$  of the interval representation supported on the consecutive nodes  $[i, j]$  is*

$$\mu_V(i, j) = r_V(i, j) - r_V(i-1, j) - r_V(i, j+1) + r_V(i-1, j+1).$$

The collection  $\{\mu_V(i, j)\}_{0 \leq i \leq j \leq m}$  is nonnegative and determines  $V$  up to isomorphism.

*Proof.* If

$$V \cong \bigoplus_{a \leq b} k_{[a, b]}^{\oplus \mu_V(a, b)},$$

then

$$r_V(i, j) = \sum_{a \leq i, b \geq j} \mu_V(a, b).$$

Möbius inversion on the product of the two finite chains gives the displayed second-difference formula. Non-negativity follows because the result equals an interval multiplicity. The type- $A_{[m]}$  interval decomposition is unique, so the multiplicities determine the isomorphism class.  $\square$

**Corollary 4.26** (Endpoint-complete barcode Return). *Under the hypotheses of Theorem 4.22, the finite rank table of  $\text{Res}_\Gamma(P)$ , together with the atom dictionary and terminal policy, reconstructs the complete decorated barcode of  $P$ .*

*Proof.* Apply Theorem 4.25 and translate each node interval back to the union of its atoms.  $\square$

**Remark 4.27** (Rank table is not a chain-level Return). The corollary returns the persistence-module isomorphism class and its decorated barcode. It does not return a filtered chain complex, a chosen chain map, a derived realization, or an external-domain object.

## 4.7 Finite projective and Ext signature

**Definition 4.28** (Projective and simple probes). For  $0 \leq i \leq m$ , let  $P_i$  be the indecomposable projective representation supported on the terminal segment  $[i, m]$ , and let  $S_i$  be the simple representation at node  $i$ . For  $i < m$ , there is a canonical projective resolution

$$0 \longrightarrow P_{i+1} \longrightarrow P_i \longrightarrow S_i \longrightarrow 0,$$

while  $S_m = P_m$  is projective.

**Definition 4.29** (Enriched finite projective/Ext signature). For  $V \in \text{Rep}_k(A_{[m]})$ , define

$$\Sigma_{\Gamma}^{\text{proj/Ext}}(V)$$

to consist of:

- (i) the spaces  $\text{Hom}_{\text{Rep}_k(A_{[m]})}(P_i, V)$ ;
- (ii) the precomposition maps induced by  $P_{i+1} \rightarrow P_i$ ;
- (iii) the local spaces  $\text{Hom}_{\text{Rep}_k(A_{[m]})}(S_i, V)$ ;
- (iv) the groups  $\text{Ext}_{\text{Rep}_k(A_{[m]})}^1(S_i, V)$ ;
- (v) the identification and artifact data relating these spaces to the grid diagram.

The word *enriched* includes the actual linear maps, not only their dimensions.

**Proposition 4.30** (Local projective/Ext exact sequence). *For  $i < m$ , there is a natural exact sequence*

$$0 \longrightarrow \text{Hom}_{\text{Rep}_k(A_{[m]})}(S_i, V) \longrightarrow V_i \longrightarrow V_{i+1} \longrightarrow \text{Ext}_{\text{Rep}_k(A_{[m]})}^1(S_i, V) \longrightarrow 0.$$

Moreover,

$$\text{Hom}_{\text{Rep}_k(A_{[m]})}(P_i, V) \cong V_i, \quad \text{Ext}_{\text{Rep}_k(A_{[m]})}^p(-, -) = 0 \quad (p \geq 2).$$

*Proof.* Apply  $\text{Hom}_{\text{Rep}_k(A_{[m]})}(-, V)$  to the projective resolution in Definition 4.28. Projectivity of  $P_i$  and  $P_{i+1}$  gives exactness and vanishing beyond degree one. The Yoneda evaluation isomorphism identifies  $\text{Hom}(P_i, V)$  with  $V_i$ , and the induced map is the quiver arrow  $V_i \rightarrow V_{i+1}$ . The linearly oriented  $A_{[m]}$  path algebra is hereditary, so all higher Ext groups vanish.  $\square$

**Theorem 4.31** (Finite enriched-signature reconstruction). *The enriched projective part of  $\Sigma_{\Gamma}^{\text{proj/Ext}}(V)$  reconstructs  $V$  functorially:*

$$V_i \cong \text{Hom}_{\text{Rep}_k(A_{[m]})}(P_i, V),$$

*and the quiver arrow  $V_i \rightarrow V_{i+1}$  is the precomposition map induced by  $P_{i+1} \rightarrow P_i$ . Consequently, the full enriched signature reconstructs the finite diagram and, under an endpoint-complete grid, the persistence barcode. The simple Hom and  $\text{Ext}^1$  terms are natural kernel/cokernel consistency data; their dimensions alone are not a complete signature.*

*Proof.* The first two assertions are the Yoneda identifications in Proposition 4.30. They recover every object and arrow of the linear diagram, hence  $V$ . The barcode conclusion follows from Corollary 4.26. The final statement follows because forgetting the actual projective probe maps can identify nonisomorphic diagrams with the same local dimensions.  $\square$

**Remark 4.32** (Two different Ext degrees). The quiver groups  $\text{Ext}_{\text{Rep}_k(A_{[m]})}^1(S_i, V)$  measure local arrow cokernels in a hereditary finite diagram category. The inherited groups  $\text{Ext}_{D^b(k\text{-mod})}^n(A_n, k)$  use the monitored persistence degree  $n$  and a separate realization into  $D^b(k\text{-mod})$ . They are not the same Ext theory. In particular, the hereditary vanishing of quiver Ext in degrees  $p \geq 2$  does not trivialize the inherited higher degree- $n$  edge.

## 4.8 Repaired degree-one input

**Definition 4.33** (Degree-one repaired profile). For  $F \in \text{FiltCh}(k)$ , a right-open window  $W = [u, u')$ , and a threshold  $\tau > 0$ , the degree-one repaired profile consumed by this chapter is

$$\text{Post}_{1,W,\tau}(F) := \text{Eval}_{1,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_1(F)).$$

The statement  $\text{Eval}_{1,W,\tau}(F) = 0$  means that this post-threshold barcode profile is the zero persistence profile.

**Definition 4.34** (Degree-one admissible representative). An admissible representative  $C_{\tau,W}F$  is *degree-one compatible* if its Chapter 3 representative record contains an actual profile isomorphism

$$\rho_{1,F,W,\tau} : \mathbf{P}_1(C_{\tau,W}F) \xrightarrow{\sim} \text{Eval}_{1,W,\tau}(F).$$

The source, target, equivalence strength, and reconstructible artifact for  $\rho_{1,F,W,\tau}$  are part of the record. A bare symbol  $\cong$  without such evidence is not an eligibility certificate. When this holds, the notation

$$\text{PH}_1(C_{\tau,W}F) = 0$$

is used only as shorthand for the vanishing of that realized repaired profile.

**Remark 4.35** (No global collapse interpretation). The notation  $C_{\tau,W}F$  does not mean  $(C_\tau F)|_W$ , and no global exact functor  $C_\tau$  is used. The only profile entering this chapter is the repaired windowed profile

$$T_\tau^{\text{bar}}(W_W \mathbf{P}_1(F)).$$

## 4.9 Eligible realization records

**Definition 4.36** (Realization datum). A *realization datum* for a degree-one compatible representative  $C_{\tau,W}F$  consists of:

(i) a declared eligible domain

$$\mathcal{E} \subseteq \text{FiltCh}(k)$$

containing  $C_{\tau,W}F$ ;

(ii) a genuine functor, fixed at the declared scope level,

$$\mathcal{R} : \mathcal{E} \longrightarrow D^b(k\text{-mod});$$

(iii) the realized object

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F);$$

(iv) immutable policy, version, and artifact identifiers for  $\mathcal{E}$  and  $\mathcal{R}$ .

The functor may be partial through the declared domain  $\mathcal{E}$ , but it is not an objectwise rule chosen after the desired Ext verdict is known.

**Definition 4.37** (Residual edge extractor). A *residual degree-one edge extractor* is a policy, invariant under isomorphism on the declared scope, with values in vector spaces,

$$\text{Edge}_1 : \text{Prof}_1^{\text{rep}} \longrightarrow \text{Vect}_k$$

on the declared groupoid or category of repaired degree-one profiles and admissible profile isomorphisms, satisfying

$$\text{Edge}_1(0) = 0.$$

If no profile morphism category has been declared, this is read as an assignment invariant under barcode-profile isomorphism. The extractor is fixed independently of the objectwise gate verdict. It need not be faithful or zero-reflecting unless a stronger bridge theorem explicitly proves that property.

**Definition 4.38** (Edge-realization certificate). An *edge-realization certificate* is attached to the declared data

$$(F; W, \tau, C_{\tau, W}F, \mathcal{R}, \text{Edge}_1).$$

It is an actual declared isomorphism

$$\epsilon_{F, W, \tau} : H^{-1}(\mathcal{R}(C_{\tau, W}F)) \xrightarrow{\sim} \text{Edge}_1(\text{Eval}_{1, W, \tau}(F)).$$

Its source, target, dependence on the representative isomorphism  $\rho_{1, F, W, \tau}$ , and reconstructible proof artifact are recorded. The only objectwise implication consumed by the Core bridge is

$$\text{Eval}_{1, W, \tau}(F) = 0 \implies H^{-1}(\mathcal{R}(C_{\tau, W}F)) = 0.$$

An objectwise certificate does not by itself establish naturality in  $F$ . Whenever naturality or transport across morphisms is claimed, the certificates must form a compatible natural family on the declared eligible domain.

**Definition 4.39** (Eligible realization record). An *eligible realization record* for  $(F; W, \tau)$  consists of:

- (i) a repaired evaluation record for  $\text{Eval}_{1, W, \tau}(F)$ ;
- (ii) a degree-one compatible admissible representative  $C_{\tau, W}F$ , its Chapter 3 representative record, and the actual isomorphism  $\rho_{1, F, W, \tau}$ ;
- (iii) a realization datum  $(\mathcal{E}, \mathcal{R})$ , membership  $C_{\tau, W}F \in \mathcal{E}$ , and the realized object  $A_{F, W, \tau}$ ;
- (iv) the amplitude certificate
$$A_{F, W, \tau} \in D^{[-1, 0]}(k\text{-mod});$$
- (v) the fixed residual-edge policy  $\text{Edge}_1$  and its policy identifier;
- (vi) an edge-realization certificate  $\epsilon_{F, W, \tau}$ , with objectwise or natural-family scope explicitly declared;
- (vii) a field coefficient declaration for  $k$ ;
- (viii) a non-circularity entry confirming that  $\mathcal{E}$ ,  $\mathcal{R}$ , and  $\text{Edge}_1$  were fixed independently of the desired pass/reject verdict;
- (ix) representative-equivalence and realization-equivalence audit data;
- (x) immutable artifact references sufficient to reconstruct the representative, realization, amplitude, and edge certificates;
- (xi) the applicable claim-status and manifest entries under the canonical Chapter 8 audit contract.

A windowed object  $(F; W, \tau)$  is called *eligible for the Ext-edge* only when the required record is complete and consistent. An invoked but incomplete, circular, or ill-typed record yields an `undefined` categorical clause, not a passing clause and not a mathematical counterexample.

**Remark 4.40** (Eligibility is stronger than having a representative). An admissible representative alone is not enough to invoke  $\text{Ext}^1$ . The  $\text{Ext}$ -edge also needs a scope-level realization functor, an amplitude certificate, a fixed residual-edge policy, and an actual edge-realization certificate. The canonical derived-category edge proved below does not construct that interface certificate. Consequently, the one-way bridge is a conditional Core/interface theorem, not an unconditional identification of persistent homology with derived  $\text{Ext}$ .

**Remark 4.41** (Gap certification and non-scalar obligations are separate). The eligible realization record does not replace the Chapter 3 comparison record. For audited B- $\text{Gate}^+$  use, every `metric_gap_certified` comparison still requires an evidenced pre-threshold discrepancy bound and the typed inequality

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B})$$

measured on the relevant two-sided pre-threshold profile family  $\mathfrak{B}$ . A comparison in `exact` mode instead carries its exact proof and is not automatically a robustness certificate. In mode `theorem_controlled_nonmetric`, the record cites the exact named theorem, hypotheses, source and target predicates, and transport ceiling; no metric radius is inferred.

Conversely, numerical clearance cannot manufacture an admissible representative, a realization functor, an amplitude certificate, an edge-realization certificate, or a complete manifest. Those are non-scalar proof obligations. The present chapter discharges only the categorical  $\text{Ext}^1$ -edge once exact repaired topological vanishing and full eligibility have been established.

#### 4.10 Degree- $n$ repaired input and standard eligibility

**Definition 4.42** (Degree- $n$  repaired profile). Fix an integer  $n \geq 1$ . The degree- $n$  repaired profile is

$$P_{n,W,\tau}(F) := \text{Post}_{n,W,\tau}(F) := \text{Eval}_{n,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_n(F)).$$

It is the exact profile whose vanishing may be established directly, by a complete Chapter 3 detector, or by a valid metric zero-transport chain.

**Definition 4.43** (Degree- $n$  compatible representative). An admissible representative  $C_{\tau,W}F$  is *degree- $n$  compatible* if its Chapter 3 record contains an actual profile isomorphism

$$\rho_{n,F,W,\tau} : \mathbf{P}_n(C_{\tau,W}F) \xrightarrow{\sim} \text{Eval}_{n,W,\tau}(F).$$

The monitored degree, source and target identities, equivalence strength, and immutable proof artifact are part of the data. Compatibility in degree  $n$  does not imply compatibility in another degree.

**Definition 4.44** (Degree-indexed realization datum). A degree-indexed realization datum consists of a declared domain  $\mathcal{E}_n \subseteq \text{FiltCh}(k)$ , a genuine scope-level functor

$$\mathcal{R}_n : \mathcal{E}_n \longrightarrow D^b(k\text{-mod}),$$

and the independently meaningful realized object

$$A_{n,F,W,\tau} := \mathcal{R}_n(C_{\tau,W}F).$$

The domain, functor, version, coefficient field, and artifact identities are fixed before evaluation of the desired zero or nonzero verdict. Different values of  $n$  may use the same realization only when that common scope and its compatibility are explicitly declared.

**Definition 4.45** (Higher residual edge extractor). For a declared repaired-profile class  $\mathcal{P}_n$ , a degree- $n$  residual edge extractor is a scope-level assignment

$$E_n : \mathcal{P}_n \longrightarrow \text{Vect}_k^{\text{fd}}$$

that is invariant under the declared profile isomorphisms and satisfies

$$E_n(0) = 0.$$

When morphism-level naturality is invoked,  $E_n$  is a functor on the specified morphism category. Objectwise invariance is not silently promoted to naturality, faithfulness, conservativity, or zero-reflection.

**Definition 4.46** (Higher edge-realization certificate). A degree- $n$  edge-realization certificate is an artifact-backed isomorphism

$$\epsilon_{n;F,W,\tau} : H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} E_n(P_{n,W,\tau}(F)).$$

The record declares whether this certificate is objectwise or belongs to a natural family. If naturality is claimed, for every allowed morphism the square formed by  $H^{-n}(\mathcal{R}_n(-))$ ,  $E_n(\text{Eval}_{n,W,\tau}(-))$ , and the corresponding  $\epsilon_n$  commutes.

**Definition 4.47** (Standard  $n$ -eligible realization record). For  $n \geq 1$ , a standard  $n$ -eligible realization record for  $(F; W, \tau)$  contains:

- (i) the exact degree- $n$  repaired profile and its Chapter 3 record;
- (ii) a degree- $n$  compatible representative and the actual  $\rho_{n;F,W,\tau}$ ;
- (iii) a degree-indexed realization datum and membership  $C_{\tau,W}F \in \mathcal{E}_n$ ;
- (iv) the conservative standard amplitude certificate

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod}),$$

or a separately proved admissible replacement explicitly registered at its exact strength;

- (v) a fixed higher edge extractor  $E_n$  and its policy identifier;
- (vi) the higher edge-realization certificate  $\epsilon_{n;F,W,\tau}$ ;
- (vii) zero compatibility  $P = 0 \Rightarrow E_n(P) = 0$ ;
- (viii) the objectwise or natural-family scope of every item;
- (ix) non-circularity, representative-change, and realization-change records;
- (x) complete claim, artifact, and manifest references.

An invoked but incomplete, circular, stale, or ill-typed record yields `undefined`. It does not yield a false theorem, a passing clause, or a zero Ext group.

**Remark 4.48** (Degree-one specialization). Definitions 4.36–4.39 are retained as the canonical v19 degree-one package. They agree with the preceding definitions at  $n = 1$ , with

$$\mathcal{R}_1 = \mathcal{R}, \quad E_1 = \text{Edge}_1, \quad A_{1;F,W,\tau} = A_{F,W,\tau}.$$

No change in the strength of the inherited degree-one theorem is intended.

**Remark 4.49** (Amplitude is an eligibility policy). The amplitude interval  $[-n, 0]$  makes the realization route auditable and isolates the intended negative edge. Over a field, however, the canonical algebraic identification of  $\text{Ext}^n(A, k)$  with the dual of  $H^{-n}(A)$  holds for every bounded  $A$ . The amplitude policy and the algebraic theorem must therefore not be conflated.



## 4.11 Field-coefficient higher Ext-edges

In this chapter,  $k\text{-mod}$  denotes finite-dimensional  $k$ -vector spaces. The superscript in  $D^{[-1,0]}$  refers to the standard cohomological  $t$ -structure.

**Theorem 4.50** (Canonical higher Ext-edge over a field). *Let  $n \geq 0$  and let  $A \in D^b(k\text{-mod})$ , where  $k$  is a field. There is a canonical isomorphism, natural contravariantly in  $A$ ,*

$$\text{Ext}^n(A, k) := \text{Hom}_{D^b(k\text{-mod})}(A, k[n]) \xrightarrow{\sim} \text{Hom}_k(H^{-n}(A), k).$$

*Proof.* The standard hyper-Ext spectral sequence gives

$$E_2^{p,q} = \text{Ext}_k^p(H^{-q}(A), k) \implies \text{Ext}^{p+q}(A, k).$$

Because  $k\text{-mod}$  is semisimple,  $\text{Ext}_k^p(V, k) = 0$  for every finite-dimensional  $V$  and every  $p > 0$ . Hence the spectral sequence is concentrated in the column  $p = 0$ , has no nonzero differential or extension ambiguity, and its total-degree  $n$  term is canonically

$$E_2^{0,n} = \text{Hom}_k(H^{-n}(A), k).$$

The spectral sequence and its edge identification are functorial in  $A$ , which gives the stated naturality.  $\square$

**Corollary 4.51** (Higher edge vanishing and reflection). *For  $A \in D^b(k\text{-mod})$  and  $n \geq 0$ ,*

$$\text{Ext}^n(A, k) = 0 \iff H^{-n}(A) = 0.$$

*Proof.* Theorem 4.50 identifies the Ext group with the linear dual of the finite-dimensional vector space  $H^{-n}(A)$ . A finite-dimensional vector space over a field has zero dual if and only if it is zero.  $\square$

**Lemma 4.52** (Formality over a field; noncanonical splitting). *Let  $A \in D^b(k\text{-mod})$ , where  $k$  is a field. Then there exists an isomorphism*

$$A \cong \bigoplus_{j \in \mathbb{Z}} H^j(A)[-j]$$

*in  $D^b(k\text{-mod})$ . In general this direct-sum splitting depends on choices and is not used to prove the naturality of the Ext-edge identification below.*

*Proof.* Every short exact sequence of  $k$ -vector spaces splits, so every bounded complex is quasi-isomorphic to its cohomology complex with zero differential. Choosing splittings gives the stated derived isomorphism. The choice-dependence is precisely why the natural edge theorem is proved instead from the canonical truncation triangle.  $\square$

**Theorem 4.53** (Canonical two-term Ext<sup>1</sup>-edge identification). *Let  $A \in D^{[-1,0]}(k\text{-mod})$ . Then there is a canonical isomorphism, natural in  $A$ ,*

$$\text{Ext}^1(A, k) := \text{Hom}_{D^b(k\text{-mod})}(A, k[1]) \xrightarrow{\sim} \text{Hom}_k(H^{-1}(A), k).$$

*Proof.* The standard  $t$ -structure gives the canonical truncation triangle

$$H^{-1}(A)[1] \longrightarrow A \longrightarrow H^0(A) \xrightarrow{+1} H^{-1}(A)[2].$$

Apply the contravariant cohomological functor  $\mathrm{Hom}_{D^b(k\text{-mod})}(-, k[1])$ . The resulting long exact sequence contains the natural exact segment

$$\begin{aligned} \mathrm{Hom}_{D^b}(H^0(A), k[1]) &\longrightarrow \mathrm{Hom}_{D^b}(A, k[1]) \\ &\longrightarrow \mathrm{Hom}_{D^b}(H^{-1}(A)[1], k[1]) \longrightarrow \mathrm{Hom}_{D^b}(H^0(A), k[2]). \end{aligned}$$

Because  $k\text{-mod}$  is semisimple,

$$\mathrm{Hom}_{D^b}(H^0(A), k[1]) = \mathrm{Ext}_k^1(H^0(A), k) = 0$$

and

$$\mathrm{Hom}_{D^b}(H^0(A), k[2]) = \mathrm{Ext}_k^2(H^0(A), k) = 0.$$

Hence the middle arrow is an isomorphism. Finally,

$$\mathrm{Hom}_{D^b}(H^{-1}(A)[1], k[1]) \cong \mathrm{Hom}_k(H^{-1}(A), k).$$

The truncation triangle and the long exact sequence are functorial in  $A$ , so the resulting isomorphism is natural.  $\square$

**Corollary 4.54** (Naturality square). *For every morphism  $f : A \rightarrow A'$  in  $D^{[-1,0]}(k\text{-mod})$ , the square*

$$\begin{array}{ccc} \mathrm{Ext}^1(A', k) & \longrightarrow & \mathrm{Ext}^1(A, k) \\ \downarrow \sim & & \downarrow \sim \\ \mathrm{Hom}_k(H^{-1}(A'), k) & \longrightarrow & \mathrm{Hom}_k(H^{-1}(A), k) \end{array}$$

*commutes, where the horizontal maps are induced contravariantly by  $f$  and by  $H^{-1}(f)$ , respectively.*

*Proof.* This is the naturality assertion of Theorem 4.53 written as a commutative square.  $\square$

**Corollary 4.55** (Edge vanishing criterion). *If  $A \in D^{[-1,0]}(k\text{-mod})$  and  $H^{-1}(A) = 0$ , then*

$$\mathrm{Ext}^1(A, k) = 0.$$

*Proof.* Apply Theorem 4.53.  $\square$

**Remark 4.56** (Why the field hypothesis matters). Over a general coefficient ring, the truncation argument yields an exact sequence whose outer terms include  $\mathrm{Ext}^1(H^0(A), k)$  and  $\mathrm{Ext}^2(H^0(A), k)$ . These groups need not vanish, so the displayed natural isomorphism is not a theorem in that generality without additional projectivity, injectivity, or homological-dimension hypotheses.

**Remark 4.57** (Splitting versus naturality). Lemma 4.52 explains the semisimple shape of the answer, but a chosen direct-sum decomposition of  $A$  need not be natural. Theorem 4.53 obtains the same group canonically from truncation and therefore supports representative and realization functoriality.

## 4.12 Higher one-way discharge and zero-reflection boundary

**Theorem 4.58** (Eligible higher Ext-edge identification). *Let  $n \geq 1$  and suppose  $(F; W, \tau)$  has a complete standard  $n$ -eligible realization record. Then there is an artifact-backed isomorphism*

$$\mathrm{Ext}^n(A_{n;F,W,\tau}, k) \xrightarrow{\sim} \mathrm{Hom}_k(E_n(\mathrm{Eval}_{n,W,\tau}(F)), k).$$

*It is natural on exactly the morphism scope on which both the realization family and the edge-realization certificates are natural.*

*Proof.* Apply Theorem 4.50 to  $A_{n;F,W,\tau}$ , then compose with the dual of the declared isomorphism

$$H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} E_n(\text{Eval}_{n,W,\tau}(F)).$$

The claimed naturality is the composite of the naturality of the derived edge and the declared naturality of the edge-realization family.  $\square$

**Theorem 4.59** (Higher one-way repaired discharge). *Let  $n \geq 1$  and let  $(F; W, \tau)$  be standard  $n$ -eligible. Then*

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

*Proof.* Set  $P = \text{Eval}_{n,W,\tau}(F)$ . Zero compatibility gives  $E_n(P) = E_n(0) = 0$ . Theorem 4.58 then identifies the Ext group with  $\text{Hom}_k(0, k) = 0$ .  $\square$

**Definition 4.60** (Natural higher zero-reflection package). A standard  $n$ -eligible record has a *natural zero-reflection package* on a declared repaired-profile class  $\mathcal{P}_n$  when:

- (i)  $E_n$  is fixed independently of the desired verdict;
- (ii)  $E_n$  is natural on the declared morphism class;
- (iii) the edge-realization certificates form a natural family;
- (iv) a proved theorem establishes

$$E_n(P) = 0 \implies P = 0 \quad (P \in \mathcal{P}_n);$$

- (v) representative-change and realization-change compatibility pass;
- (vi) every completeness, stage, endpoint, family-membership, and artifact obligation used by the zero-reflection theorem passes independently.

This definition instantiates the Chapter 1 natural zero-reflection upgrade.

**Theorem 4.61** (Zero-reflecting higher vanishing equivalence). *Let  $n \geq 1$ . Suppose  $(F; W, \tau)$  is standard  $n$ -eligible, its repaired profile  $P = \text{Eval}_{n,W,\tau}(F)$  belongs to  $\mathcal{P}_n$ , and the record has a natural higher zero-reflection package on  $\mathcal{P}_n$ . Then*

$$\boxed{\text{Eval}_{n,W,\tau}(F) = 0 \iff \text{Ext}^n(A_{n;F,W,\tau}, k) = 0}.$$

*This is an equivalence of vanishing predicates on the declared scope.*

*Proof.* The forward implication is Theorem 4.59. Conversely, if the Ext group vanishes, Theorem 4.58 and finite-dimensional dual reflection imply  $E_n(P) = 0$ . The separately proved zero-reflection theorem then gives  $P = 0$ .  $\square$

**Proposition 4.62** (Functoriality of a fixed stencil detector). *Fix a window, a threshold  $\tau$ , and a finite stencil  $T$  in the valid Chapter 3 detector domain. On any declared category of persistence modules and actual persistence morphisms on that domain, the assignment*

$$P \longmapsto D_{T,\tau}^{\text{stc}}(P) = \bigoplus_{t \in T} \text{im}(P(t) \rightarrow P(t + \tau))$$

*defines a functor to  $\text{Vect}_k^{\text{fd}}$ .*

*Proof.* For a persistence morphism  $f : P \rightarrow Q$ , naturality of the structure maps gives, for each  $t \in T$ , a commutative square

$$\begin{array}{ccc} P(t) & \longrightarrow & P(t + \tau) \\ \downarrow f_t & & \downarrow f_{t+\tau} \\ Q(t) & \longrightarrow & Q(t + \tau). \end{array}$$

Hence  $f_{t+\tau}$  sends the image of the upper horizontal map into the image of the lower horizontal map and induces a linear map between those image spaces. Taking the finite direct sum over  $T$  gives  $D_{T,\tau}^{\text{stc}}(f)$ . Identity and composition are inherited componentwise from the persistence morphisms.  $\square$

**Corollary 4.63** (Complete fixed-stencil higher vanishing equivalence). *Let  $n \geq 1$  and let  $(F; W, \tau)$  be standard  $n$ -eligible on a declared profile class  $\mathcal{P}_n$ . Assume:*

(i) *the independently meaningful edge extractor is the fixed-stencil functor*

$$E_n(P) = D_{T,\tau}^{\text{stc}}(P);$$

(ii) *the higher edge-realization certificates form a natural family;*

(iii) *a Chapter 3 completeness theorem, with all stage, endpoint, coverage, family, and artifact obligations discharged, proves*

$$D_{T,\tau}^{\text{stc}}(P) = 0 \iff P = 0 \quad (P \in \mathcal{P}_n);$$

(iv) *representative-change and realization-change compatibility pass.*

Then

$$\boxed{D_{T,\tau}^{\text{stc}}(\text{Eval}_{n,W,\tau}(F)) = 0 \iff \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.}$$

*Proof.* Proposition 4.62 supplies the required naturality of the fixed detector. The Chapter 3 completeness theorem supplies zero-reflection. The remaining hypotheses complete the natural zero-reflection package, so Theorem 4.61 applies.  $\square$

**Remark 4.64** (Detector completeness is not edge realization). A Chapter 3 completeness theorem proves that a detector reflects zero on a specified profile class. It does not construct the independently meaningful realized object  $A_n(F)$ , nor the isomorphism  $H^{-n}(A_n(F)) \cong E_n(P)$ . Conversely, an edge-realization certificate does not prove detector completeness. Both obligations are required for the zero-reflecting higher equivalence.

**Remark 4.65** (No object-level  $\text{PH}_n$ -to- $\text{Ext}^n$  isomorphism). Theorem 4.61 identifies only the truth values of two zero predicates. A persistence profile, its homology, an extracted finite-dimensional edge, and an  $\text{Ext}$  group live in different typed layers. No isomorphism  $\text{PH}_n \cong \text{Ext}^n$  is asserted.

### 4.13 The one-way bridge $\text{PH}_1 \Rightarrow \text{Ext}^1$

**Theorem 4.66** (Conditional one-way repaired bridge). *Let  $(F; W, \tau)$  be eligible for the  $\text{Ext}$ -edge in the sense of Definition 4.39. If*

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

*then*

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

*Equivalently, when the representative realizes the repaired degree-one profile,*

$$\text{PH}_1(C_{\tau,W}F) = 0 \implies \text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

*Proof.* Set

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F).$$

By eligibility,

$$A_{F,W,\tau} \in D^{[-1,0]}(k\text{-mod}).$$

The edge-realization certificate gives

$$H^{-1}(A_{F,W,\tau}) \cong \text{Edge}_1(\text{Eval}_{1,W,\tau}(F)).$$

If  $\text{Eval}_{1,W,\tau}(F) = 0$ , then by zero-compatibility of the edge extractor,

$$\text{Edge}_1(\text{Eval}_{1,W,\tau}(F)) = \text{Edge}_1(0) = 0.$$

Hence

$$H^{-1}(A_{F,W,\tau}) = 0.$$

Corollary 4.55 now gives

$$\text{Ext}^1(A_{F,W,\tau}, k) = 0,$$

which is the asserted conclusion.  $\square$

**Remark 4.67** (Discharge theorem, not equivalence). Theorem 4.66 is a conditional discharge theorem for the categorical clause of B-Gate<sup>+</sup>. Its unconditional mathematical input is the canonical two-term Ext-edge; its interface hypothesis is the eligible realization record. It states that repaired topological vanishing is sufficient to discharge the eligible Ext<sup>1</sup>-edge. It does not state that categorical vanishing reconstructs the repaired degree-one barcode profile, nor does it prove the edge-realization certificate from persistence data.

**Remark 4.68** (No use of search, spectral, or CCC reverse principles). The bridge uses only the eligible realization record and the two-term field-coefficient Ext-edge. The realization and residual-edge policies must be fixed before the objectwise verdict is evaluated. The proof does not use CCC authorization, spectral-gap hypotheses, Bridge Programs, validity maps, advisory invariants, Search-layer outputs, or AI-generated confidence values.

## 4.14 Gate integration

**Definition 4.69** (Higher Ext-edge discharge record). For an invoked degree  $n \geq 1$ , a higher Ext-edge discharge record contains:

- (i) the exact repaired-profile zero statement, including the detector or comparison theorem by which it was established;
- (ii) the complete standard  $n$ -eligible record;
- (iii) the theorem identifier for Theorem 4.59;
- (iv) the higher derived edge and edge-realization artifacts;
- (v) the categorical conclusion  $\text{Ext}^n(A_{n;F,W,\tau}, k) = 0$ ;
- (vi) when a reverse inference is consumed, the complete natural zero-reflection package and the theorem identifier for Theorem 4.61;
- (vii) immutable Claim, detector, representative, realization, and artifact references sufficient for replay.

Metric transport, when used, is cited through the Chapter 2–3 repaired-zero chain; the higher Ext theorem itself does not create a metric certificate.

**Definition 4.70** (Typed degree- $n$  Ext-clause status). For every invoked  $n \geq 1$ , the categorical clause has exactly one atomic status:

`not_invoked, undefined, pass, reject.`

Here `undefined` means that an invoked typing, representative, realization, amplitude, edge, naturality, completeness, claim, or artifact obligation is missing or ill-typed. The status `reject` requires evidence that the well-typed Ext group is nonzero; missing evidence is never reported as rejection.

**Proposition 4.71** (Higher B-Gate<sup>+</sup> categorical discharge). *For every invoked degree  $n \geq 1$ , a complete standard  $n$ -eligible record and*

$$\text{Eval}_{n,W,\tau}(F) = 0$$

*discharge the corresponding categorical clause:*

$$\text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

*Proof.* This is Theorem 4.59 read as an atomic gate clause. □

**Definition 4.72** (Aggregate higher categorical gate status). For a finite monitored degree set  $I_{\text{mon}}$ , the higher categorical gate passes exactly when every invoked degree- $n$  Ext clause passes and every excluded degree is justified as `not_invoked`. One rejected or required undefined degree prevents aggregate pass. No cancellation or numerical compensation is permitted.

**Definition 4.73** (Ext-edge discharge record). An *Ext-edge discharge record* for  $(F; W, \tau)$  contains:

- (i) the repaired evaluation artifact establishing the exact post-threshold statement  $\text{Eval}_{1,W,\tau}(F) = 0$ ;
- (ii) the comparison mode and, only when that exact vanishing was transported from another object or run, the corresponding exact or two-sided metric comparison record;
- (iii) the complete eligible realization record of Definition 4.39;
- (iv) the theorem identifier for Theorem 4.66;
- (v) the canonical Ext-edge identification and its naturality data;
- (vi) the resulting categorical conclusion

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0;$$

- (vii) immutable artifact and claim-register references sufficient for replay.

If repaired topological vanishing is asserted under perturbation, the record also cites the relevant two-sided threshold-gap certificate, evidenced pre-threshold discrepancy bound, and scalarized metric budget. Exact mode instead cites the exact equality or profile-isomorphism proof.

**Definition 4.74** (Typed Ext-edge clause status). The categorical clause attached to  $(F; W, \tau)$  has one of the following audit meanings:

- (i) `not_invoked`: the run manifest explicitly places the Ext-clause outside scope;

- (ii) **undefined**: the clause is invoked but a required representative, eligibility, amplitude, edge, artifact, claim-status, or manifest record is absent or ill-typed;
- (iii) **pass**: the clause is well-typed and  $\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0$ , whether by Theorem 4.66 or by a separately verified direct computation;
- (iv) **reject**: the clause is well-typed and the eligible  $\text{Ext}^1$ -group is evidenced to be nonzero.

The statuses **undefined** and **not\_invoked** are distinct; neither is a fabricated zero group.

**Proposition 4.75** (B-Gate<sup>+</sup> categorical discharge). *Let  $(F; W, \tau)$  be eligible for the Ext-edge. If the repaired topological clause*

$$\text{Eval}_{1, W, \tau}(F) = 0$$

*holds, then the categorical  $\text{Ext}^1$ -clause of B-Gate<sup>+</sup> is satisfied:*

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0.$$

*Proof.* This is exactly Theorem 4.66 read as a gate-discharge statement. □

**Remark 4.76** (What remains after categorical discharge). The Ext-edge discharge is only one atomic B-Gate<sup>+</sup> clause. A full audited pass still requires, independently, the exact topological statement

$$\text{Eval}_{1, W, \tau}(F) = 0,$$

complete and consistent manifest/claim-status data, and every comparison obligation in its declared mode. In particular, a metric transport through hard deletion requires

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B})$$

together with an evidenced discrepancy bound, while exact mode requires its exact proof. The pass also requires a Type IV scope status equal to

$$\text{not\_invoked} \quad \text{or} \quad \text{certified\_zero}.$$

If tower semantics are invoked, the comparison morphism, terminal-tameness, and defect diagnostics must be present; **undefined** and **obstructed** fail the gate. No passing categorical clause cancels another rejected or required undefined clause.

## 4.15 Representative and realization invariance

**Definition 4.77** (Degree- $n$  realization-equivalent representatives). Two standard  $n$ -eligible representatives are degree- $n$  realization-equivalent when a declared derived isomorphism

$$\alpha_n : A_{n; F, W, \tau} \xrightarrow{\sim} A'_{n; F, W, \tau}$$

is supplied and the two higher edge-realization certificates commute with  $H^{-n}(\alpha_n)$  and the declared profile identification.

**Proposition 4.78** (Higher representative invariance). *For degree- $n$  realization-equivalent eligible representatives, the Ext groups are identified contravariantly along  $\alpha_n$ , and both the one-way discharge and any declared natural zero-reflecting equivalence have the same truth value for the two representatives.*

*Proof.* Apply  $\text{Hom}_{D^b}(-, k[n])$  to  $\alpha_n$ . Naturality of Theorem 4.50 identifies this map with dualizing  $H^{-n}(\alpha_n)$ . Compatibility of the edge certificates and, when invoked, of the zero-reflection package transports all hypotheses and conclusions.  $\square$

**Definition 4.79** (Realization-equivalent representatives). Two eligible representatives  $C_{\tau, W}F$  and  $C'_{\tau, W}F$  are called *realization-equivalent* if their realized objects are isomorphic in the derived category:

$$\mathcal{R}(C_{\tau, W}F) \cong \mathcal{R}(C'_{\tau, W}F) \quad \text{in } D^b(k\text{-mod}),$$

and the edge-realization certificates are compatible with that isomorphism.

**Proposition 4.80** (Representative invariance of Ext discharge). *Let the realization-equivalent eligible representatives be*

$$C_{\tau, W}F \quad \text{and} \quad C'_{\tau, W}F.$$

*Then the declared realization isomorphism induces an identification of the categorical groups, and the truth value of*

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0$$

*is independent of the representative choice.*

*Proof.* The declared isomorphism

$$\alpha : \mathcal{R}(C_{\tau, W}F) \xrightarrow{\sim} \mathcal{R}(C'_{\tau, W}F)$$

induces an isomorphism under the contravariant functor  $\text{Hom}_{D^b}(-, k[1])$ . This identification is canonical relative to the recorded  $\alpha$ ; no independence from an unrecorded choice of realization is asserted. By Corollary 4.54, it agrees with dualizing the induced map on  $H^{-1}$ . Compatibility of the edge-realization certificates therefore transports the discharge data, and vanishing is invariant.  $\square$

**Remark 4.81** (Profile invariance is weaker). Profile-level representative invariance from Chapter 3 does not by itself imply realization-level invariance. For Ext-use, the proof object must record realization-equivalence or must commit to a specific representative. This prevents the  $\text{Ext}^1$ -verdict from depending on an untracked representative choice.

#### 4.16 Interaction with the $E_1$ detector

**Proposition 4.82** (Complete detector higher Ext discharge). *Let  $n \geq 1$ , suppose  $(F; W, \tau)$  is standard  $n$ -eligible, and let a Chapter 3 detector certificate on the exact profile stage prove*

$$D(\text{Eval}_{n, W, \tau}(F)) = 0 \implies \text{Eval}_{n, W, \tau}(F) = 0.$$

*If the detector value is certified zero, then*

$$\text{Ext}^n(A_{n, F, W, \tau}, k) = 0.$$

*If the detector is also the declared natural zero-reflecting edge  $E_n$ , then its zero value is equivalent to Ext vanishing under Theorem 4.61.*

*Proof.* Detector completeness gives exact repaired-profile vanishing. Apply Theorem 4.59. The final assertion is the zero-reflecting theorem specialized to the declared detector edge.  $\square$



**Proposition 4.83** (Operational  $E_1$ -detector discharge). *Suppose  $(F; W, \tau)$  is eligible for the Ext-edge and the canonical, non-truncated operational detector of Chapter 3 satisfies*

$$E_1(F; W, \tau) = 0.$$

*Then*

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0.$$

*Proof.* By the canonical detector rule of Chapter 3,  $E_1 = 0$  already excludes both finite and infinite residual bars and therefore implies

$$\text{Eval}_{1, W, \tau}(F) = 0.$$

The conclusion follows from Theorem 4.66. □

**Remark 4.84** (No stability or budget role for  $E_1$ ). The preceding proposition uses the canonical  $E_1$  only as a vanishing detector. It does not make  $E_1$  a bottleneck-stable quantity, a threshold-gap certificate, a metric-ledger component, or a replacement for

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B}).$$

A truncated detector has its own policy and does not inherit the proposition without a separate proof.

#### 4.17 Non-converse discipline

**Proposition 4.85** (Higher reverse inference requires zero-reflection). *Under standard  $n$ -eligibility alone,*

$$\text{Ext}^n(A_{n; F, W, \tau}, k) = 0$$

*implies only*

$$E_n(\text{Eval}_{n, W, \tau}(F)) = 0.$$

*It implies repaired-profile vanishing only when a separate zero-reflection theorem for  $E_n$  is available on the declared profile class.*

*Proof.* The first implication follows from Theorem 4.58 and finite-dimensional dual reflection. The passage from  $E_n(P) = 0$  to  $P = 0$  is exactly the additional zero-reflection hypothesis and is not contained in eligibility. □

**Definition 4.86** (Synthetic detector object). For a finite-dimensional detector output  $D(P)$  and  $n \geq 1$ , define the shifted encoded object

$$A_{D, n}^{\text{det}}(P) := D(P)[n] \quad \text{in } D^b(k\text{-mod}).$$

This object is called *synthetic* unless it is independently identified, through a natural realization theorem, with a purpose-faithful external or Core realization.

**Proposition 4.87** (Encoded detector duality). *For every finite-dimensional detector output  $D(P)$ ,*

$$\text{Ext}^n(A_{D, n}^{\text{det}}(P), k) \cong \text{Hom}_k(D(P), k).$$

*Consequently the encoded Ext group vanishes if and only if  $D(P) = 0$ .*

*Proof.* The shifted object has  $H^{-n}(D(P)[n]) = D(P)$  and no other cohomology. Apply Theorem 4.50 and finite-dimensional dual reflection. □

**Remark 4.88** (Synthetic duality non-promotion rule). Proposition 4.87 is a valid algebraic encoding theorem, but by itself it merely rewrites detector information in a derived category. It is classified as *Detector Duality* or *Encoded Ext Theorem*; it is not the principal higher Bridge, does not prove a realization theorem, and does not establish  $\mathrm{PH}_n \cong \mathrm{Ext}^n$ .

**Remark 4.89** (General coefficient rings are outside the v21 Core). Over a general coefficient ring  $R$ , the spectral sequence

$$E_2^{p,q} = \mathrm{Ext}_R^p(H^{-q}(A), R) \implies \mathrm{Ext}_R^{p+q}(A, R)$$

need not collapse. Higher differentials, extension data, and nonzero outer terms may obstruct an edge isomorphism. Any ring-valued higher bridge belongs to a separately proved Technical Extension or successor release.

**Proposition 4.90** (Failure of the reverse bridge for a natural nonfaithful edge). *Under Definitions 4.36–4.39, the implication*

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \implies \mathrm{Eval}_{1,W,\tau}(F) = 0$$

*is false in general, even when the realization and edge extractor are fixed functorially at scope level.*

*Proof.* Choose  $0 < \tau < L$ , choose  $\delta > 0$ , set

$$W = [0, L + \delta), \quad t_* = L + \frac{\delta}{2},$$

and let  $F$  be a finite filtered complex with a degree-one generator born at 0 and a degree-two generator born at  $L$  whose boundary kills the degree-one generator. Its degree-one persistence barcode on  $W$  is the single interval  $[0, L)$ . Since  $L > \tau$ , hard threshold deletion retains that interval, so

$$\mathrm{Eval}_{1,W,\tau}(F) \cong k_{[0,L)} \neq 0.$$

Take this filtered model as the degree-one compatible representative.

Fix, independently of the verdict, the point-evaluation edge policy and realization functor

$$\mathrm{Edge}_1(P) := P(t_*), \quad \mathcal{R}_{t_*}(C) := H_1(C^{t_*})[1].$$

Both are functorial on the evident declared domain, and  $\mathcal{R}_{t_*}(C) \in D^{[-1,-1]}(k\text{-mod}) \subset D^{[-1,0]}(k\text{-mod})$ . The representative isomorphism induces the natural edge certificate

$$H^{-1}(\mathcal{R}_{t_*}(C_{\tau,W}F)) = H_1((C_{\tau,W}F)^{t_*}) \xrightarrow{\sim} \mathrm{Eval}_{1,W,\tau}(F)(t_*) = \mathrm{Edge}_1(\mathrm{Eval}_{1,W,\tau}(F)).$$

Because  $t_* > L$ , both sides are zero. Therefore

$$\mathrm{Ext}^1(\mathcal{R}_{t_*}(C_{\tau,W}F), k) = 0,$$

while the repaired persistence profile itself remains nonzero. Thus a point-evaluation edge can miss a transient bar, and reverse recovery requires a separate zero-reflection or faithfulness theorem.  $\square$

**Remark 4.91** (What a reverse theorem would require). A reverse bridge would need a separately proved zero-reflection condition, for example

$$\mathrm{Edge}_1(P) = 0 \implies P = 0$$

on the declared repaired-profile class, together with compatibility showing that the realization edge is exactly that extractor. Such faithfulness is a new bridge theorem with new hypotheses; it cannot be inferred from Ext vanishing, amplitude, or eligibility alone.

#### 4.18 Cone handoff to Chapter 7

**Definition 4.92** (Actual finite-diagram morphism record). An actual finite-diagram morphism record contains two objects  $V, V' \in \text{Rep}_k(A_{[m]})$ , an actual natural transformation

$$f : V \longrightarrow V',$$

all component maps  $f_i : V_i \rightarrow V'_i$ , the commuting arrow squares, source and target grid identities, and immutable artifacts. A bottleneck matching, equality of rank tables, or objectwise barcode correspondence is not an actual diagram morphism.

**Definition 4.93** (Linear cone handoff record). Given an actual finite-diagram morphism  $f : V \rightarrow V'$ , regard  $V, V'$  as complexes concentrated in cohomological degree zero and form

$$\text{Cone}(f) \in D^b(\text{Rep}_k(A_{[m]})).$$

The Chapter 7 cone handoff record is

$$\text{ConeHandoff}_\Gamma(f) := (\Gamma, V, V', f, \text{Cone}(f), \\ \text{variance, coefficients, strength, artifacts}).$$

It records the actual chain-level representative used to form the cone.

**Proposition 4.94** (Cone zero criterion in the finite linear derived category). *For an actual morphism  $f : V \rightarrow V'$  in  $\text{Rep}_k(A_{[m]})$ ,*

$$f \text{ is an isomorphism} \iff \text{Cone}(f) \simeq 0 \text{ in } D^b(\text{Rep}_k(A_{[m]})).$$

*Proof.* In any derived category, a morphism is an isomorphism exactly when its cone is a zero object. Here the objects are concentrated in degree zero, so the derived isomorphism agrees with isomorphism in the abelian category.  $\square$

**Remark 4.95** (Boundary of the cone result). The proposition is a standard local fact for the finite linear category. Chapter 7 owns general finite-category generators, cone probes, branching diagrams, and descent. No cone is formed from a distance value or a matching that lacks an actual morphism.

#### 4.19 Core-P to Core-DG interface record

**Definition 4.96** (Finite linear P2DG $_\Gamma$  interface). For a declared grid-complete source profile  $P$ , define

$$\text{P2DG}_\Gamma(P) := \text{Res}_\Gamma(P) \in \text{Rep}_k(A_{[m]})$$

when  $P$  is supplied as an actual persistence module. For a barcode isomorphism class, define the objectwise branch by

$$\text{P2DG}_\Gamma^{\text{obj}}(B) := \text{GridRep}_\Gamma(B).$$

The interface record contains the Chapter 1 independent fields: action mode, exactness profile, reflection profile, information profile, preserved/discarded data, proof artifacts, and prohibited stronger readings.

**Definition 4.97** (Finite linear P2DG record). A complete record contains:

- (i) the Chapter 2 P2DGInput and any Chapter 3 Cert2ReplInput record;

- (ii) a route kind `object_route` or `morphism_route`;
- (iii) the actual source persistence module or the explicit declaration that only a barcode isomorphism class is available;
- (iv) for a morphism route, an actual persistence morphism, one common grid complete for its source and target, and the induced natural transformation after restriction;
- (v) the grid completeness and encoding strength records, and any probe-choice or refinement comparison invoked across grid versions;
- (vi) the finite quiver object and, when applicable, the actual morphism action;
- (vii) barcode reconstruction and finite-signature artifacts;
- (viii) the two-axis realization classification;
- (ix) any  $n$ -eligible derived realization and Ext-edge record;
- (x) the route-supported Return set and the requested exact Return strength: observation, profile isomorphism class, decorated barcode, quiver object, quiver morphism, or Ext vanishing predicate;
- (xi) immutable dependencies and verifier status.

A quiver-morphism Return is admissible only on a morphism route with an actual source morphism and functorial restriction. An Ext-vanishing Return is admissible only when the separate  $n$ -eligible realization route passes; it is not a consequence of finite-grid equivalence alone.

**Theorem 4.98** (P-to-DG finite linear handoff). *On the ambient endpoint-complete  $\Gamma$ -constructible category,*

$$\text{P2DG}_\Gamma : \text{Pers}_k^\Gamma(W) \xrightarrow{\sim} \text{Rep}_k(A_{[m]})$$

*is an exact equivalence and therefore is full, faithful, conservative, zero-reflecting, and lossless at persistence-module strength. When the source is restricted to repaired profiles at a fixed positive threshold, the same functor has those properties onto the repaired essential image, which may be a proper full subcategory of  $\text{Rep}_k(A_{[m]})$ . For a barcode-only source, the objectwise branch is lossless on the declared finite decorated barcode but has no asserted morphism fullness or functoriality. On the actual-module branch, a quiver morphism is returned only from an actual persistence morphism on a common complete grid; equality of object invariants does not manufacture that morphism. Neither branch is a full faithful embedding from all constructible persistence modules on all grids into the general  $\text{Core}_{\text{DG}}$ .*

*Proof.* The actual-module statement is Theorem 4.15 together with Theorem 4.22. The barcode-only statement follows from the atom-by-atom interval correspondence, while the absence of a declared morphism action prevents a stronger functorial conclusion. The final boundary follows because the source subcategory, window, grid, and linear target are fixed and finite.  $\square$

**Theorem 4.99** (Finite profile-to-quiver-to-Return closure). *Let  $P = \text{Eval}_{n,W,\tau}(F)$  be a finite repaired profile with a complete endpoint grid and P2DG record. Then Chapter 4 closes the internal finite route*

$$P \longrightarrow \text{P2DG}_\Gamma(P) \longrightarrow (r(i, j), \mu(i, j), \Sigma_\Gamma^{\text{proj/Ext}}) \longrightarrow \mathcal{B}(P)$$

*at decorated persistence-profile strength. If, in addition, a complete standard  $n$ -eligible realization record is supplied, the same source zero predicate also closes the conditional branch*

$$P = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

*Under a separately proved natural zero-reflection package, the latter branch is an equivalence of vanishing predicates.*

*Proof.* The diagram construction is Theorem 4.98. Rank and barcode reconstruction are Theorem 4.25 and Corollary 4.26. The finite signature statement is Theorem 4.31. The conditional Ext branch is Theorem 4.59, upgraded by Theorem 4.61 when its additional hypotheses hold.  $\square$

**Remark 4.100** (Internal Return only). The closure theorem returns a persistence profile, decorated barcode, finite linear diagram, finite signature, or typed Ext-vanishing predicate. It is not an external-domain Return theorem. External semantics and Return belong to Chapter 9.

## 4.20 Chapter 4/Chapter 7 boundary

Chapter 4 owns:

one parameter, finite atom chain, and  $A_{[m]}$ ,  
finite barcode/rank reconstruction,  
persistence-specific finite signatures.

Chapter 7 owns:

arbitrary finite categories or posets and branching diagrams,  
generator reconstruction and general cone probes,  
object/morphism descent and global objects.

A linear P2DG result is not promoted to the general finite-category theorem by relabeling the indexing category.

## 4.21 Canonical audit-manifest extension

**Definition 4.101** (Eligibility manifest extension). The Chapter 4 eligibility entry extends the Chapter 1 manifest and the Chapter 3 evaluation, detector, and representative records. For every invoked degree  $n$ , it records at least:

- (i) the pre-threshold and repaired degree- $n$  profile identities;
- (ii) the exact route by which repaired-profile vanishing was established: direct profile proof, complete detector, exact comparison, or metric zero-transport;
- (iii) the detector stage, endpoint convention, stencil or birth/germ policy, completeness theorem, family scope, and artifact identity whenever a detector is consumed;
- (iv) the representative identifier and actual  $\rho_{n,F,W,\tau}$ ;
- (v) the realization domain  $\mathcal{E}_n$ , functor  $\mathcal{R}_n$ , object  $A_{n;F,W,\tau}$ , versions, and immutable artifacts;
- (vi) the amplitude certificate or exact registered replacement;
- (vii) the fixed edge extractor  $E_n$ , its objectwise or natural scope, and the edge-realization certificate  $\epsilon_{n;F,W,\tau}$ ;
- (viii) zero compatibility and, when a converse is used, the complete natural zero-reflection package;
- (ix) representative-change and realization-change compatibility;
- (x) the typed degree- $n$  Ext-clause status and aggregate categorical status;

- (xi) the theorem or direct-computation identifiers supporting pass or reject;
- (xii) the exact/metric comparison records independently required by the topological zero statement;
- (xiii) a synthetic-versus-natural classification for every shifted detector object or encoded Ext result;
- (xiv) the endpoint set, atom grid, singleton/open-atom convention, and terminal sentinel policy;
- (xv) the grid completeness and encoding strength records;
- (xvi) the finite quiver object, action mode, exactness, reflection, and information profiles;
- (xvii) rank-table, Möbius multiplicity, projective/Ext-signature, and barcode reconstruction artifacts whenever those Returns are invoked;
- (xviii) actual finite-diagram morphisms and cone handoff records whenever a cone or isomorphism criterion is consumed;
- (xix) the complete P2DG record and its exact Return ceiling;
- (xx) all Claim, CM, TB, proof, hash, and provenance links needed for independent replay.

For  $n = 1$ , this contains all fields of the inherited v19 eligibility entry.

**Remark 4.102** (Manifest status and failure routing). An invoked Ext-clause with missing eligibility data is undefined; it is not a mathematical counterexample and not `not_invoked` unless scope exclusion was explicitly declared. Under Chapter 8, an incomplete manifest yields audit undefinedness unless another evidenced clause already forces rejection, while an inconsistent manifest yields Type V rejection. Only a complete and consistent manifest can support a categorical pass.

## 4.22 Bridge package

**Theorem 4.103** (Eligible Ext-edge bridge package). *In the v21 dual Core, the following conservative package is available.*

- (i) *The complete v19 degree-one eligible bridge remains valid at its original strength.*
- (ii) *For every  $n \geq 0$  and bounded derived object over the field,*

$$\mathrm{Ext}^n(A, k) \cong \mathrm{Hom}_k(H^{-n}(A), k)$$

*canonically and naturally.*

- (iii) *A standard  $n$ -eligible record supplies an independently meaningful realized object, conservative amplitude typing, a fixed edge extractor, and an artifact-backed edge-realization certificate.*
- (iv) *Under zero compatibility,*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \implies \mathrm{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

- (v) *Under the separately proved natural zero-reflection package, the preceding implication upgrades to an equivalence of vanishing predicates.*

- (vi) *A complete fixed-stencil, birth, germ, or other Chapter 3 detector may supply zero-reflection only on its certified profile class and stage.*
- (vii) *A shifted detector object gives an encoded Ext duality but remains synthetic without an independent natural realization theorem.*
- (viii) *On an endpoint-complete finite atom grid, restriction gives an exact equivalence with the linear quiver category  $\text{Rep}_k(A_{[m]})$ .*
- (ix) *The finite rank table and Möbius inversion reconstruct the complete decorated barcode at the declared terminal policy.*
- (x) *The enriched projective/Ext signature reconstructs the finite quiver object, while un-enriched Ext dimensions alone are not complete.*
- (xi) *An actual finite-diagram morphism admits a cone handoff to Chapter 7; a metric matching does not.*
- (xii) *None of these clauses replaces metric, Type IV, transient-defect, manifest, claim-status, finite-to-infinite, or external Return-Theorem obligations.*

*Proof.* Items (i)–(vii) are the inherited and higher eligible Ext package: Theorems 4.66, 4.50, 4.59, and 4.61, together with the Chapter 3 detector specializations and Proposition 4.87. The finite linear-interface items are Theorems 4.15, 4.25, 4.31, and Proposition 4.94. The remaining boundary follows from the independent contracts of Chapters 1–3, 5, and 7–9.  $\square$

**Remark 4.104** (Package status). Theorem 4.103 is a conditional closure package. It separates unconditional field-coefficient derived algebra from declared realization hypotheses and from separately proved detector zero-reflection. It does not promote eligibility data, detector completeness, synthetic encoding, or a gate record into an external-domain theorem.

### 4.23 Explicit non-claims

The following are not asserted in this chapter.

- (i) No unconditional object-level equivalence  $\text{PH}_n \cong \text{Ext}^n$  is asserted in any degree.
- (ii) No reverse implication from  $\text{Ext}^n$ -vanishing to repaired-profile vanishing is asserted without a separately proved natural zero-reflection package.
- (iii) No higher edge extractor is assumed faithful, conservative, or zero-reflecting merely because it is functorial or zero-compatible.
- (iv) No objectwise edge certificate is a natural transformation without a proved natural-family compatibility theorem.
- (v) No realization, edge extractor, stencil, or profile class may be chosen adaptively after inspecting the desired verdict.
- (vi) No invoked Ext clause is typed without the required representative, realization, amplitude policy, edge certificate, and manifest data.
- (vii) No missing eligibility, detector completeness, naturality, or artifact is interpreted as zero; the status is undefined.

- (viii) No finite detector supplies a reverse bridge outside its exact source stage, endpoint convention, coverage or birth/germ scope, and certified profile class.
- (ix) No soundness-only detector is treated as complete.
- (x) No synthetic shifted detector object is promoted to the principal natural higher Bridge merely because its encoded Ext group is dual to the detector output.
- (xi) No general-ring, sheaf, stack, motivic, dg, or spectral-sequence extension inherits the field theorem without separate hypotheses and proof.
- (xii) No standard amplitude certificate is said to be mathematically necessary for the field-coefficient higher Ext isomorphism; it is the conservative eligibility regime used by this release.
- (xiii) No global collapse functor  $C_\tau$ , identity  $C_{\tau,W}F = (C_\tau F)|_W$ , global exactness, or Serre localization is used.
- (xiv) No threshold-gap, metric-zero transport, detector rank stability, or crop theorem is reproved by the higher Ext edge.
- (xv) No categorical discharge cancels a failed detector, metric, Type IV, transient-defect, manifest, or claim-status clause.
- (xvi) No  $E_1$ , stencil, birth, germ, or other detector is a metric ledger component merely because its output is finite-dimensional.
- (xvii) No point-evaluation edge is assumed to detect transient bars; the inherited degree-one counterexample remains valid.
- (xviii) No representative-profile equivalence automatically implies realization equivalence in any degree.
- (xix) No arithmetic Ext group, Selmer group, PDE obstruction, motivic Ext, or external categorical invariant is automatically the Core group  $\text{Ext}^n(A_n(F), k)$ .
- (xx) No external mathematical conclusion follows without the separately proved interface and Return Theorem of Chapter 9.
- (xxi) No Search artifact, AI confidence value, spectral warning, ranking, or advisory invariant discharges a categorical clause.
- (xxii) No natural realization is automatically purpose-faithful; provenance and semantic fidelity are independent record axes.
- (xxiii) No endpoint set read from the complete target barcode is advertised as an independent discovery or compression certificate.
- (xxiv) No finite grid is endpoint-complete merely because it contains some critical values; singleton endpoint atoms, adjacent open atoms, and terminal policy are required.
- (xxv) No sampling theorem on an incomplete grid reconstructs endpoint decoration or cofinal tail data.
- (xxvi) No exactness of grid restriction makes hard threshold deletion exact.



- (xxvii) No repaired positive-threshold source class is asserted essentially surjective onto the whole finite quiver category; the ambient and repaired-image statements are distinct.
- (xxviii) No objectwise barcode encoding is promoted to a functor on morphisms without a declared morphism action.
- (xxix) No two complete grid refinements are identified as literally equal diagrams without a probe-change, subdivision, contraction, or common step-realization comparison.
- (xxx) No equality of dimension vectors, adjacent ranks, or Ext dimensions implies quiver isomorphism or supplies an actual quiver morphism.
- (xxxi) No quiver  $\text{Ext}_{\text{Rep}_k(A_{[m]})}^1$  is identified with the inherited degree- $n$  field Ext edge.
- (xxxii) No hereditary vanishing in  $\text{Rep}_k(A_{[m]})$  trivializes the higher field-valued Ext edge.
- (xxxiii) No cone is formed from a bottleneck matching, rank-table equality, or objectwise correspondence without an actual morphism.
- (xxxiv) No linear  $A_{[m]}$  reconstruction theorem is a theorem for arbitrary finite categories, branching posets, or descent data.
- (xxxv) No P2DG record returns a filtered chain complex, external object, or derived equivalence beyond its declared Return strength.

#### 4.24 Chapter 4 completion condition

Chapter 4 is complete for a declared finite route only when:

- (i) the source repaired profile and its stage are fixed;
- (ii) construction provenance and semantic fidelity are recorded on independent axes;
- (iii) an endpoint-complete atom grid and terminal policy are certified;
- (iv) any probe-choice or grid-refinement comparison consumed by the route is supplied at its exact strength;
- (v) the P2DG action, exactness, reflection, and information profiles are recorded;
- (vi) the finite quiver object is constructed;
- (vii) the decorated barcode is reconstructible from finite rank/Möbius data;
- (viii) the finite projective/Ext signature is typed at its exact strength;
- (ix) any cone use has an actual morphism;
- (x) any higher Ext conclusion has a complete  $n$ -eligible realization record;
- (xi) the Return strength is limited to the proved internal output.

Thus the chapter closes

finite persistence profile  $\longrightarrow$  finite linear quiver object  $\longrightarrow$  barcode and/or typed Ext Return

without claiming general finite-category globalization or external Return.

## 4.25 Provenance and status

The constructible persistence input, repaired evaluation, finite detector certificates, and representative records are consumed from Chapters 1–3. The inherited degree-one bridge is the v19 theorem package and the higher field-coefficient edge is inherited from v20. The new v21 mathematics is the endpoint-complete finite atom grid, exact linear-quiver encoding, Möbius barcode reconstruction, enriched finite projective/Ext signature, cone handoff, and finite P2DG closure. The persistence-to-derived edge remains an explicit realization hypothesis; synthetic and target-shaped encodings remain separately classified.

**Status labels for this chapter.** *New v21 Core Interface definitions and results:* Definitions 4.3, 4.5, 4.6, 4.8, 4.9, 4.10, 4.12, 4.13, 4.20, 4.21, 4.24, 4.28, 4.29, 4.92, 4.93, 4.96, and 4.97; Lemma 4.14; Theorems 4.15, 4.22, 4.25, 4.31, 4.98, and 4.99; Corollary 4.26; Propositions 4.16, 4.30, and 4.94; Policy 25; Remarks 4.11, 4.17, and 4.19.

*Inherited v19 Core definitions and results:* Definitions 4.33, 4.34, 4.36, 4.37, 4.38, 4.39, 4.73, 4.74, and 4.79; Lemma 4.52; Theorem 4.53; Corollaries 4.54 and 4.55; Theorem 4.66; Propositions 4.75, 4.80, 4.83, and 4.90.

*Inherited v20 Core derived results:* Theorem 4.50, Corollary 4.51, Theorem 4.58, and Theorem 4.59.

*Inherited conditional v20 natural Bridge results:* Definition 4.60, Theorem 4.61, Corollary 4.63, Proposition 4.62, Proposition 4.82, and Proposition 4.78.

*Higher eligibility, gate, and manifest records:* Definitions 4.42, 4.43, 4.44, 4.45, 4.46, 4.47, 4.69, 4.70, 4.72, 4.77, and 4.101; Proposition 4.71.

*Synthetic/supporting algebra, not principal natural Bridge:* Definition 4.86 and Proposition 4.87.

*Conservative closure package:* Theorem 4.103.

*Explicit exclusions:* Subsection 4.23.

## 5 Chapter 5: Towers, Type IV, and Finite-to-Infinite Reduction

**Chapter 5 contract (v21 tower and finite-to-infinite constitution).** This chapter preserves the complete v20 terminal-generic Type IV and transient- defect package and adds only theorem-controlled routes from finite tower data to an internal infinite or apex conclusion. Its governing prohibition is:

A finite prefix, a sequence of barcode isomorphism classes, equality of numerical invariants, or a bottleneck matching is never by itself an infinite-tail theorem, a limit object, or an apex comparison morphism.

Every tower is typed by variance and semantic role. Direct systems, inverse systems, mixed diagrams, filtered towers, and completed towers are distinct. Actual transition morphisms and composition compatibility are mandatory. The evaluation operation

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F))$$

does not automatically transport a source tower to a repaired tower-profile system, because hard threshold deletion is not globally functorial.

For a valid direct comparison morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty}),$$

the terminal pair  $(\mu_{i,W,\tau}, \nu_{i,W,\tau})$  measures cofinal kernel and cokernel defect. A separately declared defect threshold  $\tau_{\text{def}} > 0$  and complete Chapter 3 detector records measure long noncofinal defects. Terminal zero and long-transient zero do not imply full isomorphism unless the remaining short defects are separately excluded.

The v21 finite-to-infinite lanes are theorem-specific. They include eventual constancy with explicit apex agreement, finite-length or Artinian stabilization, Noetherian ascending-image reduction, countable Mittag–Leffler control, typed finite-state tail certification, and a finite-state stable-image inverse- limit theorem. No lane is substituted for another. In particular, Noetherianity alone does not stabilize arbitrary descending image chains, and Mittag–Leffler vanishing does not imply apex agreement.

An internal apex Return concludes only the internal predicate, object, isomorphism, inverse-limit, derived-limit, or apex-agreement strength proved by the selected lane. An arithmetic, analytic, geometric,  $p$ -adic  $L$ -function, characteristic-ideal, or other external Return still requires the separate realization and Return theorem of Chapters 8–9.

### 5.1 Scope, relocation, and standing regime

This chapter is the v21 Core-P tower and finite-to-infinite layer after the metric, certificate-complexity, finite-quiver, representative, and higher Ext-edge constructions of Chapters 1–4. It retains the v20 failure and Type IV discipline as an immutable lower baseline and adds conservative finite-to-infinite theorems at explicitly typed hypotheses. It consumes the following previously established data:

- (i) the repaired windowed evaluation

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F));$$

- (ii) the distinction between the pre-threshold profile

$$B_{i,W}(F) := W_W \mathbf{P}_i(F)$$

and the post-threshold repaired profile  $\text{Eval}_{i,W,\tau}(F)$ ;

- (iii) single-profile threshold-gap readiness records and, whenever metric transport through hard deletion is used, two-sided threshold-gap comparison certificates from Chapters 2–3; exact comparisons instead carry their exact equality or profile-isomorphism evidence;
- (iv) admissible representative and realization records from Chapters 3–4;
- (v) the degreeewise one-way categorical discharge

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0$$

in every invoked  $n$ -eligible regime of Chapter 4;

- (vi) the Chapter 3 source stages

`kernel_defect,`      `cokernel_defect,`

together with detector soundness, completeness, endpoint, coverage, family, metric, and artifact requirements.

The purpose of Chapter 5 is:

- (i) to type direct, inverse, mixed, filtered, and completed tower semantics;
- (ii) to require actual transition morphisms and composition compatibility;
- (iii) to separate direct colimit sources, inverse-limit sources, and declared apex objects;
- (iv) to give a typed, nonexclusive failure taxonomy of Types I–V;
- (v) to define terminal-generic kernel/cokernel diagnostics on bounded and right-unbounded windows;
- (vi) to separate terminal/cofinal defects, long transient interior defects, and shorter residual defect noise;
- (vii) to prove exact finite-to-infinite lanes under eventual constancy, finite-length, stable-image, Mittag–Leffler, or finite-state hypotheses;
- (viii) to prevent undefined, finite-prefix-only, or inapplicable evidence from being reported as an infinite or apex Return;
- (ix) to distinguish internal apex Return from external mathematical Return.

**Remark 5.1** (Relocation of the previous Chapter 5 material). The previous Chapter 5 material on Window Logic, Overlap Gate, Restart/Summability, and local-to-global certificate assembly is not discarded. In the v21 ordering it belongs after the present tower-and-diagnostics chapter and is therefore reserved for Chapter 6. The present chapter does not assert Restart or gluing theorems.

**Remark 5.2** (Metric repair is consumed, not reproved). The sharp threshold-gap-safe stability theorem is not reproved here. Whenever a tower artifact or comparison record is built from perturbed repaired profiles, the relevant two-sided threshold-gap-certified records must already have been supplied under Chapters 1–3 and Appendix A. If the metric ledger is  $\mathfrak{d}_{\text{met}} \in \mathbf{V}$ , the gate-scoped real safety condition is

$$\varepsilon_{\mathbf{B}}(\mathfrak{d}_{\text{met}}) := \text{val}_{\mathbf{B}}(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B}) = \min_{B \in \mathfrak{B}} \text{clear}_{\tau}(B),$$

for the actual finite comparison family  $\mathfrak{B}$  of pre-threshold profiles. The older notation  $\Sigma\delta < \text{gap}_{\tau}$  is permitted only as an explicitly declared abbreviation for this scalarized real inequality. The metric bound protects the repaired source or apex comparison named in its record. It does not automatically induce any bound on a kernel or cokernel defect profile.

## 5.2 Tower type, variance, and semantic authority

**Definition 5.3** (Typed tower variance). A Chapter 5 tower record declares exactly one primary variance type:

`direct_system,      inverse_system,      mixed_diagram.`

It may additionally carry one or both structural modifiers

`filtered_tower,      completed_tower.`

They have the following meanings.

(i) A direct system is a functor  $(\mathbb{N}, \leq) \rightarrow C$ , written

$$X_0 \longrightarrow X_1 \longrightarrow X_2 \longrightarrow \cdots .$$

(ii) An inverse system is a functor  $(\mathbb{N}, \leq)^{\text{op}} \rightarrow C$ , written

$$X_0 \longleftarrow X_1 \longleftarrow X_2 \longleftarrow \cdots .$$

(iii) A mixed diagram is a functor from a declared small indexing category whose arrows have more than one variance direction. It is not called a direct or inverse tower until a specific cofinal subdiagram and variance are selected by theorem.

(iv) A filtered tower has filtered objects and filtration-preserving actual transition maps. Its repaired evaluations do not inherit transition maps automatically.

(v) A completed tower is an inverse system together with a declared completion or apex object and an actual comparison map whose direction is fixed by the construction.

Variance is part of theorem identity. Reversing arrows, replacing a direct system by an inverse system, or changing the comparison-map direction creates a different claim.

**Definition 5.4** (Tower semantic role). Every tower record declares the external or internal meaning of each stage and one source-authority level:

`categorical,    theorem_identified,    implementation_only.`

A categorical source is an actual limit or colimit in the declared category. A theorem-identified source is an object proved isomorphic to that limit or colimit under a named theorem. An implementation-only source may support an implementation claim but not a categorical or external limit claim.

**Definition 5.5** (Actual-transition requirement). For every pair of comparable indices, the tower contains actual morphisms  $d_{n,m}$  in the declared category, with

$$d_{n,n} = \text{id}, \quad d_{m,r} \circ d_{n,m} = d_{n,r}$$

in the direct convention, or the variance-correct inverse analogue. A list of objects, dimensions, ranks, barcode multisets, bottleneck matchings, or pairwise isomorphism classes is not a tower.

**Definition 5.6** (Evaluation-lift record). Let  $F_\bullet$  be a filtered or persistence tower. An evaluation-lift record for  $(i, W, \tau)$  supplies actual morphisms between the repaired profiles

$$M_{n;i,W,\tau} = \text{Eval}_{i,W,\tau}(F_n)$$

and proves identity and composition compatibility. The record is obtained only from an explicit construction or a declared  $\tau$ -split or exact- admissible regime. Applying Eval objectwise does not itself create this record.

**Policy 26** (No variance laundering). *A theorem for direct colimits does not apply to inverse limits by reversing notation. A theorem for inverse systems does not produce a direct source. Mixed diagrams require their own reduction theorem. Every use records the variance, source object, comparison direction, and exact theorem consumed.*

### 5.3 Typed failure landscape

**Definition 5.7** (v21 inherited failure types and transient-defect refinement). Let a Core claim be represented by a declared run record on a window  $W$ , collapse threshold  $\tau$ , and finite set of monitored degrees  $I_{\text{mon}}$ . The five inherited failure types are nonexclusive. The v20 transient-defect verdict is an additional typed refinement attached to a valid tower artifact; it is not assigned a new Roman numeral and is not renamed terminal-generic Type IV.

**Type I. Topological or persistence failure.** For at least one invoked  $n \in I_{\text{mon}}$ ,

$$\text{Eval}_{n,W,\tau}(F) \neq 0.$$

A certified complete Chapter 3 detector with value nonzero supplies a witness. An incomplete, inapplicable, or wrong-stage detector leaves the clause undefined; it does not prove Type I.

**Type II. Categorical failure in an eligible regime.** For at least one invoked  $n$ -eligible realization,

$$\text{Ext}^n(A_n(F), k) \neq 0.$$

Missing eligibility, edge realization, required naturality, or artifact data makes the degreewise clause undefined, not categorically obstructed.

**Type III. Detector, operational, metric, comparison, or functorial instability.** A required finite-level operation or safety certificate is absent, undefined, wrong-stage, or evidenced to fail. This includes failure of a repaired evaluation record, representative compatibility, crop budget, window-edge clearance, two-sided threshold-gap certificate, detector applicability or completeness, defect-threshold declaration, profile system morphism, source-semantic authority, terminal tameness, declared comparison budget, finite-to-infinite reduction theorem, tail certificate, derived-limit record, or apex agreement. Missing evidence is normally undefined; an evidenced false required condition is reject.

**Type IV. Tower-level terminal-generic obstruction.** A valid comparison artifact exists and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0).$$

On a right-unbounded window this is an infinite-bar defect; on a bounded window it is a defect cofinal to the declared right endpoint. A certified nonzero transient interior detector with  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$  is recorded as a separate transient-defect obstruction and does not become terminal-generic Type IV by renaming.

**Type V. Claim-status, manifest, provenance, or promotion failure.** A run cites a rejected or deprecated claim, suppresses a required missing marker, reports undefined as pass or zero, overstates source provenance, relabels a detector certificate across profile stages, treats a sampled zero as complete, uses a Search/AI/advisory artifact as proof, treats a finite observation as an infinite theorem, or otherwise makes the canonical manifest incomplete or inconsistent.

When the governing claim requires absence of long transient defects, a valid transient detector with value `nonzero` is a mathematical gate obstruction even though the terminal Type IV status may remain `certified_zero`. Detector failure and detected defect are distinct: the former is Type III; the latter is a well-defined negative mathematical verdict for the stronger transient-defect clause.

**Remark 5.8** (The taxonomy is nonexclusive). A single run may exhibit several failure types simultaneously. For example, a missing two-sided gap record is Type III; if the manifest nevertheless reports a certified pass, the same run also has Type V failure. The type number is not a severity ranking.

**Definition 5.9** (Failure-check order). The Core audit order is:

- (1) claim and manifest well-formedness (Type V);
- (2) metric, operational, representative, and artifact well-formedness (Type III);
- (3) topological and eligible categorical verdicts (Types I–II);
- (4) invoked tower diagnostics (Type IV);
- (5) final B-Gate<sup>+</sup> conjunction and budget decision.

A later numerical zero may not repair an earlier undefined or ill-typed record.

**Remark 5.10** (Undefined is not zero). The absence of a tower comparison artifact, the failure of terminal tameness, or the absence of an actual morphism between the declared source and target profiles does not imply  $\mu_{\text{Collapse}} = \nu_{\text{Collapse}} = 0$ . It means that the Type IV diagnostic is undefined. Reporting that state as  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$  is a Type V manifest error.

**Remark 5.11** (No scalar repair of non-scalar obligations). A larger metric ledger cannot manufacture a repaired tower-profile system, an actual comparison morphism, a terminal-tameness certificate, a kernel/cokernel artifact, a valid claim status, or a complete and consistent manifest. These are typed proof obligations, not numerical error terms. Consequently  $\text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau$  is necessary for metric safety when invoked, but it cannot turn undefined into `certified_zero`.

## 5.4 Structural and audited collapse admissibility

**Definition 5.12** (Type IV scope status). Every Core run records exactly one of the following Type IV scope statuses:

`not_invoked`,      `undefined`,      `certified_zero`,      `obstructed`.

They mean respectively:

- (i) no terminal-generic kernel/cokernel comparison diagnostic is part of the claim. A direct or inverse tower may still be in scope through a finite-to-infinite, Roos, or apex-agreement lane;
- (ii) a terminal-generic Type IV clause is invoked, but its comparison morphism, ambient abelian profile category, source authority, terminal-tameness data, or defect artifacts are incomplete or ill-typed;

- (iii) a Type IV comparison artifact valid for the claim's declared source semantics exists, and its diagnostic pair satisfies

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0);$$

- (iv) such an artifact exists and  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$ .

**Definition 5.13** (Structural collapse admissibility). Fix a finite monitored degree set  $I_{\text{mon}}$ . A declared object  $(F; W, \tau, I_{\text{mon}})$  is *structurally collapse-admissible* if:

- (i)  $\text{Eval}_{n, W, \tau}(F) = 0$  for every invoked  $n \in I_{\text{mon}}$ ;
- (ii) every invoked degree- $n$  Ext clause has a complete  $n$ -eligibility record and is discharged;
- (iii) the Type IV scope status is either `not_invoked` or `certified_zero`;
- (iv) if absence of long transient defects is required, the full defect profiles, defect threshold, and complete kernel/cokernel detector records exist and the aggregate transient verdict is `certified_absent`;
- (v) no Type V claim-status defect is present in the structural record.

**Definition 5.14** (Audited collapse admissibility). A structurally collapse-admissible run is *audited collapse-admissible* if, in addition:

- (i) every required repaired comparison declares exactly one comparison mode

$$\begin{aligned} &\text{exact, metric\_gap\_certified, theorem\_controlled\_nonmetric,} \\ &\quad \text{not\_compared} \end{aligned}$$

- (ii) every metric comparison transported through hard deletion has a two-sided threshold-gap certificate; every exact comparison carries its exact equality or profile-isomorphism proof; and every `theorem_controlled_nonmetric` comparison cites the named theorem, hypotheses, source and target, and exact nonmetric conclusion;
- (iii) every crop and window-edge effect used by the proof is controlled by the appropriate exact theorem or metric ledger component;
- (iv) whenever metric mode is invoked, the ledger aggregation law, scalarization  $\text{val}_{\mathcal{B}}$ , actual discrepancy upper bound, and strict reserve
 
$$\text{val}_{\mathcal{B}}(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B})$$
 are established;
- (v) every detector is applied at its declared Chapter 3 source stage, and all endpoint, coverage, family, excess, completeness, and artifact obligations pass;
- (vi) every in-scope non-scalar proof obligation is present and well-typed;
- (vii) the canonical Chapter 8 manifest is complete and consistent;
- (viii) no Type III operational or metric failure remains.



**Proposition 5.15** (Eligible monitored vanishing discharges the invoked Type II clauses). *Fix a finite set  $I_{\text{mon}}$ . For every invoked  $n \in I_{\text{mon}}$ , assume a complete  $n$ -eligibility record in the sense of Chapter 4. If*

$$\text{Eval}_{n,W,\tau}(F) = 0 \quad \text{for every invoked } n \in I_{\text{mon}},$$

*then*

$$\text{Ext}^n(A_n(F), k) = 0 \quad \text{for every invoked } n \in I_{\text{mon}}.$$

*Consequently the invoked Type II clauses are discharged, but Type III, Type IV, transient-defect, Type V, and budget clauses remain independent.*

*Proof.* Apply the Chapter 4 higher one-way discharge theorem degreewise and then use the aggregate higher categorical status rule. The independence of the remaining clauses is part of the Chapter 1 B-Gate<sup>+</sup> contract.  $\square$

## 5.5 Windowed tower-profile systems

Let

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow \cdots$$

be a declared tower of filtered or persistence objects and let  $F_\infty$  be a declared apex, terminal object, or comparison target when one is supplied.

**Definition 5.16** (Direct-system source object). For a valid direct repaired tower-profile system, the direct source is a declared object

$$\text{TowerSrc}_{i,W,\tau}^{\rightarrow} := \text{ColimProfile}_{i,W,\tau}(F_\bullet)$$

with provenance

$$\text{genuine\_colimit}, \quad \text{verified\_stabilization}, \quad \text{implementation\_artifact}.$$

Only the first is definitionally a colimit. The second receives colimit strength only from a recorded stabilization-to-colimit theorem. The third is restricted to implementation scope.

**Definition 5.17** (Inverse-system source object). For a valid inverse system  $X_0 \leftarrow X_1 \leftarrow \cdots$ , the inverse source is

$$\text{TowerSrc}^{\leftarrow} := \varprojlim_n X_n$$

when the categorical inverse limit exists in the declared category. A finite-dimensional stable-image model or another finite presentation may be used only after a theorem supplies an isomorphism to this inverse limit. The record distinguishes

$$\text{genuine\_inverse\_limit}, \quad \text{theorem\_identified\_model}, \\ \text{implementation\_artifact}$$

**Definition 5.18** (Declared apex and comparison direction). A declared apex consists of an object  $X_\infty$ , semantic provenance, and an actual comparison map in one explicitly fixed direction:

$$\text{TowerSrc} \longrightarrow X_\infty, \quad \text{or} \quad X_\infty \longrightarrow \text{TowerSrc}.$$

The map direction is not normalized away: kernel and cokernel diagnostics depend on it. An apex name, completion notation, or compatible finite family without an induced map does not establish apex agreement.

**Definition 5.19** (Windowed repaired tower objects). For a monitored degree  $i$ , window  $W$ , and threshold  $\tau$ , write

$$M_{n;i,W,\tau} := \text{Eval}_{i,W,\tau}(F_n).$$

These objects alone do not form a directed system, because barcode-level threshold deletion is not a global functor on arbitrary persistence morphisms.

**Definition 5.20** (Repaired tower-profile system). A *repaired tower-profile system* for  $(i, W, \tau)$  consists of:

- (i) the repaired objects  $M_{n;i,W,\tau}$ ;
- (ii) actual persistence-profile morphisms

$$\psi_{n,m} : M_{n;i,W,\tau} \longrightarrow M_{m;i,W,\tau} \quad (n \leq m);$$

- (iii) the identities  $\psi_{n,n} = \text{id}$  and composition law

$$\psi_{m,r} \circ \psi_{n,m} = \psi_{n,r};$$

- (iv) provenance showing that the morphisms are either explicitly constructed or supplied canonically by a declared  $\tau$ -split/exact-admissible regime.

A sequence of barcode multisets or bottleneck matchings without these morphisms is not a repaired tower-profile system.

**Definition 5.21** (Collapsed source profile). Given a repaired tower-profile system, a source profile

$$\text{ColimProfile}_{i,W,\tau}(F_\bullet)$$

is declared with exactly one provenance status:

$$\text{genuine\_colimit}, \quad \text{verified\_stabilization}, \quad \text{implementation\_artifact}.$$

In the first case it is a genuine colimit in the declared constructible profile category. In the second it is a certified stabilized profile; it is identified with a categorical colimit only when a separate stabilization-to-colimit theorem is recorded. In the third it is an implementation object whose relation to the tower is part of the artifact and is not silently promoted either to a categorical colimit or to a verified stabilization.

Every claim also declares the source semantics it requires. A source artifact is admissible for that claim only when its provenance has sufficient semantic authority. In particular, an `implementation_artifact` may support an implementation-scoped comparison, but it cannot discharge a claim about a genuine tower colimit merely by being named `ColimProfile`.

**Definition 5.22** (Collapsed apex profile). The repaired apex profile is

$$\text{ApexProfile}_{i,W,\tau}(F_\infty) := \text{Eval}_{i,W,\tau}(F_\infty),$$

provided the repaired evaluation record for  $F_\infty$  is declared.

**Remark 5.23** (No automatic colimit–collapse interchange). No identity of the form

$$\lim_{\longrightarrow n} \text{Eval}_{i,W,\tau}(F_n) \cong \text{Eval}_{i,W,\tau}(F_\infty)$$

or

$$\lim_{\rightarrow n} T_{\tau}^{\text{bar}}(M_n) \cong T_{\tau}^{\text{bar}}\left(\lim_{\rightarrow n} M_n\right)$$

is available in the general Core. Likewise, no identity

$$\lim_{\leftarrow n} \text{Eval}_{i,W,\tau}(F_n) \cong \text{Eval}_{i,W,\tau}\left(\lim_{\leftarrow n} F_n\right)$$

or derived analogue is assumed. Any such comparison is a separate theorem or actual comparison artifact, with variance, exactness, and source semantics recorded.

## 5.6 Tower comparison artifacts

**Definition 5.24** (Direct Type IV tower comparison artifact). A *direct Type IV tower comparison artifact* for  $(i, W, \tau)$  consists of the following declared data:

- (i) a direct repaired tower-profile system;
- (ii) a source profile  $\text{ColimProfile}_{i,W,\tau}(F_{\bullet})$  with provenance status;
- (iii) a repaired apex profile  $\text{ApexProfile}_{i,W,\tau}(F_{\infty})$ ;
- (iv) an actual morphism in the declared constructible profile category on  $W$ ,
$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_{\bullet}) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_{\infty});$$
- (v) the construction or proof establishing that  $\phi_{i,W,\tau}$  is well-typed;
- (vi) the claim-required source semantics and evidence that the source provenance has sufficient semantic authority for that claim;
- (vii) the repaired evaluation, comparison-mode, crop-budget, and representative records used in constructing the source, target, or comparison; every metric comparison transported through hard deletion carries a two-sided threshold-gap certificate, every exact comparison carries its exact proof, and every `theorem_controlled_nonmetric` comparison cites the named theorem, hypotheses, source, target, and exact nonmetric conclusion;
- (viii) the terminal mode and terminal-tameness certificate used by the diagnostic;
- (ix) an explicitly declared abelian constructible profile category on  $W$ , closed under the kernels and cokernels used here, together with kernel and cokernel computation artifacts or their categorical proof.
- (x) when transient detection is invoked, immutable full defect-profile identifiers sufficient to register new Chapter 3 detector records at `kernel_defect` and `cokernel_defect`; no pre-existing detector record is relabeled across stages.

**Definition 5.25** (Inverse apex comparison record). An inverse apex comparison record contains:

- (i) an actual countable inverse system and its declared inverse-limit source or theorem-identified finite model;
- (ii) the Roos operator and the required  $\lim_{\leftarrow}^1$  status;

(iii) a declared apex and an actual comparison map in its constructional direction,

$$X_\infty \longrightarrow \varprojlim_n X_n \quad \text{or} \quad \varprojlim_n X_n \longrightarrow X_\infty;$$

(iv) the ambient abelian category in which kernel and cokernel are formed;

(v) separate system-validity, derived-limit, and apex-agreement fields;

(vi) if terminal-generic Type IV is additionally invoked for this map, a declaration that the chosen map direction is the diagnostic direction, together with terminal-tameness and full defect-profile artifacts.

The inverse record is not obtained by replacing  $\varinjlim$  with  $\varprojlim$  in the inherited direct artifact. Its Roos and apex fields remain independent even when the same kernel/cokernel calculus is later applied to the actual apex map.

**Remark 5.26** (A bottleneck matching is not the comparison morphism). A bottleneck matching controls metric discrepancy between barcodes. It is not, by itself, a natural transformation of persistence modules and therefore does not define the kernel or cokernel needed by Type IV. The tower artifact must contain an actual morphism  $\phi_{i,W,\tau}$  or a canonical construction of one. Moreover, the morphism must live in the declared abelian profile category and its source provenance must match the semantic strength required by the claim.

**Definition 5.27** (Exact-admissible tower artifact). A tower comparison artifact is *exact-admissible* if its profile-system morphisms, source construction, comparison map, kernels, and cokernels are supplied by a declared regime in which the retained-long-bar assignment is functorial and preserves every exact sequence used by the diagnostic.

**Remark 5.28** (Explicit artifacts remain allowed). Exact admissibility is sufficient but not necessary. An explicitly constructed morphism between declared repaired source and apex profiles is also admissible, provided all typing, source-semantics, provenance, ambient-abelian-category, kernel/cokernel, and terminal-tameness obligations are met. Artifact validity is always relative to the claim's declared source semantics; a lower-authority implementation source does not become a categorical colimit by successful kernel/cokernel computation.

## 5.7 Inverse systems, Roos defects, and apex agreement

**Definition 5.29** (Countable inverse-system record). In the inverse branch, fix a ring  $R$  and an actual countable system in the abelian category  $R\text{-Mod}$ ,

$$X_0 \xleftarrow{d_0} X_1 \xleftarrow{d_1} X_2 \xleftarrow{\quad} \cdots$$

The record contains all objects, maps, compositions, coefficient conventions, level semantics, and the claim for which the system is used. Products and kernels/cokernels below are those of  $R\text{-Mod}$ . No statement in this lane is automatically transferred to an arbitrary abelian, Grothendieck, derived, or higher category.

**Definition 5.30** (Roos operator and first derived-limit defect). For a countable inverse-system record define

$$\Delta_X : \prod_{n \geq 0} X_n \longrightarrow \prod_{n \geq 0} X_n, \quad \Delta_X((x_n)_n) = (x_n - d_n x_{n+1})_n.$$

Then

$$\varprojlim_n X_n = \ker \Delta_X, \quad \varprojlim_n^1 X_n := \text{coker } \Delta_X.$$

The second object is the countable Roos first-edge defect. In  $R\text{-Mod}$  it canonically computes the first right-derived inverse limit  $R^1\varprojlim X_n$ . This identification is restricted to the countable module lane declared above; it is not a definition of  $\varprojlim$  in another ambient category. Its vanishing is independent of apex agreement.

**Definition 5.31** (Mittag–Leffler condition). The inverse system is Mittag–Leffler if, for every fixed  $n$ , the descending chain

$$\text{im}(X_{n+1} \rightarrow X_n) \supseteq \text{im}(X_{n+2} \rightarrow X_n) \supseteq \cdots$$

stabilizes.

**Theorem 5.32** (Countable Mittag–Leffler vanishing). *For a countable inverse system of modules satisfying the Mittag–Leffler condition,*

$$\varprojlim_n^1 X_n = 0.$$

*This theorem clears the first derived-limit defect; it does not construct an apex comparison or prove that an apex map is an isomorphism.*

*Proof.* The countable Roos exact sequence identifies  $\varprojlim_n^1 X_n$  with the cokernel of  $\Delta_X$ . Under the Mittag–Leffler condition, the standard successive-lifting argument inside the stabilized image chains makes  $\Delta_X$  surjective. Hence its cokernel vanishes.  $\square$

**Corollary 5.33** (Finite-dimensional and finite-length ML branches). *An inverse system of finite-dimensional  $k$ -vector spaces is Mittag–Leffler. More generally, an inverse system in which every fixed  $X_n$  has finite composition length is Mittag–Leffler. Therefore  $\varprojlim_n^1 X_n = 0$  in both countable module regimes.*

*Proof.* For fixed  $n$ , the image chain is a descending chain of subobjects of  $X_n$ . Finite dimension or finite composition length permits only finitely many strict decreases. Apply Theorem 5.32.  $\square$

**Remark 5.34** (No uniform finite cutoff from pointwise ML). The stabilization bound may depend on  $n$ . Pointwise Mittag–Leffler therefore supplies no single finite prefix that verifies the whole inverse tower. A finite verifier requires an eventual-surjectivity bound, a finite-state rule, a certified uniform length bound together with an effective transition theorem, or another explicit tail certificate.

**Definition 5.35** (Inverse apex defect). Let  $X_\infty$  be a declared inverse-tower apex and let

$$\alpha : X_\infty \longrightarrow \varprojlim_n X_n \quad \text{or} \quad \beta : \varprojlim_n X_n \longrightarrow X_\infty$$

be the actual comparison map supplied by the construction. Its apex defect is

$$D_{\text{apex}}(f) := \ker f \oplus \text{coker } f.$$

The verdict `apex_agreement` means  $D_{\text{apex}}(f) = 0$ . It is logically independent of  $\varprojlim_n^1 X_n = 0$ .

**Definition 5.36** (Inverse-limit audit profile). An inverse-limit run records three independent status fields:

$$\text{InvAudit} = (\text{Sys}, R^1\text{Lim}, \text{Apex}).$$

Here

- (i) `Sys` is `undefined`, `pass`, or `reject`, according to whether the actual inverse system and its source semantics are missing, valid, or evidenced invalid;

- (ii)  $R^1\text{Lim}$  is `not_invoked`, `undefined`, `cleared`, or `obstructed`;
- (iii) `Apex` is `not_invoked`, `undefined`, `agreement`, or `disagreement`.

No component repairs another. In particular, `cleared` in the second field is not an apex verdict, and `agreement` in the third field is not a derived-limit theorem.

**Theorem 5.37** (Derived inverse-limit and apex separation). *Neither implication*

$$\varprojlim_n^1 X_n = 0 \implies D_{\text{apex}}(f) = 0, \quad D_{\text{apex}}(f) = 0 \implies \varprojlim_n^1 X_n = 0$$

*holds without additional hypotheses. Consequently an inverse finite-to-infinite record maintains system validity, the Roos defect, and the apex defect as separate fields.*

*Proof.* For the first nonimplication, take the constant zero inverse system. It is Mittag–Leffler, so  $\varprojlim^1 = 0$ , but the comparison  $0 \rightarrow A$  to any nonzero declared apex has nonzero cokernel.

For the converse, take the inverse system

$$\mathbb{Z} \xleftarrow{\times 2} \mathbb{Z} \xleftarrow{\times 2} \mathbb{Z} \xleftarrow{\times 2} \dots$$

Its inverse limit is zero, while its first derived inverse limit is nonzero. Declare the categorical inverse limit 0 itself as apex and use the identity comparison. Apex agreement holds although  $\varprojlim^1 \neq 0$ . Thus the two fields are logically independent.  $\square$

**Remark 5.38** (Direct Type IV versus inverse derived-limit audit). The inherited  $\mu, \nu$  Type IV diagnostic is canonically attached to the actual direct source-to-apex profile morphism of the v20 branch. An inverse system instead carries system validity,  $\varprojlim^1$ , and an actual apex defect. The same terminal kernel/cokernel calculus may be applied to an inverse apex comparison only after its direction, ambient abelian profile category, and terminal-tameness hypotheses are explicitly declared; it is not inferred from inverse-limit notation.

## 5.8 Terminal-generic dimension

The old shorthand “generic dimension equals infinite-bar multiplicity” is correct only on a right-unbounded window. On a bounded window, zero extension outside the window makes the ordinary  $t \rightarrow +\infty$  dimension identically zero and would render the diagnostic vacuous. The following window-relative definition removes that ambiguity.

**Definition 5.39** (Terminal-tame profile). Let  $W = [u, \nu)$ , where  $\nu \in \mathbb{R} \cup \{+\infty\}$ , and let  $M$  be a constructible persistence profile on  $W$ . We call  $M$  *terminal-tame on  $W$*  if:

- (i) every fiber near the terminal end is finite-dimensional;
- (ii) if  $\nu < +\infty$ , there exists  $s < \nu$  such that every structure map  $M(t) \rightarrow M(t')$  with  $s \leq t \leq t' < \nu$  is an isomorphism;
- (iii) if  $\nu = +\infty$ , there exists  $s \in \mathbb{R}$  such that every structure map  $M(t) \rightarrow M(t')$  with  $s \leq t \leq t'$  is an isomorphism.

**Definition 5.40** (Terminal-generic dimension). For a terminal-tame profile  $M$  on  $W = [u, \nu)$ , define

$$\text{tgdim}_W(M) := \begin{cases} \lim_{t \uparrow \nu} \dim_k M(t), & \nu < +\infty, \\ \lim_{t \rightarrow +\infty} \dim_k M(t), & \nu = +\infty. \end{cases}$$

When  $\nu = +\infty$ , we may also write  $\text{gdim}(M)$ .

**Proposition 5.41** (Well-definedness of terminal-generic dimension). *For every terminal-tame profile  $M$  on  $W = [u, v)$ , the limit in Definition 5.40 exists and is a finite nonnegative integer.*

*Proof.* By terminal tameness, there is a terminal subinterval on which every structure map is an isomorphism. Hence the finite-dimensional fibers on that subinterval all have the same dimension. The displayed dimension function is therefore eventually constant as  $t \uparrow v$  when  $v < +\infty$ , or as  $t \rightarrow +\infty$  when  $v = +\infty$ . Its eventual value is finite and is exactly the asserted limit.  $\square$

**Definition 5.42** (Cofinal bar in a window). A bar  $I$  in a barcode profile on  $W = [u, v)$  is *cofinal in  $W$*  if:

- (i)  $\sup I = v$  when  $v < +\infty$ ; or
- (ii)  $I$  is unbounded above when  $v = +\infty$ .

For bounded  $W$ , a cofinal bar is a boundary-reaching bar relative to  $W$ , not a global infinite bar.

**Remark 5.43** (Interval decomposition and terminal tameness are distinct inputs). In the declared constructible one-parameter regime, the monitored profile is interval-decomposable. Terminal tameness is an additional right-end condition; constructibility on bounded subintervals alone does not silently supply a common terminal tail. Proposition 5.44 uses both facts.

**Proposition 5.44** (Terminal dimension counts cofinal bars). *For a terminal-tame, interval-decomposable object  $M$  on  $W$ , the terminal dimension equals the multiplicity of cofinal bars:*

$$\mathrm{tgdim}_W(M) = \#\{\text{bars cofinal in } W\},$$

where multiplicity is counted. On a right-unbounded window, the right-hand side counts the infinite bars.

*Proof.* Choose a terminal tail on which all structure maps are isomorphisms. A bar contributes one dimension to every fiber on that tail exactly when it is cofinal in  $W$ . Every noncofinal bar dies before the terminal tail and contributes zero there. Summing the surviving interval summands gives the stated equality.  $\square$

**Remark 5.45** (Interpretation of earlier  $\mathrm{gdim}$  notation). The  $\mathrm{gdim}$ -notation used schematically in Chapters 1–2 is interpreted by this chapter as the terminal-generic dimension appropriate to the declared window. For  $W = [u, +\infty)$  it is the ordinary eventual dimension and counts infinite bars. For bounded  $W = [u, v)$  the canonical local diagnostic is  $\mathrm{tgdim}_W$ , which counts bars reaching  $v$  from the left.

**Proposition 5.46** (Terminal tameness passes to defects). *Let  $\phi : M \rightarrow N$  be a morphism in a declared abelian constructible profile category on  $W$ , closed under the pointwise kernels and cokernels used by the diagnostic. If  $M$  and  $N$  are terminal-tame, then  $\ker \phi$  and  $\mathrm{coker} \phi$  are terminal-tame.*

*Proof.* On a common terminal tail, the structure maps of  $M$  and  $N$  are isomorphisms and the naturality squares for  $\phi$  commute. The induced structure maps on pointwise kernels and cokernels are therefore isomorphisms on the same tail. Finite-dimensionality is inherited pointwise.  $\square$

## 5.9 Defect profiles and Type IV diagnostics

**Definition 5.47** (Defect profiles). For a valid tower comparison artifact, define

$$\mathrm{Defect}_{i,W,\tau}^{\ker} := \ker(\phi_{i,W,\tau}), \quad \mathrm{Defect}_{i,W,\tau}^{\mathrm{coker}} := \mathrm{coker}(\phi_{i,W,\tau}).$$

These are objects of the declared constructible profile category on  $W$ .

**Definition 5.48** (Degreewise Type IV diagnostics). Assume the defect profiles are terminal-tame. Define

$$\mu_{i,W,\tau} := \text{tgdim}_W(\text{Defect}_{i,W,\tau}^{\text{ker}}), \quad \nu_{i,W,\tau} := \text{tgdim}_W(\text{Defect}_{i,W,\tau}^{\text{coker}}).$$

Thus  $\mu$  measures terminal-generic kernel defect and  $\nu$  measures terminal-generic cokernel defect.

**Definition 5.49** (Total collapse diagnostics). For a fixed declared window–threshold pair  $(W, \tau)$  and a finite declared monitored-degree set  $I_{\text{mon}}$ , define

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad \nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

Deprecated Latin- $u$  cokernel notation has no normative use in v21.0. If the monitored-degree scope is infinite, neither total is defined without a separately proved aggregation or summability theorem and an explicit aggregation policy.

**Definition 5.50** (Type IV diagnostic record). A Type IV diagnostic record contains the window, threshold, monitored degrees, declared tower scope, and exactly one status

`not_invoked,      undefined,      certified_zero,      obstructed.`

Its remaining required fields depend on that status.

- (i) For `not_invoked`, the record contains an explicit scope rationale showing that no terminal-generic kernel/cokernel comparison diagnostic belongs to the claim. Direct or inverse tower, limit, colimit, Roos, stable-image, or apex-agreement lanes may remain active. No numerical  $\mu, \nu$  values are asserted.
- (ii) For `undefined`, the record contains either the identifier of a partial or invalid artifact, or an explicit missing-artifact marker, together with the list of missing or ill-typed obligations. No numerical  $\mu, \nu$  values are asserted as Core diagnostics.
- (iii) For `certified_zero` or `obstructed`, the record contains:
  - (a) a valid tower comparison artifact identifier, source-profile provenance status, claim-required source semantics, and semantic-authority check;
  - (b) the terminal mode `bounded_terminal` or `right_unbounded`;
  - (c) terminal-tameness certificates for the defect profiles;
  - (d) kernel and cokernel defect-profile artifacts;
  - (e) all degreewise and total  $\mu, \nu$  values over the fixed  $(W, \tau)$  and declared finite set  $I_{\text{mon}}$ ;
  - (f) the full-interior-defect status, recorded separately from terminal-generic and thresholded transient verdicts, with exactly one value

`not_checked,      certified_absent,      certified_present;`

- (g) when transient detection is invoked, the separately declared defect threshold  $\tau_{\text{def}} > 0$ , detector component `full_profile` or `interior_noncofinal`, source stage, detector information class, certificate status, value status, and the aggregate long-transient status

`not_invoked,      undefined,      certified_absent,      certified_present.`



For the full-interior-defect field, `certified_absent` requires proof that both entire noncofinal defect profiles vanish. An explicit nonzero noncofinal witness is required for `certified_present`. By contrast, long-transient `certified_absent` means only that no monitored noncofinal defect bar has length greater than  $\tau_{\text{def}}$ .

The status is `certified_zero` exactly when all invoked inputs are valid for the claim's declared source semantics and  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ . It is `obstructed` exactly when those inputs are valid and  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$ . Both verdicts are relative to the declared source semantics and monitored scope. An absent artifact, insufficient-provenance source, or source-semantics mismatch is represented by a missing/invalid-obligation marker and yields `undefined`, never zero. The terminal Type IV status is determined only by  $(\mu, \nu)$ . A long-transient verdict is recorded in its own field and does not change `certified_zero` into `obstructed` when  $(\mu, \nu) = (0, 0)$ .

**Definition 5.51** (Type IV obstruction and Type IV-free status). Relative to a valid tower comparison artifact whose source semantics match the claim:

- (i) the run has a *Type IV obstruction* if  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$ ;
- (ii) the run is *Type IV-free in the monitored terminal-generic sense* if  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ . Both statements are relative to the fixed  $(W, \tau)$ , the finite monitored-degree set, and the source semantics recorded by the artifact.

**Proposition 5.52** (Characterization of certified Type IV-free status). *A tower record is certified Type IV-free if and only if:*

- (i) *the tower comparison artifact is valid;*
- (ii) *its kernel and cokernel defect profiles are terminal-tame;*
- (iii)  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ .

*Proof.* This is the defining conjunction in Definitions 5.50 and 5.51. The validity and terminal-tameness clauses are necessary to ensure that the displayed zero is a defined Core diagnostic rather than a missing value.  $\square$

**Remark 5.53** (Zero diagnostics are not a full isomorphism theorem). The equality

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

says that no defect bar reaches the terminal end of the declared window. It does not imply

$$\ker \phi_{i,W,\tau} = 0 \quad \text{or} \quad \text{coker } \phi_{i,W,\tau} = 0.$$

Finite interior defect bars may remain. A full isomorphism conclusion requires a separate no-interior-defect proof establishing that the entire kernel and cokernel profiles vanish, not merely their terminal-generic parts.

**Example 5.54** (Zero terminal diagnostics with a nonisomorphism). Let  $W = [0, 10)$  and consider the morphism

$$0 \longrightarrow k_{[2,3)}.$$

Its cokernel is  $k_{[2,3)}$ , so the morphism is not an isomorphism. However, the cokernel has no bar cofinal in  $W$ , hence

$$\text{tgdim}_W(k_{[2,3)}) = 0.$$

Thus the terminal-generic diagnostic is zero while a finite interior defect remains.

**Example 5.55** (Pure kernel, pure cokernel, and mixed Type IV modes). On a window  $W = [u, v)$ , let  $K_W$  denote one cofinal interval bar:  $k_{[u, v)}$  when  $v < +\infty$  and  $k_{[u, +\infty)}$  when  $v = +\infty$ . Then:

$$\begin{aligned} K_W \rightarrow 0 & \text{ has } (\mu, \nu) = (1, 0), \\ 0 \rightarrow K_W & \text{ has } (\mu, \nu) = (0, 1), \\ K_W \oplus 0 \rightarrow 0 \oplus K_W & \text{ has } (\mu, \nu) = (1, 1) \end{aligned}$$

for the evident zero maps. These are the pure-kernel, pure-cokernel, and mixed terminal-generic Type IV modes. The mixed map is explicitly the zero morphism from  $K_W \oplus 0$  to  $0 \oplus K_W$ .

**Remark 5.56** (Finite-layer checks do not determine Type IV). Finite-layer repaired evaluations do not logically determine the kernel or cokernel of an undeclared source-to-apex comparison. For example, a zero source profile together with a declared apex containing a cofinal bar produces a pure cokernel defect once the comparison  $0 \rightarrow K_W$  is declared. This is an artifact-level mismatch witness; it is not a claim that an arbitrary apex is a categorical colimit.

## 5.10 Cofinal/interior decomposition and transient defect detectors

**Definition 5.57** (Cofinal and interior barcode profiles). Let  $M$  be an interval-decomposable constructible profile on  $W = [u, v)$ . Define the *cofinal profile*  $\text{Cof}_W(M)$  to be the barcode subprofile consisting of bars cofinal in  $W$ , and define the *interior profile*  $\text{Int}_W(M)$  to be the barcode subprofile consisting of all noncofinal bars, with multiplicity. At barcode-profile isomorphism strength,

$$M \cong \text{Cof}_W(M) \oplus \text{Int}_W(M).$$

This decomposition is canonical on barcode isomorphism classes. It is not asserted to be a functor on arbitrary persistence morphisms.

**Proposition 5.58** (Terminal/interior separation). *If  $M$  is terminal-tame and interval-decomposable, then*

$$\text{tgdim}_W(M) = \#\mathcal{B}(\text{Cof}_W(M))$$

*with multiplicity, and*

$$\text{tgdim}_W(M) = 0 \iff \text{Cof}_W(M) = 0 \iff M \cong \text{Int}_W(M).$$

*Proof.* The first assertion is Proposition 5.44. A finite direct sum of cofinal interval bars has zero terminal-generic dimension exactly when it is empty. The displayed decomposition then gives the final equivalence.  $\square$

**Definition 5.59** (Defect threshold and detector component). A transient-defect run declares a threshold

$$\tau_{\text{def}} > 0$$

independently of the collapse threshold  $\tau$ . It also declares one component for each detector application:

$$\text{full\_profile} \quad \text{or} \quad \text{interior\_noncofinal}.$$

The full-profile component reads  $K_{i, W, \tau}$  or  $Q_{i, W, \tau}$ . The interior component reads

$$K_{i, W, \tau}^\circ := \text{Int}_W(K_{i, W, \tau}), \quad Q_{i, W, \tau}^\circ := \text{Int}_W(Q_{i, W, \tau}).$$

Changing  $\tau_{\text{def}}$ , the component, the endpoint convention, or the detector source stage creates a new detector claim and a new artifact.

**Definition 5.60** (Kernel and cokernel defect detectors). Let  $D$  be a Chapter 3 finite detector with all applicability and completeness data declared at source stage `kernel_defect` or `cokernel_defect`. Define

$$D_{\tau_{\text{def}}}^{\text{ker,full}} := D_{\tau_{\text{def}}}(K_{i,W,\tau}), \quad D_{\tau_{\text{def}}}^{\text{coker,full}} := D_{\tau_{\text{def}}}(Q_{i,W,\tau}),$$

and, when the barcode-level interior component is declared,

$$D_{\tau_{\text{def}}}^{\text{ker},\circ} := D_{\tau_{\text{def}}}(K_{i,W,\tau}^{\circ}), \quad D_{\tau_{\text{def}}}^{\text{coker},\circ} := D_{\tau_{\text{def}}}(Q_{i,W,\tau}^{\circ}).$$

For a stencil detector,  $D$  is  $D_{T,\tau_{\text{def}}}^{\text{stc}}$ ; birth, germ, and certified family-mesh detectors are also admissible at their exact registered strength.

**Theorem 5.61** (Complete transient-defect characterization). *Assume a valid tower comparison artifact and full constructible defect profiles. Let the kernel and cokernel detector certificates be complete at the declared defect stages and components. Then*

$$\begin{aligned} D_{\tau_{\text{def}}}^{\text{ker}} = 0 &\iff T_{\tau_{\text{def}}}^{\text{bar}}(K^{\text{det}}) = 0, \\ D_{\tau_{\text{def}}}^{\text{coker}} = 0 &\iff T_{\tau_{\text{def}}}^{\text{bar}}(Q^{\text{det}}) = 0, \end{aligned}$$

where  $K^{\text{det}}, Q^{\text{det}}$  denote the declared full or interior components. Nonzero detector value is equivalent to the existence of a retained bar in the corresponding component.

*Proof.* The interval-image, active-start, birth, germ, coverage, and mesh arguments of Chapter 3 apply to the supplied constructible defect profile with  $\tau_{\text{def}}$  in place of the collapse threshold. The declared complete certificate therefore reflects zero of the corresponding hard threshold deletion. This theorem is the explicit authorization of those arguments at source stage `kernel_defect` or `cokernel_defect`; it is not a transport of a pre-threshold detector artifact. The component, endpoint, family, coverage, and artifact data remain hypotheses.  $\square$

**Remark 5.62** (Defect-stage registration). Theorem 5.61 is the stage-specific result anticipated by Definition 3.40. A detector proof or artifact registered at either stage

`windowed_prethreshold,`      `repaired_postthreshold`

cannot be relabeled as a kernel/cokernel certificate. The full defect profiles and all certificate data are reconstructed and registered anew.

**Corollary 5.63** (Full and interior detectors coincide under terminal zero). *If*

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0),$$

*then  $K_{i,W,\tau} \cong K_{i,W,\tau}^{\circ}$  and  $Q_{i,W,\tau} \cong Q_{i,W,\tau}^{\circ}$  at barcode-profile strength. Consequently a complete full-profile defect detector and the corresponding interior detector have the same zero/nonzero verdict.*

*Proof.* Apply Proposition 5.58 to the kernel and cokernel defect profiles and then detector invariance under profile isomorphism from Chapter 3.  $\square$

**Definition 5.64** (Aggregate long-transient verdict). For fixed  $(W, \tau, \tau_{\text{def}}, I_{\text{mon}})$ , the aggregate long-transient verdict is exactly one of

`not_invoked,`   `undefined,`   `certified_absent,`   `certified_present.`

It is `certified_absent` exactly when every required kernel and cokernel detector certificate is complete and every value is zero. It is `certified_present` when every detector needed to interpret the witness is well-typed and at least one required value is nonzero. Missing or wrong-stage evidence yields `undefined`, never absence.

**Theorem 5.65** (Joint terminal/transient defect classification). *Assume a valid tower comparison artifact, terminal-tame defect profiles, and complete interior defect detectors at  $\tau_{\text{def}}$ . Then exactly the following mathematical cases occur degreewise and after finite monitored aggregation:*

- (i) *if  $(\mu, \nu) \neq (0, 0)$ , a terminal/cofinal Type IV defect is present; the interior detector independently records whether a long noncofinal defect is also present;*
- (ii) *if  $(\mu, \nu) = (0, 0)$  and the long-transient verdict is `certified_present`, there is no terminal defect but there is a noncofinal kernel or cokernel bar of length greater than  $\tau_{\text{def}}$ ;*
- (iii) *if  $(\mu, \nu) = (0, 0)$  and the long-transient verdict is `certified_absent`, there is no terminal defect and no noncofinal defect bar longer than  $\tau_{\text{def}}$ . Shorter nonzero defects may remain.*

*No case with an undefined detector certificate is assigned a mathematical zero verdict.*

*Proof.* Terminal/cofinal presence is characterized by Proposition 5.58. Long noncofinal presence or absence is characterized by Theorem 5.61. The conclusions are independent conjunctions of these two classifications.  $\square$

**Definition 5.66** (Short-defect exclusion certificate). A short-defect exclusion certificate at  $\tau_{\text{def}}$  proves that every nonzero bar in the declared interior kernel and cokernel profiles has length strictly greater than  $\tau_{\text{def}}$ . It is a non-scalar structural theorem or artifact; it is not inferred from detector zero or from terminal-generic zero.

**Theorem 5.67** (Conditional full-isomorphism upgrade). *Let  $\phi_{i,W,\tau}$  be a valid morphism in the declared abelian profile category. Suppose:*

- (i)  $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$ ;
- (ii) *complete interior kernel and cokernel detectors at  $\tau_{\text{def}}$  are certified zero;*
- (iii) *a short-defect exclusion certificate holds.*

*Then*

$$\ker \phi_{i,W,\tau} = 0, \quad \text{coker } \phi_{i,W,\tau} = 0,$$

*and hence  $\phi_{i,W,\tau}$  is an isomorphism.*

*Proof.* Terminal zero excludes all cofinal kernel and cokernel bars. Complete interior detector zero excludes every interior bar of length greater than  $\tau_{\text{def}}$ . The short-defect certificate excludes every remaining nonzero interior bar. Therefore both full defect profiles vanish. A morphism in an abelian category with zero kernel and zero cokernel is an isomorphism.  $\square$

**Example 5.68** (Zero terminal and transient diagnostics with a short residual defect). Let  $W = [0, 10)$ , choose  $\tau_{\text{def}} > 0$ , and consider

$$0 \longrightarrow k_{[2, 2+\tau_{\text{def}}/2)}.$$

The terminal pair is zero, and every complete defect detector at threshold  $\tau_{\text{def}}$  is zero because the only defect bar is deleted by that threshold. The morphism is nevertheless not an isomorphism. Thus the short-defect exclusion in Theorem 5.67 is essential.

**Remark 5.69** (No automatic metric transport of defect profiles). A bottleneck or interleaving bound between source profiles and a bound between apex profiles do not, by themselves, produce a bottleneck bound or an interleaving between their kernels or cokernels. Metric transport of a defect-detector verdict requires actual comparison data between the defect profiles and an independently proved kernel/cokernel stability theorem. Only then may the Chapter 2 threshold-gap and zero-separation machinery be applied at stage `kernel_defect` or `cokernel_defect`.

### 5.11 Eventual constancy and exact apex Return

**Definition 5.70** (Eventual-isomorphism certificate). An eventual-isomorphism certificate for a direct or inverse tower consists of an index  $N$  and proof that every transition map on the tail  $n \geq N$  is an isomorphism, including compatibility with the declared tower structure. Agreement of objects or numerical invariants on a finite prefix is not such a certificate.

**Theorem 5.71** (Eventually constant direct-system Return). *Let  $X_0 \rightarrow X_1 \rightarrow \cdots$  be a direct system in a category admitting its sequential colimit. If every transition after  $N$  is an isomorphism, then the canonical map*

$$X_N \xrightarrow{\sim} \varinjlim_n X_n$$

*is an isomorphism. If a declared apex  $X_\infty$  is equipped with a compatible isomorphism  $X_N \xrightarrow{\sim} X_\infty$ , the direct source- to-apex comparison is an isomorphism.*

*Proof.* The tail is naturally isomorphic to the constant diagram on  $X_N$ . Removing a finite initial segment is cofinal for a sequential direct limit, so the colimit is  $X_N$ . The apex assertion follows by composition with the declared compatible isomorphism.  $\square$

**Theorem 5.72** (Eventually constant inverse-system Return). *Let  $X_0 \leftarrow X_1 \leftarrow \cdots$  be a countable inverse system of modules. If every transition after  $N$  is an isomorphism, then*

$$\varprojlim_n X_n \cong X_N, \quad \varprojlim_n^1 X_n = 0.$$

*Any declared apex equipped with a compatible isomorphism to  $X_N$  is exactly Returned from the tower at internal object-isomorphism strength.*

*Proof.* A compatible sequence is uniquely determined by its value in  $X_N$ , and any value in  $X_N$  extends uniquely along the invertible tail and then to the finite prefix. The tower is Mittag-Leffler, so the first derived limit vanishes. Compose with the declared apex isomorphism.  $\square$

### 5.12 Finite length, Noetherian images, and stable-image reduction

**Theorem 5.73** (Finite-length chain stabilization). *Let  $M$  have finite composition length  $\ell(M)$ . Every strictly increasing or strictly decreasing chain of subobjects of  $M$  has at most  $\ell(M)$  strict inclusions. Hence every monotone chain stabilizes.*

*Proof.* A strict inclusion changes composition length by at least one, while all subobject lengths lie between 0 and  $\ell(M)$ .  $\square$

**Corollary 5.74** (Stable image of a finite-length endomorphism). *For an endomorphism  $T : M \rightarrow M$  of a finite-length object, the chains*

$$M \supseteq \operatorname{im} T \supseteq \operatorname{im} T^2 \supseteq \cdots, \quad 0 \subseteq \ker T \subseteq \ker T^2 \subseteq \cdots$$

*stabilize. For any common stabilization exponent  $e$ , the restriction*

$$T|_{\operatorname{im} T^e} : \operatorname{im} T^e \xrightarrow{\sim} \operatorname{im} T^e$$

*is an automorphism.*

*Proof.* Apply Theorem 5.73. At stable image,  $T$  is surjective. A surjective endomorphism of a finite-length object is injective, hence an automorphism.  $\square$

**Theorem 5.75** (Noetherian ascending-image reduction). *Let  $M$  be Noetherian and suppose a direct tower is equipped with compatible maps  $X_n \rightarrow M$  whose images form an ascending chain. Then there is an index  $N$  such that*

$$\text{im}(X_N \rightarrow M) = \text{im}(X_n \rightarrow M) \quad (n \geq N).$$

*Thus every conclusion depending only on the stable image in  $M$  reduces to one finite stage once the embedding and monotonicity hypotheses are certified.*

*Proof.* The images are an ascending chain of subobjects of a Noetherian object, so the ascending-chain condition gives stabilization.  $\square$

**Remark 5.76** (Noetherian is not Artinian). Noetherianity alone does not stabilize arbitrary descending image chains in an inverse system. Such a claim requires an Artinian, finite-length, finite-dimensional, eventual-surjectivity, finite-state, or other separately proved condition. This boundary prevents an invalid “Noetherian stable image” argument in the inverse direction.

**Theorem 5.77** (Finite-generation finite-window zero reduction). *Let  $A = \bigoplus_{d \geq 0} A_d$  be a nonnegatively graded ring and  $M = \bigoplus_{d \geq 0} M_d$  a graded  $A$ -module with a certified generation bound  $D$ , meaning that  $M$  is generated by  $\bigoplus_{0 \leq d \leq D} M_d$ . Then*

$$M = 0 \iff M_d = 0 \text{ for all } 0 \leq d \leq D.$$

*For a graded morphism  $f : M \rightarrow N$ , if  $\ker f$  and  $\text{coker } f$  have certified generation bound  $D$ , then  $f$  is an isomorphism exactly when  $f_d$  is an isomorphism for  $0 \leq d \leq D$ .*

*Proof.* Every homogeneous element is generated from the displayed finite degree window. The morphism statement follows by applying the zero statement to the kernel and cokernel.  $\square$

**Remark 5.78** (Unknown generation bound). Finite generation without a certified degree bound does not license a predetermined finite window. The bound is a non-scalar theorem input and is part of the finite-to-infinite certificate.

### 5.13 Finite-state tail certification

**Definition 5.79** (Deterministic finite-state tail system). A deterministic tail system is a finite set  $S$ , a transition  $T : S \rightarrow S$ , an initial state  $s_0$ , and a forward-invariant terminal set  $Z \subseteq S$ . The orbit is  $s_n = T^n(s_0)$ . State equality includes all typed data used by the theorem; equality of dimensions or numerical summaries alone is insufficient.

**Theorem 5.80** (Finite-state terminality theorem). *Let  $|S| = N$ . Exactly one of the following is certified within  $s_0, \dots, s_N$ :*

- (i) *some state lies in  $Z$ , after which every state remains terminal;*
- (ii) *two nonterminal states repeat, after which the orbit is periodic outside  $Z$  and never reaches  $Z$ .*

*Hence eventual terminality is decided by at most  $N + 1$  states.*

*Proof.* If no terminal state occurs, the pigeonhole principle gives a repeated state. Determinism repeats the subsequent orbit, and forward invariance of  $Z$  prevents later entry. If a terminal state occurs, forward invariance gives permanent terminality.  $\square$

**Corollary 5.81** (Finite-state eventual-constancy lane). *Suppose terminal states are exactly those after which every tower transition is an isomorphism. The finite-state terminality certificate either proves an eventually constant Return and activates Theorem 5.71 or 5.72, or exhibits a nonterminal cycle and rejects that specific eventual-constancy lane.*

**Definition 5.82** (Typed finite-state linear inverse tower). A typed finite-state linear inverse tower over a field  $k$  consists of a finite state set  $S$ ; a finite-dimensional based space  $V_s$  for each state; a deterministic transition  $\sigma : S \rightarrow S$ ; a matrix

$$D_s : V_{\sigma(s)} \longrightarrow V_s;$$

and an initial state. It generates

$$V_0 \xleftarrow{d_0} V_1 \xleftarrow{d_1} \cdots .$$

The first repeated state supplies an entry  $N$  and period  $r$ .

**Definition 5.83** (Periodic monodromy and stable image). For a certified entry-period pair define the period monodromy

$$T = d_N d_{N+1} \cdots d_{N+r-1} : V_N \longrightarrow V_N.$$

With  $e = \dim_k V_N$ , define

$$W_{\text{st}} := \text{im}(T^e) = \bigcap_{q \geq 0} \text{im}(T^q).$$

The equality and the fact that  $T|_{W_{\text{st}}}$  is an automorphism follow from finite-dimensional image stabilization. The definition is relative to the certified periodic entry and the specified identification  $V_{N+r} = V_N$ .

**Theorem 5.84** (Exact finite-state inverse-limit Return). *For a typed finite-state linear inverse tower, evaluation at the certified periodic entry induces an isomorphism*

$$\text{ev}_N : \varprojlim_n V_n \xrightarrow{\sim} W_{\text{st}}, \quad \varprojlim_n^1 V_n = 0.$$

*The isomorphism is canonical relative to the certified entry-period record and the based state identifications. It is natural only for morphisms of finite-state towers commuting with those identifications and all transition matrices. Both conclusions are determined by the finite state table; the inverse-limit object at the entry is computed by the single monodromy matrix  $T$ . In particular,*

$$\varprojlim_n V_n = 0 \iff T^e = 0 \iff T \text{ is nilpotent.}$$

*Proof.* A compatible sequence satisfies  $x_N = T^q x_{N+qr}$  for every  $q$ , hence  $x_N \in \bigcap_q \text{im } T^q = W_{\text{st}}$ . The same argument at every period boundary shows  $x_{N+qr} \in W_{\text{st}}$ . Since  $T$  is invertible on  $W_{\text{st}}$ , these boundary values are forced by

$$x_{N+qr} = T^{-q} x_N.$$

All intermediate values in each period are then forced by the actual transition matrices, and the finite prefix is forced by composition toward level 0. Conversely, these formulas construct a compatible sequence from every  $x_N \in W_{\text{st}}$ . They are inverse to evaluation. The tower consists of finite-dimensional vector spaces, so Corollary 5.33 gives  $\varprojlim_n^1 = 0$ .  $\square$

**Theorem 5.85** (Finite-state apex-agreement criterion). *Let  $A$  be a finite-dimensional declared apex with a compatible family  $\alpha_n : A \rightarrow V_n$ . Under Theorem 5.84, the induced map*

$$\alpha : A \longrightarrow \varprojlim_n V_n$$

*is an isomorphism if and only if the entry component, viewed through the forced inclusion supplied by compatibility,*

$$\alpha_N : A \longrightarrow W_{\text{st}},$$

*is an isomorphism. Compatibility already implies  $\alpha_N(A) \subseteq W_{\text{st}}$ . Thus apex agreement is verified by one finite linear map at the periodic entry.*

*Proof.* Under the evaluation isomorphism, the composite  $A \rightarrow \varprojlim_n V_n \rightarrow W_{\text{st}}$  is exactly  $\alpha_N$ . □

**Remark 5.86** (Finite-state lane exceeds eventual constancy). The periodic transition maps need not be isomorphisms on the whole state space. The inverse limit retains exactly the recurrent stable-image sector. Therefore Theorem 5.84 strictly extends the eventual-constancy lane while remaining finite and exact.

## 5.14 Finite-to-infinite reduction records and soundness

**Definition 5.87** (Finite-to-infinite reduction record). A finite-to-infinite reduction record contains at least:

- (i) tower variance, ambient category, and actual transition maps;
- (ii) the finite witness and its immutable artifact identity;
- (iii) the exact reduction theorem and all theorem hypotheses;
- (iv) the exceptional finite index range and the tail-coverage statement;
- (v) any eventual-isomorphism, generation-bound, stable-image, Mittag–Leffler, surjective-tail, finite-state, or recurrence certificate used;
- (vi) the direct colimit or inverse-limit source and its semantic authority;
- (vii) every derived-limit defect required by the selected lane;
- (viii) the declared apex, actual comparison map, direction, and apex- agreement evidence;
- (ix) the exact route-admissible internal Return capability set, its maximal supported strengths, and a unique Return ceiling only when one exists;
- (x) explicit failure boundaries describing what the certificate does not control;
- (xi) Chapter 8 verifier, dependency, status, provenance, and replay data.

A finite witness without the reduction theorem and tail coverage is only a finite observation.

**Theorem 5.88** (Finite-to-infinite soundness theorem). *Let a finite-to-infinite record be complete and consistent. If its finite witness passes, every hypothesis of the named reduction theorem passes, the tail-coverage and exceptional-index obligations pass, and every required limit, derived-limit, and apex comparison clause passes, then every Return strength in the registered route-admissible supported set holds. No strength outside that set is obtained. If a required hypothesis is missing, the requested conclusion is *undefined*; if a well-typed hypothesis or witness is false, the selected reduction lane is *reject*.*



*Proof.* The mathematical conclusion is the named reduction theorem applied to its verified hypotheses and actual source/apex maps. The status clauses follow from the v21 rule that missing evidence is undefined and evidenced falsity is rejection. No stronger conclusion is introduced by the record wrapper.  $\square$

**Definition 5.89** (Internal apex Return capability profile). Let  $\mathfrak{S}_{\text{int}}$  be the partially ordered family of typed internal Return strengths, including records whose semantic families are

tail\_predicate, internal\_object, internal\_isomorphism,  
inverse\_limit\_object, derived\_limit\_clear, apex\_agreement.

Each token denotes a fully typed Chapter 1 Return-strength record, not merely a label. Every required route component  $e$  supplies a downward-closed supported set  $\mathfrak{S}_e \subseteq \mathfrak{S}_{\text{int}}$ . The internal Return capability profile is

$$\mathfrak{S}_{\text{route}} = \bigcap_{e \in E_{\text{required}}} \mathfrak{S}_e.$$

The record may therefore have several incomparable maximal supported strengths. The phrase *Return ceiling* is permitted only when  $\mathfrak{S}_{\text{route}}$  has a proved unique greatest element. In particular, `derived_limit_clear` and `apex_agreement` are independent; a route may prove either, both, or neither.

**Policy 27** (External Return separation). *An internal apex Return is not an external-domain theorem. Arithmetic control,  $p$ -adic completion semantics,  $p$ -adic  $L$ -functions, characteristic ideals, PDE regularity, geometric realization, and other source-specific conclusions remain in HT, UB, Problem Interface, or Chapter 9 until separate external realization and Return theorems are proved.*

## 5.15 Artifact-coherent stable bands

**Definition 5.90** (Artifact-coherent Type IV stable band). Fix a window  $W$ , monitored degrees, a tower-profile construction, and a terminal mode. An interval  $J \subset (0, +\infty)$  is an *artifact-coherent Type IV stable band* if:

- (i) for every  $\tau \in J$ , a valid tower comparison artifact  $\phi_{i,W,\tau}$  is supplied by one declared construction rule;
- (ii) the claim-required source semantics, source provenance, semantic authority, and terminal mode do not change silently across  $J$ ;
- (iii) all defect profiles are terminal-tame;
- (iv) every repaired comparison used by the artifact declares a comparison mode. In metric mode, the record carries the ledger  $\mathfrak{d}_{\text{met},\tau}$ , scalarization  $\text{val}_{\mathfrak{B}}$ , evidenced discrepancy bound, and strict inequality

$$\text{val}_{\mathfrak{B}}(\mathfrak{d}_{\text{met},\tau}) < \text{gap}_{\tau}(\mathfrak{B}_{\tau});$$

in exact mode it carries the exact equality or profile-isomorphism proof and makes no automatic positive-radius robustness claim; in `theorem_controlled_nonmetric` mode it carries the named theorem, hypotheses, source, target, and exact nonmetric conclusion;

- (v) the associated Type IV record has status `certified_zero` for every  $\tau \in J$ .

**Remark 5.91** (Stable band is an artifact family, not pointwise wishful thinking). Pointwise numerical zeros obtained from unrelated comparison maps do not define a stable band. The artifact family, source provenance, terminal convention, and comparison policy must remain auditable throughout the band. If a single budgeted

perturbation is claimed uniformly on  $J$ , the manifest must provide a common ledger  $\mathbf{d}_{\text{met},J}$  and verify the uniform reserve condition

$$\text{val}_B(\mathbf{d}_{\text{met},J}) < \inf_{\tau \in J} \text{gap}_\tau(\mathfrak{B}_\tau).$$

A pointwise family of positive reserves does not by itself imply that this infimum is positive.

**Proposition 5.92** (No terminal-generic Type IV obstruction on a stable band). *If  $J$  is an artifact-coherent Type IV stable band, then every  $\tau \in J$  is Type IV-free in the monitored terminal-generic sense relative to its declared artifact.*

*Proof.* By the stable-band definition, every threshold in  $J$  has a valid diagnostic record whose status is

`certified_zero`.

The certified-zero characterization then applies pointwise. □

**Definition 5.93** (Artifact-coherent joint defect-clean band). An interval  $J \subset (0, +\infty)$  is a joint defect-clean band at defect threshold  $\tau_{\text{def}}$  if it is an artifact-coherent Type IV stable band and, for every  $\tau \in J$ :

- (i) the same defect threshold, detector stages, endpoint convention, component policy, and completeness rule are used;
- (ii) complete kernel and cokernel interior detectors are certified zero;
- (iii) the detector artifacts and source-semantic authority remain auditable across  $J$ .

**Proposition 5.94** (Uniform terminal and long-transient cleanliness). *On an artifact-coherent joint defect-clean band, every threshold is free of monitored terminal defects and of monitored noncofinal defects longer than  $\tau_{\text{def}}$ . Full isomorphism does not follow without a uniform short-defect exclusion theorem.*

*Proof.* Apply Proposition 5.92 and Theorem 5.65 pointwise on the band. □

## 5.16 Integration with B-Gate<sup>+</sup>

**Definition 5.95** (Type IV and transient-defect gate discharge). The terminal Type IV and optional transient-defect clauses of B-Gate<sup>+</sup> are discharged as follows.

- (i) if no terminal-generic direct comparison diagnostic is consumed, the Type IV field is `not_invoked`; inverse-system audit fields remain independent;
- (ii) if a direct comparison diagnostic is invoked, its Type IV record must be valid, its source provenance must have sufficient authority, and its status must be `certified_zero`;
- (iii) if the claim additionally requires absence of long transient defects, the aggregate long-transient verdict at the declared  $\tau_{\text{def}}$  must be `certified_absent`;
- (iv) if full comparison-map isomorphism is required, either direct full defect-profile vanishing or the hypotheses of Theorem 5.67 must be certified.

The statuses `undefined`, `obstructed`, and a required long-transient `certified_present` fail their respective gate clauses. Terminal Type IV status and transient status remain separate fields.

**Remark 5.96** (Meaning of  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$  in the gate). The shorthand

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

is used only in the invoked-tower case after the diagnostic has been shown to be defined. A no-tower claim uses `not_invoked`, not a fabricated numerical zero.

**Proposition 5.97** (Audited admissibility implies the structural and safety clauses of B-Gate<sup>+</sup>). *If a run is audited collapse-admissible in the sense of Definition 5.14, then:*

- (i) *its topological clause passes;*
- (ii) *its invoked categorical clause passes;*
- (iii) *its Type IV clause is either correctly out of scope or certified zero;*
- (iv) *every required long-transient clause is certified absent, and any required full-isomorphism upgrade has its short-defect exclusion or direct full-defect proof;*
- (v) *every invoked metric threshold-gap and strict budget clause passes, and every exact comparison carries its exact proof;*
- (vi) *no Type III or Type V defect remains in those records.*

*Hence the corresponding B-Gate<sup>+</sup> record passes for every clause marked required by the run's declared Core scope. This does not by itself yield any domain-specific external conclusion; such a conclusion still requires its separately proved Interface or Bridge theorem.*

*Proof.* Each item is one of the defining clauses of structural or audited collapse admissibility. The final statement is the B-Gate<sup>+</sup> conjunction fixed in Chapter 1, with the Type IV scope refinement of Definition 5.95. □

**Definition 5.98** (Failure routing record). A failure routing record contains:

- (i) every applicable failure type among I–V;
- (ii) the first failed audit stage in Definition 5.9;
- (iii) the exact failed clause and either the relevant artifact identifier or an explicit missing-artifact marker;
- (iv) whether the failure is mathematical, operational, diagnostic, transient-defect, or manifest-level;
- (v) the terminal Type IV status and the independent long-transient status when a tower artifact is in scope;
- (vi) the permitted repair action, if any, without changing the claim status.

No scalar score replaces this typed routing record.

### 5.17 Failure-and-diagnostics contract

**Definition 5.99** (v21 tower, failure, and finite-to-infinite contract). The normative v21 Chapter 5 contract is the following.

- (i) Types I–V are nonexclusive and are checked in the declared audit order.
- (ii) Type II is meaningful only degreewise in an eligible Chapter 4 regime.
- (iii) Type III includes threshold-gap, detector applicability and completeness, crop-budget, representative, profile-system, artifact-typing, source-semantic, and terminal-tameness failures.
- (iv) Type IV is defined only relative to a valid comparison morphism and measures terminal-generic defect profiles through  $(\mu, \nu)$ .
- (v) Long transient defects are evaluated separately on full or noncofinal kernel/cokernel profiles at a declared  $\tau_{\text{def}} > 0$ .
- (vi) Type V includes false pass declarations, deprecated claims, stage relabeling, suppressed records, provenance overstatement, and undefined-as-zero substitutions.
- (vii) On bounded windows, Type IV counts defects cofinal to the right endpoint; on right-unbounded windows, it counts infinite-bar defects.
- (viii)  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$  together with transient certified absence excludes only terminal defects and noncofinal defects longer than  $\tau_{\text{def}}$ .
- (ix) Full isomorphism requires full kernel/cokernel vanishing or a separately proved short-defect exclusion upgrade.
- (x) Deprecated Latin- $u$  cokernel notation has no normative use;  $\nu_{i,W,\tau}$  and  $\nu_{\text{Collapse}}$  are canonical.
- (xi) An out-of-scope terminal-generic direct diagnostic records `not_invoked`; an invoked direct diagnostic requires a valid `certified_zero` record to pass. Inverse-system audit fields are evaluated independently.
- (xii) Search, AI, spectral, or advisory proxies cannot replace the actual comparison artifact or complete defect detector.
- (xiii) A finite-to-infinite conclusion requires a named reduction theorem, finite witness, tail coverage, source authority, and apex agreement at the strength actually consumed.
- (xiv) Direct, inverse, mixed, filtered, and completed tower semantics are never merged by notation.
- (xv) Internal apex Return and external mathematical Return are separate typed conclusions.

**Remark 5.100** (Contract, not a mixed theorem bundle). Definition 5.99 is a typing and audit contract. Its mathematical content is supplied by the preceding definitions and propositions; it is not presented as a theorem combining definitions, policies, and proved implications into one undifferentiated statement.

## 5.18 Minimal manifest consequences

**Remark 5.101** (Manifest hierarchy and canonical audit schema). The entries below extend the Chapter 1 Minimal Manifest and the Chapter 3 evaluation/representative records with failure-routing and Type IV data. They are not a competing manifest definition. Chapter 8 supplies the canonical v21 audit manifest into which these entries are embedded. An incomplete invoked Type IV entry yields `undefined`; an inconsistent entry is a Type V rejection.

**Definition 5.102** (Type IV manifest entry). Every run records the Type IV scope status. The remaining manifest fields are status-dependent.

- (i) For `not_invoked`: the scope rationale and the claim clause declaring the terminal-generic Type IV diagnostic out of scope. Other direct or inverse finite-to-infinite fields may remain invoked.
- (ii) For `undefined`: the tower-profile, source, apex, or comparison identifiers that exist, together with an explicit missing/invalid-obligation marker for every absent mandatory object, source-semantics mismatch, insufficient-provenance source, absent ambient abelian category, failed typing, or failed terminal-tameness obligation. No Core numerical diagnostic is serialized as zero.
- (iii) For `certified_zero` or `obstructed`: the tower-profile system identifier; source and apex identifiers; source provenance; claim-required source semantics and semantic-authority verdict; valid comparison morphism artifact identifier; ambient abelian profile category; window, collapse threshold, monitored degrees, and evaluation order; terminal mode and terminal-tameness artifacts; full, cofinal, and interior defect-profile artifacts; all degree-wise and total  $\mu, \nu$  values; diagnostic status; and the separate full-interior-defect status. When transient detection is invoked, the entry also records  $\tau_{\text{def}}$ , detector component, exact Chapter 3 source stage, information class, endpoint convention, stencil or birth/germ data, coverage/mesh/excess evidence, certificate and value statuses, aggregate long-transient verdict, and any short-defect exclusion certificate.

If metric perturbation enters the artifact construction, the entry also records  $\mathfrak{d}_{\text{met}}$ ,  $\text{val}_B$ , the real budget  $\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}})$ , and the gate-scoped comparison family  $\mathfrak{B}$ .

**Definition 5.103** (Failure manifest entry). Every failed or conditionally accepted run records:

- (i) the applicable failure types;
- (ii) the first failed audit stage;
- (iii) the failed claim, theorem, gate, or artifact reference;
- (iv) whether the result is rejected, repairable, conditional, or out of scope;
- (v) the version boundary, including any successor-release candidate used in analysis but excluded from proof.

## 5.19 Core promotion boundary

The following branches remain outside the v21 Chapter 5 Core unless a later release supplies a separate theorem and migration record:

- (i) arbitrary unbounded derived inverse limits;
- (ii) unrestricted towers in arbitrary Grothendieck or higher categories;

- (iii) arithmetic-specific control theorems and source-specific apex constructions;
- (iv)  $p$ -adic  $L$ -function, measure, determinant, Fitting-ideal, or characteristic-ideal Return;
- (v) noncommutative Iwasawa, Selmer, PDE, analytic, or geometric tower conclusions not reduced by an independently proved interface;
- (vi) any HT or UB theorem whose exact hypotheses, variance, coefficient category, and route-admissible Return capability set have not been migrated and re-proved in the Main architecture.

These may remain valid Technical Extension, Problem Interface, HT, UB, or Chapter 9 components. Reuse by a Bridge does not by itself promote them to Core-P or Core-DG.

## 5.20 Explicit non-claims

The following are not asserted in Chapter 5.

- (i) No sequence of repaired barcode objects is automatically a directed system.
- (ii) No bottleneck matching is automatically a persistence-module morphism.
- (iii) No automatic comparison
 
$$\varinjlim_n \text{Eval}_{i,W,\tau}(F_n) \longrightarrow \text{Eval}_{i,W,\tau}(F_\infty)$$
 is asserted.
- (iv) No exactness or global functoriality of  $T_\tau^{\text{bar}}$  is used.
- (v) No undefined diagnostic is interpreted as zero.
- (vi) No bounded-window terminal bar is called a global infinite bar.
- (vii) No equality  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$  is promoted to full comparison-map isomorphism without a separate no-interior-defect theorem.
- (viii) No Type IV result is accepted without an actual comparison morphism, a source whose provenance has sufficient authority for the claim, a declared abelian profile category, terminal-tameness data, and defect-profile artifacts; their absence or mismatch is recorded explicitly and yields `undefined`.
- (ix) No scalar ledger slack repairs a missing morphism, missing terminal-tameness certificate, invalid claim status, or incomplete manifest.
- (x) No Type II verdict is issued without eligibility.
- (xi) No scalar potential, Search artifact, AI confidence, spectral warning, or advisory invariant replaces Types I–V, B-Gate<sup>+</sup>, or the Type IV artifact.
- (xii) No Restart/Summability, Čech gluing, or local-to-global certificate theorem is proved here; those are deferred to Chapter 6.
- (xiii) No successor-release candidate, including any stronger NBT transport package, is used as a v20.0 theorem.
- (xiv) No zero terminal-generic pair implies absence of finite noncofinal defects.

- (xv) No complete transient detector at  $\tau_{\text{def}}$  excludes defect bars of length at most that threshold.
- (xvi) No full-profile defect detector isolates transient defects in the presence of cofinal bars unless an interior decomposition or separate witness is supplied.
- (xvii) No detector certificate registered at a pre-threshold or repaired profile stage is relabeled as a kernel- or cokernel-defect certificate; each defect-stage certificate is constructed and registered anew.
- (xviii) No closeness of source and apex profiles automatically yields closeness of kernel or cokernel defect profiles.
- (xix) No long-transient certified absence is a full-isomorphism theorem without direct full-defect vanishing or a short-defect exclusion certificate.
- (xx) No finite prefix, however long, determines an arbitrary tower tail without a quantified recurrence, stabilization, generation, or finite-state theorem.
- (xxi) No direct-system theorem is applied to an inverse system by reversing arrows, and no mixed diagram is assigned a limit without a variance-specific reduction.
- (xxii) No objectwise repaired evaluation supplies transition morphisms automatically.
- (xxiii) No identity between evaluation of a limit and limit of evaluations is asserted without a comparison theorem.
- (xxiv) Noetherianity alone does not stabilize arbitrary descending inverse-image chains.
- (xxv) Pointwise Mittag–Leffler stabilization does not provide a uniform finite cutoff for an arbitrary tower.
- (xxvi) Vanishing of  $\varprojlim^1$  does not imply apex agreement, and apex agreement does not imply vanishing of  $\varprojlim^1$ .
- (xxvii) An inverse-system audit does not automatically invoke the inherited direct Type IV diagnostic; terminal kernel/cokernel diagnostics require an actual inverse apex comparison record and a declared diagnostic direction.
- (xxviii) No partially ordered Return capability profile is replaced by a scalar weakest-link value or by a unique ceiling unless a greatest supported strength is proved.
- (xxix) No finite-state theorem is invoked unless state equality contains all typed object and transition data used by the recurrence.
- (xxx) No stable-image model is called the inverse limit without an explicit isomorphism theorem.
- (xxxi) No internal apex Return is promoted to an external arithmetic, analytic, geometric, or characteristic-ideal Return without Chapters 8–9 interfaces.
- (xxxii) No arbitrary Grothendieck-category derived inverse limit, unbounded derived tower, source-specific control theorem,  $p$ -adic  $L$ -function Return, or characteristic-ideal Return is promoted to Core by this chapter.

## 5.21 Chapter 5 completion criterion

**Policy 28** (Chapter 5 completion gate). *Chapter 5 is complete only when every invoked finite-to-infinite route has:*

- (i) *a typed tower and actual transition morphisms;*
- (ii) *a declared direct or inverse source with sufficient semantic authority;*
- (iii) *a named finite-to-infinite theorem and finite witness;*
- (iv) *explicit exceptional-index and tail-coverage data;*
- (v) *every required Type IV, transient, Roos, Mittag–Leffler, stable-image, and apex-agreement field;*
- (vi) *a componentwise route-admissible internal Return set, with all maximal supported strengths recorded and a ceiling only when uniquely defined;*
- (vii) *an explicit external Return exclusion or a downstream Chapter 9 route.*

*No route remains in which agreement on finitely many stages, a sampled zero, a dimension plateau, or a barcode matching is silently promoted to an infinite-tail or apex conclusion.*

## 5.22 Provenance and status

Chapter 5 depends on Chapter 1 for the v21 dual-Core constitution, finite-to-infinite discipline, failure landscape, and B-Gate<sup>+</sup>; Chapter 2 and the inherited Appendix A for repaired persistence and metric safety; Chapter 3 for detector stages and certificate complexity; and Chapter 4 for finite linear reconstruction and degreewise eligible Ext discharge. The inherited v19–v20 mathematical baseline is the window- relative terminal-generic Type IV and transient-defect package. The v21 conservative evolution adds typed tower variance, direct and inverse source semantics, eventual-constancy Return, finite-length and Noetherian reduction, the countable Roos and Mittag–Leffler branch, exact finite-state stable-image Return, and the finite-to-infinite record with its internal apex Return capability profile. Chapter 8 is the canonical verifier and manifest layer; Chapters 8–9 govern any external Return.

**New v21 tower and finite-to-infinite theorems.** Theorems [5.32](#), [5.37](#), [5.71](#), [5.72](#), [5.73](#), [5.75](#), [5.77](#), [5.80](#), [5.84](#), [5.85](#), and [5.88](#); Corollaries [5.33](#), [5.74](#), and [5.81](#).

**New v21 tower and Return records.** Definitions [5.3](#), [5.4](#), [5.5](#), [5.6](#), [5.16](#), [5.17](#), [5.18](#), [5.29](#), [5.30](#), [5.31](#), [5.35](#), [5.36](#), [5.25](#), [5.70](#), [5.79](#), [5.82](#), [5.83](#), [5.87](#), and [5.89](#); Policies [26](#), [27](#), and [28](#).

**Inherited v19 Core mathematical propositions.** Propositions [5.15](#), [5.41](#), [5.44](#), [5.46](#), [5.52](#), [5.92](#), and [5.97](#).

**New v20 transient-defect results.** Proposition [5.58](#), Theorem [5.61](#), Corollary [5.63](#), Theorem [5.65](#), Theorem [5.67](#), and Proposition [5.94](#).

**Core definitions and contracts.** Definitions [5.7](#), [5.9](#), [5.12](#), [5.13](#), [5.14](#), [5.20](#), [5.21](#), [5.24](#), [5.39](#), [5.40](#), [5.48](#), [5.49](#), [5.50](#), [5.51](#), [5.57](#), [5.59](#), [5.60](#), [5.64](#), [5.66](#), [5.90](#), [5.93](#), [5.95](#), and [5.99](#).



**Operational examples and cautions.** Examples [5.54](#), [5.55](#), and [5.68](#); Remarks [5.10](#), [5.11](#), [5.43](#), [5.101](#), [5.26](#), [5.23](#), [5.45](#), [5.53](#), [5.62](#), [5.69](#), and [5.96](#).

**Explicit exclusions.** Section [5.20](#).

## 6 Chapter 6: Windows, Overlap, Certificate Atlases, and Restart

**Chapter 6 contract (v21 certificate-atlas layer).** This chapter assembles local Core-P and interface certificates across finite or countable window families. Its strongest unconditional conclusion is a typed *record-level* atlas verdict. It does not construct a global filtered, persistence, quiver, dg, or derived object and does not prove effective object descent.

Every consumed comparison uses exactly one canonical mode

```
exact,      metric_gap_certified,
theorem_controlled_nonmetric,    not_compared.
```

Exact mode requires actual identifications or morphisms at the claimed strength. Metric mode uses the sharp Chapter 2 threshold-gap theorem. Theorem-controlled nonmetric mode transports only the named theorem's exact conclusion. None of the three affirmative modes is silently promoted to another.

Pairwise overlap never implies triple, higher-simplex, or homotopy coherence. Strict cocycles, homotopy-coherent cells, metric-reference records, and nonmetric theorem records are distinct data types. Retained-bar coverage is a separate universal obligation: every globally retained feature must have a certified local or boundary-bridge witness. Local zero, pairwise overlap, positive Restart reserve, and summable exposure do not individually establish a covered-range zero verdict.

Restart transports only the evidence explicitly inherited by a typed Restart datum. It records invalidated evidence and requires an actual target-zero route whenever zero is continued. Countable claims require an enumeration and hidden-tail-prevention record; a finite prefix is never an infinite theorem.

Direct Type IV diagnostics and the Chapter 5 inverse-system audit triple are independent optional fields. Source-profile comparison does not transport kernel, cokernel, Roos, or apex defects without a separate theorem. The final handoff of this chapter is only a complete DescentInput record for Chapter 7. Global object existence, uniqueness, zero, isomorphism, and Return strength remain Chapter 7 responsibilities.

### 6.1 Scope, dependencies, and relocation discipline

This chapter works in the v21 dual-Core constitution established by Chapters 1–5 and the inherited Foundation Appendix A. It preserves every v20 theorem and label below while adding a certificate-atlas and typed descent- handoff layer. Its inputs are:

- (i) the pre-threshold window profile

$$\mathcal{B}_{i,W}(F) := W_W \mathbf{P}_i(F);$$

- (ii) the repaired post-threshold evaluation

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(\mathcal{B}_{i,W}(F));$$

- (iii) the sharp two-sided threshold-gap theorem with closed band

$$[\tau - \varepsilon, \tau + \varepsilon];$$

- (iv) the convention

$$\text{gap}_\tau = \text{clear}_\tau$$

as a ledger-facing alias;

- (v) the representative and eligible-realization disciplines of Chapters 3–4;

- (vi) the direct Type IV status and terminal-generic diagnostics of Chapter 5;
- (vii) the independent inverse-system audit triple  $\text{InvAudit} = (\text{Sys}, \text{R}^1\text{Lim}, \text{Apex})$  whenever an inverse-limit or apex claim is in scope;
- (viii) the Chapter 1 four-mode comparison calculus and route-admissible Return-strength discipline.

The purpose of this chapter is to define exactly when local certificate records may be compared, continued, and assembled without reintroducing an unsupported global theorem. It develops:

- (i) right-open partition and overlap-cover logic;
- (ii) sharp two-sided Overlap Gate certificates;
- (iii) finite Čech-depth coherence at the *record* level;
- (iv) conditional  $\kappa$ -admissible Restart;
- (v) the correct weighted summability condition for countable restart;
- (vi) countable certificate atlases under explicit local-finiteness and coherence hypotheses;
- (vii) strict-cocycle, homotopy-coherent, metric-reference, and theorem-controlled nonmetric simplex records as distinct types;
- (viii) detector-aware local-to-global coverage and its sharp failure modes;
- (ix) detector-safe Restart handoff and transient-defect participation;
- (x) a typed DescentInput interface to, but not a proof of, object-level descent;
- (xi) a formal separation of local zero, global record zero, and global object zero.

**Remark 6.1** (Relocation of the former Chapter 6). The former v18.1 Chapter 6 material on the  $\delta$ -ledger, Audit Mode, proposer–verifier separation, and Core sovereignty is not discarded. In the v19 main-text order it is relocated to Chapter 7, after the present chapter has fixed the window, overlap, restart, and gluing records that the audit chapter must consume. The present chapter uses ledger values but does not redefine the final audit-verdict semantics.

**Remark 6.2** (No automatic local-to-global theorem). Neither a family of local B-Gate<sup>+</sup> passes nor pairwise small bottleneck distance automatically produces a global persistence object, a global filtered representative, or a global theorem. This chapter certifies only the compatibility and propagation of declared certificate records unless an independent object-level gluing theorem is supplied.

## 6.2 Conclusion levels and nonpromotion discipline

**Definition 6.3** (Local zero family). For a declared atlas scope  $A_0$ , a *local zero family* is the typed statement

$$\forall \alpha \in A_0, \quad Z_\alpha = 0,$$

where every  $Z_\alpha$  is the exact local predicate named by its certificate record. The family records the source stage, window, threshold, detector or direct proof, and verifier status. It contains no global quantifier over features unless a separate coverage theorem supplies one.

**Definition 6.4** (Global record verdict and global record zero). A *global record verdict* is a verifier output on a certificate atlas, with status

pass,      reject,      undefined.

A verdict is called *global record zero* when it passes a declared covered-range zero clause by citing:

- (i) a declared global profile or other globally quantified record target;
- (ii) complete local zero certificates;
- (iii) exact, metric-safe, or theorem-controlled interfaces at their exact authorized strength;
- (iv) retained-feature coverage, including boundary and tail coverage;
- (v) every required overlap and higher-coherence record.

This is a theorem-backed predicate in the record system. It is not a construction of a global object from local objects.

**Definition 6.5** (Global object zero). A *global object zero* statement is an assertion

$$E_{\text{glob}} \simeq 0$$

in a declared target category, after an effective object-descent theorem has constructed  $E_{\text{glob}}$  from actual local objects and descent data. It is outside the conclusion strength of Chapter 6.

**Theorem 6.6** (No object descent from record coherence). *Neither a local zero family nor a global record-zero verdict implies global object zero without a separately proved effective descent theorem and an actual descended object in the declared target category.*

*Proof.* The first two statements are predicates about registered local or global records. They need not contain local objects, transition morphisms, cocycles, or an effectivity theorem. These data are necessary to define the third statement. Hence the implication is not well-typed without the additional Chapter 7 hypotheses.  $\square$

### 6.3 Right-open domains, partitions, and overlap covers

**Definition 6.7** (Right-open Core window). A *Core window* is a right-open interval

$$W = [u, v), \quad -\infty < u < v \leq +\infty.$$

It is *bounded* if  $v < +\infty$ , and *right-unbounded* if  $v = +\infty$ . All local profiles in this chapter are evaluated directly on the declared window before threshold deletion.

**Definition 6.8** (Right-open partition cover). A family

$$\mathcal{W}_{\text{part}} = \{W_k = [u_k, u_{k+1})\}_{k \in K}$$

is a *right-open partition cover* of a declared range  $X \subseteq \mathbb{R}$  if:

- (i) the windows are pairwise disjoint under the right-open endpoint convention;
- (ii)  $\bigcup_{k \in K} W_k = X$ ;
- (iii) the indexing order is part of the data whenever sequential continuation or Restart is used.

The former term “MECE windowing” may be retained as an operational alias, but the mathematical object is a right-open partition cover.

**Definition 6.9** (Overlap-admitting cover). A family

$$\mathcal{W}_{\text{ov}} = \{W_\alpha\}_{\alpha \in A}$$

of right-open windows is an *overlap-admitting cover* of  $X$  if

$$X = \bigcup_{\alpha \in A} W_\alpha$$

and every nonempty finite intersection used by the certificate logic is explicitly indexed and recorded. For a finite index set  $\sigma = \{\alpha_0, \dots, \alpha_p\}$ , write

$$W_\sigma := W_{\alpha_0} \cap \dots \cap W_{\alpha_p}.$$

**Remark 6.10** (Partition and overlap covers have different jobs). A right-open partition cover is used for sequential Restart and accumulated-budget control. An overlap-admitting cover is used for compatibility and Čech-type coherence. A disjoint partition has no nontrivial overlap on which an Overlap Gate could be evaluated; conversely, an overlap cover does not by itself provide an ordered continuation path.

## 6.4 Local certificate records

**Definition 6.11** (Local certificate record). For a window  $W_\alpha$ , a local certificate record contains, as applicable:

- (i) immutable source, degree, coefficient, window, and run identifiers;
- (ii) the pre-threshold and repaired post-threshold profiles;
- (iii) the threshold, endpoint convention, and threshold-clearance record;
- (iv) detector information model, source stage, witness set, completeness, cost, and value/certificate statuses;
- (v) representative and finite-grid or P2DG records when invoked;
- (vi) direct Type IV data only when an actual direct comparison morphism is in scope;
- (vii) the independent inverse audit triple when an inverse-system claim is in scope;
- (viii) full, cofinal, interior, and transient defect records when required;
- (ix) every consumed comparison mode and its exact proof, metric certificate, or named nonmetric theorem;
- (x) claim, dependency, verifier, provenance, and replay artifacts.

Absent in-scope data yield `undefined`; out-of-scope data are marked `not_invoked`. The record is a bundle of typed evidence, not a local mathematical object unless an actual object field is separately supplied.

**Definition 6.12** (Local record status). A local record has exactly one verifier status

$$\text{pass}, \quad \text{reject}, \quad \text{undefined}.$$

A passing local zero clause records both certificate validity and the exact mathematical zero predicate. A missing completeness theorem is undefined, not zero; an evidenced nonzero required-zero predicate is reject.

**Definition 6.13** (Direct common-window profile). Let local objects  $F_\alpha$  and  $F_\beta$  be declared on windows  $W_\alpha$  and  $W_\beta$ , and suppose  $W_{\alpha\beta} := W_\alpha \cap W_\beta \neq \emptyset$ . Their pre-threshold common-window profiles are

$$\mathcal{B}_{i,\alpha\beta}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad \mathcal{B}_{i,\alpha\beta}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

For a declared common threshold  $\tau_{\alpha\beta}$ , the corresponding post-threshold profiles are

$$\text{Eval}_{i,\alpha\beta}^\alpha := T_{\tau_{\alpha\beta}}^{\text{bar}}(\mathcal{B}_{i,\alpha\beta}^\alpha), \quad \text{Eval}_{i,\alpha\beta}^\beta := T_{\tau_{\alpha\beta}}^{\text{bar}}(\mathcal{B}_{i,\alpha\beta}^\beta).$$

**Remark 6.14** (No crop-after-collapse substitution). The profiles in Definition 6.13 are computed directly on the common window. They are not replaced by cropping a profile already collapsed on  $W_\alpha$  or  $W_\beta$ , unless a separate window-adapted commutation certificate is supplied. This prevents the false identity

$$W_{W_{\alpha\beta}} T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_{W_{\alpha\beta}}$$

from entering the overlap proof through notation. When such a certificate is invoked, it must use the necessary-and-sufficient condition of Chapter 2 and Appendix A: every bar retained before cropping, including every infinite bar, is either disjoint from the common window or remains retained after nonempty clipping. A finite-bar-only check is insufficient.

## 6.5 Metric scalarization of the ledger

The threshold-gap theorem is stated in the real-valued bottleneck metric. Consequently, an abstract quantale-valued ledger cannot be compared directly with  $\text{clear}_\tau$  unless a metric scalarization has been declared.

**Definition 6.15** (Bottleneck scalarization). Let

$$(\mathbb{V}, \oplus, 0_{\mathbb{V}}, \preceq)$$

be the declared commutative ledger quantale. A *bottleneck scalarization* is a monotone, subadditive map

$$\text{val}_B : \mathbb{V} \longrightarrow [0, +\infty]$$

satisfying

$$\text{val}_B(0_{\mathbb{V}}) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

It converts a metric-ledger element into a real bottleneck-discrepancy upper bound. The manifest must also identify the actual profile comparison to which that upper bound applies.

**Definition 6.16** (Scalarized metric ledger). For a gate-scoped metric ledger record

$$\begin{aligned} \mathbf{d}_{\text{met}} = & \delta^{\text{alg}} \oplus \delta^{\text{disc}} \oplus \delta^{\text{meas}} \oplus \delta^{\text{rep,met}} \oplus \delta^{\text{real,met}} \oplus \delta^{\text{cmp}} \\ & \oplus \delta^{\text{crop}} \oplus \delta^{\text{cont}} \oplus \delta^{\text{ov}} \oplus \delta^{\text{rst}}, \end{aligned}$$

define its scalarized bottleneck budget by

$$\varepsilon_B(\mathbf{d}_{\text{met}}) := \text{val}_B(\mathbf{d}_{\text{met}}).$$

Every displayed component is a metric-discrepancy contribution only. In particular,  $\delta^{\text{rep,met}}$  and  $\delta^{\text{real,met}}$  may measure proved profile distortion introduced by representative or realization transport; they do not encode the existence, typing, or compatibility of a representative or realization. Only components actually consumed by the audited comparison are included. Each component retains its provenance and may not be silently reused by an unrelated gate. Crop, overlap, continuation, and restart contributions remain separately addressable even when  $\text{val}_B$  eventually combines them into one real upper bound.

**Definition 6.17** (Non-scalar continuation and coherence obligations). The following are typed proof obligations rather than metric-ledger components:

- (i) existence and typing of local representatives and realization data;
- (ii) actual overlap, face, continuation, and tower comparison morphisms when such morphisms are invoked;
- (iii) threshold reconciliation,  $\kappa_k$ -admissibility, terminal tameness, and Type IV certificates;
- (iv) higher-simplex compatibility and face coherence;
- (v) claim status, manifest completeness, consistency, and artifact readability.

No increase in  $\varepsilon_B$ , no larger threshold gap, and no summability estimate can discharge a missing obligation in this list.

**Remark 6.18** (Meaning of  $\Sigma\delta < \text{gap}_\tau$ ). Whenever this chapter writes

$$\Sigma\delta < \text{gap}_\tau,$$

it means the gate-scoped typed inequality

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) = \text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B}).$$

An unspecialized quantale element is not compared directly with the real number  $\text{clear}_\tau$ , and a budget certified for one profile family  $\mathfrak{B}$  is not silently spent again on another family.

**Remark 6.19** (Sharp threshold band versus scalarization tightness). The word *sharp* in this chapter refers to the optimal half-width  $\varepsilon$  of the two-sided forbidden band

$$[\tau - \varepsilon, \tau + \varepsilon]$$

when  $\varepsilon$  is a valid bottleneck-distance upper bound. It does not assert that the chosen scalarization  $\text{val}_B$  is numerically tight. If  $\text{val}_B(\mathfrak{d}_{\text{met}})$  overestimates the true bottleneck error, the resulting certificate is conservative even though the underlying threshold theorem remains sharp.

**Definition 6.20** (Gate-scoped threshold reserve). Let  $\mathfrak{B} = \{B_j\}_{j \in J}$  be the finite family of pre-threshold profiles consumed by one declared local, overlap, or restart gate. Define

$$\text{gap}_\tau(\mathfrak{B}) := \min_{j \in J} \text{clear}_\tau(B_j),$$

and, for its scalarized metric budget  $\varepsilon$ , define the *threshold reserve*

$$\text{res}_\tau(\mathfrak{B}; \varepsilon) := \text{gap}_\tau(\mathfrak{B}) - \varepsilon.$$

The gate has positive reserve precisely when

$$\varepsilon < \text{gap}_\tau(\mathfrak{B}).$$

This reserve belongs to the declared gate and profile family; it is not a globally reusable balance.

**Remark 6.21** (The gap is measured before thresholding). The family  $\mathfrak{B}$  in Definition 6.20 consists of pre-threshold profiles. The post-threshold evaluations are used for the compatibility verdict, but they do not define the threshold clearance.

## 6.6 Sharp two-sided Overlap Gate

**Definition 6.22** (Threshold reconciliation record). If two local records use different thresholds  $\tau_\alpha$  and  $\tau_\beta$ , an overlap comparison is well-typed only after declaring a common overlap threshold  $\tau_{\alpha\beta}$  and directly re-evaluating both local objects on  $W_{\alpha\beta}$  at that threshold. The data establishing this choice are called a *threshold reconciliation record*. No comparison between differently thresholded outputs is silently interpreted as an Overlap Gate.

**Definition 6.23** (Overlap comparison mode). Every consumed overlap inference records exactly one canonical comparison mode:

exact,      metric\_gap\_certified,      theorem\_controlled\_nonmetric,      not\_compared.

The modes mean:

- (i) **exact**: an actual equality, isomorphism, or morphism at the claimed strength is supplied on the directly evaluated common window;
- (ii) **metric\_gap\_certified**: a pre-threshold metric bound and the two-sided threshold-gap and zero-separation hypotheses required by the consumed conclusion are supplied;
- (iii) **theorem\_controlled\_nonmetric**: a named theorem with exact hypotheses, source, target, variance, and conclusion transports a nonmetric predicate or structure;
- (iv) **not\_compared**: no comparison conclusion is consumed.

Exactness does not imply positive-radius robustness. Metric closeness does not create a morphism. A non-metric theorem supplies only its printed conclusion. The last mode cannot discharge a required overlap clause.

**Definition 6.24** (Exact overlap certificate). An *exact overlap certificate* on  $(i, W_{\alpha\beta}, \tau_{\alpha\beta})$  contains:

- (i) the directly evaluated common-window pre-threshold and repaired profiles;
- (ii) the common threshold and threshold-reconciliation record;
- (iii) the exact overlap conclusion requested by the governing claim, denoted  $C_{\alpha\beta,i}^{\text{req}}$ ;
- (iv) exactly one exact relation type:

- (a) **identification**: an actual equality or isomorphism

$$\text{Eval}_{i,\alpha\beta}^\alpha \xrightarrow{\sim} \text{Eval}_{i,\alpha\beta}^\beta;$$

- (b) **directed\_morphism**: an actual morphism with declared source, target, and variance.

- (v) a theorem or direct proof that the registered exact relation entails  $C_{\alpha\beta,i}^{\text{req}}$  at precisely the claimed strength;
- (vi) immutable source, target, proof, and artifact identifiers.

A directed morphism does not discharge identification, symmetric agreement, zero reflection, or descent compatibility unless the entailment proof establishes that stronger requested conclusion. Thus the zero morphism between two arbitrary profiles is not an overlap pass merely because it is an actual morphism. No positive threshold-gap reserve follows merely from either exact relation.



**Definition 6.25** (Two-sided overlap metric certificate). Fix a monitored degree  $i$ , an overlap  $W_{\alpha\beta}$ , a common threshold  $\tau_{\alpha\beta}$ , and a scalarized budget  $\varepsilon_{\alpha\beta,i} \geq 0$ . A *two-sided overlap metric certificate* consists of:

(i) a pre-threshold bottleneck estimate

$$d_B(\mathcal{B}_{i,\alpha\beta}^\alpha, \mathcal{B}_{i,\alpha\beta}^\beta) \leq \varepsilon_{\alpha\beta,i};$$

(ii) two-sided sharp clearance

$$\text{clear}_{\tau_{\alpha\beta}}(\mathcal{B}_{i,\alpha\beta}^\alpha) > \varepsilon_{\alpha\beta,i}, \quad \text{clear}_{\tau_{\alpha\beta}}(\mathcal{B}_{i,\alpha\beta}^\beta) > \varepsilon_{\alpha\beta,i};$$

(iii) the ledger provenance whose bottleneck scalarization yields  $\varepsilon_{\alpha\beta,i}$ ;

(iv) the direct common-window and threshold-reconciliation records required above.

**Definition 6.26** (Theorem-controlled nonmetric overlap certificate). A theorem-controlled nonmetric overlap certificate contains a named theorem  $\mathcal{T}$ , its version and proof artifact, the exact source and target records on the common window, every hypothesis and variance field, the precise conclusion  $C_{\mathcal{T}}$ , and the governing requested conclusion  $C_{\alpha\beta,i}^{\text{req}}$ . The record includes a checked implication

$$C_{\mathcal{T}} \implies C_{\alpha\beta,i}^{\text{req}}$$

or declares that the two conclusions are identical. The verifier may consume only  $C_{\mathcal{T}}$  and its registered weakenings. No bottleneck bound, transition morphism, cocycle, zero reflection, or descent datum is inferred unless it is part of that theorem's conclusion.

**Theorem 6.27** (Sharp typed overlap theorem). *For a well-typed pairwise overlap record, the admissible conclusion is mode-specific:*

- (i) *exact mode returns the registered equality, isomorphism, or morphism;*
- (ii) *metric-gap-certified mode returns the sharp repaired discrepancy bound of Proposition 6.28, and repaired-zero equivalence only when the Chapter 2 zero-separation hypothesis is also present;*
- (iii) *theorem-controlled nonmetric mode returns exactly  $C_{\mathcal{T}}$ ;*
- (iv) *not-compared mode returns no overlap conclusion.*

*No cross-mode promotion is valid without a separate theorem.*

*Proof.* The exact and nonmetric clauses are the contents of their registered proof artifacts. The metric clause is the sharp threshold-gap theorem, together with the separate metric zero-transport theorem when zero reflection is claimed. The final clause is definitional. The source and target types of the four branches differ, so no additional conclusion follows formally.  $\square$

**Proposition 6.28** (Sharp overlap transfer). *Under a two-sided overlap metric certificate,*

$$d_B(\text{Eval}_{i,\alpha\beta}^\alpha, \text{Eval}_{i,\alpha\beta}^\beta) \leq \varepsilon_{\alpha\beta,i}.$$

*In the one-parameter constructible regime and under the infinite-bar convention of Appendix A, the same bound may be read using  $d_{\text{int}}$ .*

*Proof.* Clause (i) of Definition 6.25 supplies the input distance bound. Clause (ii) says that neither pre-threshold barcode contains a finite bar whose length lies in the closed band

$$[\tau_{\alpha\beta} - \varepsilon_{\alpha\beta,i}, \tau_{\alpha\beta} + \varepsilon_{\alpha\beta,i}].$$

The sharp threshold-gap-safe stability theorem of Chapter 2 and Appendix A therefore applies and yields the post-threshold bottleneck bound. The bottleneck–interleaving isometry then gives the interleaving formulation in the declared constructible regime.  $\square$

**Definition 6.29** (Pairwise Overlap Gate). The *pairwise Overlap Gate* on  $W_{\alpha\beta}$  passes if, for every monitored degree  $i$ :

- (i) the governing claim declares one requested overlap conclusion  $C_{\alpha\beta,i}^{\text{req}}$  and exactly one comparison mode;
- (ii) in mode `exact`, a valid exact overlap certificate exists and its registered relation entails  $C_{\alpha\beta,i}^{\text{req}}$ ;
- (iii) in mode `metric_gap_certified`, a two-sided overlap metric certificate exists, Proposition 6.28 holds, and the resulting metric or zero-transport conclusion entails the requested conclusion;
- (iv) in mode `theorem_controlled_nonmetric`, the named theorem, hypotheses, exact source and target, printed conclusion, and its entailment of the requested conclusion are verified;
- (v) mode `not_compared` is used only when the governing claim consumes no overlap inference;
- (vi) all local representative or eligible-realization clauses used on the overlap have compatible provenance records;
- (vii) when an actual direct tower comparison is in scope, direct Type IV has status `certified_zero`; otherwise it is explicitly `not_invoked`;
- (viii) when an inverse-system claim is in scope, each required field of `InvAudit` independently passes;
- (ix) when the governing overlap claim also requires absence of long transient kernel or cokernel defects, the Chapter 5 aggregate transient status is `certified_absent`;
- (x) `undefined` is not treated as zero, `obstructed` is a terminal failure, and `certified_present` is a transient-defect failure when absence was required;
- (xi) every required crop, comparison, exact-proof, and threshold-reconciliation artifact is present.

The gate therefore verifies a requested relation, not merely the existence of some map or some theorem on the same overlap.

**Remark 6.30** (Two layers of the Overlap Gate). The Overlap Gate has two mathematically different layers:

- (i) the discrepancy verdict compares the *post-threshold* evaluations;
- (ii) threshold safety is certified on the *pre-threshold* profiles.

The second layer is not recoverable from the first. In particular, small post-threshold distance alone does not certify that threshold classification was stable. An exact overlap certificate may bypass metric transport for the declared exact comparison, but it does not supply positive-radius threshold robustness.

**Remark 6.31** (Pairwise compatibility is not higher coherence). Passing every pairwise Overlap Gate does not by itself establish compatibility on triple or higher overlaps. Higher coherence must be declared through the nerve records below whenever the cover has nontrivial higher intersections.

**Theorem 6.32** (No higher coherence from pairwise overlap). *Pairwise exact isomorphisms need not satisfy a triple cocycle, pairwise metric bounds do not define composable maps, and pairwise theorem-controlled relations need not possess higher compatibility. Therefore every consumed triple, higher simplex, or homotopy-coherent conclusion requires its own registered cell or theorem.*

*Proof.* For exact maps, the two composites on a triple overlap are additional data and may differ. Metric bounds have no composition law on morphisms because no morphisms are supplied. A theorem-controlled relation is confined to its printed arity and hypotheses. Hence none of the pairwise branches determines the required higher datum.  $\square$

## 6.7 Finite Čech depth and record-level coherence

**Definition 6.33** (Čech depth). An overlap-admitting cover has *Čech depth at most  $K$*  if every intersection of  $K + 2$  distinct windows is empty. Equivalently, its Čech nerve has simplicial dimension at most  $K$ . Thus all nonempty overlap records occur on at most  $(K + 1)$ -fold intersections.

**Definition 6.34** (Simplex coherence mode). Every nonempty nerve simplex used by a coherence claim declares exactly one mode:

strict\_cocycle,      homotopy\_coherent\_cell,  
metric\_reference,      theorem\_controlled\_nonmetric,  
not\_compared.

Strict-cocycle mode consumes actual isomorphisms and literal cocycle identities. Homotopy-coherent mode consumes actual one-morphisms, specified invertible higher cells, and all boundary equations through the invoked simplicial degree. Metric-reference mode supplies discrepancy bounds but no maps. Theorem-controlled nonmetric mode supplies only the named theorem’s conclusion. Not-compared is allowed only when no simplex conclusion is used.

**Definition 6.35** (Exact simplex cocycle record). Let  $\sigma = \{\alpha_0, \dots, \alpha_p\}$  index a nonempty overlap  $W_\sigma$ , and fix a monitored degree and a common threshold. An *exact simplex cocycle record* contains actual repaired-profile isomorphisms

$$g_{\alpha\beta}^\sigma : \text{Eval}_{i,\sigma}^\alpha \xrightarrow{\sim} \text{Eval}_{i,\sigma}^\beta \quad (\alpha, \beta \in \sigma)$$

with immutable artifacts and the identities

$$g_{\alpha\alpha}^\sigma = \text{id}, \quad g_{\beta\alpha}^\sigma = (g_{\alpha\beta}^\sigma)^{-1}, \quad g_{\beta\gamma}^\sigma \circ g_{\alpha\beta}^\sigma = g_{\alpha\gamma}^\sigma.$$

Restriction to a face must agree with the previously declared face cocycle. These identities are mathematical equalities of actual morphisms, not merely small-distance statements.

**Definition 6.36** (Model-certified homotopy-coherent simplex-cell record). Let actual local objects  $E_\alpha$  live in a declared coherence model  $\mathfrak{M}_{\text{coh}}$ , such as a specified dg enhancement, bicategory, simplicial category, or concrete presentation of an  $(\infty, 1)$ -category. The model certificate records:

- (i) the object, one-morphism, and higher-cell types used by the record;

- (ii) units, composition, whiskering, face restriction, and the equality or homotopy relation in which boundary equations are checked;
- (iii) a finite encoding and verifier for every cell actually consumed, or a named theorem that verifies the entire finite simplex schema;
- (iv) the precise simplicial degree through which coherence is invoked.

Relative to this model, a homotopy-coherent simplex record contains actual overlap equivalences  $g_{\alpha\beta}$ , invertible two-cells

$$c_{\alpha\beta\gamma} : g_{\beta\gamma} g_{\alpha\beta} \Rightarrow g_{\alpha\gamma},$$

and every higher coherence cell required through the invoked simplex degree, together with all model-specific boundary identities and face restrictions. The record is affirmative only when the model verifier or named theorem certifies those equations. A generic label “homotopy coherent”, or an abstract  $(\infty, 1)$ -category with no auditable presentation, yields `undefined` rather than an atlas pass.

**Proposition 6.37** (Homotopy-coherent certification is model-relative). *A homotopy-coherent simplex record may be consumed by a finite Chapter 6 atlas verdict only relative to its declared  $\mathfrak{M}_{\text{coh}}$  and the registered boundary-equation verifier. Changing the enhancement, cell model, or equality notion invalidates the certificate unless a comparison theorem transports every consumed cell and equation.*

*Proof.* The source, target, composition, and boundary of a higher cell are defined in the ambient coherence model. Without that model the equations are not typed; a change of model changes their interpretation unless an explicit transport theorem is supplied.  $\square$

**Definition 6.38** (Theorem-controlled nonmetric simplex record). A theorem-controlled nonmetric simplex record specifies one theorem uniformly on the simplex and all its faces, with exact source, target, hypotheses, variance, and conclusion. It may certify a predicate-level compatibility or another named nonmetric relation. It supplies no transition morphism, cocycle, or descent cell unless the theorem explicitly constructs them.

**Definition 6.39** (Metric-reference simplex record). For the same simplex  $\sigma$ , a *metric-reference simplex record* contains a declared pre-threshold reference profile  $R_{i,\sigma}$  on  $W_\sigma$ , one common threshold  $\tau_\sigma$ , and for every  $\alpha \in \sigma$  a certified bound

$$d_B(\mathcal{B}_{i,\sigma}^\alpha, R_{i,\sigma}) \leq \varepsilon_{\alpha,\sigma,i},$$

together with a two-sided threshold-gap certificate for the pair  $(\mathcal{B}_{i,\sigma}^\alpha, R_{i,\sigma})$  at that same budget. The record also contains the reference-profile identity, scalarization, threshold reconciliation, and face-restriction provenance.

The resulting repaired comparisons are metric statements only. No map  $\text{Eval}_{i,\sigma}^\alpha \rightarrow \text{Eval}_{i,\sigma}^\beta$ , no cocycle, and no object-level descent datum is inferred from the reference profile.

**Proposition 6.40** (Pairwise bound from a metric simplex reference). *Under a metric-reference simplex record, for every  $\alpha, \beta \in \sigma$ ,*

$$d_B(\text{Eval}_{i,\sigma}^\alpha, \text{Eval}_{i,\sigma}^\beta) \leq \varepsilon_{\alpha,\sigma,i} + \varepsilon_{\beta,\sigma,i}.$$

*Proof.* Apply the Chapter 2 threshold-gap-safe theorem to each local-to-reference comparison. This gives repaired-profile bounds

$$d_B(\text{Eval}_{i,\sigma}^\alpha, T_{\tau_\sigma}^{\text{bar}}(R_{i,\sigma})) \leq \varepsilon_{\alpha,\sigma,i}$$

and the analogous bound for  $\beta$ . The triangle inequality gives the claim.  $\square$

**Remark 6.41** (Metric coherence is not an approximate cocycle). A family of pairwise distance bounds does not carry composable transition maps. The phrase *metric coherence* in this chapter means coherence of scope, reference identity, threshold, face records, and certified bounds. It does not mean that an exact cocycle exists up to an unspecified error.

**Definition 6.42** (Simplex compatibility record). Let  $\sigma = \{\alpha_0, \dots, \alpha_p\}$  index a nonempty overlap  $W_\sigma$ . A *simplex compatibility record* consists of:

- (i) direct pre-threshold profiles, or actual local objects when the object-level homotopy branch is used, for every participant on  $W_\sigma$ ;
- (ii) a common threshold and threshold-reconciliation record;
- (iii) the exact simplex-level conclusion requested by the governing claim;
- (iv) one simplex coherence mode from Definition 6.34;
- (v) in mode `strict_cocycle`, an exact simplex cocycle record;
- (vi) in mode `homotopy_coherent_cell`, a declared  $\mathfrak{M}_{\text{coh}}$ , actual maps and higher cells, and a successful model-specific boundary verifier;
- (vii) in mode `metric_reference`, a metric-reference simplex record and every local-to-reference threshold-gap artifact;
- (viii) in theorem-controlled nonmetric mode, the uniform theorem record and its face-compatible conclusion;
- (ix) an entailment check showing that the mode-specific output proves the requested simplex conclusion and an explicit runtime comparison mode for every pairwise or face comparison consumed by the record; no consumed comparison may use `not_compared`;
- (x) a face-compatibility statement saying that restriction from  $\sigma$  to any face agrees with the already declared face record at the registered conclusion strength;
- (xi) the direct Type IV status when a direct comparison diagnostic is in scope, the independent inverse-audit fields when an inverse system is in scope, and every required transient-defect artifact;
- (xii) all immutable source, target, proof, scalarization, model, and artifact references.

A theorem-controlled predicate or metric-reference bound is record-level coherence only at its declared conclusion strength; neither is object-level descent data.

**Definition 6.43** (Čech-coherent certificate atlas). A family of local certificate records on  $\mathcal{W}_{\text{ov}}$  is a *Čech-coherent certificate atlas* if every nonempty simplex of the nerve carries a simplex compatibility record and all face restrictions are mutually compatible.

**Definition 6.44** (Certificate atlas). A *certificate atlas* is the bundle

$$\mathfrak{A} = (\mathcal{W}, \{\mathcal{L}_\alpha\}, \{O_\sigma\}, C, \mathcal{R}, \mathcal{H}),$$

where  $\mathcal{W}$  is a finite or countable window family,  $\mathcal{L}_\alpha$  are local certificate records,  $O_\sigma$  are pairwise and higher-simplex records,  $C$  is retained-feature coverage when a covered-range predicate is claimed,  $\mathcal{R}$  is Restart data when continuation is claimed, and  $\mathcal{H}$  is the complete manifest and replay index. Fields not in scope are marked `not_invoked`; missing required fields are `undefined`.

**Definition 6.45** (Atlas verdict). The atlas verdict is the noncompensating conjunction of all required local, overlap, higher-coherence, coverage, Restart, direct Type IV, inverse-audit, transient-defect, and manifest clauses. It is pass exactly when every required clause passes; reject when a well-typed required mathematical clause is refuted; and undefined when an invoked obligation is missing or ill-typed and no already-evidenced rejecting clause decides the run.

**Proposition 6.46** (Finite-depth verification bound). *If the cover has Čech depth at most  $K$ , then the coherence portion of the certificate atlas requires no records above simplicial degree  $K$ .*

*Proof.* By Definition 6.33, there are no nonempty intersections indexed by simplices of degree larger than  $K$ . Hence there are no higher overlap domains on which additional compatibility records could be evaluated.  $\square$

**Theorem 6.47** (Finite-cover record-level gluing). *Let  $\mathcal{W}_{\text{ov}}$  be a finite overlap-admitting cover of finite Čech depth. Assume:*

- (i) *every window carries a valid local certificate record;*
- (ii) *every nonempty nerve simplex carries a compatibility record in one affirmative simplex mode, and the mode-specific output entails the requested simplex conclusion;*
- (iii) *every homotopy-coherent simplex is certified in one declared  $\mathfrak{M}_{\text{coh}}$  with successful boundary verification;*
- (iv) *every required direct Type IV scope is `certified_zero`, and an out-of-scope direct diagnostic is `not_invoked`; every invoked inverse-audit field independently passes; whenever long-transient cleanliness is required, its aggregate status is `certified_absent`;*
- (v) *every metric-mode comparison satisfies its sharp two-sided threshold reserve, every exact-mode comparison carries its exact proof, and every theorem-controlled comparison carries its named theorem and entailment record;*
- (vi) *all face-compatibility statements and artifact references are audit-readable.*

*Then the family is a Čech-coherent certificate atlas at exactly the record-level conclusion strengths requested by its simplex records.*

*Proof.* The local records provide the vertices of the nerve. Strict-cocycle records supply literal cocycle identities. Model-certified homotopy records supply verified higher cells and boundary equations. Metric-reference records supply face-compatible discrepancy conclusions without maps, and theorem-controlled records supply only their checked printed conclusions. The entailment fields prevent a weaker mode output from passing a stronger requested clause. Finite depth and finiteness of the cover make the invoked record set finite. The remaining diagnostic and artifact clauses exclude undefined inputs. Hence the family satisfies Definition 6.43 at its declared typed strengths.  $\square$

**Theorem 6.48** (Finite certificate-atlas soundness). *For a finite cover, if all required fields of Definition 6.44 are present and every governing local, overlap, higher-coherence, coverage, Restart, diagnostic, and manifest theorem passes, then the atlas verdict passes. If the detector-globalization clause is invoked and its hypotheses pass, the verdict may additionally be marked global record zero. Neither conclusion constructs or proves zero of a global object.*

*Proof.* The atlas verdict is the noncompensating conjunction of the cited clauses. Finite-cover and finite-depth hypotheses make the invoked record set finite. The global record-zero refinement is exactly the output of the retained-bar coverage theorem below. Object construction is absent from all hypotheses.  $\square$

**Remark 6.49** (The theorem glues records, not mathematical objects). Theorem 6.47 does not assert that the filtered objects  $F_\alpha$ , their representatives, or their derived realizations glue to a global object. It produces an auditable certificate atlas whose local verdicts and overlap records are coherent. Object-level descent requires an additional theorem in an explicitly declared category.

## 6.8 Detector-aware local-to-global coverage

**Definition 6.50** (Global profile and local restriction interface). Fix a declared covered range  $X$ , a monitored degree  $i$ , and one global threshold  $\tau > 0$ . A *global profile interface* consists of a constructible pre-threshold profile  $B_{i,X}$  and, for every atlas window  $W_\alpha$ , the exact restricted profile

$$R_i^\alpha := W_{W_\alpha} B_{i,X}.$$

The observed local pre-threshold profile  $B_i^\alpha$  is connected to  $R_i^\alpha$  by exactly one affirmative canonical comparison mode:

- (i) `exact`: an actual profile isomorphism  $R_i^\alpha \xrightarrow{\sim} B_i^\alpha$  is supplied;
- (ii) `metric_gap_certified`: an artifact-supported bound

$$d_B(R_i^\alpha, B_i^\alpha) \leq \varepsilon_\alpha$$

is supplied, both profiles satisfy the Chapter 2 two-sided threshold-gap condition at  $\varepsilon_\alpha$ , and

$$2\varepsilon_\alpha \leq \tau.$$

Thus repaired zero is preserved and reflected. The operational manifest may retain the stricter reserve  $2\varepsilon_\alpha < \tau$ ;

- (iii) `theorem_controlled_nonmetric`: a named theorem, with all hypotheses and artifacts, proves the exact zero-preservation and zero-reflection implications required between  $T_\tau^{\text{bar}}(R_i^\alpha)$  and  $T_\tau^{\text{bar}}(B_i^\alpha)$ .

Mode `not_compared` cannot support a global inference. The interface is declared input data; pairwise local compatibility does not construct  $B_{i,X}$ .

**Definition 6.51** (Certified local detector atlas). Relative to a global profile interface, a *certified local detector atlas* assigns to every  $W_\alpha$ :

- (i) a finite witness set  $T_\alpha \subseteq (W_\alpha)_\tau$ , or the canonical witness family of a birth or right-germ detector;
- (ii) a Chapter 3 detector at source stage `windowed_prethreshold`;
- (iii) `endpoint`, `critical-set`, `family`, `completeness`, and `artifact records`;
- (iv) the `exact`, `metric-gap-certified`, or `theorem-controlled nonmetric zero-reflecting` branch of Definition 6.50;
- (v) one mathematical detector value, `zero` or `nonzero`, and an independent certificate status.

A detector value may be used globally only at the strength of its local completeness theorem and interface branch.

**Definition 6.52** (Global retained-bar coverage). A certified local detector atlas has *global retained-bar coverage* if for every bar  $I$  of  $B_{i,X}$  retained by  $T_\tau^{\text{bar}}$ , including every genuinely infinite retained bar, there exist an index  $\alpha$  and a witness  $t \in T_\alpha$  such that

$$t \in I \cap W_\alpha, \quad t + \tau \in I \cap W_\alpha.$$

For birth or right-germ detectors, the same clause is read using the corresponding canonical witness of the retained interval. The coverage artifact records how the universal quantifier over retained global bars is proved.

**Theorem 6.53** (Finite retained-bar coverage theorem). *Assume the total registered stencil witness set*

$$T_{\mathfrak{A}} := \bigcup_{\alpha} T_{\alpha}$$

*is finite, as occurs for a finite atlas of finite local stencils. For the exact restricted global profiles  $R_i^\alpha$ , let the atlas stencil be the finite direct sum of all registered local structure-map probes. Then:*

- (i) *a nonzero atlas stencil always witnesses a globally retained bar;*
- (ii) *under global retained-bar coverage,*

$$D_{\mathfrak{A},\tau}(B_{i,X}) = 0 \quad \Longleftrightarrow \quad T_\tau^{\text{bar}}(B_{i,X}) = 0;$$

- (iii) *the coverage bound is family-relative sharp: if an admissible active- start region is omitted by the entire finite witness family, the corresponding one-bar profile is globally nonzero after thresholding while every atlas stencil value is zero.*

*For birth or right-germ witness families, the corresponding soundness and zero-completeness conclusions hold under their exhaustive certified endpoint sets from Chapter 3. The omitted-region sharpness clause is stated for finite stencils.*

*Proof.* For a structure-map probe, Chapter 3 identifies its image dimension with the number of interval summands whose active-start region contains the sample. This gives soundness. Coverage ensures that every globally retained interval contributes to at least one local probe, proving completeness. If an admissible active-start region is omitted, choose an interval  $I = [p, r + \tau)$  when the omitted region is  $[p, r)$ , with the stipulated domain and endpoint conditions. It survives globally and is invisible to every registered sample. Birth and germ variants use the corresponding Chapter 3 characterization theorem.  $\square$

**Corollary 6.54** (Countable detector-family zero criterion). *For a countable family of finite local detectors with global retained-bar coverage,*

$$T_\tau^{\text{bar}}(B_{i,X}) = 0 \quad \Longleftrightarrow \quad D_\alpha^{\text{loc}}(B_i^\alpha) = 0 \text{ for every } \alpha$$

*holds as a universally quantified semantic statement, provided every local interface is zero-preserving and zero-reflecting at the consumed strength. This countable family is not a single finite detector certificate unless a finite exhaustive witness theorem, a uniform schema certificate, or a Chapter 5 finite-to-infinite reduction proves the infinite conjunction from finite evidence.*

*Proof.* The same soundness and coverage argument applies componentwise. The final sentence follows because a countable conjunction of finite detector values is not itself a finite artifact without a theorem reducing or uniformly certifying its tail.  $\square$



**Theorem 6.55** (Detector-atlas local-to-global zero theorem). *Let a global profile interface, a certified local detector atlas, and global retained-bar coverage be given. Then*

$$T_\tau^{\text{bar}}(B_{i,X}) = 0 \iff D_\alpha^{\text{loc}}(B_i^\alpha) = 0 \text{ for every } \alpha.$$

*Here every metric branch satisfies its zero-transport hypotheses, every nonmetric branch proves the consumed zero implications, and every local detector is complete on its declared local profile class. For a countable atlas the equivalence is semantic; a finite verifier pass additionally requires one of the finite-certification routes in Corollary 6.54.*

*Proof.* Assume first that the global repaired profile is zero. Window restriction cannot increase the length of a finite bar, and no infinite bar is present in a globally zero repaired profile. Hence  $T_\tau^{\text{bar}}(R_i^\alpha) = 0$  for every  $\alpha$ . Exact, metric-safe, or theorem-controlled local zero transport gives  $T_\tau^{\text{bar}}(B_i^\alpha) = 0$ , and local detector completeness gives detector zero.

Conversely, suppose every local detector is zero and that a globally retained bar  $I$  exists. Global retained-bar coverage supplies  $\alpha$  and a witness  $t$  with  $t, t + \tau \in I \cap W_\alpha$ . Therefore the exact restricted profile  $R_i^\alpha$  is nonzero after thresholding. Exact, metric-safe, or theorem-controlled zero reflection transfers this nonzero repaired predicate to  $B_i^\alpha$ . Completeness of the local detector then forces a nonzero detector value, a contradiction.  $\square$

**Remark 6.56** (Coverage is the global proof obligation). The preceding theorem is valid even when coverage is verified by reading the complete global barcode. Such a post hoc verification, however, supplies no proof-compression advantage. A nontrivial finite-certificate claim must obtain coverage independently from geometry, endpoint lattices, bounded birth data, a family theorem, or another registered reduction.

**Example 6.57** (Local zero and exact overlap coherence do not imply global zero). Fix  $\tau > 0$ , let  $X = [0, 2\tau)$ , and let

$$B_{i,X} = \{[0, 2\tau)\}.$$

Cover  $X$  by

$$W_1 = [0, \tau), \quad W_2 = [\tfrac{1}{2}\tau, \tfrac{3}{2}\tau), \quad W_3 = [\tau, 2\tau).$$

Every restricted bar has length exactly  $\tau$ , so every local repaired profile is zero. All nonempty overlap repaired profiles are also zero and therefore admit exact cocycle records. Nevertheless the global bar has length  $2\tau > \tau$  and survives globally. The missing hypothesis is global retained-bar coverage: no window contains a  $\tau$ -span strictly inside the retained global bar after the hard  $\leq \tau$  deletion convention is applied locally.

## 6.9 $\kappa$ -admissible Restart

Let

$$\mathcal{W}_{\text{part}} = \{W_k = [u_k, u_{k+1})\}_{k \geq 0}$$

be an ordered right-open partition cover. For each  $k$ , let  $\mathfrak{B}_k$  be the finite pre-threshold profile family used by the local gate and choose a finite working gap

$$0 < g_k \leq \text{gap}_{\tau_k}(\mathfrak{B}_k).$$

If the exact clearance is  $+\infty$ , the manifest chooses a finite audit cap  $g_k$  for Restart arithmetic. Let  $\varepsilon_k < +\infty$  be the gate-scoped scalarized local budget and define the finite local reserve

$$r_k := g_k - \varepsilon_k \in \mathbb{R}.$$

A local gate has positive working reserve exactly when  $r_k > 0$ . The use of a finite certified lower bound avoids undefined extended-real expressions in the Restart recursion.

**Definition 6.58** (Restart datum). A Restart datum from state  $S_k$  to  $S_{k+1}$  contains:

- (i) the immutable predecessor state and target state;
- (ii) the transition artifact and exact target scope;
- (iii) the evidence proposed for inheritance, with a theorem authorizing each inherited field;
- (iv) the evidence invalidated by changed windows, thresholds, endpoint conventions, information models, representatives, tower maps, or artifacts;
- (v) fresh evidence required at the target;
- (vi) the comparison mode, reserve data, and target-zero route when invoked;
- (vii) verifier, provenance, and replay records.

Evidence not explicitly authorized for inheritance is invalidated by default. Invalidated evidence cannot be repaired by a positive numerical reserve.

**Proposition 6.59** (Restart evidence monotonicity). *A certified Restart may retain only evidence in the intersection of the predecessor's proved scope and the transition theorem's authorized image. Every other predecessor field must be invalidated or freshly re-proved at the target.*

*Proof.* A proof artifact establishes a statement only for its registered source, target, hypotheses, and scope. Changing any of those data removes the typing needed for reuse unless a transition theorem transports the statement.  $\square$

**Definition 6.60** (Continuation artifact). A continuation artifact from  $W_k$  to  $W_{k+1}$  records:

- (i) the source and target local record identifiers;
- (ii) the source and target thresholds;
- (iii) any endpoint-state, initialization, representative, or comparison data used to pass from the source record to the target record;
- (iv) a finite scalarized continuation loss  $\eta_k \geq 0$ , not already counted in the local target budget;
- (v) a declared degradation factor  $\kappa_k \in (0, 1]$ ;
- (vi) a proof object or independently checkable artifact for the reserve inequality below;
- (vii) explicit confirmation that missing morphisms, representatives, eligibility records, Type IV data, or manifest fields have not been encoded numerically inside  $\eta_k$ .

**Definition 6.61** ( $\kappa_k$ -admissible continuation). A continuation artifact is  $\kappa_k$ -admissible if it certifies

$$r_{k+1} \geq \kappa_k r_k - \eta_k.$$

The number  $\kappa_k$  is part of the certified continuation data; its existence is not inferred from window adjacency or from threshold deletion alone.

**Remark 6.62** (Restart is conditional, not automatic). There is no Core theorem asserting that every admissible-looking continuation has some universal  $\kappa > 0$ . Window restriction may change bar lengths, thresholds may change, and the next pre-threshold profile may be unrelated to the preceding one. A Restart claim is licensed only by an explicit  $\kappa_k$ -admissible continuation artifact. This is the v19 downgrade of the former unconditional Restart lemma.

**Remark 6.63** (Restart is a reserve-transport claim). A  $\kappa$ -Restart assertion transports a finite positive threshold reserve and is therefore stronger than an isolated exact equality at one transition. An exact continuation may be recorded as exact comparison data, but it does not become a  $\kappa_k$ -admissible Restart step unless finite working gaps, local budgets,  $r_k, r_{k+1}$ , and the certified recurrence

$$r_{k+1} \geq \kappa_k r_k - \eta_k$$

are all supplied.

**Definition 6.64** (Restart transition status). Each transition has exactly one of the statuses

not\_invoked,      undefined,      certified,      failed.

They mean:

- (i) **not\_invoked**: the governing scope explicitly declares that no continuation or inherited reserve claim is being made;
- (ii) **undefined**: Restart is in scope, but a required continuation artifact, scalarized budget, finite working reserve, or  $\kappa_k$ -certificate is missing or ill-typed;
- (iii) **certified**: the artifact is complete and the  $\kappa_k$ -admissible inequality is verified;
- (iv) **failed**: the inputs are defined but the declared  $\kappa_k$ -inequality is false.

The transition status certifies the continuation law only. A successful Restart step additionally requires a passing target local record, in particular positive target reserve and no rejected or undefined non-reserve clause.

**Theorem 6.65** (Finite-horizon Restart bound). *Assume the transitions  $0 \rightarrow 1, \dots, n-1 \rightarrow n$  are respectively  $\kappa_0, \dots, \kappa_{n-1}$ -admissible. Then*

$$r_n \geq \left( \prod_{j=0}^{n-1} \kappa_j \right) r_0 - \sum_{m=0}^{n-1} \left( \prod_{j=m+1}^{n-1} \kappa_j \right) \eta_m,$$

where an empty product is 1.

*Proof.* The case  $n = 1$  is Definition 6.61. Assume the formula holds for  $n$ . Then

$$r_{n+1} \geq \kappa_n r_n - \eta_n.$$

Substituting the induction hypothesis and distributing  $\kappa_n$  gives

$$r_{n+1} \geq \left( \prod_{j=0}^n \kappa_j \right) r_0 - \sum_{m=0}^n \left( \prod_{j=m+1}^n \kappa_j \right) \eta_m.$$

Thus the formula follows by induction. □

**Corollary 6.66** (Non-degrading continuation). *If  $\kappa_k = 1$  for  $0 \leq k < n$ , then*

$$r_n \geq r_0 - \sum_{m=0}^{n-1} \eta_m.$$

*Consequently, if*

$$\sum_{m=0}^{n-1} \eta_m < r_0,$$

*all reserves through level  $n$  are positive.*

**Definition 6.67** (Normalized Restart loss). Set

$$K_0 := 1, \quad K_n := \prod_{j=0}^{n-1} \kappa_j \quad (n \geq 1).$$

The normalized loss through level  $n$  is

$$L_n^{\text{norm}} := \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}}.$$

**Corollary 6.68** (Normalized reserve criterion). *Under the hypotheses of Theorem 6.65,*

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\text{norm}}.$$

*Hence  $L_n^{\text{norm}} < r_0$  implies  $r_n > 0$ .*

*Proof.* Factor  $K_n$  from the lower bound in Theorem 6.65 and use

$$\prod_{j=m+1}^{n-1} \kappa_j = \frac{K_n}{K_{m+1}}.$$

□

**Remark 6.69** (Ordinary summability is not enough). The condition  $\sum_k \eta_k < \infty$  does not by itself guarantee positive reserves when the factors  $\kappa_k$  can contract. The correct inherited-reserve condition is normalized by the accumulated degradation:

$$\sum_{k \geq 0} \frac{\eta_k}{K_{k+1}} < r_0.$$

If  $K_n \rightarrow 0$ , even a certified infinite run need not possess a positive uniform lower bound on  $r_n$ .

**Example 6.70** (Ordinary summability can coexist with reserve failure). Let  $r_0 = 1$ ,  $\kappa_k = \frac{1}{2}$ , and  $\eta_k = 2^{-k-2}$ . Then

$$\sum_{k \geq 0} \eta_k = \frac{1}{2} < r_0,$$

but  $K_{k+1} = 2^{-k-1}$ , so

$$\frac{\eta_k}{K_{k+1}} = \frac{1}{2}$$

and the normalized loss diverges. If equality holds in the Restart recurrence, then

$$\frac{r_n}{K_n} = 1 - \frac{n}{2},$$

so the reserve is nonpositive from a finite stage onward. Thus ordinary summability is not a substitute for normalized reserve-safety.

**Example 6.71** (Pointwise positive reserve need not be uniform). Let  $\eta_k = 0$ ,  $\kappa_k = \frac{1}{2}$ , and take equality in every Restart step. Then

$$r_n = 2^{-n}r_0 > 0$$

for every finite  $n$ , but  $\inf_n r_n = 0$ .

## 6.10 Countable Restart and summability

**Definition 6.72** ( $\kappa$ -normalized summability). A countable certified Restart run is  $\kappa$ -normalized summable if

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < \infty.$$

It is *reserve-safe* if, in addition,

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

**Definition 6.73** (Countable Restart specification and hidden-tail prevention). A countable Restart claim declares an enumeration of all states and transitions in scope and one of two certification levels:

- (i) *countably\_specified*: every indexed transition is quantified in the theorem statement and carries a reconstructible record;
- (ii) *finitely\_certified*: a finite witness and a Chapter 5 finite-to-infinite reduction theorem prove tail coverage, including exceptional indices and the failure boundary.

A finite checked prefix without the second branch is not hidden-tail prevention and cannot support an infinite verdict.

**Theorem 6.74** (Conditional countable Restart principle). *Let  $\{W_k\}_{k \geq 0}$  be an ordered right-open partition cover. Assume:*

- (i) *the countable Restart specification and hidden-tail-prevention record are complete;*
- (ii) *the initial local record has  $r_0 > 0$ ;*
- (iii) *every invoked transition has status *certified*;*
- (iv) *the run is reserve-safe in the sense of Definition 6.72;*
- (v) *every target window has a complete local record whose required non-reserve clauses pass, including representative, eligibility, and Type IV clauses when invoked;*
- (vi) *no transition or target record is *undefined*.*

*Then every inherited reserve  $r_n$  is positive, and the sequence of local records forms an auditable sequential certificate record on the partition cover.*

*Proof.* For every finite  $n$ , reserve-safety implies

$$L_n^{\text{norm}} \leq \sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Corollary 6.68 therefore gives  $r_n > 0$ . The declared continuation artifacts and complete target records make every transition reconstructible. Thus the ordered family is an auditable sequential certificate record.  $\square$

**Corollary 6.75** (Pointwise reserve and uniform reserve are distinct). *Under the hypotheses of Theorem 6.74, set*

$$L_{\infty}^{\text{norm}} := \sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}}, \quad c := r_0 - L_{\infty}^{\text{norm}} > 0.$$

*Then every finite stage satisfies*

$$r_n \geq K_n c > 0.$$

*Consequently, pointwise positivity of all  $r_n$  follows from reserve-safety. A positive uniform lower bound follows if  $\inf_n K_n > 0$ ; it does not follow merely from reserve-safety when  $K_n \rightarrow 0$ .*

*Proof.* Corollary 6.68 gives

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\text{norm}} \geq r_0 - L_{\infty}^{\text{norm}} = c.$$

Since every  $K_n > 0$ , the pointwise bound follows. Taking infima gives a uniform positive bound only when  $\inf_n K_n > 0$ .  $\square$

**Remark 6.76** (What the countable theorem does not say). Theorem 6.74 does not construct a single global filtered object or assert convergence of the local objects. It certifies the sequential inheritance of positive threshold reserve and the auditability of the record chain.

## 6.11 Detector-aware Restart handoff

**Definition 6.77** (Detector-aware Restart handoff). A Restart transition that consumes a repaired-zero predicate records, in addition to reserve transport, exactly one target-zero route:

- (i) a fresh complete target-zero certificate, supplied by a direct proof or a complete target-window detector;
- (ii) an exact zero-reflecting identification from a certified source profile to the target profile;
- (iii) a Chapter 2 metric-safe zero-transport record, including the post-threshold zero-separation condition;
- (iv) a theorem-controlled nonmetric record whose named conclusion contains the required source-zero to target-zero implication, with all hypotheses and variance fields verified.

The route identifies the source and target stages, threshold, endpoint convention, requested zero predicate, and immutable artifact. A directed map without zero reflection, or a  $\kappa_k$ -admissible reserve inequality alone, is not a zero-transport theorem.

**Theorem 6.78** (Restart zero-transport theorem). *Suppose the predecessor zero predicate is certified and a Restart datum is well-typed. The target zero predicate is certified only through one of the four routes in Definition 6.77:*

- (i) a fresh direct target-zero proof or a complete target detector with zero value;
- (ii) an actual exact zero-reflecting identification;
- (iii) a Chapter 2 metric zero-transport record with both threshold-gap and zero-separation hypotheses;
- (iv) a theorem-controlled nonmetric implication whose exact conclusion is target zero.

*A positive  $\kappa$ -reserve by itself is neither necessary nor sufficient for target zero; it is an independent safety clause unless a separately named theorem explicitly uses it inside one of these zero-transport routes.*

*Proof.* Each listed route contains an explicit implication to the target zero predicate. The reserve recurrence contains only real inequalities and no map or logical implication between source and target zero predicates. Hence it cannot establish target zero by itself.  $\square$

**Theorem 6.79** (Restart record-transport theorem). *Let the predecessor local record pass. If the Restart datum is complete, the continuation artifact is certified, every inherited field satisfies Proposition 6.59, every invalidated or fresh target field is re-proved, one target-zero route of Theorem 6.78 passes when zero is required, and all remaining target clauses pass, then the target local record passes.*

*Proof.* The transition theorem authorizes exactly the inherited fields. Fresh proofs supply every other target field, and the target-zero theorem supplies the zero predicate when invoked. The target verdict is the noncompensating conjunction of these clauses.  $\square$

**Definition 6.80** (Boundary-bridge coverage for a partition). For an ordered partition  $\{W_k\}$ , let  $\{V_k\}$  be declared bridge windows meeting both sides of the boundary between  $W_k$  and  $W_{k+1}$ . The augmented family

$$\mathcal{W}_{\text{aug}} := \{W_k\}_k \cup \{V_k\}_k$$

has *boundary-bridge coverage* at  $\tau$  when every globally retained bar admits a certified  $\tau$ -span witness in some partition or bridge window. Each bridge window carries its own direct evaluation and complete detector certificate.

**Corollary 6.81** (Detector-safe partition globalization). *Assume a global profile interface for the augmented partition/bridge family, boundary-bridge coverage, and zero complete detectors on every partition and bridge window. Then the global repaired profile is zero.*

*Proof.* The augmented family is a certified local detector atlas with global retained-bar coverage. Apply Theorem 6.55.  $\square$

**Remark 6.82** (Reserve transport does not see crossing bars). A positive Restart reserve controls threshold classification inside the profiles named by the continuation record. It does not prove that a long bar crossing a partition boundary is detected in either disjoint cell. Fresh target detection or a zero-transport theorem controls the target predicate; boundary-bridge coverage controls the global predicate. These are distinct obligations.

## 6.12 Countable overlap atlases

**Definition 6.83** (Audit-locally finite cover). A countable overlap cover  $\{W_n\}_{n \geq 1}$  is *audit-locally finite* if every bounded subinterval of the declared range meets only finitely many windows and only finitely many recorded nerve simplices.

**Definition 6.84** (Countable atlas enumeration and hidden-tail record). A countable atlas records a complete enumeration of windows, nonempty nerve simplices, face maps, and required local clauses, and states why no unindexed tail datum can affect the claimed verdict. It declares one verification level:

- (i) `countably_specified`: the infinite family is defined and all clauses are universally quantified, but no finite verifier certificate is claimed merely from enumeration;
- (ii) `finitely_certified`: finite evidence proves the entire countable family through either `uniform_schema`—a finite generator and proof that every generated record satisfies the required clauses—or `finite_to_infinite_reduction`—a Chapter 5 reduction with exceptional indices, tail coverage, and failure boundary.

For a global record-zero claim, hidden-tail prevention additionally requires a quantified retained-feature coverage theorem, a Chapter 5 tail reduction, or an independently proved cofinal/geometric tail theorem. Audit-local finiteness controls bounded queries only; it does not prove enumeration completeness, tail coverage, or finite verifiability. A checked finite prefix is neither verification level.

**Theorem 6.85** (Countable semantic record-coherence principle). *Let  $\{W_n\}_{n \geq 1}$  be a countable, audit-locally finite overlap cover of finite Čech depth. Suppose the countable atlas is completely specified, every local and simplex clause holds in its requested mode and strength, every face restriction is compatible, and all invoked diagnostic clauses pass. Then the family is a countably indexed Čech-coherent certificate atlas as a mathematical record system. No summability assumption is required for this coherence conclusion. The conclusion is not by itself a finite verifier pass.*

*Proof.* The universally quantified local and simplex records give compatibility on every nonempty nerve simplex, while finite Čech depth bounds the arity of each record and audit-local finiteness bounds every bounded query. Summability concerns aggregate metric exposure, not whether the face-compatible record system exists. Finite verification is a separate information-model claim.  $\square$

**Definition 6.86** (Summable simplex exposure). Let  $N_{\text{met}}$  be the countable set of recorded nerve simplices whose simplex coherence mode is `metric_reference`. For  $\sigma \in N_{\text{met}}$ , let

$$e_\sigma := \sum_{\alpha \in \sigma} \varepsilon_{\alpha, \sigma}$$

be the declared scalarized exposure of all local-to-reference comparisons consumed on that simplex. More generally, a manifest may use any nonnegative  $e_\sigma$  that explicitly dominates every metric comparison charged to  $\sigma$ , provided its aggregation rule is fixed in advance. Define the unordered nonnegative sum by

$$\sum_{\sigma \in N_{\text{met}}} e_\sigma := \sup_{S \subseteq N_{\text{met}}, S \text{ finite}} \sum_{\sigma \in S} e_\sigma.$$

The atlas has *summable simplex exposure* when this quantity is finite. This definition is independent of enumeration. Strict-cocycle simplices require exact proof artifacts and are not assigned fictitious metric exposure merely to enter this sum.

Summability does not replace the separate local-to-reference sharp inequalities

$$\varepsilon_{\alpha, \sigma} < \text{gap}_{\tau_\sigma}(\mathcal{B}_{i, \sigma}^\alpha, R_{i, \sigma})$$

for every consumed metric comparison, nor any non-scalar coherence obligation. Repeated use of the same underlying discrepancy is charged according to the declared no-double-counting policy rather than silently reused for free.

**Theorem 6.87** (Budgeted countable record-atlas principle). *Let  $\{W_n\}_{n \geq 1}$  be a countable, audit-locally finite overlap cover of finite Čech depth. Assume:*

- (i) *the countable atlas enumeration and hidden-tail record is complete;*
- (ii) *the hypotheses of Theorem 6.85 hold;*
- (iii) *every homotopy-coherent simplex is model-certified and every mode-specific output entails its requested conclusion;*
- (iv) *the metric-reference simplex exposures are summable;*



- (v) if a global detector-zero conclusion is invoked, the atlas carries a global profile interface, complete local detector certificates, zero-reflecting local interfaces, global retained-bar coverage, zero local detector values, and a complete tail-coverage record.

Then the atlas is semantically Čech-coherent and its total declared metric-reference exposure is finite. If the verification level is *finitely\_certified*, the finite verifier may return pass at record-coherence strength. If it is only *countably\_specified*, the conclusion remains a universally quantified semantic atlas and is not promoted to a finite audit pass merely by enumeration. The same distinction applies to global record zero. No global object conclusion follows.

*Proof.* Semantic coherence follows from Theorem 6.85. Summability records finite aggregate metric exposure but creates no maps or cells. In the *finitely\_certified* branch, the uniform-schema or Chapter 5 reduction artifact proves all infinitely indexed clauses from finite evidence, so a finite verifier can replay the verdict. In the *countably\_specified* branch, the hypotheses describe a valid infinite record system but provide no finite certificate for the infinite conjunction. The detector conclusion follows semantically from Theorem 6.55; its finite audit promotion requires the same finite tail certification.  $\square$

**Remark 6.88** (Summability is not coherence). Simplex-budget summability is an audit-exposure condition. It neither creates missing face maps nor proves higher compatibility. Conversely, a coherent atlas may fail the chosen global exposure policy if its scalarized simplex budgets are not summable. The two obligations are logically independent. Coherence is governed by Theorem 6.85; summability is additionally required only for the budgeted conclusion of Theorem 6.87.

**Remark 6.89** (No global pass from summability alone). A summable atlas is not automatically a single global B-Gate<sup>+</sup> pass. The conclusion is a coherent atlas of local and overlap certificates. A single global pass requires an independently declared global target (profile or object), a global window, and a global gate record at the claimed strength.

**Remark 6.90** (Canonical Type IV notation in Chapter 6). Whenever Chapter 6 cites aggregate Type IV diagnostics, the canonical pair is

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

Deprecated Latin-*u* diagnostic notation has no normative use. The coordinate *u* in a window  $[u, v)$  is unrelated and is retained as a domain endpoint.

### 6.13 Object-descent eligibility and Chapter 7 handoff

**Definition 6.91** (Object-level descent interface). An object-level descent claim requires a declared target category with restriction functors, actual local objects, actual transition equivalences, strict or homotopy-coherent cells, and a named effective-descent theorem. Metric-reference and predicate-only theorem records do not supply these data. This definition is a boundary declaration; Chapter 6 does not invoke the effectivity theorem.

**Definition 6.92** (Object-descent eligibility record). An object-descent eligibility record is the tuple

$$\text{DescentInput} = (\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, C, \mathfrak{M}_{\text{coh}}, \mathcal{T}_{\text{eff}}, \mathcal{A}),$$

where:

- (i)  $E_\alpha$  are actual local objects in a declared category;
- (ii)  $g_{\alpha\beta}$  are actual overlap equivalences with source, target, and variance;

- (iii)  $c_{\alpha\beta\gamma}$  are strict cocycle equalities or actual invertible two-cells;
- (iv)  $h_\sigma$  are all higher cells required by the target model;
- (v)  $\mathcal{W}$  is the cover with its nerve and face maps;
- (vi)  $C$  records object-level cover and restriction compatibility, not merely retained-bar coverage;
- (vii)  $\mathfrak{M}_{\text{coh}}$  fixes the enhancement, cell types, composition, boundary equations, and verifier used to interpret  $g, c, h$ ;
- (viii)  $\mathcal{T}_{\text{eff}}$  names the proposed effective-descent theorem and every hypothesis, without asserting that it has yet been applied;
- (ix)  $\mathcal{A}$  contains immutable artifacts and verifier statuses.

The record has status `ready` only when all fields required by the target category and coherence model are complete and mutually consistent. A metric-only or predicate-only atlas is not descent-ready unless separate actual maps and cells are supplied.

**Theorem 6.93** (Object-descent handoff theorem). *A complete and consistent object-descent eligibility record determines a well-typed `DescentInput` for Chapter 7. Its Chapter 6 conclusion is only*

`descent_input_ready`.

*It does not imply existence, uniqueness, zero, isomorphism, or any Return property of a global object.*

*Proof.* Completeness supplies each component of the displayed tuple with matching source, target, face, and artifact identities. Thus the input is well-typed. The record contains only a named effectivity target and its hypotheses; it does not apply the effective-descent theorem or construct its output.  $\square$

**Remark 6.94** (Record gluing and object descent are separate). Theorem 6.55 proves a predicate about a declared global profile. It neither constructs that profile from local data nor descends filtered or derived objects. Definition 6.91 records the additional route needed for an object-level theorem.

## 6.14 Integration with B-Gate<sup>+</sup> and Type IV

**Definition 6.95** (Local certificate status). A local window record has status

`pass`,     `reject`,     `undefined`

according to Definition 6.12. Direct Type IV and the inverse audit triple are evaluated only when their respective scopes are invoked. Undefined records cannot participate as successful vertices of an overlap atlas or as successful targets of Restart.

**Proposition 6.96** (Compatibility does not repair a local failure). *Passing an Overlap Gate does not convert a locally rejected or undefined record into a local pass. Likewise, a certified Restart transition does not discharge a missing local B-Gate<sup>+</sup>, representative, or Type IV requirement on the target window.*

*Proof.* Overlap and Restart records certify compatibility or continuation between local records. They do not alter the internal gate clauses of either endpoint record. Hence local validity remains a separate necessary condition.  $\square$

**Proposition 6.97** (Typed Type IV participation). *Whenever Type IV is in scope on a local window, overlap, or continuation target:*

- (i) *certified\_zero permits the record to proceed;*
- (ii) *obstructed forces rejection;*
- (iii) *undefined prevents certification and is routed as Type III operational undefinedness; it becomes Type V only if the missing state is concealed, serialized as zero or pass, or makes the manifest inconsistent;*
- (iv) *not\_invoked is admissible only when the governing policy explicitly places Type IV outside the scope of that record.*

*Proof.* This is the typed Type IV discipline of Chapter 5 applied to the local, overlap, and restart records defined in this chapter.  $\square$

**Proposition 6.98** (Inverse-audit participation). *If a local, overlap, Restart, or atlas claim invokes an inverse system, it carries the Chapter 5 triple*

$$\text{InvAudit} = (\text{Sys}, R^1\text{Lim}, \text{Apex})$$

*with independent statuses. Direct Type IV may simultaneously be not\_invoked. Vanishing of  $R^1\text{lim}$  does not discharge apex agreement, and apex agreement does not discharge the derived-limit field. No overlap or Restart budget transports either field without a separate theorem on the actual inverse systems and apex maps.*

*Proof.* This is the direct/inverse separation and derived-limit/apex independence proved in Chapter 5, applied componentwise to the present records.  $\square$

**Proposition 6.99** (Transient-defect participation and nontransport). *Suppose a local, overlap, simplex, Restart, or atlas claim requires absence of long transient kernel or cokernel defects. Then its Chapter 5 aggregate transient status must be certified\_absent on the same monitored scope. Status certified\_present forces failure of that stronger claim, and status undefined prevents certification.*

*An exact or metric comparison of the underlying source or repaired profiles does not transport this transient status. Such transport requires actual comparison data for the kernel and cokernel defect profiles and a separately proved defect-stability or exact-identification theorem.*

*Proof.* The first assertion is the Chapter 5 terminal/transient separation applied to the record types of this chapter. The second follows because kernel and cokernel are defined from the actual tower comparison morphism. Object-level or barcode-level closeness of its source and target does not determine the defect profiles without additional morphism-level comparison data.  $\square$

## 6.15 Failure routing

**Definition 6.100** (Chapter 6 failure routing). *Failures in this chapter are routed as follows; the types are not mutually exclusive.*

- (i) *an unstable threshold comparison or failed sharp gap condition is Type III;*
- (ii) *a false crop-commutation substitution, failed overlap comparison, failed exact cocycle, failed metric-reference record, failed detector coverage, or failed  $\kappa_k$ -inequality is Type III;*

- (iii) a nonzero terminal-generic tower defect is Type IV; a certified long-transient defect fails any claim that required transient cleanliness, while missing transient evidence remains Type III undefinedness;
- (iv) a missing required mathematical or operational artifact yields an `undefined` clause and is Type III at the operational level. It is additionally Type V only when the missing-artifact marker is omitted or concealed, an unsupported pass or zero is reported, or the manifest is incomplete or inconsistent;
- (v) a local topological obstruction remains Type I, and an eligible categorical obstruction remains Type II.

**Remark 6.101** (Why missing  $\kappa$  is not evidence for  $\kappa = 1$ ). Absence of a degradation certificate does not justify setting  $\kappa_k = 1$ ,  $\eta_k = 0$ , or declaring the transition harmless. It yields status `undefined` and therefore no certified Restart.

## 6.16 Minimal manifest consequences

**Proposition 6.102** (Window/overlap/Restart manifest extension). *An audit-readable manifest using this chapter extends the Chapter 1 base manifest and the Chapter 3 profile/representative extension by recording, at minimum:*

- (i) *the cover type: `partition` or `overlap`;*
- (ii) *every window and every nonempty recorded overlap;*
- (iii) *the direct common-window evaluation policy;*
- (iv) *the local and overlap thresholds, including threshold-reconciliation records;*
- (v) *the pre-threshold profiles or their immutable artifact identifiers;*
- (vi) *the clearance and gap values used by each comparison;*
- (vii) *the ledger quantale and bottleneck scalarization;*
- (viii) *the comparison mode of every consumed local, overlap, and simplex comparison, together with every exact proof artifact or scalarized metric budget as applicable;*
- (ix) *every scalarized local, overlap, crop, and continuation budget actually consumed by a metric-mode inference;*
- (x) *every pairwise and higher simplex compatibility record, including strict, homotopy-coherent, metric-reference, or theorem-controlled mode;*
- (xi) *every strict-cocycle morphism and cocycle check, or every metric-reference identity and local-to-reference certificate, as applicable;*
- (xii) *the Čech depth and audit-local-finiteness declaration, when applicable;*
- (xiii) *when detector globalization is invoked: the global profile identity, local restriction interfaces, detector stages, witness sets, completeness artifacts, local zero-transport branches, retained-bar coverage theorem, and local/global detector verdicts;*
- (xiv) *every Restart datum, inherited and invalidated evidence field, continuation artifact,  $\kappa_k$ ,  $\eta_k$ , reserve  $r_k$ , status, target-zero route, and boundary-bridge coverage record;*

- (xv) *all representative and eligible-realization records used by local clauses;*
- (xvi) *every typed direct Type IV scope status and, independently, every invoked inverse-audit triple and apex artifact;*
- (xvii) *whenever transient cleanliness is invoked, every full/interior defect component, detector record, and aggregate transient status;*
- (xviii) *any object-descent eligibility record, the complete DescentInput, and the proposed Chapter 7 effectivity target;*
- (xix) *all local, overlap, restart, detector-globalization, and atlas verdicts;*
- (xx) *the failure-routing record for every rejected or undefined clause.*

*Proof.* Each item is an input to, or a reconstructibility condition for, one of the definitions and theorems of this chapter. Omitting any invoked item prevents an independent verifier from reproducing the corresponding compatibility, restart, or gluing verdict.  $\square$

**Remark 6.103** (Canonical manifest location). Proposition 6.102 is an extension schema, not a competing canonical manifest. Chapter 7 integrates the Chapter 1 base fields, the Chapter 3 evaluation fields, the present window/overlap/Restart fields, and the remaining audit fields into the canonical v20 audit manifest with statuses complete, incomplete, and inconsistent.

## 6.17 Window, overlap, and Restart package

**Theorem 6.104** (Conditional window/overlap/Restart closure package). *Within the v20 conservative extension of the repaired Core, the following package is available.*

- (i) *Overlap profiles are evaluated directly on the common window before threshold deletion.*
- (ii) *Metric ledger data are compared with threshold clearance only after a declared bottleneck scalarization; the sharpness of the threshold band does not assert tightness of that scalarization.*
- (iii) *Overlap comparison uses the canonical four modes: exact, metric-gap-certified, theorem-controlled non-metric, and not-compared. Each returns only its authorized conclusion.*
- (iv) *Pairwise Overlap Gates do not replace higher Čech coherence.*
- (v) *Finite-depth overlap data produce a coherent certificate atlas only at the record level.*
- (vi) *Restart is conditional on a declared  $\kappa_k$ -admissible continuation artifact.*
- (vii) *Finite-horizon reserves obey the product-weighted inequality of Theorem 6.65.*
- (viii) *Countable Restart requires  $\kappa$ -normalized reserve-safety; ordinary summability alone is insufficient, and pointwise positive reserves do not automatically have a positive uniform infimum.*
- (ix) *Strict simplex coherence uses actual cocycle maps; homotopy coherence uses actual higher cells; metric-reference and theorem-controlled modes remain non-morphism record relations.*
- (x) *Local detector zeros globalize only through a declared global profile interface and retained-bar coverage theorem; exact overlap coherence alone does not suffice.*

- (xi) *Restart reserve transport does not transport detector zero and does not detect bars crossing a disjoint partition boundary without a target-zero route and boundary-bridge coverage.*
- (xii) *Countable overlap atlases additionally require audit-local finiteness, finite Čech depth, simplex coherence, exact proof artifacts on strict-cocycle simplices, and positive reserves plus summable exposure on metric-reference simplices.*
- (xiii) *Terminal Type IV cleanliness and long-transient cleanliness are separately typed and are not transported by source-profile comparison alone.*
- (xiv) *Local zero, global record zero, and global object zero are distinct conclusion types with no automatic promotion.*
- (xv) *A complete object-descent eligibility record yields only a typed DescentInput handoff to Chapter 7.*
- (xvi) *Undefined artifacts or statuses are never promoted to successful compatibility, Restart, detector, or gluing verdicts.*

*Proof.* Items (i)–(iii) follow from Definitions 6.13, 6.15, 6.23, 6.24, and Proposition 6.28. Items (iv)–(v) follow from Remark 6.31 and Theorem 6.47. Items (vi)–(viii) follow from Definitions 6.61, Theorem 6.65, and Theorem 6.74. The exact/metric coherence distinction follows from Definitions 6.35 and 6.39. Detector globalization and its no-go boundary are Theorem 6.55 and Example 6.57. Detector-aware Restart follows from Definition 6.77 and Corollary 6.81. The countable statement is Theorem 6.87. Terminal/transient participation follows from Propositions 6.97 and 6.99. The conclusion-strength separation follows from Definition 6.91. The final nonpromotion rule follows from the typed status disciplines of Definitions 6.64 and 6.95.  $\square$

**Remark 6.105** (Closure package, not an additional mechanism). Theorem 6.104 is a closure package collecting the proved transfer bounds and the explicitly conditional record theorems of this chapter. It does not turn operational definitions into a new unconditional local-to-global theorem, and it does not supply any missing non-scalar artifact.

## 6.18 Explicit non-claims

The following are not asserted in this chapter.

- (i) No unconditional Restart lemma is asserted.
- (ii) No universal or automatically existing  $\kappa > 0$  is asserted.
- (iii) No missing continuation artifact is interpreted as  $\kappa = 1$  or zero loss.
- (iv) No ordinary summability condition  $\sum_k \eta_k < \infty$  is claimed to preserve positive reserve under arbitrary contracting  $\kappa_k$ .
- (v) No pairwise Overlap Gate is claimed to imply higher Čech coherence.
- (vi) No exact overlap comparison is automatically a positive-radius robustness certificate, and no exact proof is forced into a fictitious metric budget merely for ledger summation.
- (vii) No comparison of differently thresholded outputs is made without threshold reconciliation and direct common-threshold re-evaluation.

- (viii) No crop-after-collapse comparison replaces direct common-window evaluation without a commutation certificate.
- (ix) No quantale-valued ledger is compared directly with a real threshold clearance without a declared scalarization.
- (x) No claim is made that the bottleneck scalarization is numerically tight merely because the two-sided threshold band is sharp.
- (xi) No metric reserve, large gap, or summability estimate repairs a missing representative, morphism, compatibility record, Type IV certificate, claim status, or manifest field.
- (xii) No pointwise conclusion  $r_n > 0$  for every finite  $n$  is promoted to a uniform bound  $\inf_n r_n > 0$  without an additional lower bound on  $K_n$ .
- (xiii) No coherent certificate atlas is automatically a global filtered object, global persistence module, or global B- $\text{Gate}^+$  pass.
- (xiv) No metric-reference simplex record is an exact cocycle, transition-map family, or descent datum.
- (xv) No family of local detector-zero verdicts, even with exact pairwise overlap coherence, implies a global repaired-zero predicate without a declared global profile interface and retained-bar coverage.
- (xvi) No Restart reserve inequality transports a detector-zero predicate or detects a bar crossing a partition boundary without a target-zero route and boundary-bridge coverage.
- (xvii) No summable simplex exposure creates detector completeness, face morphisms, higher coherence, or global retained-bar coverage.
- (xviii) No source-profile overlap or Restart comparison transports kernel or cokernel defect profiles without a separately proved defect comparison theorem.
- (xix) No direct terminal Type IV zero is identified with long-transient cleanliness, and no transient `certified_absent` verdict is promoted to full defect vanishing without the Chapter 5 short-defect exclusion hypotheses.
- (xx) No global profile is reconstructed merely from pairwise or simplex records; when a global profile is used by the detector theorem, it is declared input data.
- (xxi) No record-level gluing theorem is an effective object-level descent theorem.
- (xxii) No local zero family is a global record-zero theorem without retained-feature coverage.
- (xxiii) No global record-zero verdict is global object zero.
- (xxiv) No pairwise map, metric bound, or predicate relation supplies triple or higher coherence.
- (xxv) No homotopy-coherent label is accepted without actual cells and boundary identities.
- (xxvi) No theorem-controlled nonmetric comparison is promoted to a metric bound or morphism.
- (xxvii) No finite Restart prefix is a countable Restart theorem without hidden-tail prevention.
- (xxviii) No inherited Restart evidence survives a scope change without an authorizing transport theorem.

- (xxix) No direct Type IV status substitutes for the inverse-system audit triple, or conversely.
- (xxx) No complete DescentInput proves effectivity before Chapter 7 applies its theorem.
- (xxxi) No countable certificate atlas supplies a finite-to-infinite reduction or an external Return Theorem.
- (xxxii) No undefined Type IV, transient, Restart, detector, representative, or overlap status is treated as zero or pass.
- (xxxiii) No actual directed morphism passes an overlap clause unless its registered consequence entails the conclusion requested by that clause.
- (xxxiv) No generic phrase “homotopy coherent” is an affirmative certificate without a declared coherence model and verified boundary equations.
- (xxxv) No countably specified enumeration is promoted to a finite audit pass without a uniform-schema certificate or a Chapter 5 finite-to-infinite reduction.
- (xxxvi) No countable family of finite local detectors is represented as one finite detector merely by writing an infinite direct sum.
- (xxxvii) No summability hypothesis is treated as necessary for semantic record coherence; it is an additional aggregate-exposure policy.
- (xxxviii) No Search-side map, score, or AI proposal supplies a missing overlap, continuation, detector, coverage, descent, or gluing certificate.

## 6.19 Provenance and status

The sharp overlap-transfer step is an application of the threshold-gap-safe stability theorem of Chapter 2 and Appendix A. The representative and eligible-realization clauses are inherited from Chapters 3–4. The typed direct terminal Type IV, inverse-audit, and transient-defect clauses are inherited from Chapter 5. The v19 inheritance consists of direct common-window evaluation, exact/metric overlap modes,  $\kappa$ -admissible Restart, normalized reserve bounds, and finite/countable record-atlas logic. The v20 contribution is preserved in full. The v21 extension adds the canonical four-mode comparison calculus, strict versus homotopy-coherent simplex cells, explicit local/global-record/global-object zero types, Restart evidence invalidation and hidden-tail prevention, independent inverse-audit participation, and the typed DescentInput handoff to Chapter 7.

**Status labels for this chapter.** *Core and conditional mathematical results.* Propositions [6.28](#), [6.37](#), [6.40](#), [6.46](#), [6.96](#), [6.97](#), [6.99](#), and [6.102](#). Theorems [6.47](#), [6.55](#), [6.85](#), [6.65](#), [6.74](#), [6.87](#), and [6.104](#). Corollaries [6.54](#), [6.66](#), [6.68](#), [6.75](#), and [6.81](#).

*Counterexamples and sharpness witnesses.* Examples [6.57](#), [6.70](#), and [6.71](#).

*Core, interface, and operational record definitions.* Definitions [6.7](#), [6.8](#), [6.9](#), [6.13](#), [6.15](#), [6.16](#), [6.17](#), [6.20](#), [6.22](#), [6.23](#), [6.24](#), [6.25](#), [6.29](#), [6.33](#), [6.34](#), [6.35](#), [6.39](#), [6.42](#), [6.43](#), [6.50](#), [6.51](#), [6.52](#), [6.91](#), [6.60](#), [6.61](#), [6.64](#), [6.67](#), [6.72](#), [6.77](#), [6.80](#), [6.83](#), [6.86](#), [6.95](#), and [6.100](#).

*Operational cautions, nonpromotion rules, and integration remarks.* Remarks [6.1](#), [6.2](#), [6.10](#), [6.14](#), [6.18](#), [6.19](#), [6.21](#), [6.30](#), [6.31](#), [6.41](#), [6.49](#), [6.56](#), [6.94](#), [6.62](#), [6.63](#), [6.69](#), [6.76](#), [6.82](#), [6.88](#), [6.89](#), [6.90](#), [6.101](#), [6.103](#), and [6.105](#).



*New v21 certificate-atlas and handoff layer.* Definitions [6.3](#), [6.4](#), [6.5](#), [6.11](#), [6.12](#), [6.26](#), [6.36](#), [6.38](#), [6.44](#), [6.45](#), [6.92](#), [6.58](#), [6.73](#), and [6.84](#); Proposition [6.59](#) and [6.98](#); Theorems [6.6](#), [6.27](#), [6.32](#), [6.48](#), [6.53](#), [6.93](#), [6.78](#), and [6.79](#).

*Explicit exclusions.* Subsection [6.18](#).

## 7 Chapter 7: Finite Diagram and Globalization Core

**Chapter 7 contract (v21 Core-DG).** This chapter is the mathematical center of the v21 Finite Diagram and Globalization Core. It receives actual finite diagrams, finite-grid P2DG outputs, and Chapter 6 records with status `descent_input_ready`. It may promote them, under explicit finite hypotheses, to diagram-object zero, morphism isomorphism, exact object reconstruction, effective finite descent, and global-object zero.

For a small category  $P$  with finite object and morphism sets, the basic abelian category is

$$\text{Diag}_P(k) := \text{Fun}(P, \text{Vect}_{\text{fd}}(k)),$$

and the bounded derived category is

$$D_P^b(k) := D^b(\text{Diag}_P(k)).$$

The finite projective generator

$$G_P := \bigoplus_{p \in \text{Ob } P} k[\text{Hom}_P(p, -)]$$

provides a zero-reflecting enriched probe. Its full module structure, differential, and composition action are retained; scalar dimensions and ranks are not treated as complete reconstruction data.

The canonical object-descent lane is finite Alexandrov sheaf descent with a declared upper-set topology and covariant variance. Strict bounded-complex descent is proved degreewise. Derived-only or homotopy-coherent descent is consumed only through a named model-certified effectivity theorem. Record coherence, local zero, global record zero, and global object zero remain separate conclusion types.

The chapter does not assert that every finite category has finite global dimension, that  $G_P$  thickly generates all of  $D_P^b(k)$ , or that an arbitrary homotopy-coherent datum is effective. The generator theorem used here is the stronger-for-detection but logically different statement that  $\mathbf{R}\text{Hom}(G_P, E) \simeq 0$  reflects the zero object.

### 7.1 Scope, inputs, and Core-DG boundary

Fix a field  $k$  and a small category  $P$  with finite object set and finite morphism set. Unless explicitly stated otherwise, all diagrams take values in finite-dimensional  $k$ -vector spaces and all complexes are bounded. The chapter consumes:

- (i) the dual-Core and Core-Interface discipline of Chapter 1;
- (ii) finite linear P2DG outputs and actual morphism routes from Theorem 4.98;
- (iii) cone handoff records from Chapter 4;
- (iv) finite-to-infinite outputs only at their registered internal Return strength from Chapter 5;
- (v) a complete Chapter 6 object-descent handoff `DescentInput` from Theorem 6.93.

**Definition 7.1** (Core-DG admissible input). A *Core-DG admissible input* declares exactly one route:

- (i) `direct_diagram`: an actual object or morphism in  $\text{Diag}_P(k)$  or  $D_P^b(k)$ ;
- (ii) `p2dg`: a complete Core-P to Core-DG interface record whose output lies in the declared finite diagram category;

- (iii) `descent_input`: a complete `DescentInput` whose effectivity target is one of the descent lanes proved or separately admitted below.

The route records object and morphism variance, coefficient field, category presentation, relations, exactness profile, information profile, artifacts, and prohibited stronger readings.

**Remark 7.2** (Finite does not mean semisimple). The category algebra of a finite category is finite-dimensional, but it need not be semisimple and need not have finite global dimension. Consequently, the regular projective generator always detects zero, while equality  $\text{Perf}(\Lambda_P) = D^b(\Lambda_P\text{-mod}_{\text{fd}})$  requires an additional finite-global-dimension hypothesis.

## 7.2 Finite categories, finite posets, and orientation

**Definition 7.3** (Finite category presentation). A finite category presentation record contains:

- (i) the finite set  $\text{Ob } P$ ;
- (ii) the finite hom-sets  $\text{Hom}_P(p, q)$ ;
- (iii) identity morphisms and a complete composition table;
- (iv) every relation among generated morphisms;
- (v) the chosen orientation and, when relevant, the separately identified opposite category  $P^{\text{op}}$ .

A directed graph without its relations and composition law is not yet the category consumed by this chapter.

**Definition 7.4** (Finite-poset category). For a finite poset  $(P, \leq)$ , the associated category has one morphism  $p \rightarrow q$  exactly when  $p \leq q$ . The opposite category corresponds to the reversed order. A variance reversal changes the diagram category and is never treated as notation-only.

**Definition 7.5** (Upper Alexandrov convention). For a finite poset  $P$ , let  $X_P$  be the Alexandrov space whose open sets are upper sets. Its minimal open neighbourhood is

$$U_p := \{q \in P \mid p \leq q\}.$$

If  $p \leq q$ , then  $U_q \subseteq U_p$ , and sheaf restriction gives a map

$$\mathcal{E}(U_p) \longrightarrow \mathcal{E}(U_q).$$

Thus the declared sheaf model is covariant in  $P$ . Lower-set topology, cosheaves, or contravariant conventions require a separate record and are not silently identified with this lane.

## 7.3 The abelian finite-diagram category

**Definition 7.6** (Finite diagram category). Define

$$\text{Diag}_P(k) := \text{Fun}(P, \text{Vect}_{\text{fd}}(k)).$$

Objects are finite-dimensional covariant diagrams and morphisms are natural transformations.

**Theorem 7.7** (Abelian structure of finite diagrams). *The category  $\text{Diag}_P(k)$  is a  $k$ -linear abelian category. Kernels, cokernels, images, coimages, finite limits, and finite colimits are computed objectwise, and exactness is objectwise.*

*Proof.* The functor category into an abelian category is abelian. For a natural transformation  $f : E \rightarrow F$ , define  $(\ker f)(p) = \ker(f_p)$  and similarly for cokernels. Naturality of  $f$  induces the structure maps on these objects, and the universal properties are verified objectwise. The remaining abelian axioms and exactness follow from  $\text{Vect}_{\text{fd}}(k)$ .  $\square$

**Theorem 7.8** (Complexes of diagrams are diagrams of complexes). *There is a canonical isomorphism of categories*

$$\text{Ch}^b(\text{Diag}_P(k)) \cong \text{Fun}\left(P, \text{Ch}_{\text{fd}}^b(k)\right).$$

*Under this identification, cohomology objects, chain maps, chain homotopies, and mapping cones are computed objectwise.*

*Proof.* A bounded complex in a functor category is exactly a sequence of functors and natural differentials. Evaluating at each object of  $P$  gives a bounded complex, and naturality gives a functor into the complex category. The construction is reversible degree by degree. All listed operations are formed objectwise.  $\square$

**Definition 7.9** (Bounded derived finite-diagram category). Set

$$D_P^b(k) := D^b(\text{Diag}_P(k)).$$

A morphism in  $D_P^b(k)$  is represented through the usual localization of bounded complexes at quasi-isomorphisms. An actual chain map and its artifact are required whenever this chapter forms a concrete cone or performs strict descent.

## 7.4 Representable projectives and the finite generator

**Definition 7.10** (Linearized representable). For  $p \in P$ , define

$$G_p(q) := k[\text{Hom}_P(p, q)],$$

with structure maps induced by postcomposition. This is the linearized covariant representable  $k[P(p, -)]$ .

**Proposition 7.11** (Finite Yoneda evaluation). *For every  $E \in \text{Diag}_P(k)$ , there is a natural isomorphism*

$$\text{Hom}_{\text{Diag}_P(k)}(G_p, E) \xrightarrow{\sim} E(p).$$

*Consequently each  $G_p$  is projective.*

*Proof.* A natural transformation from  $G_p$  to  $E$  is determined by the image of the basis vector  $\text{id}_p$ , and every vector in  $E(p)$  determines such a transformation by functoriality. Evaluation at  $p$  is exact because exactness in the functor category is objectwise; hence  $\text{Hom}(G_p, -)$  is exact and  $G_p$  is projective.  $\square$

**Definition 7.12** (Finite projective generator). Define

$$G_P := \bigoplus_{p \in \text{Ob } P} G_p.$$

The direct sum is finite.

**Proposition 7.13** (Generator property). *The object  $G_P$  is a projective generator of  $\text{Diag}_P(k)$ . In particular,*

$$\text{Hom}(G_P, E) = 0 \iff E = 0.$$

*Proof.* Projectivity follows from the finite direct sum. By Proposition 7.11,

$$\mathrm{Hom}(G_P, E) \cong \bigoplus_{p \in P} E(p).$$

This vector space is zero exactly when every component of the diagram is zero.  $\square$

**Theorem 7.14** (DG-1: generator zero-reflection). *For every  $E \in D_P^b(k)$ ,*

$$E \simeq 0 \iff \mathbf{R}\mathrm{Hom}_{\mathrm{Diag}_P(k)}(G_P, E) \simeq 0.$$

*Because  $G_P$  is projective, the derived probe is represented by the ordinary Hom complex  $\mathrm{Hom}(G_P, E^\bullet)$ .*

*Proof.* For every integer  $r$ , projectivity gives

$$H^r \mathbf{R}\mathrm{Hom}(G_P, E) \cong \mathrm{Hom}(G_P, H^r(E)).$$

The generator property reflects zero of each cohomology diagram. Hence the probe is acyclic exactly when every  $H^r(E)$  vanishes, which is equivalent to  $E \simeq 0$  in the bounded derived category.  $\square$

**Remark 7.15** (Detection is not thick generation). Theorem 7.14 does not assert that every object of  $D_P^b(k)$  belongs to the thick subcategory generated by  $G_P$ . That stronger statement holds when the associated finite-dimensional algebra has finite global dimension; without it,  $G_P$  thickly generates the perfect subcategory but still detects zero in all of  $D_P^b(k)$ .

## 7.5 Cone detection and morphism isomorphism

**Theorem 7.16** (DG-2: cone-isomorphism criterion). *For a morphism  $f : E \rightarrow E'$  in  $D_P^b(k)$ ,*

$$f \text{ is an isomorphism in } D_P^b(k) \iff \mathrm{Cone}(f) \simeq 0.$$

*If an actual chain map represents  $f$ , its ordinary mapping cone is a representative of  $\mathrm{Cone}(f)$ .*

*Proof.* This is the defining exact-triangle criterion in a triangulated category:  $E \xrightarrow{f} E' \rightarrow \mathrm{Cone}(f) \rightarrow E[1]$ . The first map is an isomorphism exactly when the third object is zero.  $\square$

**Corollary 7.17** (Generator detection of morphism isomorphism). *For  $f : E \rightarrow E'$  in  $D_P^b(k)$ ,*

$$f \text{ is an isomorphism} \iff \mathbf{R}\mathrm{Hom}(G_P, \mathrm{Cone}(f)) \simeq 0.$$

*Equivalently,  $\mathbf{R}\mathrm{Hom}(G_P, f)$  is a quasi-isomorphism of enriched probe complexes.*

*Proof.* Combine Theorems 7.16 and 7.14; exactness of  $\mathbf{R}\mathrm{Hom}(G_P, -)$  on distinguished triangles gives the equivalent formulation.  $\square$

## 7.6 Finite Morita realization and enriched probes

**Definition 7.18** (Finite category algebra and handedness). Define

$$\Lambda_P := \text{End}_{\text{Diag}_P(k)}(G_P)^{\text{op}}.$$

The algebra is finite-dimensional, and  $\Lambda_P\text{-mod}_{\text{fd}}$  denotes finite-dimensional *left* modules. If  $a \in \text{End}(G_P)$ , write  $a^{\text{op}} \in \Lambda_P$ . The two module actions used below are fixed by

$$a^{\text{op}} \cdot f := f \circ a \quad (f \in \text{Hom}(G_P, E)),$$

and

$$x \cdot a^{\text{op}} := a(x) \quad (x \in G_P(q), q \in P).$$

Thus  $\text{Hom}(G_P, E)$  is a left  $\Lambda_P$ -module and  $G_P$  is a right  $\Lambda_P$ -object. For a left module  $M$ , the object  $G_P \otimes_{\Lambda_P} M$  is computed objectwise from the right modules  $G_P(q)$ . Reversing any one of these conventions changes the module side and is not a notation-only modification.

**Definition 7.19** (Enriched projective probe). For  $E \in \text{Diag}_P(k)$ , define

$$\text{Probe}_P^{\text{enr}}(E) := \text{Hom}(G_P, E) \in \Lambda_P\text{-mod}_{\text{fd}},$$

with the left action of Definition 7.18. For  $E \in D_P^b(k)$ , define

$$\text{Probe}_P^{\text{enr}}(E) := \mathbf{R}\text{Hom}(G_P, E) \in D^b(\Lambda_P\text{-mod}_{\text{fd}}).$$

The probe retains the full module action, differential, composition, kernels, cokernels, and extension data available in the module or derived-module category.

**Definition 7.20** (Scalar probe). A *scalar probe* retains only selected numbers such as

$$\dim E(p), \quad \text{rank } E(a), \quad \dim \text{Ext}^r(G_P, E).$$

Unless a separate completeness theorem is proved for a restricted family, such numbers are invariants only and are not object- or morphism-reconstruction data.

**Theorem 7.21** (DG-3: finite Morita reconstruction). *With the handedness fixed above, the functors*

$$\text{Hom}(G_P, -) : \text{Diag}_P(k) \longrightarrow \Lambda_P\text{-mod}_{\text{fd}}$$

and

$$G_P \otimes_{\Lambda_P} - : \Lambda_P\text{-mod}_{\text{fd}} \longrightarrow \text{Diag}_P(k)$$

*are mutually quasi-inverse exact equivalences. Consequently their termwise extensions induce an exact equivalence*

$$D_P^b(k) \xrightarrow{\sim} D^b(\Lambda_P\text{-mod}_{\text{fd}}),$$

*under which the forward functor is represented by  $\mathbf{R}\text{Hom}(G_P, -) = \text{Hom}(G_P, -)$ .*

*Proof.* The object  $G_P$  is a finite projective generator by Proposition 7.13; hence it is a progenerator. The standard Morita theorem, with  $\Lambda_P = \text{End}(G_P)^{\text{op}}$ , gives the displayed adjoint quasi-inverse pair with exactly the left/right actions fixed in Definition 7.18. Concretely, the unit is an isomorphism on finite free  $\Lambda_P$ -modules and the counit is an isomorphism on the summands  $G_P$ . Finite presentations and exactness extend these isomorphisms to every finite-dimensional module and diagram. An exact equivalence of abelian categories extends termwise to bounded complexes, preserves quasi-isomorphisms, and therefore induces the bounded derived equivalence.  $\square$

**Remark 7.22** (Morita handedness is part of the artifact). A reconstruction artifact records whether it uses left  $\Lambda_P$ -modules, right  $\text{End}(G_P)$ -modules, or an explicitly converted equivalent convention. A module action without this declaration is insufficient to reconstruct the arrow orientation of the source diagram.

**Theorem 7.23** (Exact enriched-probe reconstruction). *For every diagram  $E$  and morphism  $f : E \rightarrow E'$ , the enriched data*

$$\text{Probe}_P^{\text{enr}}(E), \quad \text{Probe}_P^{\text{enr}}(E'), \quad \text{Probe}_P^{\text{enr}}(f)$$

*reconstruct  $E, E'$ , and  $f$  up to unique isomorphism through the Morita quasi-inverse. The analogous statement holds in  $D_P^b(k)$  at derived isomorphism strength.*

*Proof.* Apply the quasi-inverse of Theorem 7.21 to the modules and module morphism, or to their bounded derived objects. Full faithfulness reconstructs the morphism, not merely its isomorphism class as an unlabelled linear map.  $\square$

**Example 7.24** (Execution example: reconstruction of a linear diagram). Let  $P = (0 \rightarrow 1)$ , and let

$$E = (k^2 \xrightarrow{A} k^3), \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The representables are

$$G_0 = (k \xrightarrow{\text{id}} k), \quad G_1 = (0 \rightarrow k).$$

The vector spaces  $\text{Hom}(G_0, E) \cong E(0)$  and  $\text{Hom}(G_1, E) \cong E(1)$ , together with the action of the algebra element corresponding to the arrow  $0 \rightarrow 1$ , recover the matrix  $A$ . Hence  $G_P \otimes_{\Lambda_P} \text{Probe}_P^{\text{enr}}(E) \cong E$ . Dropping the algebra action would retain only the dimensions  $(2, 3)$  and would not reconstruct  $A$ .

**Example 7.25** (Scalar ranks are not complete). Assume  $|k| \geq 3$ . Let  $P$  have two parallel arrows  $a, b : 0 \rightrightarrows 1$ . For  $\lambda \in k \setminus \{0, 1\}$ , consider

$$E = (k \xrightarrow{1} k), \quad E_\lambda = (k \xrightarrow{\lambda} k).$$

Both vertex dimensions are  $(1, 1)$ , and every arrow has rank one. An isomorphism would be given by nonzero scalars  $s_0, s_1$  satisfying  $s_1 = s_0$  from the first arrow and  $s_1 = \lambda s_0$  from the second, which is impossible. Thus scalar vertex and arrow-rank probes are not complete, while the enriched module action distinguishes the diagrams.

## 7.7 Finite Alexandrov realization

**Definition 7.26** (Finite-dimensional Alexandrov sheaf category). For the upper Alexandrov space  $X_P$ , let  $\text{Sh}_{\text{fd}}(X_P; k)$  be the category of sheaves of  $k$ -vector spaces whose stalks are finite-dimensional.

**Theorem 7.27** (DG-4: finite Alexandrov realization). *For a finite poset  $P$  with the convention of Definition 7.5, there is an exact equivalence*

$$\text{Sh}_{\text{fd}}(X_P; k) \xrightarrow{\sim} \text{Diag}_P(k).$$

*It sends a sheaf  $\mathcal{E}$  to the diagram  $p \mapsto \mathcal{E}(U_p)$ . A quasi-inverse sends a diagram  $E$  to the sheaf*

$$\tilde{E}(U) := \left\{ (s_p)_{p \in U} \in \prod_{p \in U} E(p) \mid E(p \leq q)s_p = s_q \text{ for } p \leq q \text{ in } U \right\}.$$

*Proof.* Every point has a minimal open neighbourhood. A sheaf is therefore determined by its values on the  $U_p$  and the restriction maps among them. The displayed compatible-family formula satisfies the sheaf axiom and recovers those values. The two constructions are mutually inverse and preserve kernels and cokernels objectwise.  $\square$

**Corollary 7.28** (Derived Alexandrov realization). *The exact equivalence induces*

$$D^b(\mathrm{Sh}_{\mathrm{fd}}(X_P; k)) \xrightarrow{\sim} D_P^b(k).$$

**Remark 7.29** (Variance boundary). Changing to lower Alexandrov opens, cosheaves, or contravariant diagrams changes the functor category. A source that does not record specialization order, sheaf/cosheaf convention, covariance, and restriction direction is not admissible for the Alexandrov equivalence.

## 7.8 Local systems and strict finite descent

**Definition 7.30** (Finite open-cover descent datum). Let  $X_P = \bigcup_{i=1}^N U_i$  be a finite cover by Alexandrov open sets. A strict sheaf descent datum consists of:

(i) local sheaves  $\mathcal{E}_i \in \mathrm{Sh}_{\mathrm{fd}}(U_i; k)$ ;

(ii) actual isomorphisms

$$g_{ij} : \mathcal{E}_j|_{U_{ij}} \xrightarrow{\sim} \mathcal{E}_i|_{U_{ij}};$$

(iii)  $g_{ii} = \mathrm{id}$ ,  $g_{ji} = g_{ij}^{-1}$ , and on every triple overlap

$$g_{ij} \circ g_{jk} = g_{ik};$$

(iv) immutable source, target, restriction, cocycle, and artifact data.

Pairwise existence without the cocycle is not a descent datum.

**Theorem 7.31** (DG-5: effective finite Alexandrov descent). *Every strict finite open-cover descent datum in Definition 7.30 is effective. There exists a sheaf  $\mathcal{E} \in \mathrm{Sh}_{\mathrm{fd}}(X_P; k)$ , unique up to unique isomorphism compatible with the descent data, together with isomorphisms*

$$\mathcal{E}|_{U_i} \xrightarrow{\sim} \mathcal{E}_i.$$

*For an open set  $U \subseteq X_P$ , one explicit construction is*

$$\mathcal{E}(U) = \left\{ (s_i) \in \prod_i \mathcal{E}_i(U \cap U_i) \left| s_i|_{U \cap U_{ij}} = g_{ij}(s_j|_{U \cap U_{ij}}) \right. \right\}.$$

*Proof.* The compatible-family assignment is a sheaf because the local sheaf gluing axioms hold component-wise and the cocycle makes the pairwise compatibility conditions consistent on triple overlaps. Projection to each local factor gives the required local isomorphism. Any two descended sheaves admit a unique isomorphism because morphisms of sheaves are determined on an open cover and the local identifications agree on overlaps.  $\square$



**Example 7.32** (Execution example: explicit finite descent). Let  $P = \{a, b, c\}$  with  $a < c$  and  $b < c$ . The upper Alexandrov opens

$$U_a = \{a, c\}, \quad U_b = \{b, c\}$$

cover  $X_P$  and intersect in  $\{c\}$ . On  $U_a$ , take the diagram  $k \xrightarrow{\text{id}} k$ . On  $U_b$ , take  $k^2 \xrightarrow{(1 \ 0)} k$ . Identify the two fibers over  $c$  by the identity. Theorem 7.31 constructs the global diagram

$$E(a) = k, \quad E(b) = k^2, \quad E(c) = k,$$

with structure maps  $\text{id}$  and  $(1 \ 0)$ . Restricting this diagram recovers both local inputs exactly.

**Definition 7.33** (Strict bounded-complex descent datum). A strict bounded-complex descent datum consists of bounded local complexes  $E_i^\bullet$ , actual chain isomorphisms on overlaps, and literal cocycle equalities compatible with differentials. All complexes share one finite cohomological amplitude bound.

**Theorem 7.34** (Strict bounded-complex descent). *Every strict bounded-complex descent datum on a finite Alexandrov open cover descends to a bounded global complex. Its terms are obtained by applying Theorem 7.31 degreewise, and its differential is the unique descended morphism induced by the local differentials.*

*Proof.* Degreewise effectivity gives global sheaves  $E^r$ . Compatibility of the local differentials with the overlap chain isomorphisms makes them morphisms of descent data, so they descend uniquely to  $d^r : E^r \rightarrow E^{r+1}$ . The identity  $d^{r+1}d^r = 0$  holds locally and therefore globally. The common finite amplitude gives boundedness.  $\square$

**Definition 7.35** (Model-certified derived descent lane). A derived-only or homotopy-coherent descent input is admissible only when the record declares:

- (i) a dg, model, or stable enhancement  $\mathfrak{M}_{\text{coh}}$ ;
- (ii) actual local objects, equivalences, higher cells, and boundary equations in that model;
- (iii) a named effectivity theorem valid for the declared cover and object class;
- (iv) a verifier and artifact constructing the descended object.

Perfectness being local does not by itself prove effectivity of an arbitrary derived descent datum.

**Theorem 7.36** (Chapter 6 to Chapter 7 descent application). *Let a Chapter 6 record have status `descent_input_ready`. If its `DescentInput` instantiates either the strict sheaf lane of Theorem 7.31, the strict complex lane of Theorem 7.34, or a complete model-certified lane of Definition 7.35, then the named effectivity theorem constructs a global object. Otherwise the descent conclusion remains `undefined`, not false and not effective.*

*Proof.* The Chapter 6 handoff supplies typed inputs but no effectivity. In each admitted lane, the corresponding theorem consumes those inputs and constructs the global object. Outside those lanes no theorem in this chapter has the required source type.  $\square$

## 7.9 Finite Čech totalization

**Definition 7.37** (Finite Čech computation datum). Let  $\mathcal{U} = \{U_i\}_{i=1}^N$  be a finite open cover, let  $E \in D^b(\mathrm{Sh}_{\mathrm{fd}}(X_P; k))$ , and choose a bounded complex  $E^\bullet$  representing  $E$ . A *finite Čech computation datum* supplies a bounded complex of sheaves  $A^\bullet$  and a quasi-isomorphism of complexes

$$E^\bullet \xrightarrow{\sim} A^\bullet,$$

and, for every nonempty finite intersection  $U_{i_0 \dots i_p}$  and every cochain degree  $q$ , a certificate that  $A^q|_{U_{i_0 \dots i_p}}$  is acyclic for the section functor. The direct lane takes  $A^\bullet = E^\bullet$  when the terms themselves satisfy this condition; the replacement lane records the bounded acyclic replacement and its artifact.

Define

$$C^{p,q}(\mathcal{U}, A) := \bigoplus_{i_0 < \dots < i_p} \Gamma(U_{i_0 \dots i_p}, A^q),$$

with the alternating Čech differential horizontally and the complex differential vertically. Its finite total complex is  $\mathrm{Tot} C^{\bullet, \bullet}(\mathcal{U}, A)$ .

**Theorem 7.38** (Finite acyclic-cover totalization). *For every finite Čech computation datum,*

$$\mathrm{Tot} C^{\bullet, \bullet}(\mathcal{U}, A) \xrightarrow{\sim} \mathbf{R}\Gamma(X_P, E)$$

in  $D^b(k\text{-mod})$ .

*Proof.* For each term  $A^q$ , the augmented Čech resolution is exact as a sheaf resolution, and the declared intersection-wise section acyclicity makes the resulting Čech complex compute derived global sections. Applying this termwise produces a bounded double complex whose totalization computes the hypercohomology of  $A^\bullet$ . The registered quasi-isomorphism  $E^\bullet \simeq A^\bullet$  identifies this hypercohomology with  $\mathbf{R}\Gamma(X_P, E)$ . Finiteness of the cover and boundedness of  $A^\bullet$  make the total complex bounded and finite in each degree.  $\square$

**Remark 7.39** (Cohomology-sheaf acyclicity is not silently substituted). Acyclicity of the cohomology sheaves  $H^r(E)$  is not, by itself, the same certificate as section acyclicity of the terms used in the displayed naive Čech double complex. A route based on cohomology-sheaf acyclicity requires its own hypercohomology spectral-sequence comparison or a certified acyclic replacement. The two hypotheses are not called equivalent by notation.

**Remark 7.40** (Totalization is not automatic descent). A Čech total complex computes a declared derived invariant under its computation datum. It does not replace the actual transition maps, cocycle, effectivity theorem, or descended-object artifact required for object descent.

## 7.10 Global zero and global isomorphism certificates

**Proposition 7.41** (Joint conservativity of finite Alexandrov open restriction). *Let  $P$  be a finite poset and let  $X_P = \bigcup_i U_i$  be a finite cover by upper Alexandrov opens. The family of exact restriction functors*

$$(-)|_{U_i} : D^b(\mathrm{Sh}_{\mathrm{fd}}(X_P; k)) \longrightarrow D^b(\mathrm{Sh}_{\mathrm{fd}}(U_i; k))$$

*is jointly conservative.*

*Proof.* If every restriction of  $E$  is zero, then every cohomology sheaf  $H^r(E)$  restricts to zero. Every point lies in some  $U_i$ , so every stalk of every  $H^r(E)$  is zero. Hence all cohomology sheaves vanish and  $E \simeq 0$ .  $\square$

**Theorem 7.42** (DG-6a: global zero certificate). *Let  $P$  be a finite poset, let  $X_P = \bigcup_i U_i$  be a finite upper-open cover, and let  $E \in D^b(\text{Sh}_{\text{fd}}(X_P; k))$ . The object may be given directly or constructed by an admitted descent lane whose output is in this same category. For each  $U_i$ , let  $G_i$  be the finite projective generator of the equivalent local diagram category. Then*

$$E \simeq 0 \iff \mathbf{RHom}(G_i, E|_{U_i}) \simeq 0 \text{ for every } i.$$

A local certificate may replace the full probe only when its own zero-reflection and source-stage theorem passes.

*Proof.* The forward implication is functorial. Conversely, local generator zero-reflection gives  $E|_{U_i} \simeq 0$  for all  $i$ . Apply joint conservativity from Proposition 7.41.  $\square$

**Theorem 7.43** (DG-6b: global isomorphism certificate). *In the same finite Alexandrov lane, let  $f : E \rightarrow E'$  be an actual global morphism, or the morphism obtained by effective descent of compatible local morphisms. Then*

$$f \text{ is an isomorphism} \iff \mathbf{RHom}(G_i, \text{Cone}(f|_{U_i})) \simeq 0 \text{ for every } i.$$

*Proof.* Restriction is exact and commutes with cones. Local probe zero implies that  $\text{Cone}(f)|_{U_i} \simeq 0$  for all  $i$ . Joint conservativity gives  $\text{Cone}(f) \simeq 0$ , and Theorem 7.16 gives the conclusion. The converse is immediate.  $\square$

**Remark 7.44** (Model-certified globalization requires conservativity). A model-certified effectivity theorem outside the finite Alexandrov lane does not automatically inherit Theorems 7.42 and 7.43. The record must separately supply exact restriction functors, their joint conservativity, local zero-reflecting probes, and compatibility of restriction with cones. Mere effectivity constructs a global object but does not prove that the selected local observations detect its zero object or its isomorphisms.

**Proposition 7.45** (No local-zero twisting in the admitted descent lane). *In a jointly conservative effective finite descent lane, actual local zero objects cannot descend to a nonzero global object. The familiar twisting phenomenon concerns local trivializations of a nonzero model, not local zero.*

*Proof.* A descended object whose restrictions are all zero is zero by the registered joint-conservativity theorem; in the canonical finite Alexandrov lane this is Proposition 7.41.  $\square$

## 7.11 Model-relative global certificate complexity

**Definition 7.46** (Family-relative global certificate cost model). Fix a global input family  $\mathcal{F}_{\text{global}}$ , a target predicate  $\Theta_{\text{global}}$ , and one Chapter 3 oracle-closed information, query, encoding, and verifier model  $\mathfrak{M} = (\mathcal{I}, \mathcal{Q}, \mathcal{E}, \mathcal{V})$ . A global certificate *scheme* must be correct uniformly on  $\mathcal{F}_{\text{global}}$ ; its cost is the Chapter 3 family-level worst-case vector

$$c_{\mathcal{F}_{\text{global}}}^{\mathfrak{M}}(\Pi) = (q, b, t, a) \in (\mathbb{N} \cup \{+\infty\})^4.$$

Local, overlap, coherence, effectivity, generator, cone, and replay schemes must use compatible access policies. Object-dependent information not present in  $\mathcal{I}$ , not obtained through charged queries, or supplied as uncharged advice is unavailable. Shared evidence, duplicate queries, content-addressed artifacts, and composition overhead are charged according to one fixed replayable merge policy.

**Theorem 7.47** (DG-7: model-relative global certificate upper bound). *Suppose a finite atlas and effective-descent route are certified uniformly on  $\mathcal{F}_{\text{global}}$  in the fixed model  $\mathfrak{M}$ . Let  $c_i$  be the family-level local scheme costs, and let  $c_{\text{ov}}$ ,  $c_{\text{coh}}$ ,  $c_{\text{eff}}$ ,  $c_{\text{gen}}$ , and  $h_{\text{merge}}$  be the separately encoded family-level overlap, coherence, effectivity, generator/cone, and merge-overhead costs. Then the composed global scheme  $\Pi_{\text{global}}$  satisfies*

$$c_{\mathcal{F}_{\text{global}}}^{\mathfrak{M}}(\Pi_{\text{global}}) \preceq \sum_i c_i + c_{\text{ov}} + c_{\text{coh}} + c_{\text{eff}} + c_{\text{gen}} + h_{\text{merge}}.$$

*If every term on the right is finite, the Pareto frontier*

$$\text{MinCert}_{\mathcal{F}_{\text{global}}}^{\mathfrak{M}}(\Theta_{\text{global}})$$

*contains an element dominated by that right-hand side. This is not a universal bound independent of  $\mathfrak{M}$ , the input family, the cover, or the encoding and reuse policy.*

*Proof.* For each input, execute the compatible local schemes, add the overlap and coherence verifiers, run the named effectivity construction, and apply the enriched generator or cone verifier. Oracle closure prevents hidden object-dependent information from entering the construction. Taking the worst case over the common input family and applying the fixed merge policy gives the displayed achievable vector. The Chapter 3 well-posedness theorem for the finite-cost Pareto frontier supplies a minimal achievable vector below every finite achievable vector, hence below the displayed bound.  $\square$

**Remark 7.48** (No scalar or per-object collapse of global cost). A smaller query count can require a larger bit encoding or artifact footprint. Moreover, a small certificate for one selected object is not the complexity of a scheme that works uniformly on the declared family. The theorem therefore returns a family-relative vector bound and a Pareto statement, not a canonical scalar  $\text{MinCert}_{\text{global}}$ .

## 7.12 Lower-bound and nonpromotion counterexamples

**Example 7.49** (Omitted generator). Let  $P$  be the discrete category on  $p, q$ . The nonzero diagram  $E(p) = 0, E(q) = k$  is invisible to the probe  $G_p$  alone. Hence every zero-reflecting generator family must cover all object components or provide a separate theorem showing that the omitted component is determined.

**Example 7.50** (Pairwise identifications without a cocycle). Assume  $\text{char } k \neq 2$  and three local copies of the one-dimensional object meet on a triple overlap. Set

$$g_{12} = \text{id}, \quad g_{23} = \text{id}, \quad g_{13} = -\text{id}.$$

Every pair is isomorphic, but  $g_{12}g_{23} \neq g_{13}$ . These data are not a descent datum and do not construct a global object.

**Example 7.51** (Variance reversal). For the poset  $0 < 1$ , a covariant diagram is a map  $E(0) \rightarrow E(1)$ . Replacing  $P$  by  $P^{\text{op}}$  changes it to a map  $E(1) \rightarrow E(0)$ . The diagrams  $k \rightarrow 0$  and  $0 \rightarrow k$  are not the same object under a notational reversal. A sheaf/cosheaf or specialization-order record is therefore mandatory.

**Remark 7.52** (Effectivity is an independent theorem). Even a perfectly typed family of local objects and higher cells does not imply effectivity in an arbitrary target. Effectivity follows in the strict finite Alexandrov sheaf and complex lanes proved above, or in a separately named model-certified lane. Missing effectivity is undefined, not a global object.

### 7.13 Core-P to Core-DG and descent handoff package

**Definition 7.53** (General finite-diagram handoff record). A general finite-diagram handoff contains:

- (i) the finite category presentation and variance;
- (ii) actual objects and, when used, actual morphisms;
- (iii) the P2DG or descent-input ancestry;
- (iv) enriched probe and reconstruction artifacts;
- (v) cone records for every requested isomorphism conclusion;
- (vi) the admitted descent lane and effectivity theorem when globalization is requested;
- (vii) the supported downward-closed Return-strength set and every prohibited stronger reading;
- (viii) immutable verifier and replay data.

**Theorem 7.54** (Finite Diagram and Globalization Core package). *Within the declared finite Core-DG scope, the following package is available.*

- (i) *finite diagrams form an abelian category with objectwise exactness;*
- (ii) *the finite representable sum  $G_P$  is a projective zero-reflecting generator;*
- (iii) *a morphism is an isomorphism exactly when its cone is zero;*
- (iv) *the enriched generator probe, with its declared left/right module convention, gives exact abelian and bounded-derived Morita reconstruction;*
- (v) *finite-poset diagrams are equivalent to sheaves on the declared upper Alexandrov space;*
- (vi) *strict finite sheaf and bounded-complex descent are effective;*
- (vii) *model-certified derived descent is admitted only at the exact strength of its named theorem;*
- (viii) *local generator/cone certificates yield global zero/isomorphism only after a global object or effective descent and joint conservativity;*
- (ix) *global certificate complexity admits only model-relative vector and Pareto bounds;*
- (x) *no external-domain theorem follows without the later Bridge and Return layers.*

*Proof.* Items (i)–(ix) are respectively Theorems 7.7, 7.14, 7.16, 7.21, 7.27, 7.31 and 7.34, Definition 7.35, Theorems 7.42 and 7.43, and Theorem 7.47. Item (x) is the Chapter 1 Core-Interface and Return boundary.  $\square$

### 7.14 Explicit Core-DG non-claims

The following are not asserted by the v21 Chapter 7 mathematical Core.

- (i) No infinite category, infinite poset, or uncontrolled locally finite category is included.
- (ii) No arbitrary constructible sheaf, hypersheaf, stack,  $(\infty, 1)$ -categorical descent, microlocal sheaf, or wild multiparameter persistence theorem is imported.
- (iii) No unbounded complex or unbounded derived category is included.
- (iv) No general-ring zero-reflection theorem is inferred from the field-coefficient statements without separate hypotheses.
- (v) No finite category algebra is assumed to have finite global dimension.
- (vi) No zero-detection theorem is relabeled as thick generation of all bounded derived objects.
- (vii) No scalar dimension or rank probe reconstructs an arbitrary diagram, morphism, extension class, or derived object.
- (viii) No pairwise isomorphism implies a Čech cocycle.
- (ix) No local trivialization of a nonzero model implies global triviality; conversely, actual local zero cannot twist into a nonzero global object in the jointly conservative lane.
- (x) No metric-reference or theorem-controlled record creates actual descent maps or cells.
- (xi) No `descent_input_ready` record is effective before a Chapter 7 effectivity theorem is applied.
- (xii) No perfectness-locality statement alone proves arbitrary derived descent effectivity.
- (xiii) No Čech totalization theorem is used without a bounded termwise-acyclic computation datum, a certified acyclic replacement, or an explicitly proved derived comparison theorem.
- (xiv) No local zero becomes global object zero before a global object and jointly conservative restrictions exist.
- (xv) No local cone zero constructs a global morphism; the morphism must already exist or descend effectively.
- (xvi) No universal scalar global certificate bound is asserted.
- (xvii) No Appendix or successor statement enters Core-DG without complete statement, proof, hypotheses, source binding, conservative-extension analysis, forbidden directions, artifacts, and Claim Register promotion.

### 7.15 Provenance and v21 status

The finite functor-category, Yoneda, projective-generator, cone, Morita, and strict sheaf-descent results are the mathematical Core-DG package. The Alexandrov lane is fixed to finite posets, upper opens, covariant diagrams, and finite-dimensional stalks. The model-certified derived lane is a typed interface and not an unconditional theorem. The global zero and isomorphism certificates consume actual effectivity and joint conservativity. The certificate-complexity theorem is model-relative.

**Core-DG theorems.** Theorems 7.7, 7.8, 7.14, 7.16, 7.21, 7.23, 7.27, 7.31, 7.34, 7.36, 7.38, 7.42, 7.43, 7.47, and 7.54.

**Core-DG definitions and interfaces.** Definitions 7.1, 7.3, 7.4, 7.5, 7.6, 7.9, 7.10, 7.12, 7.18, 7.19, 7.20, 7.26, 7.30, 7.33, 7.35, 7.37, 7.46, and 7.53.

**Counterexamples and boundaries.** Examples 7.24, 7.25, 7.32, 7.49, 7.50, and 7.51; Remarks 7.2, 7.15, 7.29, 7.40, 7.48, and 7.52.

## 7.16 Retained v20 typed-audit compatibility layer

The material below preserves every v20 Chapter 7 label and its historical audit meaning so that inherited references remain resolvable. In the v21 architecture, canonical audit ownership moves to Chapter 8. Therefore this block is consumed only at frozen v20 compatibility strength or through an explicit v21 supersession record. If a retained v20 token registry conflicts with the canonical v21 four-mode comparison, six-stage detector, inverse-audit, or descent semantics established in Chapters 1–7, the v21 statement governs new runs and the retained statement remains historical provenance.

**Retained v20 audit compatibility contract.** This retained block preserves the canonical audit layer of the v20.0 Core for historical label and theorem-strength compatibility. It preserves the v19 atomic status algebra

pass,      reject,      undefined,      not\_invoked,

and integrates the detector, higher-obstruction, transient-defect, finite-to-infinite, atlas, and descent records introduced in Chapters 1–6. No new independent verdict algebra is created.

Every token is assigned a declared semantic sort. Runtime comparison modes, simplex-coherence modes, profile stages, interface branches, cover types, source-semantics tokens, and mathematical values are not atomic statuses and may not be serialized or consumed as though they were.

A Core verdict is a dependency-closed typed conjunction of independently evidenced clauses. Exact proof is not confused with positive-radius robustness. Numerical slack may bound a declared metric discrepancy, but it may not create detector completeness, zero reflection, a representative, morphism, realization, terminal or transient defect certificate, overlap cocycle, global retained-bar coverage, finite-to-infinite reduction, descent theorem, Return Theorem, claim status, or manifest field. The statuses reject, undefined, and justified not\_invoked remain distinct.

### 7.16.1 Scope, dependency order, and relocation

This chapter closes the audit semantics of the v20 Core after the mathematical and record-theoretic constructions of Chapters 1–6. It consumes, but does not replace, the following previously established layers:

- (i) the claim discipline, comparison-mode distinction, and B-Gate<sup>+</sup> contract of Chapter 1;
- (ii) the standard bottleneck metric, hard threshold deletion, and sharp two-sided stability theorem of Chapter 2 and Appendix A;
- (iii) the repaired-evaluation, finite detector, profile-stage, completeness, metric zero-transport, and representative disciplines of Chapter 3;

- (iv) the degree-one and higher eligible realization records, canonical  $\text{Ext}''$ -edge, one-way discharge, and conditional zero-reflection upgrade of Chapter 4;
- (v) the typed failure taxonomy, source-semantics discipline, terminal-generic Type IV diagnostics, transient-defect calculus, and finite-to-infinite reduction records of Chapter 5;
- (vi) the typed ledger scalarization, comparison-mode Overlap Gate, exact-cocycle/metric-reference simplex calculus, detector globalization, object-descent boundary, Čech atlas, and  $\kappa$ -admissible Restart discipline of Chapter 6.

The purpose of the present chapter is to determine when those inputs yield an auditable Core verdict. It therefore specifies:

- (i) which deviations may enter a numerical ledger, which comparisons are exact or metric, and which proof obligations may not be scalarized;
- (ii) the typed meanings of `pass`, `reject`, `undefined`, and `not_invoked`;
- (iii) the aggregation rule for atomic clause statuses in finite and countable scopes;
- (iv) the audit reading of  $\text{B-Gate}^+$ , Type IV, Overlap, Restart, and atlas records;
- (v) manifest completeness, inconsistency, artifact closure, semantic replayability, schema migration, and repair traceability;
- (vi) detector, higher-edge, transient, global-zero, finite-to-infinite, descent, and release-acceptance audit readings;
- (vii) Core sovereignty and proposer–verifier separation.

**Remark 7.55** (Relocation of the former Chapter 7). The former chapter entitled “Core Theorem Package and Explicit Non-Claims” is not discarded. In the v19 and v20 dependency order it followed the historical audit chapter and was the former Chapter 8. Under the v21 renumbering it is relocated to Chapter 9. The canonical v21 verifier is Chapter 8, and the unified Bridge/Return closure follows only after that verifier kernel.

**Remark 7.56** (Normative rather than external-domain content). Most statements in this chapter are definitions, typing contracts, or consequences of those contracts. They govern admissible proof objects and verdicts; they are not new theorems about an external arithmetic, geometric, PDE, computational, or cryptographic domain.

### 7.16.2 Canonical token sorts and evidence graph

**Definition 7.57** (Canonical token-sort registry). Every token occurrence consumed by a v20 audit record is assigned exactly one declared semantic sort. The canonical sorts relevant to the Main body are:

- (i) *atomic audit status*:  

$$\text{pass}, \text{reject}, \text{undefined}, \text{not\_invoked};$$
- (ii) *claim evidence status* and *invocation role*, as in Definitions 7.77 and 7.78;
- (iii) *runtime profile-comparison mode*:  

$$\text{exact}, \text{metric\_gap\_certified}, \text{theorem\_controlled\_nonmetric}, \text{not\_compared};$$



(iv) *collapse policy*:

barcode\_level,      tau\_split,      exact\_admissible;

(v) *gap-certificate status*:

gap\_certified,   gap\_failed,   gap\_pending;

(vi) *detector profile stage*:

windowed\_prethreshold,   repaired\_postthreshold,  
kernel\_defect,      cokernel\_defect,  
diagram\_probe,      descent\_coherence;

(vii) *detector mode*:

birth,   germ,   certified\_stencil,   family\_mesh;

(viii) *detector certificate status* and independent value:

detector\_certified,   detector\_failed,   detector\_pending,  
zero,   nonzero,   undefined;

(ix) *terminal Type IV status*:

not\_invoked,   undefined,   certified\_zero,   obstructed;

(x) *full/interior inspection status* and *aggregate long-transient status*:

not\_checked,   certified\_absent,   certified\_present,  
not\_invoked,   undefined,   certified\_absent,   certified\_present;

The associated detector component is `full_profile` or `interior_noncofinal`.

(xi) *terminal mode* and *source semantics*:

bounded\_terminal,   right\_unbounded,  
genuine\_colimit,   verified\_stabilization,   implementation\_artifact;

(xii) *cover type* and *simplex-coherence mode*:

partition,   overlap,  
exact\_cocycle,   homotopy\_coherent\_cell,  
metric\_reference,   theorem\_controlled\_nonmetric,  
not\_compared;

(xiii) *derived interface branch*:

metric\_zero\_transport.

It abbreviates a successful `metric_gap_certified` comparison plus the separate repaired-zero separation hypotheses; it is neither a comparison mode nor a status.

(xiv) *Restart-transition status*:

`not_invoked, undefined, certified, failed;`

(xv) *manifest status*:

`complete, incomplete, inconsistent.`

Every serialized occurrence records both its field name and token sort. Equal character strings in different sorts do not identify the underlying semantic values.

**Proposition 7.58** (No cross-sort token substitution). *A token of one semantic sort cannot discharge a field of another sort. In particular, a simplex-coherence mode, cover type, profile stage, source semantics, interface branch, detector value, or manifest status is not an atomic audit verdict merely because its spelling resembles a verdict token. A cross-sort substitution makes the affected clause *undefined*; an intentional or concealed substitution is additionally a Type V inconsistency.*

*Proof.* The fields have different source types and governing predicates. A value of one type supplies no inhabitant or proof of another type. Concealment also contradicts the declared manifest semantics.  $\square$

**Definition 7.59** (Audit clause identity). An atomic audit clause is identified by

$c = (\text{ClauseID}, \text{kind}, \text{scope}, \text{predicate}, \text{dependencies}, \text{governing\_claim}).$

Its scope includes every degree, window, threshold, detector stage, defect threshold, cover/simplex, tower source semantics, and realization policy that changes the meaning of the predicate. Two clauses differing in one of these fields are not silently merged.

**Definition 7.60** (Typed evidence record). A typed evidence record for a clause contains an EvidenceID, evidence type, exact source and target, theorem or policy identifier, all hypotheses, profile stage or semantic branch, transformation history, immutable artifact identities, and one atomic status. Numerical output, a bare isomorphism symbol, a detector value without certificate status, or a citation without claim-use preflight is not a complete evidence record.

**Definition 7.61** (Well-founded audit dependency graph). The dependency graph of a run has clauses and evidence records as typed nodes and has a directed edge from each conclusion to every prerequisite it consumes. It is *audit-well-founded* when every dependency chain reaches registered primitive evidence or an explicitly assumed interface hypothesis and no clause is used, directly or indirectly, to certify its own prerequisite. For a countable graph, the manifest additionally supplies a well-founded rank or an equivalent independently checkable termination certificate.

**Proposition 7.62** (Determinacy on a complete well-founded graph). *Fix an immutable scope and canonical token-sort registry. If its clause registry is complete, its dependency graph is audit-well-founded, and every primitive record has a determined atomic status, then recursive evaluation by Definition 7.86 produces a unique headline verdict and preserves every lower-level status and failure route.*

*Proof.* Evaluate clauses in well-founded rank order. Each clause is then a fixed function of already evaluated prerequisites. The aggregate rule is deterministic, while the manifest separately retains all component statuses.  $\square$

**Definition 7.63** (Claim supersession and conditional override record). A v20 statement that narrows, repairs, or conditionally supersedes a v19 non-claim or weaker claim carries a new claim identity and records the predecessor statement, exact successor statement, added hypotheses, proof location, permitted stronger reading, still-forbidden readings, and immutable ancestry. The predecessor source and historical verdict are not mutated.

**Proposition 7.64** (No silent claim supersession). *Absent a complete supersession record and successful claim-use preflight, the weaker predecessor reading governs. In particular, a conditional zero-reflecting vanishing equivalence does not silently become an unconditional object-level  $\text{PH}_n \cong \text{Ext}^n$  claim.*

*Proof.* The stronger use has additional hypotheses and a distinct claim identity. Without their registered discharge, the weaker-safe interpretation applies.  $\square$

### 7.16.3 Audit identity and immutable run scope

**Definition 7.65** (Audit run identity). An *audit run identity* is a tuple

$$\mathfrak{R} = \left( \text{RunID}, \text{theory\_version}, \text{schema\_version}, \right. \\ \left. \text{claim\_register\_version}, \text{artifact\_root} \right).$$

It identifies the exact theory release, manifest schema, claim-register snapshot, and content-addressed artifact root under which all verdicts in the run are interpreted.

**Definition 7.66** (Immutable audit scope). The *immutable audit scope* of a run records, at minimum:

- (i) the object or object family under audit;
- (ii) the monitored degrees and, for Type IV totals, the finite set  $I_{\text{mon}}$ ;
- (iii) the windows, thresholds, terminal modes, and evaluation order;
- (iv) the complete registry of invoked clauses and explicitly non-invoked clauses;
- (v) the comparison mode of every consumed comparison, every simplex-coherence mode, cover type, derived interface branch, and token-sort declaration;
- (vi) every detector stage, mode, information class, certificate/value pair, defect threshold, global-profile interface, coverage family, finite-to-infinite reduction, and object-descent scope when invoked;
- (vii) the claim identifiers, invocation roles, theorem or policy locations, registered strengths, and every claim-supersession relation;
- (viii) the complete clause/evidence dependency graph and the immutable identifiers and content hashes of all source, profile, representative, comparison, detector, diagnostic, reduction, descent, and verdict artifacts.

A change to any of these data defines a new audit run rather than a silent mutation of the existing run.

**Remark 7.67** (Why immutable identity matters). A previously rejected or undefined run may be repaired, but the repaired object receives a new RunID and ManifestID, optionally linked to the former run by an explicit parent or repair relation. Later evidence does not rewrite the historical meaning of the earlier verdict.

#### 7.16.4 The typed $\delta$ -ledger

**Definition 7.68** (Declared ledger algebra). A run using numerical error budgets declares a commutative unital ordered monoid

$$(\mathbf{V}, \oplus, 0_{\mathbf{V}}, \preceq)$$

whose operation is monotone in each variable. If the implementation also uses arbitrary joins and  $\oplus$  distributes over those joins, the ledger algebra may be declared a commutative quantale. The word “quantale” is not used merely for a four-tuple lacking completeness and distributivity. Every component carries a declared semantic type, provenance, gate scope, and aggregation rule.

**Definition 7.69** (Metric-bearing ledger components). The *metric-bearing ledger* is the declared element

$$\begin{aligned} \mathbf{d}_{\text{met}} = & \delta^{\text{alg}} \oplus \delta^{\text{disc}} \oplus \delta^{\text{meas}} \oplus \delta^{\text{rep,met}} \oplus \delta^{\text{real,met}} \oplus \delta^{\text{cmp}} \\ & \oplus \delta^{\text{crop}} \oplus \delta^{\text{cont}} \oplus \delta^{\text{ov}} \oplus \delta^{\text{rst}}, \end{aligned}$$

with only the components actually consumed by the audited metric comparison included. The terms represent implementation, discretization, measurement, comparison, crop, continuation, overlap, and Restart discrepancies, together with metric errors caused by representative choice or realization. A representative-related or realization-related term is included only when an actual comparison theorem supplies a quantitative bound. Existence, typing, and compatibility remain non-numerical obligations.

**Definition 7.70** (Bottleneck scalarization). A bottleneck scalarization is a monotone subadditive map

$$\text{val}_{\mathbf{B}} : \mathbf{V} \longrightarrow [0, +\infty]$$

satisfying

$$\text{val}_{\mathbf{B}}(0_{\mathbf{V}}) = 0, \quad \text{val}_{\mathbf{B}}(q \oplus q') \leq \text{val}_{\mathbf{B}}(q) + \text{val}_{\mathbf{B}}(q').$$

The scalarized metric budget is

$$\varepsilon_{\mathbf{B}}(\mathbf{d}_{\text{met}}) := \text{val}_{\mathbf{B}}(\mathbf{d}_{\text{met}}).$$

A metric gate requires this value to be finite and to be proved to dominate the actual bottleneck discrepancy consumed by that gate. A scalarization value without its metric upper-bound theorem is only a recorded number, not a certificate.

**Remark 7.71** (Sharp threshold constant versus conservative scalarization). The adjective *sharp* refers to the optimal two-sided exclusion band  $[\tau - \varepsilon, \tau + \varepsilon]$  for a valid bottleneck upper bound  $\varepsilon$ . It does not assert that a chosen scalarization is numerically tight. A conservative scalarization preserves soundness but may reduce the set of passing runs.

**Definition 7.72** (Non-scalarizable proof obligations). The following are *non-scalarizable proof obligations*:

- (i) existence, typing, and equivalence strength of a required representative;
- (ii) eligibility, amplitude, edge-realization, and realization-equivalence certificates;
- (iii) existence and typing of an actual overlap, continuation, or tower comparison morphism;
- (iv) source semantics, terminal tameness, defect profiles, and interior-defect status;
- (v) threshold reconciliation, exact comparison, simplex coherence, continuation artifacts, and  $\kappa$ -admissibility certificates;

- (vi) detector stage applicability, soundness, completeness, birth/germ exhaustivity, stencil coverage, excess-bound source, endpoint convention, and family membership;
- (vii) naturality and zero reflection of a higher edge, representative and realization-change compatibility, and the non-synthetic classification of any claimed Bridge;
- (viii) local pass, target pass, retained-bar coverage, completeness of a countable registry, higher coherence, finite-to-infinite reduction, effective object descent, and Return-Theorem evidence;
- (ix) claim status, invocation role, supersession, repair records, and version boundaries;
- (x) manifest completeness, consistency, artifact closure, and identity obligations.

No increase in a numerical budget discharges an item in this list.

**Proposition 7.73** (No scalar repair of missing evidence). *Let a required non-scalarizable proof obligation be missing or ill-typed. Then no value of  $\mathbf{d}_{\text{met}}$ , no scalarized tolerance, and no positive threshold clearance can convert the corresponding clause into a certified pass.*

*Proof.* A metric ledger bounds a declared discrepancy. It does not construct the missing mathematical or governance object. Hence the obligation remains undefined, or a contradictory governance declaration remains rejected, independently of numerical slack.  $\square$

**Definition 7.74** (Gate-scoped threshold reserve). Let  $\mathfrak{B} = \{B_j\}_{j \in J}$  be the finite family of pre-threshold profiles consumed by one comparison in `metric_gap_certified` mode. For finite  $\varepsilon_B = \text{val}_B(\mathbf{d}_{\text{met}})$ , define

$$\text{gap}_\tau(\mathfrak{B}) := \min_{j \in J} \text{clear}_\tau(B_j)$$

and

$$\text{res}_\tau(\mathfrak{B}; \mathbf{d}_{\text{met}}) := \text{gap}_\tau(\mathfrak{B}) - \varepsilon_B.$$

The metric clause has positive reserve precisely when

$$\varepsilon_B < \text{gap}_\tau(\mathfrak{B}).$$

If the scalarized budget is infinite, the metric clause does not pass. An exact comparison uses its exact proof artifact and receives no positive-radius robustness certificate merely from exactness.

**Remark 7.75** (Meaning of  $\Sigma\delta < \text{gap}_\tau$ ). The legacy notation  $\Sigma\delta < \text{gap}_\tau$  denotes only the typed real inequality of Definition 7.74. An unscaled ledger element is never compared directly with real clearance. A budget belongs to its declared gate and profile family and is not silently spent again elsewhere.

**Definition 7.76** (Ledger completeness). A ledger is *complete and non-duplicative* for a metric clause if:

- (i) every metric perturbation affecting the clause is represented exactly once or by a documented joint bound;
- (ii) no perturbation is omitted merely because it is difficult to estimate;
- (iii) no component is counted twice under different names;
- (iv) every scalarized value is traceable to a proof or immutable artifact;
- (v) the aggregation law is the one declared by the run;
- (vi) the resulting real value is proved to bound the actual consumed discrepancy.

This definition is not invoked for a purely exact comparison unless a separate metric robustness claim is also made.

### 7.16.5 Claim-status and artifact preflight

**Definition 7.77** (Claim-use status). Every registered item consumed by a run has exactly one status:

proved, conditional, operational, spec,  
deprecated, rejected, unknown, superseded.

The statuses distinguish theorem-level evidence, conditional evidence, procedural rules, specifications, historical claims, unavailable claims, superseded claims, and unclassified items.

**Definition 7.78** (Claim invocation role). Every registered item consumed by a run is also assigned exactly one role:

theorem\_evidence, operational\_rule,  
specification\_target, provenance\_only.

The admissible combinations are:

- (i) `theorem_evidence`: status `proved`, or status `conditional` with every registered condition discharged;
- (ii) `operational_rule`: status `operational`, used only to govern a typed procedure;
- (iii) `specification_target`: status `spec`, used only as a schema, target, or acceptance criterion and never as proof of attainment;
- (iv) `provenance_only`: a `deprecated`, `rejected`, `unknown`, or `superseded` item cited only for history and contributing no passing theorem clause.

The role is fixed before verdict computation and may not be selected after seeing the desired result.

**Definition 7.79** (Audit preflight). The *audit preflight* passes exactly when:

- (i) the run identity and immutable scope are complete;
- (ii) every consumed registered item has both a status and an invocation role;
- (iii) every status–role combination satisfies Definition 7.78;
- (iv) no `operational`, `spec`, `deprecated`, `rejected`, or `unknown` item is promoted to theorem evidence;
- (v) every relevant v19-to-v20 or earlier repair and every conditional claim supersession has an immutable ancestry record;
- (vi) every token inhabits the field’s declared sort, every detector is applied at a theorem-authorized stage, and every derived interface branch cites its component theorems;
- (vii) the clause registry and dependency graph are complete and audit-well-founded for the invoked finite or countable scope;
- (viii) every required artifact is present, typed, versioned, content-addressed, and linked to the run;
- (ix) no identifier, hash, version, role, status, stage, branch, or token sort makes incompatible assertions about the same input.

**Definition 7.80** (Typed preflight verdict). The preflight has exactly one status:

`pass, reject, undefined.`

It is `pass` when Definition 7.79 is satisfied. It is `reject` when a well-typed record evidences forbidden theorem use, incompatible version or hash claims, an unauthorized scope exclusion, or another definite governance contradiction. It is `undefined` when a required identity, role, status, repair record, or artifact is missing, unresolved, or ill-typed but no evidenced contradiction yet forces rejection.

**Remark 7.81** (Claim registration is not proof). A Claim Register records permitted strength and use. It does not prove an item by listing it. A proved entry points to an actual theorem and proof location; a conditional entry records all conditions; operational and specification entries retain their non-theorem roles.

**Proposition 7.82** (Preflight failure blocks pass). *If the preflight status is not `pass`, the run cannot receive a Core pass verdict.*

*Proof.* A nonpassing preflight means that admissible theorem strength, scope, version, or artifact identity is either unresolved or contradicted. In neither case can every required Core clause be certified.  $\square$

#### 7.16.6 Typed atomic clause statuses

**Definition 7.83** (Atomic audit status). Every atomic clause in the declared audit scope has exactly one status:

`pass, reject, undefined, not_invoked.`

They mean:

- (i) `pass`: the clause is well-typed, fully evidenced, and true;
- (ii) `reject`: the clause is well-typed and evidenced but false or obstructed, or a governed consistency clause is contradicted by incompatible declared evidence;
- (iii) `undefined`: the clause is required but incomplete, ill-typed, unresolved, or not computable from the declared evidence, with no evidenced contradiction yet forcing rejection;
- (iv) `not_invoked`: the governing immutable scope explicitly excludes the clause.

**Remark 7.84** (Not invoked requires an explicit scope decision). A missing clause is not automatically `not_invoked`. The scope policy must positively declare that the clause is irrelevant to the claim; otherwise it is `undefined`.

**Definition 7.85** (Canonical specialization maps). Specialized records embed into the atomic status algebra without creating a new algebra.

- (i) Gap status maps by

`gap_certified`  $\mapsto$  `pass`,    `gap_failed`  $\mapsto$  `reject`,    `gap_pending`  $\mapsto$  `undefined`.

- (ii) Detector certificate status maps by

`detector_certified`  $\mapsto$  `pass`,  
`detector_failed`  $\mapsto$  `reject`,  
`detector_pending`  $\mapsto$  `undefined`.

The independent detector value is consumed only after certificate pass. For a required zero predicate, certified zero passes, certified nonzero rejects, and value undefined is undefined. The interpretation is reversed only when the registered target predicate explicitly requires nonzero. An uncertified zero-looking output never passes.

(iii) Type IV status maps by

$$\text{certified\_zero} \mapsto \text{pass}, \text{ obstructed} \mapsto \text{reject}, \text{ undefined} \mapsto \text{undefined}.$$

The token `not_invoked` remains available only under an explicit no-direct-Type IV-comparison scope decision.

(iv) For an invoked absence-of-long-transient clause, the map is

$$\begin{aligned} \text{certified\_absent} &\mapsto \text{pass}, \\ \text{certified\_present} &\mapsto \text{reject}, \\ \text{undefined, not\_checked} &\mapsto \text{undefined}. \end{aligned}$$

A justified out-of-scope record remains `not_invoked`. This field is independent of terminal Type IV status.

(v) Chapter 3 representative status maps by

$$\text{certified} \mapsto \text{pass}.$$

Missing or ill-typed representative data map to undefined; an explicit scope exclusion remains `not_invoked`.

(vi) Chapter 4 degreewise Ext-clause statuses map identically to the atomic statuses with the same names. A one-way discharge passes only the invoked Ext-vanishing clause; a zero-reflection equivalence additionally requires its naturality and faithfulness package.

(vii) Chapter 6 Restart-transition status maps by

$$\text{failed} \mapsto \text{reject}, \text{ undefined} \mapsto \text{undefined}.$$

Transition `certified` passes only the recurrence subclause, not the complete Restart step.

(viii) Manifest status maps by

$$\text{complete} \mapsto \text{pass}, \text{ incomplete} \mapsto \text{undefined}, \text{ inconsistent} \mapsto \text{reject}.$$

(ix) Runtime comparison modes, simplex-coherence modes, profile stages, cover types, source-semantics tokens, terminal modes, detector components, detector mathematical values, and `metric_zero_transport` are not statuses. Their own typed validation produces separate atomic clauses.

**Definition 7.86** (Aggregate audit verdict). Let  $C_{\text{all}}$  be the finite or countable audit-indexed registry of potential clauses in one immutable scope, and let  $C_{\text{req}} \subseteq C_{\text{all}}$  be the invoked clauses. Every clause outside  $C_{\text{req}}$  carries a justified `not_invoked` status. For a countable scope, the manifest certifies that the registry and its dependency closure are complete and audit-readable; a finite prefix cannot support a pass. The aggregate headline verdict is:



- (i) `reject` if at least one invoked clause is `reject`;
- (ii) `undefined` if no invoked clause is `reject` and at least one is `undefined`, or if a purported `not_invoked` clause lacks valid scope justification;
- (iii) `pass` if every invoked clause is `pass` and every excluded clause is justified.

The record retains all rejected, undefined, and non-invoked clause identifiers in addition to the headline.

**Proposition 7.87** (Determinacy of complete records). *If every required clause is well-typed and evidenced and the scope registry is complete, then the aggregate verdict is either `pass` or `reject`; it is not `undefined`.*

*Proof.* Every required clause then has a truth-valued status `pass` or `reject`; apply Definition 7.86.  $\square$

**Proposition 7.88** (Undefined is neither success nor counterexample). *An `undefined` verdict is not a Core pass and does not, by itself, constitute a mathematical counterexample to the intended claim.*

*Proof.* Undefinedness records missing, ill-typed, unresolved, or unavailable evidence. It establishes neither the clause nor the negation of a well-typed mathematical statement.  $\square$

**Definition 7.89** (Comparison-mode audit record). Every consumed profile comparison declares exactly one mode:

`exact`, `metric_gap_certified`, `theorem_controlled_nonmetric`, `not_compared`.

The modes mean:

- (i) `exact`: an actual equality or persistence-profile isomorphism, together with every exact crop, restriction, source, target, and proof artifact used by the inference, is present;
- (ii) `metric_gap_certified`: an evidenced pre-threshold bottleneck upper bound, a complete scalarized metric ledger, and a two-sided threshold-gap certificate at the same budget are present;
- (iii) `not_compared`: the governing claim consumes no comparison inference between the profiles.

Exact mode proves only the declared exact relation and does not itself give positive-radius robustness. The mode `not_compared` cannot discharge a comparison, Overlap, Restart, or coherence obligation that is actually invoked.

### 7.16.7 Detector, higher-obstruction, defect, and globalization verdicts

**Definition 7.90** (Detector-clause audit verdict). An invoked detector clause is `pass` exactly when the detector record has the correct source stage and scope, every soundness and completeness obligation passes, the certificate status is `detector_certified`, the value is defined, all transport branches pass, and the certified value satisfies the registered target predicate. It is `reject` when a well-typed required certificate condition is proved false or a certified value contradicts the target predicate. It is `undefined` when stage applicability, completeness, coverage, endpoint data, transport, value, or artifacts are missing, stale, circular, or ill-typed. It is `not_invoked` only by explicit scope exclusion.

**Definition 7.91** (Degreewise higher-obstruction audit family). For every invoked  $n \in I_{\text{mon}}$ , the run records separate clauses for the repaired degree- $n$  profile, admissible representative,  $n$ -eligibility, realized object and amplitude policy, edge extractor, edge-realization isomorphism, one-way Ext discharge, and any natural zero-reflection upgrade. The one-way conclusion

$$\text{Eval}_{n,w,\tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0$$

may pass without a converse. A vanishing equivalence passes only when the registered natural zero-reflection package passes. An encoded synthetic object is classified as supporting detector duality and cannot discharge a principal natural-Bridge clause.

**Definition 7.92** (Terminal and transient defect audit). An invoked tower record carries independent atomic clauses for source-semantic authority, actual comparison morphism, ambient abelian profile category, terminal tameness, terminal Type IV status, detector component, defect threshold  $\tau_{\text{def}}$ , and long-transient status. Terminal `certified_zero` does not discharge transient absence, and transient `certified_absent` does not prove full kernel/cokernel vanishing. A full-isomorphism clause additionally requires direct full-defect vanishing or a valid short-defect-exclusion upgrade.

**Definition 7.93** (Global detector-atlas verdict). A detector atlas has three separately audited conclusion levels:

- (i) *record coherence*: local vertices and every required simplex record pass at their declared modes;
- (ii) *global zero predicate*: a declared global profile interface, exact local restrictions, complete local detectors, exact or `metric_zero_transport` zero branches, and a global retained-bar coverage theorem pass;
- (iii) *object-level descent*: an external effective-descent theorem, actual transition isomorphisms and cocycle, and a descended-object artifact pass.

Passing a lower level does not impute either higher level.

**Definition 7.94** (Finite-to-infinite reduction verdict). An invoked finite-to-infinite clause passes only when the manifest contains the actual directed-system morphisms, declared colimit/source semantics, the registered reduction theorem, its finite witness, a compatible apex cocone, the induced comparison morphism, apex agreement, and dependency-closed artifacts. A finite prefix, sampled stabilization, or eventually constant appearance without the reduction theorem is undefined, not pass. A well-typed counterexample or failed registered hypothesis is reject.

**Definition 7.95** (Object-descent and Return-Theorem verdicts). Object descent and external Return are distinct interface clauses. Descent requires the category, restriction functors, actual local objects, exact transition data, effective-descent theorem, and descended-object artifact. Return requires a proved theorem from the exact AK-side conclusion to the stated external conclusion. Record coherence, global detector zero, finite-to-infinite reduction, and Core pass do not by themselves discharge either clause.

**Proposition 7.96** (Separation of audit conclusion levels). *None of the following implications is available without the named additional clause:*

$$\begin{aligned} \text{record coherence} &\implies \text{global zero}, \\ \text{global zero} &\implies \text{object descent}, \\ \text{Core pass} &\implies \text{external theorem}. \end{aligned}$$

*The required additional clauses are respectively retained-bar coverage, effective descent, and a Return Theorem.*

*Proof.* Chapters 6 and 1 explicitly type the three conclusions differently and supply counterexamples or non-claims to the omitted implications. Chapter 9 governs the external return boundary, using the Chapter 8 verifier kernel.  $\square$

### 7.16.8 Audit reading of B-Gate<sup>+</sup>

**Definition 7.97** (B-Gate<sup>+</sup> audit clause set). For a declared object, window, threshold, monitored degree family, and claim scope, the B-Gate<sup>+</sup> registry contains every applicable clause below.

- (i) **preflight and identity:** run identity, token sorts, claim use, dependency graph, and artifacts pass;
- (ii) **evaluation typing:** every degree uses the fixed window-first repaired evaluation;
- (iii) **degreewise topological clauses:** each invoked  $\text{Eval}_{n,W,\tau}(F) = 0$  is established directly or by an admitted complete detector;
- (iv) **degreewise categorical clauses:** each invoked higher edge has complete  $n$ -eligibility and the required  $\text{Ext}^n$ -vanishing; a converse or vanishing equivalence is consumed only through natural zero reflection;
- (v) **detector clauses:** stage, endpoint, soundness, completeness, coverage, certificate/value axes, metric transport, and artifact identity pass whenever detector evidence is consumed;
- (vi) **comparison clauses:** every runtime comparison has exactly one canonical mode; exact mode carries exact evidence, metric mode carries the complete two-sided chain, and `not_compared` is used only outside comparison scope;
- (vii) **metric budget clauses:** every metric ledger is complete, nonduplicative, scalarized, proved to dominate the actual discrepancy, and has the strict reserve required by its theorem;
- (viii) **terminal and transient clauses:** the terminal Type IV clause has a passing specialized record. When tower semantics are invoked, its status is `certified_zero`; otherwise `not_invoked` requires explicit scope exclusion. Every required long-transient clause is `certified_absent`, and any full-isomorphism claim carries direct full-defect or short-defect-exclusion evidence;
- (ix) **representative and realization clauses:** all actual profile isomorphisms, equivalence strengths, amplitude policies, edge certificates, and change-of-representative compatibilities pass;
- (x) **overlap and atlas clauses:** pairwise runtime comparisons pass, and every invoked simplex uses one declared coherence mode,

`exact_cocycle` or `metric_reference`.

Face-to-simplex compatibility, detector overlap certificates, and all countable-registry conditions also pass;

- (xi) **global detector clause:** when global zero is asserted, the global profile interface and retained-bar coverage theorem pass;
- (xii) **Restart clause:** every transition recurrence, target reserve, target local record, target-zero route, boundary-bridge coverage when required, and non-reserve obligation passes;
- (xiii) **finite-to-infinite, descent, and Return clauses:** each is included only when the governing claim uses that conclusion and must pass at its own type;
- (xiv) **manifest and failure routing:** the canonical manifest is complete, consistent, artifact-closed, semantically replayable, and every non-pass atomic clause has all applicable Type I–V routes.

**Remark 7.98** (One-way bridge discharge only). Under a complete eligible realization record, the degree-one theorem 4.66 and its registered higher-degree successors may discharge the corresponding one-way categorical clauses from repaired vanishing. They do not discharge detector completeness, metric, terminal or transient defect, overlap, global coverage, Restart, reduction, descent, Return, manifest, or claim-status clauses. A converse requires the separately audited natural zero-reflection package.

**Definition 7.99** (Core audit pass). A run receives a *Core audit pass* exactly when the aggregate verdict of its required B-Gate<sup>+</sup> clause set is pass.

**Definition 7.100** (Core audit reject). A run receives a *Core audit reject* exactly when the aggregate verdict is reject. The record identifies every evidenced failed clause and its failure route.

**Definition 7.101** (Core audit undefined). A run receives a *Core audit undefined* verdict exactly when no evidenced clause forces rejection but at least one required clause remains undefined. The record identifies the missing, ill-typed, unresolved, or unavailable evidence.

**Proposition 7.102** (Soundness of the pass reading). *A Core audit pass implies, for every invoked clause, the exact structural, metric, artifact, scope, and manifest conditions listed in Definition 7.97.*

*Proof.* Every required atomic clause has status pass; unpack the definitions. □

**Proposition 7.103** (No pass by cancellation). *A passing clause cannot cancel a rejected or undefined clause. In particular, large clearance, a zero Ext group, or a zero Type IV diagnostic cannot repair a missing representative, failed exact comparison, failed metric reserve, undefined Restart transition, or forbidden claim use.*

*Proof.* The aggregate verdict is a typed conjunction, not a signed numerical sum. □

### 7.16.9 Local, overlap, Restart, and atlas audit composition

**Definition 7.104** (Local certificate verdict). A local window certificate inherits pass, reject, or undefined from its local B-Gate<sup>+</sup> audit. Only a local pass may serve as a successful atlas vertex or Restart endpoint.

**Definition 7.105** (Compatibility verdict). Pairwise overlap comparison and higher-simplex coherence are audited at different token sorts.

- (i) A pairwise record passes in runtime mode `exact` or `metric_gap_certified` only when the directly evaluated common-window profiles, threshold reconciliation, applicable exact or metric evidence, detector overlap-scope certificate when consumed, face provenance, and terminal/transient obligations pass.
- (ii) A higher-simplex record passes in `exact_cocycle` mode only with actual profile isomorphisms, inverse/identity/cocycle equations, and face-to-simplex compatibility artifacts.
- (iii) It passes in `metric_reference` mode only with one immutable reference profile, all local-to-reference metric-gap certificates, face-to-simplex compatibility, and the declared simplex exposure. This mode produces no transition morphism or cocycle.
- (iv) `metric_zero_transport` may pass only as a derived branch after the runtime metric comparison and repaired-zero separation clauses pass.
- (v) A well-typed false exact relation, metric bound, reserve, cocycle, reference comparison, face condition, detector condition, or defect condition is `reject`; missing or ill-typed data are `undefined`; `not_invoked` requires explicit scope exclusion.

**Definition 7.106** (Restart verdict). A complete Restart step receives:

- (i) `pass` exactly when the Chapter 6 transition status is `certified`, the target reserve is positive, the target local record passes, and every non-reserve artifact is present;
- (ii) `reject` when the declared  $\kappa_k$ -inequality fails, the defined target reserve is nonpositive, or the target local record rejects;
- (iii) `undefined` when a continuation artifact, finite working reserve,  $\kappa_k$ ,  $\eta_k$ , scalarized budget, endpoint certificate, target record, or other required input is missing or ill-typed;
- (iv) `not_invoked` only when sequential inheritance is explicitly outside scope.

Transition status `certified` is necessary but not sufficient for a complete Restart pass.

**Proposition 7.107** (Atlas composition rule). *A finite or countable certificate atlas is auditable as record-coherent only if every local vertex passes, every required nerve simplex is present, each simplex declares `exact_cocycle` or `metric_reference`, all consumed pairwise runtime comparisons pass, face-to-simplex compatibility passes, the declared Čech depth and audit-local-finiteness conditions pass, metric-reference exposure summability passes when invoked, all terminal/transient and Restart clauses pass, and the manifest is complete and artifact-closed. A rejected item rejects the atlas; if none rejects but one is undefined, the atlas is undefined.*

*A global detector-zero verdict is added only if Definition 7.93(ii) passes. An object-level descent verdict is added only if item (iii) passes.*

*Proof.* Apply the aggregate verdict rule to the Chapter 6 atlas clause registry. □

**Remark 7.108** (Atlas pass is not a global-object pass). A coherent atlas certifies compatibility of local records. It does not create a single global filtered object, persistence module, or global B-Gate<sup>+</sup> pass. Object-level descent and global audit require separate theorems and records.

**Remark 7.109** (Pointwise reserve is not uniform reserve). Countable Restart may give  $r_n > 0$  at every finite stage while  $r_n \rightarrow 0$  when  $K_n = \prod_{j < n} \kappa_j \rightarrow 0$ . A claim  $\inf_n r_n > 0$  requires its own lower-bound hypothesis and artifact.

#### 7.16.10 Manifest completeness, consistency, and replayability

**Remark 7.110** (Canonical manifest integration). The earlier manifest definitions form an extension chain rather than competing schemas: Chapter 1 supplies the base; Chapters 3–6 add evaluation, representative, eligibility, Type IV, overlap, Restart, and atlas fields. Definition 7.112 is their canonical semantic superset for v20.0. A chapter-local fragment may add fields, but it may not change shared meanings or final status semantics.

**Definition 7.111** (Manifest status). A manifest has exactly one status:

`complete`,      `incomplete`,      `inconsistent`.

Here `complete` means every required field and artifact is present, typed, dependency-closed, and mutually consistent; `incomplete` means at least one required field or artifact is absent or unresolved; and `inconsistent` means declared fields, versions, hashes, statuses, roles, or artifacts make incompatible assertions about the same run.

**Definition 7.112** (Canonical minimal v20 audit manifest). The canonical minimal v20 audit manifest contains, at minimum:

- (i) ManifestID, RunID, theory, SchemaID, claim-register versions, immutable scope, repair and supersession ancestry;
- (ii) the canonical token-sort registry and every field's declared sort;
- (iii) source-object identities, content hashes, artifact root, complete finite or countable clause registry, evidence graph, dependency closure, and well-foundedness witness;
- (iv) monitored degrees, finite  $I_{\text{mon}}$ , windows, cover type, terminal modes, collapse and defect thresholds, evaluation order, hard deletion, infinite-bar and endpoint conventions;
- (v) every pre-threshold and repaired-profile identity and every detector source stage;
- (vi) every detector ID, theorem, information class, mode, sample/birth/germ set, endpoint convention, soundness, completeness, coverage/mesh/excess source, certificate size, two-axis status, value, and replay artifact;
- (vii) every runtime comparison mode; in exact mode the actual relation and crop/restriction artifacts; in metric mode the protected profile family, ledger algebra, scalarization, discrepancy theorem, two-sided gap record, zero-separation hypotheses when invoked, and strict reserve;
- (viii) representative, realization, degreewise  $n$ -eligibility, amplitude, edge extractor, edge-realization, naturality, zero-reflection, and records governing changes of representative and realization, together with the synthetic or principal Bridge classification;
- (ix) Type IV scope, tower system, source-semantics requirement and authority, ambient abelian category, actual comparison morphism, terminal tameness, degreewise and total  $\mu, \nu$ , and terminal status;
- (x) full/cofinal/interior defect profile identities, detector component,  $\tau_{\text{def}}$ , transient detector record, long-transient status, and any short-defect-exclusion or direct full-defect proof;
- (xi) cover, local, overlap, threshold-reconciliation, and simplex records, with coherence mode recorded separately as  

`exact_cocycle or metric_reference;`

the entry also contains the actual cocycle data or immutable reference profiles, face-to-simplex directions, Čech depth, audit-local finiteness, and exposure sums;
- (xii) when global detector zero is invoked: global profile identity, exact local restrictions, independent global coverage witnesses, local detector samples, zero-transport branches, retained-bar coverage theorem, and local/global verdicts;
- (xiii) every continuation artifact, working gap,  $\kappa_k, \eta_k, K_n$ , normalized loss, pointwise and claimed uniform reserves, target-zero route, bridge-window coverage, transition and complete Restart verdicts;
- (xiv) every invoked finite-to-infinite reduction theorem, actual directed system, finite witness, colimit, compatible apex cocone, induced comparison, apex agreement, and reduction verdict;
- (xv) every invoked object-descent category, restriction functor, actual local objects and transition isomorphisms, effective-descent theorem, descended object, and every external Return-Theorem statement and proof;

- (xvi) all atomic statuses, specialized-to-atomic maps, headline verdicts, Type I–V routes, missing-artifact markers, and justified scope exclusions;
- (xvii) every ClaimID, document role, evidence status, invocation role, theorem/policy locator, discharged condition, permitted use, forbidden stronger reading, supersession and repair record;
- (xviii) release-acceptance clause registry when the document set is advertised as a complete v20 release;
- (xix) provenance and semantic-replay data sufficient to reconstruct every deterministic computation and every logical verdict transition.

**Definition 7.113** (Artifact closure). A canonical manifest has *artifact closure* when:

- (i) every referenced artifact has a content hash, type declaration, provenance, and immutable identifier;
- (ii) every transitive dependency needed to interpret or verify a consumed artifact is referenced or belongs to the registered theory release;
- (iii) no identifier is dangling, ambiguous, mutable under the same hash claim, or resolved differently by two fragments;
- (iv) the dependency graph contains no undeclared artifact replacement after the run identity was fixed.

Artifact closure is required for replayability but is not itself mathematical proof.

**Definition 7.114** (Semantic schema-migration record). A migration from one manifest schema to another records source and target SchemaID, a total map on every consumed field, token-sort preservation, status and scope preservation, explicit defaults only for newly out-of-scope fields, semantic-diff evidence, artifact hashes, and a proof that verdict reconstruction at the old supported strength is preserved. Changed theorem strength, scope, or evidence creates a new run rather than a schema-only migration.

**Proposition 7.115** (No lossy migration of theorem evidence). *A migration that drops a required hypothesis, dependency, token sort, missing-artifact marker, lower-level status, or forbidden stronger reading cannot preserve audit equivalence. The migrated record is *incomplete* or *inconsistent* until the loss is repaired by a new evidence-backed run.*

*Proof.* The omitted field participates in clause meaning or verdict reconstruction. Deleting it prevents reconstruction or changes the consumed claim. □

**Proposition 7.116** (Canonical manifest conformance). *A chapter-local manifest fragment is audit-admissible only if it embeds into the canonical manifest without changing the meaning of a shared field, status, role, identifier, comparison mode, or version boundary. Incompatible assignments make the canonical manifest *inconsistent*.*

*Proof.* The canonical manifest describes one immutable run; incompatible shared assignments violate that condition. □

**Definition 7.117** (Computational and semantic replayability). A manifest is *computationally replayable* if artifact closure lets an independent verifier reconstruct every deterministic profile, detector, ledger, diagnostic, dependency, and serialization computation. It is *semantically replayable* if the verifier can additionally reconstruct each clause’s source type, governing claim, hypotheses, token sort, specialization map, dependency transition, and headline aggregation. Core pass requires semantic replayability for every consumed machine-produced verdict.

**Proposition 7.118** (Replay invariance under fixed inputs). *Two semantic replays using the same immutable artifacts, token registry, claim-register snapshot, schema semantics, and verifier version yield the same atomic and headline verdicts. A disagreement evidences an implementation, schema, artifact, or manifest inconsistency and blocks pass.*

*Proof.* Under fixed inputs every clause and aggregation operation is deterministic by Proposition 7.62.  $\square$

**Remark 7.119** (Replayability is not mathematical proof by itself). Replayability establishes reproducibility of the declared computation and record graph. It does not validate a false theorem, insufficient hypothesis, or semantically inadequate artifact.

**Proposition 7.120** (Manifest effect on the headline verdict). *A manifest with status `incomplete` makes the manifest clause `undefined` unless another evidenced clause already forces rejection. A manifest with status `inconsistent` forces rejection as a Type V governance failure. Only a `complete`, artifact-closed manifest can support a Core pass.*

*Proof.* Incomplete data leave required evidence unresolved; inconsistent data contradict the single-run semantics.  $\square$

#### 7.16.11 Failure routing and preservation of diagnostic detail

**Definition 7.121** (Audit failure-routing record). Every non-pass run records all applicable, nonexclusive failure types:

- (i) **Type I:** nonzero repaired topological profile;
- (ii) **Type II:** nonzero eligible categorical obstruction;
- (iii) **Type III:** any failed or undefined mathematical or operational obligation, including metric or crop control; detector stage, applicability, soundness, or completeness; overlap or simplex coherence; global coverage; exact comparison; continuation or Restart; representative or realization data; comparison morphisms or terminal tameness; transient defects; finite-to-infinite reduction; object descent; or a Return theorem;
- (iv) **Type IV:** a valid, semantically authorized tower comparison artifact whose terminal-generic diagnostic satisfies  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$ ;
- (v) **Type V:** forbidden theorem use, incomplete or inconsistent canonical manifest, missing repair/governance record, suppressed missing-artifact marker, unsupported zero or pass declaration, misrepresented scope/version/hash/provenance, cross-sort token substitution, silent stage relabeling, lossy schema migration, silent claim supersession, unauthorized scope exclusion, or proposer output treated as proof.

Mere absence or ill-typing of mathematical evidence first yields an `undefined` Type III clause. Type V is added when the governance record is itself incomplete, inconsistent, concealed, or semantically false.

**Remark 7.122** (Headline compression does not erase details). The precedence of reject over undefined is only a headline rule. If one clause rejects and another is undefined, the manifest retains both states and both failure routes.

**Proposition 7.123** (Repair creates a new verdict record). *Resolving an undefined clause or correcting a rejected clause produces a new run and manifest identity, optionally linked to the former run by an explicit repair relation. The original immutable RunID and historical verdict are not reused or overwritten.*

*Proof.* This is the immutable-scope rule of Definition 7.66.  $\square$



### 7.16.12 Core sovereignty and proposer–verifier separation

**Definition 7.124** (Core sovereignty). Only Core-typed evidence evaluated under the registered verifier discipline may determine a Core verdict. No external proposal mechanism may overwrite, waive, or reinterpret an atomic Core clause. A human maintainer may revise theory, schema, or scope only by creating a new version or run identity.

**Definition 7.125** (Proposer and verifier). A *proposer* is any human, algorithmic, Search-side, Platform-side, Companion-side, or AI mechanism that produces candidate objects, windows, thresholds, representatives, morphisms, claims, bridge programs, rankings, maps, or proof sketches. The *verifier* is the typed Core audit discipline of this chapter and the mathematical chapters it consumes.

**Definition 7.126** (Re-entry protocol). A proposer output may influence a Core verdict only after:

- (i) normalization into a declared Core type;
- (ii) claim-status and invocation-role classification;
- (iii) production of required mathematical and artifact evidence;
- (iv) comparison-mode, ledger, and threshold-gap typing where applicable;
- (v) when the output is transformed, translated, summarized, formalized, or migrated, a semantic-diff or transformation-preservation certificate;
- (vi) manifest registration, token-sort validation, dependency closure, artifact closure, and immutable identity;
- (vii) execution of every relevant verifier clause.

**Proposition 7.127** (Search and AI outputs do not update verdicts). *A Search score, potential, terrain map, ranking, validity-map value, AI confidence, proof sketch, or agent recommendation does not alter a Core verdict merely by being produced.*

*Proof.* Such an output is proposer-side until it completes the re-entry protocol and passes the verifier.  $\square$

**Proposition 7.128** (Verifier independence). *Replacing the proposer mechanism does not change the mathematical meaning of a fixed verifier verdict computed from the same immutable Core inputs and registered theory versions.*

*Proof.* The verdict depends on registered inputs and clause semantics, not on the identity of the proposer.  $\square$

### 7.16.13 v20 release-acceptance audit

**Definition 7.129** (v20 release-acceptance record). A document set advertised as an *AK-HPDST v20.0 complete release* carries a release-level registry whose required clauses are exactly the mandatory Core mathematics, verification/governance, and external-demonstration obligations of Chapter 1 Policy 18. Each obligation has a claim identity, exact source strength, dependencies, artifacts, and one atomic status. Release status is distinct from the verdict of any individual Core run.

**Definition 7.130** (Release headline status). The release headline is *pass* only when every mandatory obligation passes and all cross-document and machine-readable records are mutually consistent. It is *reject* if a well-typed mandatory claim or consistency requirement is evidenced false. It is *undefined* when no mandatory clause rejects but at least one remains missing, pending, unregistered, or ill-typed. Optional recommendations do not affect the headline unless the release policy explicitly promotes them.

**Proposition 7.131** (No release completion by local success). *No individual detector theorem, higher-edge theorem, transient-defect theorem, Core pass, toy bridge, known-theorem calibration, or favorable external example can compensate for another missing mandatory release obligation.*

*Proof.* The release registry is a typed conjunction of independently mandatory clauses, governed by the same non-compensation rule as a run verdict.  $\square$

#### 7.16.14 Audit package

**Proposition 7.132** (Typed audit-and-sovereignty contract consequences). *The v20 Core audit discipline has the following consequences:*

- (i) *metric discrepancies are controlled by a declared ledger, scalarization, and actual upper-bound proof;*
- (ii) *non-scalar obligations cannot be bought off by numerical slack;*
- (iii) *metric reserve is gate-scoped and pre-threshold only in `metric_gap_certified` mode, while exact mode is governed by its exact proof artifact;*
- (iv) *every atomic clause has one typed status;*
- (v) *pass requires every invoked clause to pass and every exclusion to be justified;*
- (vi) *false evidence yields rejection, while unresolved evidence yields undefinedness unless another clause rejects;*
- (vii) *detector, gap, Type IV, transient, and complete Restart records embed into the same atomic status algebra through typed specialization;*
- (viii) *local, compatibility, global-zero, Restart, reduction, descent, Return, release, and external verdicts remain distinct;*
- (ix) *only a complete, consistent, artifact-closed and semantically replayable manifest can support pass;*
- (x) *failure routing preserves Types I–V and all clause-level detail;*
- (xi) *proposer outputs remain non-authoritative until typed re-entry;*
- (xii) *the theorem-bearing verifier remains sovereign.*

*Proof.* Items (i)–(iii) follow from Definitions 7.69, 7.70, 7.72, 7.74, and 7.89. Items (iv)–(vi) follow from Definitions 7.83 and 7.86. Items (vii)–(viii) follow from Definitions 7.85, 7.105, and 7.106, together with Proposition 7.107. Items (ix)–(x) follow from Definitions 7.111, 7.113, and 7.121. Items (xi)–(xii) follow from Definitions 7.124, 7.125, and 7.126.  $\square$

**Remark 7.133** (Contract consequence list, not a new external theorem). Proposition 7.132 is a closure list for the typed audit contracts of this chapter. It is not a new theorem about persistence modules, external interfaces, or a problem-specific domain.

### 7.16.15 Explicit non-claims

The following are not asserted in this chapter.

- (i) No unscaled ledger element is directly compared with  $\text{clear}_\tau$ .
- (ii) No scaled budget repairs a missing representative, morphism, realization, claim status, or manifest field.
- (iii) No equality  $\varepsilon_B = \text{gap}_\tau$  is a metric pass.
- (iv) No budget on one clause is reused elsewhere without an explicit joint-bound theorem.
- (v) No exact comparison is required to fabricate a threshold-gap reserve, and no exact equality or profile isomorphism is promoted to positive-radius robustness without a separate metric certificate.
- (vi) No `not_compared` mode discharges a comparison, overlap, Restart, or coherence obligation.
- (vii) No passing clause cancels a rejected or undefined clause.
- (viii) No `undefined` verdict is reported as pass or as a mathematical counterexample.
- (ix) No missing Type IV artifact is interpreted as  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ .
- (x) No missing  $\kappa$ -certificate is interpreted as  $\kappa = 1$  or zero loss.
- (xi) No pointwise-positive Restart reserve is interpreted as a uniform positive lower bound.
- (xii) No finite prefix of a countable audit scope is treated as a complete pass certificate.
- (xiii) No pairwise overlap pass alone proves higher Čech coherence.
- (xiv) No coherent certificate atlas creates an unrestricted global object or global B-Gate<sup>+</sup> pass.
- (xv) No zero terminal-generic Type IV diagnostic implies full comparison-map isomorphism on interior bars.
- (xvi) No deprecated Latin-*u* diagnostic notation is normative; the cokernel diagnostics are  $\nu_{i,W,\tau}$  and  $\nu_{\text{Collapse}}$ .
- (xvii) No total  $\mu/\nu$  diagnostic is aggregated across different windows, thresholds, or infinitely many monitored degrees without a separately proved aggregation or summability theorem.
- (xviii) No deprecated global non-expansiveness claim for hard threshold deletion is used.
- (xix) No band of half-width  $2\varepsilon$  is called sharp for the two-sided threshold theorem; the sharp half-width is  $\varepsilon$ .
- (xx) No Claim Register entry proves itself by registration.
- (xxi) No `operational`, `spec`, `deprecated`, `rejected`, or `unknown` item is theorem evidence; its admissible non-theorem role remains explicitly typed.
- (xxii) No Search, Platform, Companion, or AI artifact updates a Core verdict outside the re-entry protocol.
- (xxiii) No successor-release candidate or unpromoted NBT statement is treated as a v20.0 theorem.
- (xxiv) No audit pass is a problem-specific external theorem without a separately proved Interface or Return theorem.

- (xxv) No specialized detector, gap, transient, Restart, manifest, or release token creates an independent status algebra outside the canonical four atomic statuses.
- (xxvi) No runtime comparison mode is replaced by `exact_cocycle`, `metric_reference`, or `metric_zero_transport`.
- (xxvii) No detector value is consumed without a passing certificate-status axis, and no certificate pass determines the mathematical value by itself.
- (xxviii) No detector theorem is moved by relabeling among the stages

`windowed_prethreshold,   repaired_postthreshold,`  
`kernel_defect,   cokernel_defect.`

- (xxix) No terminal Type IV zero is transient absence, and no transient certified absence is full comparison-map isomorphism.
- (xxx) No local detector atlas proves global zero without an independent retained-bar coverage theorem.
- (xxxi) No record-coherent atlas proves object descent, and no Core pass proves an external conclusion without a Return Theorem.
- (xxxii) No finite prefix, observed stabilization, or finite sample proves an infinite tower statement without a finite-to-infinite reduction theorem.
- (xxxiii) No schema migration, summary, translation, formalization, or serialization is presumed semantics-preserving without a semantic-diff or preservation record.
- (xxxiv) No conditional successor claim silently erases its predecessor non-claim, hypotheses, or forbidden stronger readings.
- (xxxv) No complete-release designation is inferred from completion of only the Main-body mathematical chapters.

#### 7.16.16 Provenance and status

This chapter introduces no new barcode stability theorem, Ext computation, detector characterization, or tower invariant. Its v19 inheritance is the four-valued atomic audit calculus, immutable run identity, non-compensating ledger discipline, canonical manifest, replay and sovereignty rules. Its v20 extension is the sorted token registry and the canonical integration of finite detectors, higher obstruction edges, terminal/transient defects, detector atlases, finite-to-infinite reductions, descent/Return boundaries, semantic schema migration, and release-acceptance governance.

#### Status labels for this chapter. *Core audit definitions:*

Definitions [7.57](#), [7.59](#), [7.60](#), [7.61](#), [7.63](#), [7.65](#), [7.66](#), [7.68](#), [7.69](#), [7.70](#), [7.72](#), [7.74](#), [7.76](#), [7.77](#), [7.78](#), [7.79](#), [7.80](#), [7.83](#), [7.85](#), [7.86](#), [7.89](#), [7.90](#), [7.91](#), [7.92](#), [7.93](#), [7.94](#), [7.95](#), [7.97](#), [7.99](#), [7.100](#), [7.101](#), [7.104](#), [7.105](#), [7.106](#), [7.111](#), [7.112](#), [7.113](#), [7.114](#), [7.117](#), [7.121](#), [7.124](#), [7.125](#), [7.126](#), [7.129](#), [7.130](#).

#### *Logical and audit consequences:*

Results [7.58](#), [7.62](#), [7.64](#), [7.73](#), [7.82](#), [7.87](#), [7.88](#), [7.96](#), [7.102](#), [7.103](#), [7.107](#), [7.115](#), [7.116](#), [7.118](#), [7.120](#), [7.123](#), [7.127](#), [7.128](#), [7.131](#), [7.132](#).

*Examples and precision remarks:*

Examples/Remarks [7.55](#), [7.56](#), [7.67](#), [7.71](#), [7.75](#), [7.81](#), [7.84](#), [7.98](#), [7.108](#), [7.109](#), [7.110](#), [7.119](#), [7.122](#), [7.133](#).

*Explicit non-claims:*

Section [7.16.15](#).

## 8 Chapter 8: Typed Verification, Proof-DAG, and Governance Kernel

**Chapter 8 contract (v21 verification kernel).** This chapter is the canonical verifier and governance layer of v21. It does not add a new persistence, derived, tower, or descent theorem. It determines when the mathematical conclusions of Chapters 1–7 may be consumed as typed proof evidence, when they remain conditional or undefined, and when a target claim is blocked.

The primitive audit statuses are

pass,      reject,      undefined,      not\_invoked.

A target is consumable only through a dependency-closed target sub-DAG whose necessary predecessor nodes, edges, artifacts, comparison modes, source bindings, and non-scalar obligations all pass at the strength required by that target. Every unresolved assumption is propagated into the target conclusion; an unconditional reading is permitted only when the propagated assumption context is empty or independently discharged. Graph reachability, a complete-looking manifest, hash agreement, successful replay, formal compilation, agent consensus, or numerical reserve does not replace theorem evidence.

Finite ordinary proof dependency is acyclic. For countable or larger dependency scopes, acyclicity alone is insufficient: an independently checkable well-founded rank or a finite-to-infinite reduction is required. Mathematical recursion, induction, coinduction, or fixed-point reasoning is admitted only through a separately proved theorem node whose internal cyclic semantics is hidden behind an acyclic, well-founded external interface. Known-Theorem Recovery places the benchmark conclusion behind a quarantine cut: the frozen target statement may identify the desired consequent, but its proof, truth value, target constants, conclusion-derived witnesses, or equivalent oracles may not flow upstream into realization, detection, reduction, or Return.

Chapter 9 consumes the verifier-side closure object produced here. It may not weaken, bypass, or reinterpret this chapter’s status algebra, necessary-edge closure, non-compensation rule, source quarantine, or Return-node typing.

### 8.1 Scope, ownership, and dependency position

This chapter consumes the complete mathematical and interface record of Chapters 1–7. In particular it consumes:

- (i) the dual-Core constitution, claim taxonomy, comparison modes, Return strength order, and proof-DAG sovereignty policy of Chapter 1;
- (ii) repaired metric transport and exact zero-separation data from Chapter 2;
- (iii) finite detector schemes, certificate complexity, representative records, and no-go boundaries from Chapter 3;
- (iv) P2DG, finite-quiver reconstruction, higher-edge eligibility, and cone handoffs from Chapter 4;
- (v) direct and inverse tower records, finite-to-infinite reductions, derived-limit defects, and internal apex Returns from Chapter 5;
- (vi) overlap, higher coherence, certificate atlases, Restart, and `descent_input_ready` records from Chapter 6;
- (vii) finite diagram reconstruction, projective-generator detection, finite Morita realization, effective finite descent, and global zero/isomorphism certificates from Chapter 7.

The frozen v20 audit definitions retained inside Chapter 7 remain available only for historical compatibility. The definitions and theorems of this chapter are canonical for new v21 runs. A conflict is resolved by an explicit supersession record; no historical field is silently reinterpreted.

**Remark 8.1** (Why Chapter 8 precedes the unified Bridge chapter). A unified Bridge or Return claim must already know what counts as a valid source node, theorem node, interface edge, reduction edge, Return edge, artifact identity, failure state, and dependency closure. Therefore the verifier kernel must be fixed before Chapter 9 assembles any Bridge route.

**Remark 8.2** (Normative scope). Most results below are typing contracts and logical consequences of those contracts. They are mathematically substantive as proof-governance statements, but they do not prove an external arithmetic, geometric, analytic, PDE, or cryptographic conclusion.

## 8.2 Verifier-side datum and independent statement axes

**Definition 8.3** (Verifier-side datum). A *verifier-side datum* is a tuple

$$\mathfrak{B} = \left( \begin{array}{l} \text{RunID, RevisionID, theory\_version, SchemaID, claim\_register\_snapshot,} \\ \text{Scope, NodeRegistry, EdgeRegistry, ArtifactRegistry,} \\ \text{EvidencePolicies, VerifierProfile, FailurePolicy} \end{array} \right).$$

It is admissible only when every registry is immutable for the run, every identifier resolves to exactly one typed record, and every consumed theorem, interface, artifact, comparison, reduction, or Return is attached to the same scope or to an explicitly authorized import.

**Definition 8.4** (Verifier input admissibility). An object may enter the verifier only as one of the following:

- (i) a theorem or definition with exact statement, hypotheses, scope, proof location, and permitted conclusion strength;
- (ii) a typed interface record with declared source, target, variance, exactness, reflection, information, and artifact fields;
- (iii) an immutable artifact with content identity, type, provenance, and interpretation policy;
- (iv) a metric record with an actual discrepancy theorem and a declared scalarization;
- (v) a coherence, descent, reduction, or Return record satisfying the owner chapter's complete input contract;
- (vi) an explicit assumption node whose assumed status and downstream conditionality are visible in the target claim.

A prose assertion, confidence score, graph path, successful build, citation string, or agent recommendation is not verifier-side evidence by default.

**Definition 8.5** (Independent statement-status axes). Every statement record has independent axes

$$\text{Axis}(c) = \left( \begin{array}{l} \text{document\_role, evidence\_status, invocation\_role,} \\ \text{mathematical\_verdict, conformance\_status, workflow\_status, migration\_status} \end{array} \right).$$

Typical document roles include theorem, definition, policy, interface, specification, candidate, example, counterexample, and explicit non-claim. Evidence status distinguishes proved, conditional, operational, specification, deprecated, rejected, superseded, and unknown. Invocation role distinguishes theorem evidence, operational rule, specification target, and provenance-only use.

**Proposition 8.6** (No cross-axis promotion). *A favorable value on one statement axis does not promote another axis. In particular, workflow completion, schema conformance, audit success, replay success, human approval, agent consensus, or formal compilation does not by itself imply mathematical pass or theorem-evidence status.*

*Proof.* The axes answer different typed predicates. None supplies an inhabitant or a proof object for another predicate without a registered conversion theorem.  $\square$

**Definition 8.7** (Claim-use preflight). A theorem-level claim use passes preflight exactly when:

- (i) the exact claim identity and source statement are fixed;
- (ii) the evidence status is proved, or conditional with every condition discharged;
- (iii) the invocation role is theorem evidence;
- (iv) the target use does not exceed the registered strength;
- (v) every supersession, weakening, migration, or interpretation step is explicit;
- (vi) the proof and all transitive artifacts are available to the verifier;
- (vii) no quarantined benchmark information and no information chosen from the target conclusion has contaminated construction-side evidence.

**Proposition 8.8** (Preflight blocks unsupported theorem use). *If claim-use preflight does not pass, the claim cannot discharge a theorem, interface, reduction, descent, or Return node.*

*Proof.* The node would otherwise consume evidence at an unregistered identity, hypothesis set, role, or strength. Its required theorem dependency therefore remains undischarged.  $\square$

**Definition 8.9** (Typed obligation atom). A typed obligation atom is

$$\mathfrak{o} = \left( \begin{array}{c} \text{ObligationID, owner, scope, kind,} \\ \text{evidence\_policy, dependencies, failure\_routes, manifest\_fields} \end{array} \right).$$

Its kind is one of mathematical verdict, existence/construction, compatibility/coherence, metric-bearing safety, operational computation, or governance/manifest. The primary kind does not prevent one failure from receiving several diagnostic routes.

**Definition 8.10** (Declared obligation registry). For one immutable run, the obligation registry is a finite or admissibly countable family

$$\text{Obl}(\mathfrak{B}) = \{\mathfrak{o}_a\}_{a \in A}$$

with a required subset, a positive scope-exclusion record for every nonrequired atom, complete dependency edges, evidence identifiers, evidence policy versions, and governing claim versions. A silent change creates a new revision.

**Definition 8.11** (Certificate-readiness bundle). A certificate-readiness bundle for a target consists of the complete required obligation registry, the corresponding necessary target sub-DAG, all primitive evidence records, artifact closure, source and quarantine records, and the verifier policies needed to evaluate every atom. Readiness means the proof object is fully posed for verification; it does not mean that every atom passes.

**Remark 8.12** (Registry and readiness are not proof). An obligation registry identifies what must be checked, and a readiness bundle makes the check reconstructible. Neither establishes the satisfaction predicate of an obligation without the required evidence.



### 8.3 Atomic status algebra, scope exclusion, and aggregation

**Definition 8.13** (Canonical atomic status). Every registered verifier clause, proof-DAG node, and registered edge in the immutable target registry has exactly one atomic status

`pass, reject, undefined, not_invoked.`

Their meanings are:

- (i) `pass`: well-typed, fully evidenced, dependency-closed, and true at the declared strength;
- (ii) `reject`: well-typed evidence establishes falsity, obstruction, inconsistency, or a failed mandatory condition;
- (iii) `undefined`: required but missing, ill-typed, stale, circular, semantically unresolved, or blocked by an undischarged prerequisite;
- (iv) `not_invoked`: positively excluded by the immutable scope.

**Definition 8.14** (Scope-exclusion record). A scope-exclusion record names the excluded clause, the governing policy, the reason the clause is irrelevant, the exact source and target scopes, and the immutable fields on which the exclusion depends. An exclusion chosen after observing an inconvenient value is invalid.

**Remark 8.15** (Missing is not out of scope). The absence of a node, edge, artifact, morphism, realization, coherence cell, descent theorem, or Return theorem yields `undefined` when required. It does not yield `not_invoked` without a valid scope-exclusion record.

**Definition 8.16** (Aggregate verifier verdict). Let  $C_{\text{req}}$  be the complete registry of required atomic clauses for one target. The headline verdict is:

- (i) `reject` if some required clause rejects;
- (ii) `undefined` if none rejects and some required clause is undefined, or a purported exclusion is unjustified;
- (iii) `pass` if every required clause passes and every excluded clause has a valid exclusion record.

Every component status is retained; the headline does not erase simultaneous undefined or rejected causes.

**Theorem 8.17** (Non-compensation theorem). *No passing node, edge, metric reserve, local certificate, replay record, manifest field, or sibling branch can compensate for a rejected or required undefined clause in the target dependency closure.*

*Proof.* Definition 8.16 is a typed conjunction over required clauses, not a signed or weighted sum. A pass establishes only its own predicate and does not alter the status of another predicate.  $\square$

**Corollary 8.18** (No proof by majority or consensus). *A majority of passing checks, agreement among agents, or repeated successful executions cannot override one necessary rejected or undefined dependency.*

*Proof.* These are collections of sibling observations and are subject to Theorem 8.17.  $\square$

## 8.4 Comparison-mode compatibility

**Definition 8.19** (Canonical comparison modes). Every consumed comparison declares exactly one mode

$$\begin{array}{l} \text{exact,} \quad \text{metric\_gap\_certified,} \\ \text{theorem\_controlled\_nonmetric,} \quad \text{not\_compared.} \end{array}$$

The modes return respectively an actual declared relation, a theorem-certified metric conclusion, the exact output of a named nonmetric theorem, or no comparison conclusion.

**Definition 8.20** (Claim-mode compatibility profile). For each comparison claim  $c$ , define

$$\text{Allowed}(c) \subseteq \left\{ \begin{array}{l} \text{exact, metric\_gap\_certified,} \\ \text{theorem\_controlled\_nonmetric, not\_compared} \end{array} \right\}.$$

and, for each allowed mode  $m$ , an output-strength set  $\text{Out}(c, m)$ . The record also states the exact relation required by the target, denoted  $\text{Strength}_{\text{req}}(c)$ .

**Definition 8.21** (Comparison-mode compatibility clause). A comparison-mode compatibility clause passes exactly when:

- (i) the runtime mode belongs to  $\text{Allowed}(c)$ ;
- (ii) its source, target, stage, threshold, variance, and artifacts match the claim;
- (iii) every mode-specific theorem hypothesis passes;
- (iv) some output in  $\text{Out}(c, m)$  entails  $\text{Strength}_{\text{req}}(c)$ .

**Theorem 8.22** (No comparison-mode substitution). *A comparison record in one canonical mode cannot be consumed as another mode without a proved conversion theorem. In particular:*

- (i) *exact equality does not itself provide positive-radius robustness;*
- (ii) *metric closeness does not manufacture an actual morphism, identification, cocycle, or descent datum;*
- (iii) *a theorem-controlled nonmetric result returns only the conclusion of its named theorem;*
- (iv) *not\_compared cannot discharge an invoked comparison.*

*Proof.* The four modes have different source data, codomains, and admissible output strengths. Substitution would violate Definition 8.21 unless a separate theorem provides the required conversion.  $\square$

## 8.5 Non-scalar obligations and the metric ledger

**Definition 8.23** (Canonical non-scalar obligations). The following obligations are non-scalar by default:

- (i) an actual morphism, natural transformation, cone map, or transition family;
- (ii) an admissible representative and its exact equivalence strength;
- (iii) higher-edge eligibility, realization, amplitude, naturality, and zero-reflection;
- (iv) detector stage, completeness, endpoint coverage, family membership, and oracle-closed information policy;

- (v) finite generator completeness, enriched probe action, and cone compatibility;
- (vi) overlap cocycle, higher coherence cell, face compatibility, and ambient coherence model;
- (vii) descent effectivity, joint conservativity, and descended-object construction;
- (viii) direct or inverse tower morphisms, stable-image or ML hypotheses, and finite-to-infinite reduction;
- (ix) source authority, source binding, benchmark quarantine, and independence;
- (x) a final Return theorem at the requested target strength;
- (xi) manifest completeness, artifact closure, semantic preservation, revision lineage, and immutable scope.

**Definition 8.24** (Declared metric ledger). A metric-bearing verifier clause declares a commutative ordered unital monoid

$$(V, \oplus, 0_V, \preceq)$$

and a monotone subadditive scalarization

$$\text{val}_B : V \longrightarrow [0, +\infty]$$

with  $\text{val}_B(0_V) = 0$ . If arbitrary joins and distributivity are used, the structure may be declared a quantale; otherwise the word *quantale* is not used.

**Definition 8.25** (Gate-scoped metric budget). A gate-scoped budget record contains the primitive discrepancies consumed by one comparison, their provenance, a no-double-counting partition or joint bound, the aggregation law, the scalarized value

$$\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}}),$$

and a theorem proving that this value dominates the actual discrepancy of the named source and target. When threshold transport is invoked it also records its strict reserve against the relevant pre-threshold family.

**Theorem 8.26** (No scalar manufacture of non-scalar evidence). *No ledger value, scalarized tolerance, threshold clearance, accumulated reserve, or certificate-cost bound can create or repair a missing or failed obligation in Definition 8.23.*

*Proof.* A metric ledger bounds a declared numerical discrepancy. Each listed obligation asserts existence, typing, exact semantic compatibility, independence, or theorem validity. These predicates are not implied by the size of an unrelated metric bound.  $\square$

**Definition 8.27** (No-double-counting invariant). Each primitive metric discrepancy may contribute at most once to a selected route's aggregate budget unless a documented joint-bound theorem replaces the individual charges. Multiple graph aliases, manifest references, or agent records do not create additional independent discrepancies and do not license free reuse.

## 8.6 Proof-DAG node types and node discharge

**Definition 8.28** (Proof-DAG). A *proof-DAG* is a typed directed graph

$$\text{ProofDAG} = (N, E, s, t, \text{NodeType}, \text{EdgeType})$$

together with immutable node and edge registries, a target node, a required strength, and a dependency policy. The graph records proof dependencies; its existence, visual connectivity, or reachability is not itself proof.

**Definition 8.29** (Canonical proof-bearing node types). The canonical proof-bearing node types are:

- (i) *theorem node*: a proved statement with exact hypotheses and conclusion;
- (ii) *interface node*: a typed cross-layer realization, P2DG, DG2P, interpretation, or source/target interface;
- (iii) *artifact node*: an immutable proof, computation, profile, morphism, complex, source, or machine-readable object;
- (iv) *metric node*: a discrepancy theorem, ledger, scalarization, reserve, or zero-separation certificate;
- (v) *coherence node*: an overlap, cocycle, higher-cell, face, or compatibility result;
- (vi) *descent node*: an effectivity, joint-conservativity, descended object, or global zero/isomorphism result;
- (vii) *reduction node*: a finite-to-infinite, stable-image, ML, eventual-constancy, finite-state, or other reduction theorem;
- (viii) *Return node*: a directional AK-to-target theorem at a declared external or internal strength.

Auxiliary source-binding, formalization, interpretation, validation, replay, audit, and governance nodes may be added, but their authority is fixed by their node type and incident edges.

**Definition 8.30** (Node semantic record). Every node  $n \in N$  has a semantic record

$$\mathfrak{n} = \left( \begin{array}{l} \text{NodeID, type, statement, owner, scope, role, input\_type, output\_type, required\_strength,} \\ \text{provided\_strength\_set, assumption\_context, dependencies, verifier\_policy, artifacts, status} \end{array} \right).$$

The statement field contains the exact typed proposition or construction, not only a title. The provided-strength set is downward closed in the declared strength preorder. The assumption context records every undischarged explicit assumption on which the node depends. The owner identifies the chapter, appendix, external theorem, proof-assistant environment, or imported source governing its semantics.

**Definition 8.31** (Primitive and derived nodes). A node is *primitive* when its verifier policy consumes a registered proof, accepted axiom, independently verified computation, or explicitly assumed interface without relying on another node in the same target DAG. It is *derived* when its discharge is computed from incoming necessary edges and source-node discharges.

**Definition 8.32** (Assumption context and conditionality propagation). Every node  $n$  carries a finite or admissibly countable set  $\text{Cond}(n)$  of explicit undischarged assumption identifiers. A primitive assumption node contributes its own identifier. For a derived node, the context contains the union of the contexts of every consumed predecessor, minus only those assumptions discharged by a separately passing theorem or interface node. The semantic conclusion of  $n$  is therefore the context sequent

$$\text{Cond}(n) \vdash \text{statement}(n).$$

This notation does not assume that the ambient logic admits a single finite or countable conjunction object for the entire context. A node may be consumed as unconditional only when  $\text{Cond}(n) = \emptyset$ . Otherwise every downstream node must accept and preserve the same context or discharge it explicitly.

**Proposition 8.33** (No silent erasure of assumptions). *A passing conditional node cannot discharge an unconditional target node by omitting, renaming, or hiding an element of its assumption context.*

*Proof.* The conditional node proves only its registered sequent. Removing an undischarged premise strengthens the conclusion and is not a weakening edge. It requires a separate discharge theorem.  $\square$

**Definition 8.34** (Necessary target sub-DAG). For a target node  $z$ , the necessary target sub-DAG

$$\text{TargetDAG}(z) \subseteq \text{ProofDAG}$$

is the smallest registered subgraph containing  $z$ , every dependency node actually consumed by  $z$ , every dependency recursively consumed by those nodes, and every necessary edge among them. Advisory, alternative, rejected, deprecated, or unused candidate branches are excluded unless the target explicitly consumes them.

**Remark 8.35** (Target-locality). A broad Validity Map or research graph may contain many unresolved nodes. A target can be certified from its own complete necessary sub-DAG without certifying unrelated branches. Conversely, omitting one necessary branch from the target projection is a closure defect.

**Definition 8.36** (Node discharge certificate). A node discharge certificate records the node identity, exact target statement, input and output types, required input strengths, provided output-strength set, verifier policy, incoming necessary edges, supplied evidence, artifact identities, propagated assumption context, and derived atomic status. A node passes only when its registered sequent is established at the required strength. A conditional pass remains conditional and does not become an unconditional theorem merely because its node status is pass.

**Proposition 8.37** (Node existence does not discharge the node). *The presence of a theorem name, Bridge Program, artifact, formal file, replay result, or graph node does not discharge the represented proposition without a valid node discharge certificate.*

*Proof.* Graph presence records that an object or claim has been indexed. Discharge requires the verifier predicate of Definition 8.36.  $\square$

## 8.7 Typed edge semantics and admissible composition

**Definition 8.38** (Canonical edge semantics). Every proof-DAG edge has exactly one primary semantic type:

- (i) *logical*: the source proposition is a premise in a proof of the target proposition;
- (ii) *artifact*: the target references or requires the source artifact for identity, computation, or reconstruction;
- (iii) *comparison*: the source supplies an exact, metric, or theorem-controlled comparison consumed by the target;
- (iv) *bridge*: a proved directional theorem transports a statement between declared semantic layers;
- (v) *interpretation*: a theorem connects formal, Core, or encoded semantics to an intended informal or domain-specific statement;
- (vi) *formalization*: an informal statement is represented by a formal statement or proof-assistant object;
- (vii) *weakening*: a proved order relation licenses consumption of a stronger conclusion as a specified weaker conclusion.

Auxiliary source-binding, validation, replay, audit, provenance, and advisory edges are permitted but do not inherit logical authority from visual placement.

**Definition 8.39** (Edge semantic record). Every edge  $e$  has a record

$$e = \begin{pmatrix} \text{EdgeID}, s(e), t(e), \text{type}, \text{relation}, \text{direction}, \\ \text{input\_strength}, \text{output\_strength\_set}, \\ \text{assumption\_transport}, \text{discharge\_policy}, \\ \text{artifacts}, \text{status} \end{pmatrix}.$$

The relation field states the exact implication, map, equality, comparison bound, interpretation, or weakening licensed by the edge. The assumption-transport field states which source assumptions are preserved, discharged by a named theorem, or added as explicit target hypotheses; silent erasure is forbidden.

**Definition 8.40** (Edge compatibility). An edge is compatible when its source-node output type, target-node input type, semantic direction, comparison mode, strength, assumption-context transport, and discharge policy match. An edge whose type is weaker or different from the target’s required input is *misrouted*.

**Theorem 8.41** (Misrouted edges do not discharge dependencies). *A necessary misrouted edge leaves its target dependency undefined until it is retyped by a proved conversion or replaced by a compatible passing edge.*

*Proof.* The source output does not inhabit the target’s required semantic input. Therefore the dependency predicate is not evaluable at the requested type.  $\square$

**Definition 8.42** (Weakening edge). Let  $(\Sigma, \preceq)$  be a declared conclusion-strength preorder. A weakening edge from  $A$  at strength  $\sigma$  to  $A'$  at strength  $\sigma'$  is admissible only when a registered theorem proves  $\sigma' \preceq \sigma$  and transports the exact source statement to the exact target statement.

**Proposition 8.43** (No reversal of weakening). *A weakening edge cannot be traversed backward to obtain a stronger conclusion without an independent promotion theorem.*

*Proof.* A preorder relation  $\sigma' \preceq \sigma$  licenses strong-to-weak use only. Its converse is not implied.  $\square$

**Definition 8.44** (Edge composition policy). A path may be composed only when every adjacent edge is semantically compatible, every intermediate node supplies the exact input expected by the next edge, and the composed conclusion does not exceed the intersection of the permitted output-strength sets of the path components.

**Proposition 8.45** (Graph reachability is not theorem composition). *The existence of a directed path from node  $A$  to node  $B$  does not imply that  $A$  proves  $B$ .*

*Proof.* The path may contain several edge types whose visual concatenation does not form the required implication. Composition is licensed only by Definition 8.44.  $\square$

## 8.8 Acyclicity, well-founded dependency, and controlled recursion

**Definition 8.46** (Ordinary dependency acyclicity). The ordinary logical dependency relation of a proof-DAG contains no directed cycle. For a finite target subgraph this is equivalent to the existence of a finite topological order. For an infinite target subgraph, acyclicity alone does not exclude an infinite backward dependency chain and is therefore not a complete discharge condition.

**Definition 8.47** (Well-founded dependency rank). A necessary target sub-DAG is *dependency-well-founded* when it is finite and acyclic, or when there exists an independently checkable ordinal rank

$$\rho : N \longrightarrow \text{Ord}$$

such that every necessary dependency edge  $u \rightarrow v$  satisfies  $\rho(u) < \rho(v)$ . For an infinite graph this is equivalent, for the consumed predecessor relation, to the absence of an infinite chain of ever earlier prerequisites below a fixed target. It implies acyclicity and is strictly stronger than acyclicity in the infinite case.

**Definition 8.48** (Controlled recursive theorem node). An induction, recursion, coinduction, fixed-point, or mutually recursive proof is represented by a single theorem node whose internal semantics is supplied by a separately proved well-foundedness, contractivity, monotonicity, initial-algebra, final-coalgebra, or fixed-point theorem. The theorem node exposes an acyclic input/output interface to the surrounding DAG.

**Proposition 8.49** (Unlicensed cycle blocks certification). *A directed cycle in the necessary logical dependency graph that is not encapsulated by a controlled recursive theorem node makes the affected target undefined. If the cycle is concealed while a pass is asserted, the manifest is inconsistent and the target rejects on governance grounds.*

*Proof.* Without an external recursion theorem, every node in the cycle depends on its own conclusion and no well-founded discharge order exists. Concealment also contradicts the declared dependency structure.  $\square$

**Definition 8.50** (Countable dependency certificate). A countable necessary sub-DAG is admissible only with either:

- (i) an independently checkable rank satisfying Definition 8.47 and a uniform verifier schema covering every node and edge; or
- (ii) a Chapter 5 finite-to-infinite reduction replacing the countable closure by a finite certificate and a proved tail theorem.

A verified finite prefix is not a countable closure certificate.

**Theorem 8.51** (Determinacy on a well-founded typed DAG). *Fix immutable registries and verifier policies. If a finite or admissible countable target sub-DAG is well-founded and every primitive node and edge has a determined atomic status, then recursive evaluation in rank order produces a unique status for every derived node and for the target.*

*Proof.* Well-founded induction evaluates each derived record after all of its necessary predecessors. The node, edge, and aggregate verifier policies are deterministic functions of those already fixed inputs.  $\square$

## 8.9 Failure precedence and complete failure routing

**Definition 8.52** (Diagnostic failure precedence). Causal precedence is determined first by the dependency order of the selected target sub-DAG: an upstream failed prerequisite precedes each downstream consequence that consumes it. For simultaneous or incomparable blockers, the following list is a repair-triage order rather than a substitute for the DAG:

- (1) source identity, source authority, or source-binding mismatch;
- (2) ill-typed statement, node, edge, scope, variance, or comparison mode;

- (3) missing realization, representative, interpretation, or eligibility;
- (4) missing actual morphism, transition, cone map, or comparison artifact;
- (5) incomplete detector, generator, or finite-certificate theorem;
- (6) failed reduction, overlap, cocycle, higher coherence, or Restart compatibility, according to their actual dependency positions;
- (7) failed descent effectivity, joint conservativity, or global-object construction;
- (8) a well-typed mathematical obstruction or nonzero target predicate;
- (9) failed final Bridge or Return theorem.

The precedence relation identifies causal blockers and a deterministic repair order. It does not erase later independent failures.

**Definition 8.53** (Failure-routing record). A failure-routing record contains every nonpassing node and edge, its atomic status, broad Type I–V routes when applicable, the earliest causal blocker on each target path, all downstream nodes invalidated by that blocker, and the permitted repair action.

**Proposition 8.54** (Typing precedes obstruction). *A numerical zero or nonzero value is not interpreted as a mathematical obstruction verdict until source identity, typing, stage, morphism, and verifier authority pass.*

*Proof.* Before those clauses pass, the number is not established as the value of the claimed mathematical object. The corresponding clause is therefore undefined rather than proved obstructed or unobstructed.  $\square$

**Proposition 8.55** (Failure precedence is not status cancellation). *Selecting an earliest causal blocker for diagnosis does not remove another rejected or undefined clause from the manifest or headline aggregation.*

*Proof.* The precedence order is a repair-routing relation, whereas Definition 8.16 aggregates all required statuses.  $\square$

## 8.10 Source binding and benchmark quarantine

**Definition 8.56** (Source-statement binding). A source-statement binding is a tuple

$$\mathfrak{S} = \left( \begin{array}{l} \text{ClaimID, source\_authority, artifact, locator, exact\_statement,} \\ \text{hypotheses, permitted\_use, forbidden\_stronger\_reading} \end{array} \right).$$

A citation, title, DOI, page image, or hash without an exact statement and locator does not by itself supply theorem evidence.

**Definition 8.57** (Precursor package). A Known-Theorem Recovery route declares the exact source-side hypotheses, lemmas, constructions, normalizations, and externally permitted data that may be used before the target conclusion is reconstructed. This collection is the *precursor package*.

**Definition 8.58** (Benchmark conclusion). The benchmark conclusion is the frozen external statement against which an independently reconstructed Return is compared. It may identify the target claim and the consequent shape of the final Return theorem, but it is not an upstream proof premise.



**Definition 8.59** (Benchmark-quarantine cut). A benchmark-quarantine cut partitions a recovery target sub-DAG into a construction side and a terminal validation side. The benchmark proof, known truth value, answer table, target constants, exceptional bounds, conclusion-derived witnesses, equivalent oracles, and independent validation artifacts are prohibited from crossing into realization, detector, certificate, higher-edge, reduction, descent, or Return construction nodes.

**Definition 8.60** (Permitted quarantine crossings). Only the following may cross from the benchmark registry to the construction metadata:

- (i) immutable target identity and exact statement locator;
- (ii) the declared comparison strength required at terminal validation;
- (iii) source-side hypotheses independently listed in the precursor package;
- (iv) data explicitly authorized as independent hypotheses by the target theorem itself.

The terminal validation edge flows from the frozen construction result to the benchmark comparison node, never backward.

**Theorem 8.61** (Benchmark leakage invalidates independent recovery). *If prohibited conclusion-side information is consumed to choose or construct an upstream witness, stabilization index, exceptional bound, realization, detector, reduction, or Return proof, then the affected route cannot certify independent Known-Theorem Recovery unless a separate theorem proves that the consumed datum was independently admissible.*

*Proof.* The route is conclusion-conditioned and therefore circular relative to the registered precursor package. The quarantine independence predicate fails.  $\square$

**Proposition 8.62** (Source storage is not theorem recovery). *Storing an authoritative source statement, source-binding record, benchmark PDF, or exact known conclusion does not establish a recovery route.*

*Proof.* Recovery additionally requires purpose-faithful realization, Core certification, every necessary reduction and descent step, a final Return theorem, and an independent terminal comparison.  $\square$

**Definition 8.63** (Recovery sub-DAG). A Known-Theorem Recovery sub-DAG contains separate source-binding, precursor, realization, Core, reduction, descent, Return, quarantine, and terminal validation nodes. Its target strength is exactly the frozen benchmark strength; no stronger theorem is inferred from successful recovery.

## 8.11 Manifest, artifact identity, and semantic serialization

**Definition 8.64** (Immutable artifact identity). An immutable artifact identity is a tuple

$$\alpha = \left( \begin{array}{l} \text{ArtifactID, media\_type, canonicalization, hash\_algorithm,} \\ \text{Hash, size, provenance, semantic\_locator} \end{array} \right).$$

It identifies exact bytes or a declared canonical representation. The record states which interpretation theorem, parser, formal environment, or profile schema gives those bytes mathematical meaning.

**Proposition 8.65** (Hash identity supports identity, not truth). *Hash agreement supports a claim that two artifacts have the same canonical content under the declared hash and canonicalization policy. It does not prove that the content is mathematically true, correctly interpreted, complete, non-circular, or sufficient for a target theorem.*

*Proof.* A content hash is a function of bytes. The listed mathematical properties are semantic predicates requiring separate evidence.  $\square$

**Definition 8.66** (Cross-format semantic-equivalence record). A claim that two different byte artifacts—for example TeX, PDF, JSON, and a proof-assistant file—represent the same theorem-relevant object requires a cross-format record containing parsers or interpretation maps, normalized semantic objects, a semantic diff, and a preservation theorem at the exact consumed strength.

**Definition 8.67** (Artifact closure). A target manifest has artifact closure when every referenced artifact and every transitive artifact dependency resolves uniquely, has immutable identity and provenance, matches the consuming node’s declared type and version, and has no undeclared replacement inside the run.

**Definition 8.68** (Canonical v21 verifier manifest). The canonical verifier manifest contains at least:

- (i) ManifestID, RunID, RevisionID, theory, schema, claim-register, verifier, and artifact-root versions;
- (ii) immutable target scope, requested strength, clause registry, and all justified exclusions;
- (iii) complete node and edge registries with typed statements, semantic directions, statuses, and necessary-target projection;
- (iv) statement-axis records and claim-use preflight results;
- (v) every comparison mode, output strength, metric ledger, scalarization, and no-double-counting record;
- (vi) every representative, realization, detector, generator, morphism, coherence, descent, reduction, and Return obligation consumed by the route;
- (vii) source bindings, precursor packages, quarantine cuts, independence records, and terminal benchmark comparisons;
- (viii) immutable artifact identities, artifact closure, cross-format equivalence records, and semantic locators;
- (ix) agent action, conversion, authority, proof-assistant, audit, and replay records;
- (x) revision lineage, semantic diffs, dirty downstream closure, status inheritance decisions, and historical verdict links;
- (xi) every atomic status, headline verdict, failure route, and repair instruction sufficient for independent reconstruction.

**Definition 8.69** (Manifest status). The manifest status is exactly one of

complete,      incomplete,      inconsistent.

Complete means every required field and dependency is present, typed, artifact-closed, and mutually consistent. Incomplete means some required field or dependency is absent or unresolved. Inconsistent means two or more records make incompatible claims about the same immutable run.

**Proposition 8.70** (Manifest status specialization). *For the manifest clause,*

complete  $\mapsto$  pass,  
incomplete  $\mapsto$  undefined,  
inconsistent  $\mapsto$  reject.

*No manifest status determines the truth of a mathematical theorem by itself.*

*Proof.* The first three implications are the verifier meaning of the schema states. The final statement follows from Proposition 8.65 and the separation of statement axes.  $\square$

**Definition 8.71** (Semantic-preserving schema migration). A schema migration records source and target SchemaID, a total map on every consumed field, preservation of node and edge types, statuses, scope, required strengths, missing-artifact markers, source quarantine, and failure routes, together with a semantic diff and verdict-reconstruction test. Changed theorem strength, scope, source semantics, or evidence creates a new run rather than a schema-only migration.

**Proposition 8.72** (No lossy migration of proof evidence). *A migration that drops a required hypothesis, dependency, edge type, source locator, quarantine boundary, artifact identity, lower-level status, or forbidden stronger reading does not preserve verifier equivalence.*

*Proof.* The omitted information changes target meaning or prevents reconstruction of the original verdict. The migrated manifest is incomplete or inconsistent.  $\square$

## 8.12 Agent conversion and authority separation

**Definition 8.73** (Agent roles). The canonical roles are:

- (i) *proposer*: generates candidate statements, objects, witnesses, morphisms, proof sketches, routes, or priorities;
- (ii) *verifier*: evaluates declared mathematical and interface predicates under a registered authority and environment;
- (iii) *auditor*: checks identity, provenance, dependency closure, manifest conformance, replayability, and reconstruction;
- (iv) *proof assistant*: checks a formal statement and proof term relative to a formal environment and accepted axioms.

A human, AI system, script, platform, or formal tool may hold several roles, but each action is recorded under exactly one authority role.

**Definition 8.74** (Authority matrix). The authority matrix states which node and edge predicates each role may establish. By default:

- (i) a proposer establishes only candidate and workflow records;
- (ii) a verifier may establish the exact predicates in its registered verifier profile;
- (iii) an auditor may establish reconstructibility and governance predicates, not an unaudited mathematical theorem;
- (iv) a proof assistant establishes the formal statement represented in its environment, not the intended informal interpretation without a passing interpretation edge.

**Definition 8.75** (Proposer-to-verifier conversion certificate). A proposer output becomes verifier-side input only through a conversion certificate containing:

- (i) normalization into a declared Core, interface, proof-DAG, or artifact type;

- (ii) exact source and target statements and all hidden assumptions made explicit;
- (iii) claim role, evidence status, invocation role, and requested strength;
- (iv) immutable input and output artifacts and a semantic diff for every transformation;
- (v) source-authority, quarantine, independence, and contamination checks;
- (vi) the verifier profile and complete list of predicates still requiring verification.

**Proposition 8.76** (Agent assertion does not establish a Core node). *An agent assertion that a theorem, morphism, detector, descent datum, DAG closure, or Return holds does not discharge the corresponding node unless the proposer-to-verifier conversion and verifier-side evidence pass.*

*Proof.* The assertion is a candidate statement about the predicate. It is not the verifier evidence required by the predicate. □

**Proposition 8.77** (Agent consensus is non-compensating). *Agreement among agents, models, tools, reviewers, or proof sketches cannot repair a failed or missing verifier-side dependency.*

*Proof.* Consensus is proposer- or review-side metadata and is subject to Theorem 8.17. □

**Definition 8.78** (Formalization and interpretation pair). A proof-assistant route contains a formalization edge from the intended source statement to the exact formal statement and, when the target is informal or domain-specific, an interpretation edge from the verified formal theorem to the intended target. Both edges must pass.

**Proposition 8.79** (Formal verification does not silently prove the intended claim). *A verified proof-assistant node does not discharge an intended informal claim when the formalization or interpretation edge is undefined, rejected, or weaker than the requested target.*

*Proof.* The proof assistant certifies only the formal proposition in its environment. The semantic identification with the intended claim is a separate dependency. □

**Definition 8.80** (Agent contamination event). An agent contamination event includes benchmark leakage, fabricated source data, unsupported citation, detector-stage mutation, hidden use of a target conclusion, artifact drift, unrecorded post-processing, or semantic transformation without preservation evidence.

**Proposition 8.81** (Contamination forces status downgrade). *When contamination affects a necessary node or edge, its previous pass cannot be inherited. The affected route is undefined or rejected according to the evidenced defect until a new clean revision is verified.*

*Proof.* The earlier evidence no longer establishes independence, source identity, stage, or semantic preservation for the consumed object. □

### 8.13 Replay hierarchy and verification strength

**Definition 8.82** (Replay boundary). A replay boundary is the exact collection of input artifacts, code, environments, models, parser and verifier versions, settings, random seeds, external dependencies, and network or source snapshots reconstructed by a replay action.

**Definition 8.83** (Replay hierarchy). The canonical hierarchy contains four distinct records:

- (i) *operational replay*: the declared sequence of actions executes again under the replay boundary;
- (ii) *artifact replay*: the declared output artifacts or canonical semantic serializations are reproduced and identity-checked;
- (iii) *semantic replay*: theorem-relevant statements, scopes, edge directions, statuses, source bindings, and dependency meanings are shown to be preserved;
- (iv) *mathematical verification*: every theorem and interface obligation required by the target is proved or checked under its declared verifier policy.

**Theorem 8.84** (Replay hierarchy is strict). *In general,*

$$\text{operational replay} \Rightarrow \text{artifact replay} \Rightarrow \text{semantic replay} \Rightarrow \text{mathematical verification}.$$

*Each promotion requires an additional identity, preservation, or proof record.*

*Proof.* An execution may be nondeterministic or produce different artifacts; identical artifacts may be interpreted under different schemas; preserved semantics may encode a false theorem; and a mathematically verified statement need not reproduce the original workflow. The four predicates are therefore distinct.  $\square$

**Remark 8.85** (Build and render success). Compilation, parsing, schema validation, rendering, byte equality, and test success are valuable operational or artifact checks. They prove only their declared replay and conformance predicates unless semantic and mathematical verification are also present.

**Definition 8.86** (Semantic replay certificate). A semantic replay certificate contains the source and target schemas, statement normalizations, token-sort preservation, node and edge semantic diffs, source-binding preservation, quarantine preservation, status reconstruction, and proof that the target dependency closure has the same mathematical meaning before and after replay.

**Proposition 8.87** (Replay disagreement blocks silent reuse). *If two replays under supposedly identical immutable inputs produce different semantic nodes, edge directions, statuses, or target verdicts, the affected manifest is inconsistent until the environment, artifact, schema, or verifier difference is resolved.*

*Proof.* Determinacy of fixed verifier inputs is Theorem 8.51. A disagreement evidences that the inputs or semantics were not in fact identical.  $\square$

## 8.14 Revision, repair, and immutable history

**Definition 8.88** (Immutable revision record). A revision record is

$$\mathfrak{R}' = \left( \begin{array}{c} \text{RevisionID}', \text{RevisionID}_{\text{parent}}, \text{SemDiff}, \\ \text{changed\_artifacts}, \text{changed\_nodes}, \text{impact\_closure}, \text{reverification\_plan} \end{array} \right).$$

Every proof-relevant change creates a new revision and, when target scope or evidence changes, a new run identity.

**Definition 8.89** (Impact closure). For a set of changed nodes or edges  $S$ , let  $V(S)$  contain every changed node and the target of every changed edge. The impact closure is

$$V(S) \cup \text{Desc}_{\text{ProofDAG}}(V(S)),$$

the changed vertices together with every necessary descendant whose discharge may depend on the changed content. These descendants are marked dirty until re-evaluated or protected by a semantic-preservation theorem.

**Proposition 8.90** (Status does not automatically inherit across revisions). *A pass, reject, undefined, replay, audit, or Return status attached to one revision does not automatically transfer to another revision whose proof-relevant content or dependency closure changed.*

*Proof.* The status was computed for the earlier immutable inputs. The changed revision may alter the governing statement, evidence, or dependencies.  $\square$

**Definition 8.91** (Repair relation). A repair relation links a new revision to a historical nonpassing revision and records the exact defect, changed fields, new evidence, affected dependency closure, and the new verifier outcome. It does not overwrite the historical verdict.

**Theorem 8.92** (Immutable-history theorem). *A valid repair produces a new revision and verdict record. The historical run, artifacts, statuses, and failure routes remain immutable and independently replayable.*

*Proof.* The old verdict is a function of the old immutable verifier datum. Replacing its inputs would destroy identity and replayability. Therefore corrected content is represented by a new datum linked through Definition 8.91.  $\square$

## 8.15 Typed DAG closure and target consumption

**Definition 8.93** (Proper predecessor closure). For a target node  $z$ , let  $\text{PredDAG}(z)$  denote the proper predecessor package consisting of every node of  $\text{TargetDAG}(z)$  other than  $z$ , together with every necessary edge of  $\text{TargetDAG}(z)$ , including the edges entering  $z$ . Thus the final node is not assumed discharged merely by being part of the target projection.

**Definition 8.94** (Target derivation readiness). A target node  $z$  is *derivation-ready* at assumption context  $\Gamma$  and strength  $\sigma$  when:

- (i)  $\text{TargetDAG}(z)$  is complete, finite or admissibly countable, and dependency-well-founded;
- (ii) every node in  $\text{PredDAG}(z)$  passes at the strength consumed by its outgoing edges;
- (iii) every necessary edge is typed, compatible, directionally valid, and passes;
- (iv) the target node has a fixed semantic record and a theorem-backed, total verifier policy for the supplied predecessor tuple;
- (v) every comparison mode passes its claim-mode compatibility clause;
- (vi) every non-scalar obligation passes independently of metric budgets;
- (vii) the selected route satisfies no-double-counting and every metric node passes at its gate-scoped bound;
- (viii) source binding, source authority, precursor scope, and benchmark quarantine pass;
- (ix) artifact closure, manifest completeness, semantic replayability, and immutable revision identity pass;
- (x) the propagated predecessor assumption context is contained in  $\Gamma$ , and the target verifier policy does not erase any member of that context;
- (xi) the output-strength set registered by the target verifier policy contains  $\sigma$ .

No target pass status is assumed in this definition.

**Theorem 8.95** (Typed DAG closure theorem). *Let  $z$  be a target theorem, interface, reduction, descent, or Return node. If  $z$  is derivation-ready at  $(\Gamma, \sigma)$ , then recursive evaluation of its registered verifier policy assigns  $z$  status *pass* and licenses downstream consumption of the context sequent*

$$\Gamma \vdash \text{statement}(z)$$

*exactly at strength  $\sigma$  and at every separately registered weakening of  $\sigma$ . The conclusion is unconditional only when  $\Gamma = \emptyset$ .*

*Conversely, if any necessary predecessor node or edge is rejected, undefined, misrouted, unlicensed by scope, contaminated by benchmark leakage, outside artifact closure, weaker than the required input, or carries an assumption not preserved by the target, then  $z$  is not consumable at that context and strength.*

*Proof.* Because the necessary sub-DAG is dependency-well-founded, evaluate every proper predecessor in ordinal rank order. They pass by derivation readiness, and every necessary edge transports the exact registered type, strength, and assumption context. The theorem-backed target verifier policy is total on that supplied tuple, so its deterministic evaluation discharges the target sequent at a strength containing  $\sigma$ . Registered weakening edges license only the stated weaker uses. Empty context gives the unconditional specialization.

For the converse, the target dependency closure contains the failed, missing, misrouted, contaminated, or silently strengthened necessary item. By Theorem 8.17, no sibling pass can repair it; by Theorem 8.41, an incompatible edge cannot supply it; and by Proposition 8.33, an undischarged premise cannot be hidden. Hence derivation readiness fails.  $\square$

**Corollary 8.96** (Graph existence is not typed closure). *A complete-looking graph, a path to the target, or a manifest containing all node names does not license target consumption unless the hypotheses of Theorem 8.95 pass.*

*Proof.* Names and reachability do not establish node discharge, edge compatibility, source independence, or artifact closure.  $\square$

**Definition 8.97** (Route-admissible conclusion set). For a target route  $R$ , let each required node and edge provide a downward-closed permitted strength set  $\mathfrak{S}_x$ . Define

$$\mathfrak{S}_R := \bigcap_{x \in \text{Req}(R)} \mathfrak{S}_x.$$

A requested conclusion strength is admissible only when it belongs to  $\mathfrak{S}_R$ . The set may have several incomparable maximal elements; a unique scalar “ceiling” is not presumed.

**Definition 8.98** (Route-admissible conclusion record). For a route  $R$ , let  $\text{Cond}(R)$  be the union of all propagated assumption contexts after registered discharges. Its admissible conclusions are pairs

$$\mathfrak{A}_R := \{(\Gamma, \sigma) : \text{Cond}(R) \subseteq \Gamma, \sigma \in \mathfrak{S}_R\}.$$

An unconditional conclusion is admissible only if  $\text{Cond}(R) = \emptyset$ . Thus compatibility is two-dimensional: both conclusion strength and assumption context must survive route composition.

**Proposition 8.99** (No stronger conclusion than the route intersection). *A typed-DAG pass at one strength does not license a conclusion outside  $\mathfrak{S}_R$ , nor may it omit an assumption in  $\text{Cond}(R)$ .*

*Proof.* At least one required component does not permit that stronger conclusion or requires an undischarged assumption, so the composed route lacks a compatible output at the claimed context and strength.  $\square$

## 8.16 Chapter 9 handoff

**Definition 8.100** (Unified Bridge verifier handoff). The Chapter 8 handoff object is

$$\text{BridgeDAGInput} = \left( \begin{array}{c} \text{source\_node}, \\ \text{prebridge\_boundary\_node}, \\ \text{BridgeGoal}, \text{TargetDAG}(\text{prebridge\_boundary}), \\ \text{requested\_context}, \text{requested\_strength}, \\ \text{Cond}(R), \mathfrak{S}_R, \\ \text{node\_registry}, \text{edge\_registry}, \\ \text{source\_bindings}, \text{quarantine\_cut}, \\ \text{artifact\_closure}, \text{verifier\_verdict}, \\ \text{failure\_routing}, \text{manifest\_identity} \end{array} \right).$$

Here BridgeGoal is a specification of the Chapter 9 Bridge or Return claim, not a node already proved by Chapter 8. The target sub-DAG is rooted at the last pre-Bridge boundary node. The handoff has status `bridge_input_ready` only when that boundary node is discharged at the requested context and strength and every verifier/governance obligation required before Chapter 9 passes.

**Theorem 8.101** (Verifier handoff soundness). *A bridge\_input\_ready record guarantees that Chapter 9 receives a typed, dependency-closed, artifact-closed, source-bound, quarantine-safe pre-Bridge input with its exact assumption context and route-admissible strength set. It does not prove, include as a discharged node, or presuppose the Chapter 9 Bridge or Return theorem named by BridgeGoal.*

*Proof.* The first statement unpacks Definition 8.100, Theorem 8.95, and Definition 8.98. The final Bridge and Return nodes are owned by Chapter 9 and remain new necessary dependencies downstream of the pre-Bridge boundary.  $\square$

**Remark 8.102** (No forward semantic dependency from Chapter 8). Chapter 8 may name the input type required by Chapter 9, but none of its own verifier theorems uses a Chapter 9 result as a premise. This preserves the proof-DAG order required by Chapter 1.

## 8.17 Verifier and governance kernel package

**Theorem 8.103** (v21 typed verification and governance kernel). *Within the v21 architecture, the following package is available:*

- (i) *verifier inputs are immutable, typed, source-bound records rather than unclassified assertions;*
- (ii) *statement role, evidence, invocation, mathematical verdict, conformance, workflow, and migration are independent axes;*
- (iii) *every required clause has one canonical atomic status and `not_invoked` requires positive scope exclusion;*
- (iv) *aggregate verdicts are non-compensating typed conjunctions;*
- (v) *every comparison is consumed only at a compatible canonical mode and output strength;*
- (vi) *metric ledgers govern only proved numerical discrepancies and cannot manufacture non-scalar evidence;*



- (vii) *proof-DAG nodes and edges have exact semantic types, and reachability is not proof composition;*
- (viii) *finite ordinary dependency is acyclic, while countable dependency additionally requires a well-founded rank or a certified finite-to-infinite reduction; controlled recursion is represented through separately proved theorem nodes;*
- (ix) *failure precedence identifies causal blockers without erasing other failures;*
- (x) *Known-Theorem Recovery preserves source binding, precursor scope, benchmark quarantine, and terminal validation direction;*
- (xi) *manifests and hashes support identity and replay, not mathematical truth by themselves;*
- (xii) *proposer, verifier, auditor, and proof-assistant authorities remain separate until explicit conversion and interpretation pass;*
- (xiii) *operational, artifact, semantic, and mathematical replay are strictly separated;*
- (xiv) *repairs create new immutable revisions and preserve historical verdicts;*
- (xv) *only a derivation-ready target sub-DAG licenses consumption of its root sequent at a context and strength in the route-admissible record;*
- (xvi) *Chapter 9 receives only a typed BridgeDAGInput, not an automatic Bridge theorem.*

*Proof.* Items (i)–(ii) follow from Definitions 8.3–8.5. Items (iii)–(iv) are Definition 8.13 and Theorem 8.17. Items (v)–(vi) are Theorems 8.22 and 8.26. Items (vii)–(viii) follow from Definitions 8.28, 8.38, 8.46, and 8.47, together with Proposition 8.49. Items (ix)–(x) follow from Definitions 8.52 and 8.59, and Theorem 8.61. Items (xi)–(xiv) follow from the artifact, agent, replay, and revision sections. Item (xv) is Theorem 8.95. Item (xvi) is Theorem 8.101. □

## 8.18 Explicit non-claims

The following are not asserted in this chapter.

- (i) No manifest, hash, DAG, graph path, database row, or proof-store entry proves a theorem by existence.
- (ii) No `not_invoked` status is inferred from missing data.
- (iii) No passing sibling clause, majority vote, confidence score, or consensus compensates for a rejected or required undefined clause.
- (iv) No exact comparison is automatically robust, and no metric comparison creates an actual morphism or cocycle.
- (v) No theorem-controlled nonmetric mode returns more than its named theorem.
- (vi) No metric ledger, reserve, or certificate-cost bound creates a representative, morphism, detector completeness, generator completeness, coherence, descent, reduction, source authority, or Return theorem.
- (vii) No graph reachability is theorem composition without typed compatible edges and strength control.
- (viii) No ordinary dependency cycle is accepted as self-supporting proof.

- (ix) No finite prefix certifies a countable dependency closure without a uniform schema or finite-to-infinite theorem.
- (x) No source locator or benchmark storage establishes Known-Theorem Recovery.
- (xi) No benchmark proof, known truth value, target constant, exceptional bound, conclusion-derived witness, or validation artifact enters an upstream recovery construction without independent authorization.
- (xii) No byte hash establishes cross-format semantic equivalence or mathematical truth.
- (xiii) No schema migration drops proof-relevant fields while preserving audit equivalence.
- (xiv) No proposer output, AI assertion, human approval, agent consensus, or audit record is theorem evidence without verifier conversion.
- (xv) No verified formal statement proves the intended informal claim without a passing formalization and interpretation route.
- (xvi) No operational replay implies artifact, semantic, or mathematical verification.
- (xvii) No status silently inherits across proof-relevant revisions.
- (xviii) No repair overwrites a historical run or verdict.
- (xix) No typed-DAG closure licenses a conclusion stronger than the route-admissible strength set.
- (xx) No conditional node or route is consumed as unconditional while its propagated assumption context is nonempty.
- (xxi) No `bridge_input_ready` handoff is itself a unified Bridge or external Return theorem.
- (xxii) No Chapter 10 benchmark or validation result is imported upstream into Chapters 1–9 as proof data.

## 8.19 Provenance and status

This chapter promotes into the v21 Main body the minimum verifier kernel of the v20 typed-audit chapter, Foundation Appendices F–G, and the agent, Certificate-DAG, proof-store, quarantine, replay, and immutable-history rules of Part V. The detailed schemas and platform workflows remain in their appendices. The Main-body contribution is the canonical typed closure needed by Chapter 9.

**Canonical verifier definitions.** Definitions [8.3](#), [8.4](#), [8.5](#), [8.7](#), [8.9](#), [8.10](#), [8.11](#), [8.13](#), [8.14](#), [8.16](#), [8.19](#), [8.20](#), [8.21](#), [8.23](#), [8.24](#), [8.25](#), and [8.27](#).

**Proof-DAG and closure definitions.** Definitions [8.28](#), [8.29](#), [8.30](#), [8.31](#), [8.32](#), [8.34](#), [8.36](#), [8.38](#), [8.39](#), [8.40](#), [8.42](#), [8.44](#), [8.46](#), [8.47](#), [8.48](#), [8.50](#), [8.93](#), [8.94](#), [8.97](#), and [8.98](#).

**Governance, quarantine, manifest, agent, replay, and revision definitions.** Definitions [8.52](#), [8.53](#), [8.56](#), [8.57](#), [8.58](#), [8.59](#), [8.60](#), [8.63](#), [8.64](#), [8.66](#), [8.67](#), [8.68](#), [8.69](#), [8.71](#), [8.73](#), [8.74](#), [8.75](#), [8.78](#), [8.80](#), [8.82](#), [8.83](#), [8.86](#), [8.88](#), [8.89](#), [8.91](#), and [8.100](#).

**Core logical consequences.** Propositions [8.6](#), [8.8](#), [8.37](#), [8.33](#), [8.43](#), [8.45](#), [8.49](#), [8.54](#), [8.55](#), [8.62](#), [8.65](#), [8.70](#), [8.72](#), [8.76](#), [8.77](#), [8.79](#), [8.81](#), [8.87](#), [8.90](#), and [8.99](#); Theorems [8.17](#), [8.22](#), [8.26](#), [8.41](#), [8.51](#), [8.61](#), [8.84](#), [8.92](#), [8.95](#), [8.101](#), and [8.103](#).

**Explicit exclusions.** Subsection [8.18](#).

## 9 Chapter 9: Unified HT–UB Bridge and Return Calculus

**Chapter 9 contract.** This chapter promotes only the generic, repeatedly reusable composition kernel of Appendices HT and UB into the v21 Main body. It does not promote an Iwasawa, Serrin, Tachikawa, Leopoldt, Navier–Stokes, arithmetic, analytic, or other source-specific theorem. Such a theorem remains an external adapter, source binding, or Return theorem and must enter through a separately verified Chapter 8 proof-DAG node.

A Chapter 8 record with status `bridge_input_ready` supplies a closed pre-Bridge boundary, not a Bridge conclusion. Chapter 9 owns the realization, component composition, rigidity, replacement, and final Return nodes. Every undischarged assumption remains in the route context, every component retains its downward-closed strength set, and no local pass compensates for a missing source, coefficient, coherence, reduction, descent, rigidity, or Return edge.

The canonical logical shape is

$$\text{external source} \longrightarrow \text{Corep} \longrightarrow \text{Core}_{\text{DG}} \longrightarrow \text{HT/UB transfer} \longrightarrow \text{external Return}.$$

Every arrow is optional only through an immutable scope exclusion and only when the requested conclusion does not consume it.

### 9.1 Scope, partial internalization, and ownership

Chapter 9 consumes the final-audited Chapters 1–8. In particular:

- (i) Chapters 2–4 supply the repaired persistence, finite-detector, finite-grid, and persistence-to-diagram routes;
- (ii) Chapter 5 supplies direct/inverse tower typing, finite-state and Noetherian reduction, derived-limit control, and apex agreement;
- (iii) Chapter 6 supplies record atlases, Restart, retained-feature coverage, and the DescentInput handoff;
- (iv) Chapter 7 supplies finite diagram reconstruction, projective-generator detection, finite Morita, strict finite descent, and global zero/isomorphism certificates;
- (v) Chapter 8 supplies the verifier datum, assumption-context propagation, necessary proof-DAG closure, source quarantine, artifact identity, failure routing, and route-admissible strength set.

**Definition 9.1** (Partial internalization policy). The v21 Main internalizes an HT/UB statement only when it is:

- (i) independent of a named external problem;
- (ii) stated on explicitly typed source and target categories or object classes;
- (iii) accompanied by complete hypotheses, direction, proof or registered classical input, failure boundary, strength set, and artifact policy;
- (iv) compatible with the dual-Core constitution and Chapter 8 verifier;
- (v) conservative over the final-audited Chapters 1–8.

A domain-specific control theorem, regularity theorem, comparison theorem, special-value formula, or open-problem claim remains outside the Main body.

**Proposition 9.2** (Conservative promotion criterion). *Internalizing a generic HT/UB component under Definition 9.1 does not strengthen any Chapter 1–8 theorem. It only licenses a new typed composition when all component hypotheses and Chapter 8 predecessor obligations pass.*

*Proof.* The component conclusion is conditional on its displayed inputs and is consumed downstream only through a new Chapter 9 node. No prior theorem, status, or non-claim is changed.  $\square$

**Remark 9.3** (Appendix sovereignty). Appendix HT remains the complete research spine for HC, PE, SS, RL, and IW/Return. Appendix UB remains the complete reusable component library. Chapter 9 is the Main-body normal form of their generic composition grammar, not a replacement for domain adapters, worked cases, or successor-strength theorems.

## 9.2 Universal Bridge datum and route classifier

**Definition 9.4** (Universal v21 Bridge datum). A universal Bridge datum is

$$\text{Bridge} = (S, R, C, T, \Sigma, \Gamma, \mathcal{D}, \mathcal{A}),$$

where:

- (i)  $S$  is the source semantics: category or object class, morphisms, equivalence relation, coefficient and variance policy, and target predicate;
- (ii)  $R$  is the source realization and every coefficient, completion, normalization, or observation component needed to create a Core-readable object;
- (iii)  $C$  is the selected  $\text{Core}_P$ ,  $\text{Core}_{DG}$ , tower, atlas, and HT/UB route, represented by necessary Chapter 8/9 nodes and edges;
- (iv)  $T$  is the Return target, its category, equivalence relation, target identifier, and source-domain interpretation theorem;
- (v)  $\Sigma$  is the requested Return strength or downward-closed strength set;
- (vi)  $\Gamma$  is the complete propagated assumption context;
- (vii)  $\mathcal{D}$  is the typed defect and failure-leaf stack;
- (viii)  $\mathcal{A}$  is the immutable source, theorem, artifact, replay, quarantine, and manifest bundle.

The shorter notation  $(S, R, C, T, \Sigma, \mathcal{A})$  is permitted only when  $\Gamma$  and  $\mathcal{D}$  are explicitly recoverable from the Chapter 8 handoff and Chapter 9 manifest.

**Definition 9.5** (Route classifier). Every Bridge datum declares exactly one principal route class:

$$\begin{array}{lll} P\_route, & DG\_route, & P\_plus\_DG\_route, \\ HT\_internal\_route, & & external\_validation\_route. \end{array}$$

The classification records which chapters and components are necessary, not which vocabulary appears in the exposition. An external-validation route may validate a theorem already proved independently but may not leak the benchmark conclusion into its precursor package.

**Theorem 9.6** (Route classification and ownership). *For every well-typed Bridge datum, the principal route class and the set of necessary component owners are uniquely determined by the declared source and target types, the Core-readable intermediate objects, and the requested strength, up to replacement of a component by a separately certified theorem with the same input type and an output strength containing the consumed strength.*

*Proof.* The Chapter 8 edge registry fixes the source and target type of every consumed edge. The route class is determined by whether the necessary predecessor closure contains  $\text{Core}_P$ ,  $\text{Core}_{DG}$ , both, neither but an HT internal object, or only a quarantined validation branch. Any admissible replacement must have the same consumed interface type and sufficient output strength; otherwise the necessary-edge closure changes and defines a different route.  $\square$

**Definition 9.7** (Target-relative Bridge strength preorder). Return strength is compared only relative to a fixed Return target record, or along an explicitly registered forgetful functor between target records. On a single object-reconstruction spine one may have

$$\Sigma_0 \preceq \Sigma_{\text{pred}} \preceq \Sigma_{\text{inv}} \preceq \Sigma_{\text{obj}} \preceq \Sigma_{\text{mor}} \preceq \Sigma_{\text{ex}} \preceq \Sigma_{\text{der}} \preceq \Sigma_{\text{str}},$$

meaning respectively no theorem-bearing Return, predicate Return, a specified functorial invariant, object Return up to the declared equivalence, morphism Return, exact/Yoneda Return, bounded-derived Return, and structured Return such as monoidal, marked, integral, completed, or multiplicative Return.

This display is not a universal linear order on all partial targets. Determinants, Fitting ideals, truncations, graded pieces, stable classes, and pseudo-isomorphism classes are distinct target records and are incomparable unless a named theorem or forgetful functor relates them. Every comparison of strength therefore records its target identity and the arrow witnessing the comparison. No reverse arrow is automatic.

**Proposition 9.8** (No comparison of unrelated partial Returns). *Let  $T_1, T_2$  be two partial Return target records. In the absence of a registered target functor or theorem*

$$T_1 \longrightarrow T_2 \quad \text{or} \quad T_2 \longrightarrow T_1,$$

*neither Return strength is above the other. In particular, determinant, ideal, graded-piece, stable-class, and pseudo-isomorphism Returns do not share a canonical scalar ceiling merely because they are all called invariants.*

*Proof.* Strength comparison is the existence of a certified forgetting or reconstruction arrow between the exact target records. Without such an arrow, the outputs have different semantic types and neither entails the other.  $\square$

### 9.3 Generic component record and compatibility

**Definition 9.9** (Universal Bridge component). A Chapter 9 component is a record

$$\mathfrak{c} = (C_{\text{src}}, C_{\text{tgt}}, F, \text{Hyp}, \text{Pres}, \text{Out}, \text{Cond}(\mathfrak{c}), \mathfrak{S}_{\mathfrak{c}}, \text{Fail}, \mathcal{A}_{\mathfrak{c}}),$$

where  $F$  is the construction, Hyp its complete hypotheses, Pres the proved preservation/reflection statement, Out the exact output type,  $\mathfrak{S}_{\mathfrak{c}}$  its downward-closed strength set, and Fail its local failure leaves. A component is not certified merely because a theorem with a similar name exists.

**Definition 9.10** (Component compatibility). Components  $\mathfrak{c}_1, \mathfrak{c}_2$  are composable at strength  $\sigma$  when:

- (i) the output type and variance of  $\mathfrak{c}_1$  match the input type and variance consumed by  $\mathfrak{c}_2$ ;
- (ii) coefficient, window, amplitude, completion, equivalence, and source identities agree or are connected by separately certified components;
- (iii) every assumption is propagated or discharged by a named theorem;
- (iv) every parallel route has the required strict or homotopy-coherent cell;
- (v)  $\sigma \in \mathfrak{S}_{\mathfrak{c}_1} \cap \mathfrak{S}_{\mathfrak{c}_2}$ .

**Theorem 9.11** (Component compatibility theorem). *Compatible certified components compose to a certified component whose assumption context is the propagated union after registered discharges and whose strength set is the intersection of the component strength sets. No stronger output or smaller assumption context is licensed.*

*Proof.* Compose the proved functors or directional implications at their matched interface. Chapter 8 assumption-transport edges determine the resulting context, and the route-admissible strength rule gives the intersection.  $\square$

## 9.4 Source realization and coefficient components

**Definition 9.12** (Source realization component). A source realization component records a construction

$$R : C_{\text{source}} \longrightarrow C_{\text{readable}}$$

and independently records construction provenance, semantic fidelity, functoriality, exactness, coefficient behavior, information class, and the external predicate it preserves or reflects. Purpose-faithfulness for one predicate is not categorical faithfulness.

**Theorem 9.13** (Faithfully flat zero and isomorphism reflection). *Let  $A \rightarrow B$  be faithfully flat. For an  $A$ -module  $M$ ,*

$$M = 0 \iff M \otimes_A B = 0.$$

*For a morphism of bounded complexes of  $A$ -modules whose cohomology modules are defined in the declared category,*

$$f \text{ is a quasi-isomorphism} \iff f \otimes_A B \text{ is a quasi-isomorphism.}$$

*Thus faithfully flat base change is a zero- and cone-reflecting coefficient component at the corresponding module or bounded-derived strength.*

*Proof.* Faithful flatness reflects zero modules. Flatness gives  $H^q(C \otimes_A B) \cong H^q(C) \otimes_A B$ . Apply zero reflection to the cohomology of the cone of  $f$ .  $\square$

**Theorem 9.14** (Derived Nakayama component). *Let  $(A, \mathfrak{m}, k)$  be a commutative local ring and let  $E \in D^b(A)$  have finitely generated cohomology. Then*

$$E \simeq 0 \iff E \otimes_A^{\mathbf{L}} k \simeq 0.$$

*Consequently residue-field vanishing is zero-reflecting on this bounded finite cohomology class.*

*Proof.* If  $E \neq 0$ , choose the largest  $q$  with  $H^q(E) \neq 0$ . The hyper-Tor spectral sequence has an edge  $H^q(E) \otimes_A k$  in total degree  $q$ , and Nakayama gives this module nonzero. Hence the derived fiber is nonzero. The converse is immediate.  $\square$

**Corollary 9.15** (Finite semilocal residue atlas). *Let  $A$  be commutative semilocal with maximal ideals  $\mathfrak{m}_1, \dots, \mathfrak{m}_s$ , and let  $E \in D^b(A)$  have finitely generated cohomology. Then*

$$E \simeq 0 \iff E \otimes_A^{\mathbf{L}} k(\mathfrak{m}_j) \simeq 0 \text{ for all } j.$$

*Proof.* A nonzero finitely generated cohomology module has support containing a maximal ideal. Localize there and apply derived Nakayama.  $\square$

**Corollary 9.16** (Noetherian completion base-change component). *For a Noetherian local ring  $(A, \mathfrak{m})$ , the adic completion map  $A \rightarrow \widehat{A}$  is faithfully flat. Hence the coefficient base-change functor*

$$E \mapsto E \otimes_A^{\mathbf{L}} \widehat{A}$$

*reflects zero objects and quasi-isomorphisms; because  $\widehat{A}$  is flat, its cohomology is obtained by ordinary flat base change.*

*This component is completion base change. It is not identified with the derived  $\mathfrak{m}$ -adic completion functor  $\mathbf{L}\Lambda_{\mathfrak{m}}$  without an additional comparison theorem and the required derived-completeness or finiteness hypotheses. It also does not return integral information from a nonfaithful coefficient observation.*

**Remark 9.17** (Completion base change is not derived completion by notation). The target object  $E \otimes_A^{\mathbf{L}} \widehat{A}$  and the derived inverse limit defining  $\mathbf{L}\Lambda_{\mathfrak{m}}(E)$  may agree on an admitted Noetherian finite or derived-complete lane, but that agreement is itself a Bridge component. The word “completion” does not supply it automatically.

**Definition 9.18** (Bockstein and universal-coefficient boundary component). A Bockstein/UCT component consists of a specified coefficient triangle or short exact coefficient sequence, the induced long exact sequence or spectral sequence, all connecting maps, convergence data, and the exact edge or graded piece consumed by the route. It Returns only the displayed boundary, edge, or extension information. Vanishing after reduction modulo one coefficient is not an integral Return without a reflection or rigidity theorem.

## 9.5 Core-P and Core-DG route components

**Definition 9.19** (Core-P component). A Core-P component consumes a Chapter 2–4 object or certificate at one exact profile stage and may output repaired zero/nonzero, a finite detector witness, a complete grid diagram, barcode data, rank/Moebius data, or an eligible higher-edge conclusion. Its comparison mode, stage, endpoint, completeness, and zero-reflection fields remain independent.

**Definition 9.20** (Core-DG component). A Core-DG component consumes a finite-category diagram or bounded derived object together with the exact Chapter 7 generator, Morita, cone, descent, or globalization hypotheses. It may Return zero, isomorphism, diagram object, morphism, exact/Yoneda, bounded-derived, or descended-object information only at the strength proved by Chapter 7.

**Theorem 9.21** (P-to-DG Bridge component). *On a declared endpoint-complete finite grid  $\Gamma$ , the Chapter 4 exact interface*

$$\mathrm{Res}_{\Gamma} : \mathrm{Pers}_k^{\Gamma}(W) \xrightarrow{\sim} \mathrm{Rep}_k(A_{[m]})$$

*composes with the Chapter 7 finite projective generator. Hence, for  $M \in \mathrm{Pers}_k^{\Gamma}(W)$ ,*

$$M \simeq 0 \iff \mathrm{Res}_{\Gamma}(M) = 0 \iff \mathbf{R}\mathrm{Hom}(G_{A_{[m]}}, \mathrm{Res}_{\Gamma}(M)) \simeq 0.$$

*Actual persistence morphisms similarly pass to diagram morphisms; barcode matchings or scalar rank data do not manufacture those morphisms.*

*Proof.* Use the Chapter 4 exact equivalence and the Chapter 7 generator detection theorem. Functoriality on morphisms is part of the exact finite-grid interface.  $\square$



## 9.6 HT transfer spine

**Definition 9.22** (HT sovereign components). The Main-body HT spine contains five typed component classes:

- (i) **HC**: zero-shift filtered contractions and exact persistent homology preservation;
- (ii) **PE**: finite persistence/Ext, object, morphism, exact, cone, triangle, and separately typed enhanced reconstruction;
- (iii) **SS**: source-bound page, differential, convergence, filtration, and extension transfer;
- (iv) **RL**: inverse systems, Roos/derived-limit edge, stable image, finite-to-infinite reduction, and apex comparison;
- (v) **IW/Return**: native target comparison, equivalence relation, rigidity, defects, and final source interpretation.

The names are component classes, not automatic invocations.

**Theorem 9.23** (HC protected persistence component). *Let finite filtered chain complexes  $(T, \partial)$  and  $(C, d)$  admit maps*

$$T \xrightleftharpoons[P]{I} C, \quad PI = \text{id}_T, \quad \text{id}_C - IP = dH + Hd,$$

*with  $I, P, H$  preserving every filtration sublevel. Then the induced persistence modules  $\text{Pers}_i(T)$  and  $\text{Pers}_i(C)$  are naturally isomorphic in every degree. Ordinary unfiltered chain-homotopy equivalence is not a substitute.*

*Proof.* Restrict the contraction identities to every filtration sublevel. They give compatible homotopy equivalences and hence compatible homology isomorphisms.  $\square$

**Definition 9.24** (PE component ladder). The PE ladder records separately:

$$\Sigma_{\text{barcode}}, \quad \Sigma_{\text{obj}}, \quad \Sigma_{\text{mor}}, \quad \Sigma_{\text{ex}}, \quad \Sigma_{\text{der}}, \quad \Sigma_{\text{enhanced}}.$$

Finite-grid object or morphism Return does not imply marked  $A_\infty$ , deformation, integral-boundary, or other enhanced Return without its own source binding and rigidity theorem.

**Definition 9.25** (SS component). An SS component records actual pages  $E_r^{p,q}$ , differentials, the page from which a comparison theorem is valid, convergence, the filtration on the abutment, and extension data. It may Return concentration, an edge/five-term statement, a nonzero differential, or an abutment statement at the registered strength. Degeneration alone does not identify the unfiltered target.

**Definition 9.26** (RL component). An RL component records the inverse or direct system, actual transition maps, ambient category,  $\varprojlim$ ,  $R^1\varprojlim$  or the declared finite-state/Noetherian substitute, and apex comparison. Pagewise or finite stage agreement does not remove an undeclared derived-limit or apex layer.

**Theorem 9.27** (HT transfer-spine composition). *Let a finite Chapter 9 proof-DAG invoke any ordered subset of*

$$\{\text{HC}, \text{PE}, \text{SS}, \text{RL}, \text{Return}\}.$$

*If every local theorem, typed interface, assumption transport, coherence cell, finite-to-infinite condition, and final native comparison passes, then the selected internal target is certified at the intersection of the invoked component strength sets. No noninvoked module strength is inferred.*

*Proof.* Apply Theorem 9.11 inductively over a topological order of the finite DAG.  $\square$

## 9.7 Finite-state, Noetherian, descent, and globalization components

**Definition 9.28** (Tail component selector). Every infinite or all-degree route selects exactly one certified tail mechanism for each infinite quantifier it consumes:

eventual\_constancy,   finite\_length,   noetherian\_generation,  
mittag\_leffler,   finite\_state,   external\_reduction.

Different infinite quantifiers may require different mechanisms.

**Theorem 9.29** (Tail component soundness). *A Chapter 5 tail theorem may discharge only the infinite predicate named in its reduction record. In particular, eventual constancy, stable-image reduction,  $R^1\varprojlim = 0$ , and apex agreement remain separate conclusions unless one theorem proves their conjunction.*

*Proof.* These predicates have different source types and counterexamples separate them, as recorded in Chapter 5. □

**Definition 9.30** (Descent/globalization component). A descent component consumes a Chapter 6 descent-input record (of type DescentInput) together with a Chapter 7 effectivity lane. It records actual local objects, transition maps, cocycle or higher cells, ambient model, effectivity theorem, descended object, and the exact joint-conservativity theorem used by any global zero or isomorphism conclusion.

**Theorem 9.31** (Finite descent Bridge). *In the strict finite Alexandrov lane of Chapter 7, a finite strict descent datum produces a global object. If the restriction family is jointly conservative, local generator-zero certificates imply global object zero, and local cone-zero certificates imply that the descended global morphism is an isomorphism. Outside that lane, the same conclusion requires a separately named effective descent and joint-conservativity theorem.*

*Proof.* Apply the Chapter 7 strict descent construction and its global zero/isomorphism certificates. □

## 9.8 Monoidal, duality, and determinant components

**Theorem 9.32** (Field Kunnet component). *For bounded complexes of finite-dimensional vector spaces over a field  $k$ ,*

$$H^n(C \otimes_k D) \cong \bigoplus_{p+q=n} H^p(C) \otimes_k H^q(D).$$

*Thus tensor-product cohomology is exactly recoverable from the graded cohomology objects in this field-bounded regime. No Tor-free statement over a general ring is inferred.*

*Proof.* Every short exact sequence of vector spaces splits, so each bounded complex is homotopy equivalent to its cohomology with zero differential plus a contractible summand. Tensor and take cohomology. □

**Theorem 9.33** (Perfect duality and invertible tensor reflection). *Let  $P$  be a perfect complex over a field  $k$ . Then*

$$P \simeq 0 \iff \mathbf{R}\mathrm{Hom}_k(P, k) \simeq 0.$$

*If  $L$  is tensor-invertible in a monoidal category, then*

$$X \simeq 0 \iff X \otimes L \simeq 0,$$

*and a morphism is an isomorphism exactly when its tensor with  $L$  is an isomorphism.*

*Proof.* Finite-dimensional duality reflects nonzero cohomology. Tensoring by an invertible object has an inverse functor.  $\square$

**Definition 9.34** (Determinant-level partial Return). A determinant component Returns a determinant line, determinant class, Fitting-type invariant, or characteristic-type invariant only when its source category and determinant functor are declared and the comparison theorem is proved. Determinant agreement is  $\Sigma_{\text{inv}}$  or another explicitly ordered partial strength; it does not reconstruct the object, morphism, or exact sequence without rigidity.

## 9.9 Coherence and rigidity

**Definition 9.35** (Bridge coherence record). A Bridge coherence record contains every comparison cell required when two necessary routes reach the same object. It declares strict equality, specified isomorphism, or a higher cell in a named ambient model, together with all boundary equations and finite generator or explicit verification data. Metric proximity is not a coherence cell.

**Theorem 9.36** (Finite projective-presentation rigidity). *Let  $\mathcal{A}_{\text{fp}}$  be an abelian category with a projective generator  $G$ , and assume every object  $X$  admits a finite presentation*

$$G^{\oplus r_1} \longrightarrow G^{\oplus r_0} \longrightarrow X \longrightarrow 0.$$

*Let  $\mathcal{B}$  be an abelian category and let  $F, F' : \mathcal{A}_{\text{fp}} \rightarrow \mathcal{B}$  be right-exact additive functors. A morphism  $\eta_G : F(G) \rightarrow F'(G)$  compatible with the  $\text{End}_{\mathcal{A}_{\text{fp}}}(G)$ -actions extends uniquely to a natural transformation  $\eta : F \Rightarrow F'$ . If  $\eta_G$  is an isomorphism, then  $\eta_X$  is an isomorphism for every  $X \in \mathcal{A}_{\text{fp}}$ .*

*In particular the theorem applies to the finite-dimensional module and finite diagram categories used in Chapter 7. It is not asserted for arbitrary objects of an abelian category merely from the existence of a generator.*

*Proof.* For a chosen finite presentation of  $X$ , action compatibility determines the maps on the two finite sums of  $G$ . Right exactness identifies  $F(X)$  and  $F'(X)$  as the corresponding cokernels and therefore induces  $\eta_X$ . The standard comparison of finite presentations, using projectivity of  $G$ , shows independence and naturality. Repeating the construction with  $\eta_G^{-1}$  proves the isomorphism statement.  $\square$

**Definition 9.37** (Rigidity package). A rigidity package names the observation or reconstruction functor, its essential source class, the external equivalence relation, a finite generator or presentation theorem, normalization data, and the theorem that promotes observed equality or internal equivalence to the requested external strength. Without rigidity, the Return ceiling is the observation strength.

**Proposition 9.38** (No promotion without rigidity). *If the selected observation functor identifies two non-equivalent source objects and no rigidity theorem excludes that pair on the declared source class, no downstream route consuming only that observation can Return exact source objects or morphisms.*

*Proof.* Every downstream construction factors through the same observation and hence cannot distinguish the pair. Exact Return would contradict their declared non-equivalence.  $\square$

## 9.10 Return targets, partial Return, and replacement

**Definition 9.39** (Return target record). A Return target record contains the target category or object class, target identifier, target predicate or object, equivalence relation, requested strength, source-domain interpretation theorem, all remaining assumptions, and immutable artifacts. “The same target” means equality of this complete record, not similarity of prose.

**Definition 9.40** (Partial Return). A partial Return explicitly names the returned quotient of information, such as a predicate, invariant, truncation, graded piece, ideal, determinant, semisimplification, stable class, or pseudo-isomorphism class. It also names the exact target functor, target equivalence relation, residual defect, and the theorem preventing promotion to a stronger strength. Two partial Returns are compared only by the target-relative rule of Definition 9.7; sharing the word “invariant” does not make them comparable.

**Theorem 9.41** (Partial Return theorem). *If every necessary component of a route passes for a target functor  $Q : C_{\text{target}} \rightarrow Q$ , but the route lacks rigidity for objects of  $C_{\text{target}}$ , then the route may Return  $Q(Y)$  at the certified invariant strength and may not Return  $Y$  itself.*

*Proof.* The component composition proves the statement in  $Q$ . The absent rigidity theorem is exactly the missing reverse implication needed to recover  $Y$ .  $\square$

**Definition 9.42** (Replacement or regularized target datum). A replacement datum contains a new target identifier  $T_{\text{reg}}$ , a source-certified theorem from the audited internal conclusion to  $T_{\text{reg}}$ , the relation between  $T_{\text{reg}}$  and the original target, all pole, exceptional-zero, singular-specialization, correction, or contact data, and an explicit statement that the exact original target was not Returned at the requested strength.

**Theorem 9.43** (Regularized-target soundness). *A certified replacement datum may yield the primary result*  

$$\text{regularized\_return}$$

*for  $T_{\text{reg}}$ . It does not change a failed or undefined exact-target node into pass and does not inherit the exact target identifier or strength.*

*Proof.* The replacement theorem has a different target proposition. Its proof may be valid independently, but logical validity cannot identify two different target records without an additional theorem.  $\square$

## 9.11 Composable Chapter 9 Bridge DAG

**Definition 9.44** (Composable unified Bridge DAG). A composable Chapter 9 Bridge DAG extends a Chapter 8 BridgeDAGInput by adding only downstream nodes owned by Chapter 9: source realization, coefficient passage, Core-P/Core-DG interface, HT/UB components, reduction, descent, coherence, rigidity, replacement when invoked, and final Return. It contains:

- (i) one certified local theorem record for every necessary node;
- (ii) one type-, variance-, assumption-, and strength-correct edge record;
- (iii) every required parallel-route coherence cell;
- (iv) a complete source binding and benchmark-quarantine cut;
- (v) artifact closure and semantic replayability;
- (vi) a final Return target record and external interpretation theorem.

The final Return node is not part of the incoming Chapter 8 handoff.

**Definition 9.45** (Unified Bridge defect stack). The defect stack is

$$\mathcal{D}_{\text{Bridge}} = (D_{\text{src}}, D_{\text{real}}, D_{\text{coeff}}, D_P, D_{DG}, D_{HT}, D_{\text{tail}}, D_{\text{desc}}, D_{\text{coh}}, D_{\text{rig}}, D_{\text{repl}}, D_{\text{return}}, D_{\text{cert}}).$$

Each entry is a typed family of failure leaves. A nonzero terminal Type IV defect is recorded only after a valid comparison artifact exists. Source mismatch, ill-typing, missing realization, and missing morphism precede that diagnostic in repair order.

**Definition 9.46** (Bridge primary result). For a fixed target record, the Chapter 9 primary result is exactly one of:

`reject`, `undefined`, `obstructed`, `partial_return`, `regularized_return`, `return`.

This is a Bridge-result sort, not a replacement for the Chapter 8 atomic audit status. Every node and edge still has its four-valued atomic status.

## 9.12 Unified Bridge Closure Theorem

**Theorem 9.47** (Unified HT–UB Bridge Closure). *Let Bridge be a composable unified Bridge DAG with requested context  $\Gamma$  and strength  $\sigma$ . Suppose:*

- (1) *source binding and quarantine pass;*
- (2) *source realization and every coefficient/base-change component pass;*
- (3) *every invoked Core-P and Core-DG node passes;*
- (4) *every invoked HT/UB local theorem and interface passes;*
- (5) *every infinite quantifier has a certified finite-to-infinite or derived-limit reduction;*
- (6) *every overlap, higher cell, and route-coherence obligation passes;*
- (7) *every invoked descent and global construction passes;*
- (8) *every promotion beyond observation strength has a passing rigidity theorem;*
- (9) *the final Return theorem passes for the exact target record;*
- (10) *artifact closure, semantic replay, and immutable manifest identity pass;*
- (11) *all propagated assumptions are contained in  $\Gamma$ ;*
- (12)  *$\sigma$  belongs to the intersection of all necessary node and edge strength sets.*

*Then the final Return node is verified at sequent*

$$\Gamma \vdash \text{Return}_\sigma(S, T),$$

*and the primary result is `return`. Every registered weakening of  $\sigma$  is also available. No stronger strength and no smaller assumption context is licensed.*

*Proof.* The incoming Chapter 8 handoff supplies a closed, source-bound, quarantine-safe pre-Bridge predecessor. Apply Theorem 9.11 in a topological order of the Chapter 9 DAG. Tail, descent, coherence, and rigidity hypotheses discharge the corresponding nonlocal gaps. The final Return theorem converts the audited internal target into the exact external target. Assumption contexts propagate by Chapter 8 edge semantics, and strengths intersect by the route-admissible rule. The Chapter 8 typed-DAG closure theorem then verifies the final node.  $\square$

**Corollary 9.48** (Typed Bridge-DAG theorem). *The conjunction*

$$\left( \bigwedge_{v \in V_{\text{req}}} \text{Pass}(v) \right) \wedge \left( \bigwedge_{e \in E_{\text{req}}} \text{Closed}(e) \right)$$

*licenses a Return only after the assumptions, target identity, and requested strength satisfy Theorem 9.47.*

**Theorem 9.49** (Non-composition theorem). *The existence of an AK-side certificate and an external equivalence or theorem does not by itself imply an external Return. In general the following are independent necessary arrows:*

*source realization,      Core/HT transfer,*  
*finite-to-infinite or descent reduction,      source-domain Return.*

*Omitting any arrow leaves the corresponding Return route undefined or capped at a weaker internal strength.*

*Proof.* The arrows have different source and target types. An AK certificate has no source-domain semantics without realization and Return; a finite certificate has no infinite conclusion without reduction; an external theorem has no connection to the audited Core object without a typed interface. None is a logical consequence of the others in the general architecture.  $\square$

**Theorem 9.50** (Return-ceiling and failure-leaf theorem). *Let  $R$  be a proposed route. Its admissible Return set is*

$$\mathfrak{S}_R = \bigcap_{x \in \text{Req}(R)} \mathfrak{S}_x$$

*together with the propagated assumption context  $\text{Cond}(R)$ . If the requested strength is outside this set, the route may emit the strongest certified partial Return but not the request. If closure fails, every first causal failure is localized to a typed leaf of  $\mathcal{D}_{\text{Bridge}}$ ; incomparable independent failures are all retained. A unique scalar ceiling is asserted only when  $\mathfrak{S}_R$  has a unique maximum.*

*Proof.* This is the Chapter 8 route-intersection theorem applied to the Chapter 9 component registry, together with dependency-based failure precedence.  $\square$

### 9.13 A complete conservative external-to-external route

**Definition 9.51** (Finite filtered-chain source class). Fix a field  $k$ , a finite source-breakpoint set

$$K_{\text{src}} = \{a_0 < \cdots < a_r\},$$

a degree  $i$ , a right-open window  $W = [u, u')$ , and  $\tau > 0$ . Let  $\mathcal{F}_{K_{\text{src}}}$  be the class of finite-dimensional chain complexes with an increasing filtration that is constant on every open stratum between consecutive source breakpoints. For  $C \in \mathcal{F}_{K_{\text{src}}}$ , write

$$P_i(C)(t) = H_i(F_t C).$$

The external long-span predicate is

$$\mathcal{L}_{i,W,\tau}(C) : \quad \text{im}(H_i(F_s C) \rightarrow H_i(F_t C)) = 0 \quad \text{for all } s \leq t \text{ in } W \text{ with } t - s > \tau.$$

Endpoint decorations are read from the actual persistence module; no value at the excluded right endpoint  $u'$  is inserted.

**Proposition 9.52** (Common endpoint-complete atom grid from source breakpoints). *For the source class of Definition 9.51, set*

$$K_W = \{u\} \cup (K_{\text{src}} \cap W)$$

*and form the Chapter 4 singleton/open-stratum atom decomposition, including the bounded-terminal or cofinal-tail policy. Its enriched probe grid*

$$\Gamma_W = \{\gamma_0 < \cdots < \gamma_\ell\}$$

*is endpoint-complete simultaneously for every windowed persistent-homology profile in the source class and for every profile obtained from it by hard threshold deletion. In particular, threshold deletion removes interval summands but creates no new endpoint.*

*Proof.* The filtered chain complex and all sublevel maps are constant on each source stratum, hence persistent homology can change only at a source breakpoint or at the declared window boundary. The Chapter 4 atom decomposition separates each breakpoint singleton from its adjacent open strata and therefore retains endpoint decoration. Hard deletion only discards interval summands, so the remaining endpoints are a subset of the same finite endpoint set.  $\square$

**Lemma 9.53** (Long-span criterion). *For an interval-decomposable constructible persistence module  $P$  on  $W$ ,*

$$T_\tau^{\text{bar}}(P) = 0 \iff \text{im}(P(s) \rightarrow P(t)) = 0 \quad \text{whenever } s \leq t \text{ and } t - s > \tau.$$

*Proof.* An interval summand contributes to the image exactly when it contains both parameters. A finite interval contains two parameters more than  $\tau$  apart exactly when its length is greater than  $\tau$ ; an unbounded interval also does. Sum over the interval decomposition.  $\square$

**Theorem 9.54** (Complete HC–Core–P–Core–DG–HT/UB Return route). *Let  $C \in \mathcal{F}_{K_{\text{src}}}$ . Suppose a zero-shift filtered contraction  $T \rightleftharpoons C$  as in Theorem 9.23 is supplied, and let*

$$M = T_\tau^{\text{bar}}(W_W P_i(T)).$$

*Use the common endpoint-complete enriched atom grid  $\Gamma_W$  of Proposition 9.52, with  $\ell + 1$  ordered atoms, and set*

$$E_{\Gamma_W} = \text{Res}_{\Gamma_W}(M) \in \text{Rep}_k(A_{[\ell]}), \quad G = G_{A_{[\ell]}}.$$

*Then the following are equivalent:*

- (i) *the external predicate  $\mathcal{L}_{i,W,\tau}(C)$  holds;*
- (ii)  $T_\tau^{\text{bar}}(W_W P_i(C)) = 0$ ;
- (iii)  $M = 0$ ;
- (iv)  $E_{\Gamma_W} = 0$ ;
- (v)  $\mathbf{RHom}(G, E_{\Gamma_W}) \simeq 0$ .

*Consequently the route*

$$C \xrightarrow{\text{HC}} T \xrightarrow{\text{Corep}} M \xrightarrow{\text{CoreDG}} E_{\Gamma_W} \xrightarrow{\text{PE/UB generator}} \mathbf{RHom}(G, E_{\Gamma_W}) \xrightarrow{\text{Return}} \mathcal{L}_{i,W,\tau}(C)$$

*is a complete external-to-external Bridge at predicate strength  $\Sigma_{\text{pred}}$ . The repaired persistence module and its finite diagram remain available as certified internal object-level outputs up to their declared equivalences. They are not an object Return to the source category, and the full filtered chain complex  $C$  is not reconstructed.*

*Proof.* HC gives a natural isomorphism  $P_i(T) \cong P_i(C)$ , so (ii) and (iii) are equivalent. Lemma 9.53 gives (i) $\Leftrightarrow$ (ii). Proposition 9.52 supplies the noncircular common atom grid. The Chapter 4 finite-grid equivalence gives (iii) $\Leftrightarrow$ (iv), and the Chapter 7 projective-generator theorem gives (iv) $\Leftrightarrow$ (v). All arrows are natural on the declared source class and no infinite, descent, or coefficient component is invoked. The final nonclaim follows because the route observes only persistent homology and its reconstructed diagram.  $\square$

**Corollary 9.55** (Finite replay certificate for the complete route). *Under Theorem 9.54, a finite replay object may consist of the filtered contraction matrices, finite grid, finite diagram matrices, projective-generator action, zero certificate, source binding, and Return theorem. Its certificate complexity is model-relative and does not include an unrecorded full-barcode oracle.*

**Remark 9.56** (Why this route satisfies the v21 completion condition). The source and final predicate are stated in the external semantics of a finite filtered chain complex. The route actually traverses HC, Core-P, Core-DG, PE/UB generator reconstruction, Chapter 8 verification, and a proved external Return. It is conservative: it rederives a known finite persistence fact and does not claim a new source-domain classification or an open-problem result.

**Remark 9.57** (Endpoint and threshold boundary of the complete route). The strict span condition  $t - s > \tau$  matches the Chapter 2 convention that finite bars of length at most  $\tau$  are deleted. Replacing it by  $t - s \geq \tau$ , collapsing singleton and open atoms, or sampling the excluded right endpoint changes the target predicate and requires a new Return target record.

## 9.14 Inherited v20 closure, Interface, and Return layer

The remainder of this chapter retains the closure, Interface, finite-to-infinite, known-theorem-recovery, release-boundary, and final projection material of the v20 Chapter 8 source. Its labels are migrated deterministically from the ch8 namespace to ch9. When an inherited formulation is weaker or uses an older ownership description, the v21 definitions and theorems above, together with Chapters 1–8, are normative.

## 9.15 Final role of the chapter

**Inherited closure and Interface contract.** This chapter closes the theorem-bearing main body of AK-HPDST v21.0. It introduces no theorem about an external arithmetic, geometric, analytic, PDE, or cryptographic domain merely by closure. It does, however, place the v21 finite-certificate, higher-obstruction, defect, atlas, and finite-to-infinite results into one final inference architecture and proves the internal eventually-constant reduction theorem used by that architecture.

The chapter fixes:

- (i) the proved mathematical nucleus and its exact hypotheses;
- (ii) the conditional and operational layers;
- (iii) the rejected shortcuts and immutable non-claims;
- (iv) the canonical Problem-Interface entry, realization, reduction, and Return-Theorem obligations;
- (v) the separation between main-body closure and release-package acceptance.



**Projection supplies a Core-readable shadow. Verification certifies the shadow. Return alone carries a certified statement back to the source domain.**

The governing inference rule is

The Core proves only what follows from declared types, hypotheses, and artifacts. A finite detector may compress evidence but may not invent completeness. A finite-to-infinite theorem may transport a certified finite witness but may not be replaced by sampled stabilization. An external conclusion requires its own proved Return Theorem.

The final dependency order in v21 is

- dual-Core constitution and migration discipline (Ch. 1)
- repaired persistence and metric zero transport (Ch. 2)
- finite detectors and certificate complexity (Ch. 3)
- finite-grid reconstruction and higher obstruction (Ch. 4)
- tower reduction and terminal/transient defects (Ch. 5)
- window atlases, overlap, and Restart (Ch. 6)
- finite diagram and globalization Core (Ch. 7)
- typed verification and proof-DAG governance (Ch. 8)
- unified HT-UB Bridge and Return calculus. (Ch. 9)

**Remark 9.58** (Closure is not a new external theorem). The strongest new mathematical statement in this chapter is an internal categorical reduction theorem for an eventually constant directed system with compatible apex agreement. No problem-specific external conclusion is asserted without a separately proved source-domain Return Theorem.

**Remark 9.59** (Claim registration follows proof, not conversely). The v21 Claim Register records status, conformance, comparison policy, and permitted use. Registration proves nothing and cannot increase claim strength. If a register entry conflicts with a theorem statement, proof, explicit hypothesis, stage, or non-claim, the safer weaker reading governs until a new version records and proves the repair.

**Remark 9.60** (Main-body closure and release-package closure are distinct). This chapter closes the inference paths of the theorem-bearing main body. The designation *AK-HPDST v21.0 complete release* is controlled by the Chapter 1 release contract and the Chapter 8 release registry. It additionally requires every mandatory appendix, Claim Register, toy-verification, counterexample, realization, known-theorem-recovery, and external-demonstration obligation to pass. Main-body closure does not fabricate those artifacts.

## 9.16 Canonical status vocabulary for the release

**Definition 9.61** (Release-level claim classes). Every statement consumed by the release has exactly one evidence class:

proved, conditional, operational, spec, deprecated, rejected, unknown.

A proved item is used only under its theorem hypotheses. A conditional item becomes theorem evidence only after every registered condition is discharged. Operational and specification items remain non-theorem material.

**Definition 9.62** (Claim invocation role). Every consumed item has exactly one invocation role:

`theorem_evidence`, `operational_rule`, `specification_target`, `provenance_only`.

Evidence class, conformance status, claim-level comparison policy, and runtime comparison mode are different fields and may not be substituted for one another.

**Definition 9.63** (Safe-use order). A statement may be used only at or below its proved and registered strength. In particular:

- (i) a conditional theorem instance is unavailable until all conditions and dependencies pass;
- (ii) an operational rule does not become external mathematics;
- (iii) a specification target is not evidence of attainment;
- (iv) a pending-v21-repair or legacy-v21-only item cannot be used as v21 theorem evidence;
- (v) a successor claim supersedes a predecessor only through the immutable relation required by Chapter 8;
- (vi) ambiguity is resolved by the weaker class, role, conformance, and conclusion strength.

**Definition 9.64** (Release token-sort conformance). Every semantic token used by the release belongs to the Chapter 8 canonical sort registry. Atomic status, claim class, invocation role, runtime comparison mode, claim comparison policy, detector stage, detector mode, detector certificate status, detector value, Type IV status, transient status, cover type, simplex-coherence mode, Restart status, and manifest status remain distinct sorts. A field name, label string, identifier, or derived branch such as `metric_zero_transport` is not silently promoted to an atomic status or runtime comparison mode.

**Proposition 9.65** (Status cannot be promoted by prose). *A stronger verbal description, diagram, implementation success, numerical experiment, detector score, Search report, or AI-generated explanation does not promote the class, role, conformance, theorem strength, or conclusion level of a statement.*

*Proof.* Promotion requires a proof, complete typed artifacts, a conforming immutable registration change, and the Chapter 8 verifier. None of the listed objects supplies these requirements by itself.  $\square$

## 9.17 The proved mathematical nucleus of v21.0

**Definition 9.66** (Proved Core nucleus). The *proved Core nucleus* is the collection of theorem-level mathematical statements established in the final-audited Chapters 1–6, Appendix A, and the internal reduction theorem of this chapter, together with their proved counterexamples and exact boundaries. A theorem with hypotheses belongs to the nucleus at that conditional logical strength; a run may instantiate it only after the hypotheses pass. Operational status calculi, manifests, schemas, and acceptance policies belong to the verifier layer rather than the external mathematical nucleus.

**Proposition 9.67** (Contents of the proved Core nucleus). *The following statements belong to the proved Core nucleus at exactly the stated strength.*

- (i) *Constructible one-parameter persistence modules over the coefficient field admit the declared barcode decomposition in the working regime.*
- (ii) *The standard bottleneck metric uses endpoint sup-cost for finite bars, half-length diagonal cost, and finite matching of infinite bars only with infinite bars using birth displacement.*

- (iii) *Hard barcode threshold deletion deletes precisely finite bars of length at most  $\tau$ , retains longer finite bars and infinite bars, and is endpoint-independent, idempotent, and threshold-monotone at barcode-profile level.*
- (iv) *Positive-threshold short bars do not form a Serre subcategory in general; hence no global exact positive-threshold Serre reflector, quotient formula, or abelian localization is available in the general Core.*
- (v) *Hard positive-threshold deletion is not globally non-expansive for the standard bottleneck metric.*
- (vi) *If  $d_B(B, C) \leq \varepsilon$  and both pre-threshold barcodes avoid  $[\tau - \varepsilon, \tau + \varepsilon]$ , then*

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

*The half-width  $\varepsilon$  is sharp.*

- (vii) *Threshold clearance is measured after windowing and before deletion, and a numerical ledger enters a metric inference only through a scalarization proved to bound the actual consumed discrepancy.*
- (viii) *The repaired evaluation order is*

$$B_{i,W}(F) = W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

*Window restriction and hard deletion do not commute automatically.*

- (ix) *Admissible representatives of the same repaired profile give the same persistence-level vanishing verdict. Derived or Ext invariance requires separately declared realization-equivalence data.*
- (x) *The finite stencil detector is sound. Exhaustive birth sets, decoration-invariant right germs, certified coverage, and admissible augmented family meshes give the corresponding zero-reflecting characterizations under their exact stage, endpoint, family, and artifact hypotheses.*
- (xi) *Strict mesh separation is sharp at the equality boundary; the family-relative certificate bounds and the exact augmented-mesh stencil count hold at their declared information class. No uniform complete finite detector exists on the right-unbounded birth family without an additional escape hypothesis.*
- (xii) *The metric zero-verdict composition theorem transports a zero predicate only through the complete two-sided threshold-gap and zero-separation chain. It does not assert stability of detector rank.*
- (xiii) *For every  $n \geq 0$  and bounded complex  $A$  of finite-dimensional vector spaces,*

$$\text{Ext}^n(A, k) \cong \text{Hom}_k(H^{-n}(A), k)$$

*naturally contravariantly in  $A$ .*

- (xiv) *Under a complete standard  $n$ -eligible realization,*

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n,F,W,\tau}, k) = 0.$$

*A converse or equivalence requires an independently meaningful, natural, zero-reflecting edge package.*

- (xv) *A complete fixed-stencil detector may supply such zero reflection on its declared class when the detector functor, completeness theorem, natural edge realization, representative compatibility, and all artifacts pass. Encoded synthetic detector duality alone is not the principal natural Bridge.*

- (xvi) *For terminal-tame profiles,  $\text{tgdim}_W$  counts bars cofinal in the declared window. Applied to an actual tower comparison morphism it defines  $\mu_{i,W,\tau}$  and  $\nu_{i,W,\tau}$ . Aggregation is only over the fixed finite monitored degree set.*
- (xvii) *The terminal/interior decomposition, complete transient-defect characterization, and joint terminal/transient classification hold under their barcode and detector hypotheses. Terminal zero does not exclude finite noncofinal defects.*
- (xviii) *Terminal zero, complete interior-detector zero, and a proved short-defect exclusion imply full kernel/cokernel vanishing and hence comparison-map isomorphism. The short-residual-defect example proves the necessity of an additional short-defect hypothesis.*
- (xix) *Exact and metric-reference simplex coherence are distinct. Finite-depth records glue at record level. A global detector-zero conclusion additionally requires a declared global profile interface, complete local detectors, valid local zero transport, and retained-bar coverage. Local zero plus exact overlap coherence alone is insufficient.*
- (xx) *Certified recurrences  $r_{k+1} \geq \kappa_k r_k - \eta_k$  imply the product-weighted finite bound and normalized reserve criterion. Ordinary summability need not preserve reserve under contraction, and pointwise positive reserve need not have a uniform positive infimum.*
- (xxi) *A countable record atlas requires audit-local finiteness, finite Čech depth, complete simplex records, face-to-simplex compatibility, typed exposure summability, and a complete dependency closure. It does not create object-level descent.*
- (xxii) *For a finite small category, the finite-dimensional diagram category is abelian, its finite representable sum is a projective generator, and*

$$E \simeq 0 \iff \mathbf{R}\text{Hom}(G_P, E) \simeq 0.$$

*Cone zero detects isomorphisms; finite Morita reconstruction and the declared finite Alexandrov strict-descent lane hold at the exact hypotheses of Chapter 7.*

- (xxiii) *The eventually-constant finite-to-infinite theorem of Theorem 9.75 identifies the colimit with a stabilized finite stage and, under compatible apex agreement, with the apex.*

*Proof.* Items (i)–(ix) are the repaired v21 persistence nucleus. Items (x)–(xii) are the Chapter 3 finite-detector and metric-composition results. Items (xiii)–(xv) are the Chapter 4 higher Ext-edge and zero-reflection boundary. Items (xvi)–(xviii) are the Chapter 5 terminal/transient results. Items (xix)–(xxi) are the Chapter 6 atlas and Restart results. Item (xxii) is proved below. Every item is included with its original hypotheses and non-claim boundary.  $\square$

**Remark 9.68** (The nucleus is small by design). The nucleus is narrower than the project architecture. Its strength comes from finite certificates with explicit completeness, natural higher edges, separation of terminal and transient defects, exact local-to-global coverage, and refusal to cross an infinite or external boundary without a proved reduction or Return Theorem.

## 9.18 Conditional mathematics and operational contracts

**Definition 9.69** (Conditional Core layer). The *conditional Core layer* consists of proved implications whose run instances require separately supplied hypotheses or artifacts. It includes:

- (i) crop-collapse commutation after the exact all-retained-bars test;
- (ii) finite-detector completeness after stage, endpoint, coverage, family, mesh, and noncircular excess-source obligations;
- (iii) metric zero transport after the two-sided threshold and zero-separation chain, with one parent MetricUseID and no duplicated budget;
- (iv) realization and representative invariance beyond profile level;
- (v) eligible higher Ext-edges and natural zero-reflection packages;
- (vi) actual tower systems, colimits, and comparison morphisms; terminal tameness and complete transient-defect records;
- (vii) exact-cocycle or metric-reference simplex records, global retained-bar coverage, and object descent only through their separate theorems;
- (viii)  $\kappa$ -admissible continuation, normalized reserve-safety, and complete countable dependency closure;
- (ix) finite-to-infinite reduction and external Return Theorems.

**Definition 9.70** (Operational Core layer). The *operational Core layer* contains the binding contracts determining what counts as an auditable proof object or verdict. It includes fixed evaluation order; token sorts; claim classes, conformance, policies, and roles; comparison and simplex modes; detector stage and two-axis status; gate-scoped metric-use identity; representative, realization, tower, overlap, atlas, Restart, reduction, descent, and Return records; Type I–V routing; the four atomic statuses; immutable run identity; well-founded finite or countable evidence graphs; semantic replay; release acceptance; and Core sovereignty.

**Definition 9.71** (Canonical comparison mode). Every consumed runtime profile comparison has exactly one mode:

exact,      metric\_gap\_certified,      theorem\_controlled\_nonmetric,      not\_compared.

In mode `theorem_controlled_nonmetric`, the record names the theorem, its complete hypotheses, exact source and target, and the nonmetric conclusion licensed by that theorem. The claim-level comparison policy and the simplex-coherence modes `exact_cocycle` and `metric_reference` are different sorts. The derived branch `metric_zero_transport` references one parent metric comparison and is neither a fourth comparison mode nor a second spending of its budget.

**Proposition 9.72** (Conditional and operational layers do not self-promote). *A conditional statement without discharged hypotheses is not an available theorem instance, and an operational contract is not an external mathematical theorem.*

*Proof.* The first assertion is implication logic. The second follows because verifier contracts classify and check evidence but do not supply the missing source-domain semantics or transport theorem. □

**Remark 9.73** (Canonical manifest location). The Chapter 1 base manifest and the Chapters 3–6 extensions are conforming fragments of the canonical Chapter 8 v21 schema. An incomplete manifest gives undefined unless another evidenced clause rejects; an inconsistent manifest gives Type V reject.

## 9.19 Inherited eventually-constant specialization of Chapter 5

**Definition 9.74** (Eventually constant reduction package). Let  $\mathcal{C}$  be a category and

$$M_0 \xrightarrow{f_0} M_1 \xrightarrow{f_1} M_2 \longrightarrow \cdots$$

a directed sequence with a declared colimit  $(L, \lambda_n)$ . An *eventually constant reduction package* consists of:

- (i) an index  $N$  and proofs that  $f_n$  is an isomorphism for every  $n \geq N$ ;
- (ii) the actual directed-system morphisms and colimit cocone;
- (iii) when an apex  $A_\infty$  is used, a compatible cocone  $a_n : M_n \rightarrow A_\infty$ ;
- (iv) an apex-agreement proof that  $a_N$  is an isomorphism;
- (v) the induced comparison morphism  $\phi : L \rightarrow A_\infty$  satisfying  $\phi\lambda_n = a_n$ ;
- (vi) immutable theorem, witness, and artifact identities.

A finite prefix or sampled stabilization is not such a package.

**Theorem 9.75** (Bridge specialization: eventually constant tower with compatible apex agreement). *Under Definition 9.74, the canonical map*

$$\lambda_N : M_N \xrightarrow{\sim} L$$

*is an isomorphism. If the apex cocone and apex agreement are included, then*

$$\phi : L \xrightarrow{\sim} A_\infty$$

*is an isomorphism.*

*Proof.* For  $n \leq N$ , let  $b_n : M_n \rightarrow M_N$  be the forward composite. For  $n > N$ , the composite  $M_n \rightarrow M_N$  is an isomorphism; let  $b_n$  be its inverse. These maps form a cocone over the sequence with vertex  $M_N$ . By the universal property of  $L$ , there is a unique  $\psi : L \rightarrow M_N$  with  $\psi\lambda_n = b_n$ . In particular  $\psi\lambda_N = \text{id}_{M_N}$ . Both  $\lambda_N\psi$  and  $\text{id}_L$  agree after composition with every  $\lambda_n$ , so colimit uniqueness gives  $\lambda_N\psi = \text{id}_L$ . Hence  $\lambda_N$  is an isomorphism. For the apex comparison,  $a_N = \phi\lambda_N$ . Since  $a_N$  and  $\lambda_N$  are isomorphisms, so is  $\phi = a_N\lambda_N^{-1}$ .  $\square$

**Corollary 9.76** (Vanishing of full comparison defects under reduction). *If Theorem 9.75 is applied in the abelian profile category used by Chapter 5, then*

$$\ker \phi = 0, \quad \text{coker } \phi = 0.$$

*Consequently all degreewise terminal diagnostics vanish, every complete full or interior defect detector has zero value, and the comparison is a full isomorphism on the declared profile scope.*

*Proof.* An isomorphism in an abelian category has zero kernel and cokernel. The Chapter 5 diagnostics and detector statements then follow from their definitions and complete characterizations.  $\square$

**Example 9.77** (Finite prefixes do not determine an infinite tower). Fix  $N$ . In  $\text{Vect}_k$ , let two directed systems agree with the zero object through stage  $N$ . Let the first remain zero forever. Let the second have  $M_{N+1} = k$ , map  $0 \rightarrow k$ , and use identity maps on  $k$  thereafter. The two systems have identical data through stage  $N$ , but their colimits are respectively 0 and  $k$ . Thus no finite prefix alone determines the infinite colimit, eventual constancy, apex agreement, or comparison defect.

## 9.20 Deprecated and rejected claims closed by the repair

**Definition 9.78** (Metric-repair and architecture rejection set). The following are unavailable for v21 theorem use:

- (i) global non-expansiveness of hard positive-threshold deletion;
- (ii) a general positive-threshold Serre localization or exact reflector;
- (iii)  $\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$  or a sharp band of half-width  $2\varepsilon$ ;
- (iv) unconditional crop–collapse commutation, automatic filtered collapse, or automatic tower comparison morphisms;
- (v) detector completeness inferred from soundness, finite sampling, observed success, stage relabeling, or a circular/full-barcode source of the claimed compression bound;
- (vi) detector pointwise rank stability inferred from bottleneck stability;
- (vii) synthetic shifted detector duality promoted to a natural higher PH–Ext Bridge;
- (viii) the reverse higher bridge without an independently proved natural zero-reflection theorem;
- (ix) terminal zero interpreted as full comparison isomorphism or absence of transient/short defects;
- (x) fabricated Type IV zero in a no-tower scope;
- (xi) a universal Restart constant, ordinary summability under arbitrary contraction, or pointwise reserve promoted to uniform reserve;
- (xii) pairwise overlap promoted to higher coherence, local detector zero promoted to global zero without coverage, or record coherence promoted to object descent;
- (xiii) finite-prefix or sampled stabilization promoted to an infinite tower, colimit, continuum, or external theorem;
- (xiv) duplicated spending of one metric discrepancy through a runtime comparison and its derived zero-transport branch;
- (xv) an operational, specification, pending-repair, legacy-only, deprecated, rejected, or unknown item promoted to theorem evidence;
- (xvi) a Core pass, known-theorem calibration, toy theorem, Search output, or AI proposal promoted to an external conclusion without Return.

**Proposition 9.79** (Rejected claims cannot be repaired locally by a ledger). *No rejected structural claim in Definition 9.78 becomes valid by assigning it a larger numerical error budget.*

*Proof.* The rejected claims concern false global assertions, missing structure, circular proof sources, invalid type changes, duplicated accounting, or absent logical hypotheses. A scalar metric bound constructs none of them. □

## 9.21 The final Core closure contract

**Definition 9.80** (v21 Core closure contract). The v21 Core is the smallest versioned theorem-and-verifier package containing:

- (i) the proved nucleus of Definition 9.66;
- (ii) the conditional layer only at discharged hypotheses;
- (iii) the operational verifier layer and canonical Chapter 8 manifest;
- (iv) Appendix A at its registered metric-repair strength;
- (v) finite-detector, higher-edge, defect, atlas, and reduction theorems only at their exact stages and information classes;
- (vi) explicitly imported micro-theorems only at registered strength, role, conformance, and hypotheses;
- (vii) no external-domain semantics without a separately proved directional transport or Return Theorem.

The package is closed under valid mathematical consequence and proof-preserving versioned reuse. It is not enlarged by analogy, implementation, registration, projection, numerical success, or rhetoric.

**Proposition 9.81** (Main-body Core closure). *Within the declared v21.0 main-body scope, every Core verdict is determined by the final-audited mathematical inputs, typed artifacts, detector certificates, comparison modes, gate-scoped metric uses, evidence graph, canonical manifest, and Chapter 8 aggregation. No proposer-side object changes that verdict without completing typed re-entry.*

*Proof.* The mathematical inputs are fixed by the proved and conditional layers. The verdict calculus and manifest are fixed by Chapter 8. Core sovereignty excludes direct updates by proposer-side material.  $\square$

**Proposition 9.82** (Final-audit closure criterion). *A v21 main-body run is audit-closed exactly when every invoked clause is well-typed and has a determined pass or reject status, every exclusion is justified, every finite or countable dependency closure is complete, every conditional and non-scalar obligation is discharged, every metric use is nonduplicative and bounded, and the canonical manifest is complete, consistent, artifact-closed, and semantically replayable. A fully evidenced rejection is a closed result. A run with unresolved required evidence remains undefined.*

*Proof.* Under the listed hypotheses every atomic clause has a truth-valued status and the Chapter 8 aggregate is determined. Conversely, an unresolved required clause or incomplete dependency closure prevents a determined audit closure.  $\square$

**Definition 9.83** (Main-body and release acceptance boundary). *Main-body closure* means that Chapters 1–9 form a closed typed inference architecture. *Release acceptance* means that the Chapter 8 release registry for every mandatory Chapter 1 obligation passes. These statuses are independent: the main body may be closed while a mandatory external calibration, appendix registration, toy verification, or Return artifact is still pending.

**Remark 9.84** (This is a closure contract, not a universal completeness theorem). Closure does not claim that the Core decides every mathematical question, possesses a finite detector on every scope, realizes every external invariant, or supplies every Return Theorem. It states that every result claimed by the release must traverse the declared typed route.



## 9.22 High-dimensional projection as an Interface mechanism

The repaired AK route is

$$\begin{aligned} \text{external object} &\longmapsto \text{Core-readable shadow} \\ &\longmapsto \text{finite or structural Core evidence} \\ &\longmapsto \text{audited Core conclusion.} \end{aligned}$$

A further proved arrow is required to return to the external source semantics.

**Definition 9.85** (External source class). An *external source class* is a declared category, typed family, or solution space outside the minimal one-parameter persistence Core. Its objects, morphisms, equivalences, and target predicates are part of the source semantics and are not Core objects by default.

**Definition 9.86** (AK projection datum). An *AK projection datum* consists of a source object  $X$ , a typed route

$$\mathfrak{R}_{AK} : \mathcal{D} \longrightarrow \text{FiltCh}(k) \quad \text{or} \quad \mathfrak{R}_{AK} : \mathcal{D} \longrightarrow \text{Pers}_k^{\text{cons}},$$

the projected object  $F_X$ , the declared windows, thresholds, degrees, coefficient field, detector source stages, approximation and comparison data, constructibility evidence, purpose-faithfulness target, and immutable artifacts. Every invoked detector, realization, tower, atlas, reduction, and Return statement is separately identified with its direction, class, role, and hypotheses.

**Definition 9.87** (Core-readable projected object). A projected object  $F_X$  is *Core-readable* on its declared scope when the window-first pre-threshold and repaired profiles are defined, every invoked detector has its exact stage and certificate, every comparison has a valid mode and nonduplicative metric-use identity when needed, every representative, realization, Type IV, overlap, Restart, reduction, and claim obligation is typed, every countable dependency registry is complete, and the Chapter 8 verifier can evaluate the canonical manifest.

**Proposition 9.88** (Projection supplies input, not verdict). An *AK projection datum* determines neither a Core verdict nor an external-domain theorem by itself.

*Proof.* A Core verdict additionally requires all theorem hypotheses, artifacts, and audit clauses. An external theorem additionally requires a proved Return statement in the source semantics.  $\square$

**Remark 9.89** (Projection is structural). Projection may be geometric, arithmetic, analytic, categorical, spectral, computational, tower-theoretic, or hybrid. Its admissibility depends on exact typing and proof, not on conceptual elegance or numerical success.

## 9.23 Purpose-faithfulness and Interface entry

**Definition 9.90** (Purpose-faithful realization). A route is *purpose-faithful* for a declared external claim only when it identifies the source feature, the Core-visible feature intended to represent it, the direction of use, and the theorem or controlled mismatch hypotheses connecting them. Purpose-faithfulness is relative to that claim and is not categorical faithfulness unless separately proved.

**Definition 9.91** (Interface entry record). An *Interface entry record* contains the source object and semantics, projection datum and declared purpose, Core-readable target type, constructibility, filtration orientation, windows, thresholds, degrees, fixed evaluation order, every detector realization and completeness source, comparison and metric-use records, representative and higher-realization data, Type IV scope, tower and defect data, atlas and Restart data, finite-to-infinite package when invoked, complete finite or countable evidence graph, canonical manifest, immutable run identity, and the class, conformance, role, direction, and hypotheses of every transport statement.

**Proposition 9.92** (No admissible Interface without typed entry). *An external object has no theorem-bearing access to the Core until its Interface entry record is complete and consistent and every invoked entry clause is typed.*

*Proof.* Without a complete typed entry, the verifier cannot reconstruct the Core input. Missing or unresolved entry evidence is undefined; evidenced false or inconsistent entry claims are reject. Neither is pass.  $\square$

**Remark 9.93** (Interface failure is not automatically Core rejection). A proposed route may be nonconstructible, not purpose-faithful, insufficiently evidenced, or ill-typed. This is an Interface failure or undefined entry. It is a Core rejection only when a properly invoked Core clause is well-typed and evidenced false or the canonical manifest is inconsistent.

## 9.24 Detector, higher-obstruction, and reduction realization interfaces

**Definition 9.94** (Detector realization theorem interface). A detector realization theorem for an external source class identifies a finite external observable object  $Q_X$  and proves a natural isomorphism

$$Q_X \xrightarrow{\sim} D_{T,\tau}^{\text{stc}}(\text{Eval}_{n,w,\tau}(F_X))$$

or the corresponding birth, germ, family-mesh, kernel-defect, or cokernel-defect detector. It records the exact detector stage, endpoint convention, stencil or family, source of completeness, noncircular excess provenance, and preservation under the chosen realization. A numerical agreement without this theorem is only calibration.

**Definition 9.95** (Detector soundness Bridge). A detector soundness Bridge is a proved directional theorem from a certified nonzero detector witness to a precisely stated external obstruction or failure predicate. Soundness consumes a well-typed nonzero witness and does not require detector completeness unless the external conclusion itself uses a zero equivalence.

**Definition 9.96** (Detector completeness Bridge). A detector completeness Bridge is a proved theorem establishing the direction needed for an external zero conclusion, for example

$$D(P_X) = 0 \implies P_X = 0 \implies \mathcal{E}(X),$$

with the first implication supplied by an admitted complete detector and the second by a Return theorem. Sampling success, soundness, or full-barcode readout disguised as finite compression is insufficient.

**Definition 9.97** (Higher Ext realization theorem interface). A higher Ext realization theorem identifies an independently meaningful external or geometric object  $\mathcal{H}_n(X)$  with the natural negative cohomology edge or Ext obstruction of the eligible Core realization, through natural artifact-backed isomorphisms. Re-encoding a detector output into a shifted synthetic object may support computation but does not satisfy this principal realization interface by itself.

**Definition 9.98** (Finite-to-infinite Interface record). Whenever a finite certificate is used for an infinite tower, continuum, unbounded window, or limit object, the Interface record identifies the exact finite witness, the registered reduction theorem, actual directed-system maps, colimit semantics, compatible apex cocone, apex agreement, and induced comparison. Theorem 9.75 is one admissible reduction route when its hypotheses are met; it is not a universal substitute for problem-specific compactness, convergence, or continuation.

## 9.25 The repaired evaluation path for external data

**Definition 9.99** (Interface evaluation path). Every Core-relevant Interface run uses

$$\begin{aligned}
X &\longmapsto F_X = \mathfrak{R}_{AK}(X) \\
&\longmapsto \mathbf{P}_n(F_X) \\
&\longmapsto B_{n,W}(F_X) = W_W \mathbf{P}_n(F_X) \\
&\longmapsto \text{Eval}_{n,W,\tau}(F_X) = T_\tau^{\text{bar}}(B_{n,W}(F_X)) \\
&\longmapsto \text{direct proof or admitted detector evidence} \\
&\longmapsto \text{typed Core clauses and audit verdict.}
\end{aligned}$$

If the target is infinite, the declared reduction step follows the finite Core evidence. If the target is external, a Return Theorem follows the audited Core conclusion. No raw observable replaces a profile, detector, reduction, or Return arrow without a proved typed identification.

**Proposition 9.100** (External origin does not relax Core hypotheses). *A projected object is subject to the same constructibility, detector, comparison, representative, eligibility, Type IV, atlas, Restart, dependency, claim, and manifest conditions as an internally specified Core object.*

*Proof.* Core theorems and verifier clauses are typed by inputs and hypotheses, not by data origin or intended application.  $\square$

**Remark 9.101** (Approximation must be visible). Sampling, truncation, discretization, smoothing, numerical solution, normalization, compression, learned representation, and adaptive detector selection must be explicit. Metric effects enter one identified ledger use only when scalarizable; semantic, naturality, completeness, morphism, and Return obligations remain non-scalar.

## 9.26 Distinct conclusion levels

**Definition 9.102** (Interface outcome). The *Interface outcome* states whether the source was converted into a complete, consistent, Core-readable entry. Its statuses are

pass,      reject,      undefined,

under the declared entry contract.

**Definition 9.103** (Core outcome for projected data). The *Core outcome for projected data* is the Chapter 8 verdict about the repaired projected data under the declared scope. It may concern record coherence, profile zero, higher Ext vanishing, terminal/transient defects, or another exactly typed Core predicate.

**Definition 9.104** (Bridge outcome). A *Bridge outcome* is the registered status of a directional realization, soundness, completeness, higher-obstruction, reduction, or transport statement. Specification, toy theorem, conditional theorem, and proved transport are not interchangeable.

**Definition 9.105** (External-domain outcome). An *external-domain outcome* is a theorem or refutation stated entirely in the original semantics of  $X$ .

**Definition 9.106** (Finite-certificate outcome). A finite-certificate outcome records whether the finite detector or proof compression is applicable, certified, and sufficient for its exact Core predicate. It is not an infinite, descended, or external conclusion.

**Definition 9.107** (Reduction and Return outcomes). A *reduction outcome* records whether a finite-to-infinite theorem applies to the declared limit target. A *Return outcome* records whether a proved Core-to-source theorem applies to the exact audited Core conclusion. They are separate clauses even when both are required by one external claim.

**Proposition 9.108** (No outcome-layer collapse). *None of the following implications is available without the named additional clause:*

$$\begin{aligned} \text{Interface pass} &\Rightarrow \text{Core pass}, \\ \text{finite detector pass} &\Rightarrow \text{infinite conclusion}, \\ \text{record coherence} &\Rightarrow \text{global zero}, \\ \text{global zero} &\Rightarrow \text{object descent}, \\ \text{Core pass} &\Rightarrow \text{external theorem}. \end{aligned}$$

*The missing clauses are respectively the Core verifier, finite-to-infinite reduction, retained-bar coverage, effective descent, and Return Theorem.*

*Proof.* The antecedents and consequents inhabit different typed conclusion levels. Each omitted implication requires mathematical content not contained in its antecedent.  $\square$

## 9.27 Transport theorems, Return, and Bridge promotion

**Definition 9.109** (Interface-to-Core transport statement). An *Interface-to-Core transport statement* has direction

$$\text{external-domain structure} \Longrightarrow \text{specified Core-visible behavior}.$$

**Definition 9.110** (Core-to-Interface transport statement). A *Core-to-Interface transport statement* has direction

$$\text{specified Core-visible behavior} \Longrightarrow \text{external-domain conclusion}.$$

**Definition 9.111** (Transport theorem). A *transport theorem* is a proved directional statement with precise source and target semantics, purpose-faithful realization, Core-readable target, all detector and representative hypotheses, every comparison and metric-use record, all tower/atlas/reduction data, complete manifest, proof of the stated direction, and an explicit converse and non-claim boundary.

**Definition 9.112** (External Return Theorem). Let  $C_{AK}(X)$  be the exact Core conclusion produced for the projected object and let  $\mathcal{E}(X)$  be an external-domain predicate. A *Return Theorem* is a proved theorem

$$C_{AK}(X) \Longrightarrow \mathcal{E}(X)$$

on a declared source class, together with every realization, preservation, finite-to-infinite, descent, and mismatch hypothesis needed by the proof. A bidirectional external characterization requires a separately proved reverse direction.

**Definition 9.113** (Default status of transport statements). Every problem-specific realization, Bridge, reduction application, and Return statement is `spec` with role `specification_target` by default. It becomes theorem evidence only through explicit proof, conforming registration, artifact closure, and Chapter 8 verification. The two transport directions are independent.

**Proposition 9.114** (No silent Bridge promotion). *A natural-looking projection, successful calibration, stable experiment, finite detector, Core pass, toy bridge, Search result, or AI-generated proof sketch does not silently promote a transport specification to theorem status.*

*Proof.* None supplies all source semantics, hypotheses, naturality, preservation, reduction, manifest, and proof obligations of Definitions 9.111 and 9.112.  $\square$

**Theorem 9.115** (Composed AK Return principle). *Let  $X$  be an external object. Assume:*

- (i) *a proved typed Interface entry produces a Core-readable  $F_X$ ;*
- (ii) *the required Core theorem or certified detector establishes the exact Core predicate  $C_{\text{fin}}(X)$ ;*
- (iii) *if the target is infinite, a proved reduction theorem establishes  $C_{\text{fin}}(X) \Rightarrow C_{AK}(X)$ , while otherwise  $C_{AK} = C_{\text{fin}}$ ;*
- (iv) *a proved Return Theorem establishes  $C_{AK}(X) \Rightarrow \mathcal{E}(X)$ .*

*Then  $\mathcal{E}(X)$  holds.*

*Proof.* Compose the proved implications in their declared direction. The typed entry ensures that each intermediate predicate has the source and target required by the next implication.  $\square$

**Remark 9.116** (External Ext is not automatically the Core Ext). An arithmetic, sheaf-theoretic, deformation-theoretic, or categorical Ext group is not automatically  $\text{Ext}^n(A_n; F_X, W, \tau, k)$ . Any identification is itself higher-realization and Return mathematics and must be natural, directional, and proved.

## 9.28 Micro-theorems, toy bridges, Search, and Platform material

**Definition 9.117** (Permitted adjacent support). Appendices and systems may provide registered micro-theorems, finite detector examples, endpoint/cofinal/mesh/metric counterexamples, toy bridge theorems, known-theorem-recovery calibrations, Interface definitions, Search proposals, Platform logs, proof-assistant artifacts, or Companion templates. Such material supports the main body only at its exact class, conformance, role, hypotheses, and transformation history.

**Proposition 9.118** (Adjacent material does not enlarge the Core silently). *No appendix, Companion, Search report, Platform artifact, micro-theorem, toy theorem, or Interface schema enlarges the v21 Core merely by appearing in the project.*

*Proof.* Import requires a compatible theorem statement, proof, registered strength, discharged hypotheses, artifact closure, and all relevant verifier clauses.  $\square$

**Remark 9.119** (NBT release boundary). NBT remains a successor-release candidate program unless a later version proves and registers the required theorem package. Its possible control of apex-only Type IV cokernel modes does not strengthen v21.0.

## 9.29 Known-theorem recovery and release demonstration

**Definition 9.120** (Known-theorem-recovery record). A *known-theorem-recovery record* selects an external theorem already proved independently on a benchmark class and reconstructs its conclusion through a complete AK Interface, Core, reduction, and Return chain. It records where the independent external proof is used only for validation and where the AK chain supplies its own proof obligations.

**Proposition 9.121** (Calibration is not promotion). *Successful known-theorem recovery validates the advertised Interface and Return chain on the benchmark class but does not by itself prove a stronger external theorem, extend the source class, or discharge another release gate.*

*Proof.* The recovery theorem is scoped to its registered class and hypotheses. A stronger conclusion requires a stronger independently proved transport or Return theorem.  $\square$

**Definition 9.122** (v21 release-closure record). The final publication package contains one Chapter 8 release-acceptance record covering all mandatory Chapter 1 gates. In particular it separately records the finite detector theorems, higher Ext-edge and one-way discharge, any advertised natural zero-reflection instance, transient-defect results, finite-to-infinite theorem, CM/TB/counterexample and manifest synchronization, known-theorem recovery, and every advertised external realization and Return claim. Optional proof-assistant and compression comparisons remain optional unless promoted by policy.

**Proposition 9.123** (Main closure does not predeclare release pass). *The completion of Chapters 1–9 does not by itself set the v21 release headline to pass. The headline is computed from the actual release registry and remains undefined whenever a mandatory external, appendix, registration, or artifact obligation is pending.*

*Proof.* Chapter 8 defines release acceptance as a conjunction of independently mandatory obligations. Main-body closure is only one of those obligations.  $\square$

**Definition 9.124** (v21 release-gate reflection matrix). The final release registry reflects the Chapter 1 gates without changing their strength. The mandatory Core-mathematics block is:

- G1 birth-detector characterization;
- G2 germ or decorated-endpoint characterization;
- G3 certified-stencil and family-mesh completeness;
- G4 detector sharpness and boundary counterexamples;
- G5 right-unbounded finite-detector no-go;
- G6 metric zero-verdict composition;
- G7 family-relative minimal-certificate results;
- G8 canonical higher Ext-edge;
- G9 higher one-way repaired discharge;
- G10 an advertised non-synthetic natural zero-reflection instance on its declared class;
- G11 transient Type IV kernel/cokernel defect detection.

The mandatory verification/governance block is:

- G12 CM-v21 Core Micro-Theorem registration at exact theorem strength;
- G13 TB-v21 Toy Bridge verification;
- G14 endpoint, cofinal, mesh, metric, higher-edge, and transient-defect counterexamples;
- G15 manifest, token-sort, claim, and non-scalar-obligation consistency.

The mandatory external-demonstration block is:

G16 at least one known-theorem recovery in the declared Navier–Stokes or Iwasawa route, at the exact recovered strength and with a complete Interface–Core–Reduction–Return chain;

G17 at least one proved natural realization or finite-to-infinite reduction theorem. Theorem 9.75 supplies the internal reduction option, but any advertised external application still requires its own entry and Return record.

The strongly recommended block is:

G18 partial proof-assistant formalization;

G19 computational demonstration of certificate reduction;

G20 comparison with full-barcode decision procedures.

Each gate remains an independently identified Chapter 8 release clause.

**Proposition 9.125** (Mandatory and recommended release gates do not compensate). *A pass on one mandatory gate does not compensate for another mandatory gate that is rejected or undefined. Gates G18–G20 do not alter the headline unless a later release policy promotes them to mandatory status.*

*Proof.* The release headline is the typed conjunction defined in Chapter 8 and the Chapter 1 release contract. Optional recommendations are outside that mandatory conjunction at v21.0.  $\square$

### 9.30 Explicit non-claims of the closed main body

The closed v21.0 main body asserts none of the following.

- (i) Global non-expansiveness of hard threshold deletion.
- (ii) Positive-threshold Serre localization, a global exact reflector, or a universal quotient formula.
- (iii) Sharpness of a band of half-width  $2\varepsilon$ , or  $\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$ .
- (iv) Unconditional crop–collapse commutation or a global strict filtered collapse functor.
- (v) Automatic derived or Ext equivalence of profile representatives.
- (vi) Detector completeness from soundness, observed sampling, an unverified birth list, stage relabeling, or a circular/full-barcode source of the claimed finite proof.
- (vii) Pointwise detector-rank stability from bottleneck stability.
- (viii) A uniform complete finite detector on an unrestricted right-unbounded family without an escape theorem.
- (ix) A natural higher PH–Ext Bridge produced merely by shifting detector output into a synthetic derived object.
- (x) The reverse higher Bridge or global PH–Ext equivalence without independently proved natural zero reflection.
- (xi) An automatic tower comparison morphism from barcodes, sampled layers, or bottleneck matchings.
- (xii) Missing Type IV evidence interpreted as  $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ , or no-tower scope represented by fabricated zeros.

- (xiii) Terminal zero interpreted as full defect-profile vanishing, absence of transient defects, or full comparison isomorphism.
- (xiv) A universal  $\kappa > 0$ , default  $\kappa = 1$ , ordinary summability under contraction, or pointwise reserve promoted to uniform reserve.
- (xv) Pairwise overlap as higher Čech coherence, metric reference as an exact cocycle, or summability as a substitute for face maps.
- (xvi) Local detector zero as global zero without retained-bar coverage.
- (xvii) A coherent record atlas as a global filtered object or effective descent theorem.
- (xviii) A finite prefix, finite detector, numerical stabilization, or eventually constant appearance as an infinite theorem without a registered reduction package.
- (xix) A scalarization value as a certificate without an actual upper-bound proof, or duplicated spending of one metric use through a derived zero-transport branch.
- (xx) A numerical budget repairing a missing morphism, naturality, realization, eligibility, completeness, coherence, rank witness, status, role, or manifest field.
- (xxi) `not_compared` discharging a comparison-dependent clause, or `undefined` identified with `reject`, `pass`, `zero`, or `not_invoked`.
- (xxii) Claim status, conformance, or theorem strength created by registration, prose, implementation, Search, or AI output.
- (xxiii) A detector realization, higher Ext realization, finite-to-infinite theorem, or Return Theorem inferred from another one of those clauses.
- (xxiv) Completion base change  $E \otimes_A^L \widehat{A}$  identified with derived adic completion without a separate comparison theorem.
- (xxv) A generator-rigidity theorem applied outside a declared finite-presentation lane.
- (xxvi) Determinant, ideal, graded-piece, stable-class, and pseudo-isomorphism Returns placed in a common scalar order without a registered target map.
- (xxvii) A Core pass, toy bridge, known-theorem calibration, or natural-looking projection proving an external theorem by itself.
- (xxviii) A predeclared v21 complete-release pass before every mandatory release obligation is actually registered and verified.
- (xxix) v21.0 theorem status for NBT or any other successor-release candidate.
- (xxx) Completeness of AK theory for all mathematical problems, all realizations, all unbounded scopes, or all external source categories.

**Remark 9.126** (Non-claims are part of the result). These exclusions are not rhetorical caution. They are typed boundaries that identify the exact theorem, artifact, or later version required to cross each remaining gap. The theory is defined by the routes it proves and by the shortcuts it makes impossible to report as proof.



### 9.31 Final architecture theorem

**Proposition 9.127** (Final architecture of the v21 main body). *There are two distinct routes. A candidate route may begin with a specified Interface*

$$X \xrightarrow{\text{specified Interface}} F_X,$$

*but produces only a candidate Core input. A theorem-bearing route has the form*

$$\begin{aligned} X &\xrightarrow{\text{proved typed Interface}} F_X = \mathfrak{R}_{AK}(X) \\ &\xrightarrow{\text{window-first persistence}} B_{n,W}(F_X) \\ &\xrightarrow{\text{hard threshold deletion}} \text{Eval}_{n,W,\tau}(F_X) \\ &\xrightarrow{\text{direct theorem or admitted finite certificate}} C_{\text{fin}}(X) \\ &\xrightarrow{\text{proved reduction, when required}} C_{AK}(X) \\ &\xrightarrow{\text{typed audit, evidence graph, and manifest}} \text{Core verdict} \\ &\xrightarrow{\text{separately proved Return Theorem}} \mathcal{E}(X). \end{aligned}$$

*Every arrow has declared source and target, direction, hypotheses, theorem class, role, artifacts, and failure boundary. No arrow may be omitted, reversed, merged, or replaced by numerical plausibility.*

*Proof.* The first arrow is governed by purpose-faithful Interface entry. The next two are the repaired evaluation path. The finite evidence arrow is supplied by a Core theorem or admitted detector. Infinite targets require a reduction such as Theorem 9.75. Chapter 8 governs the verdict. The final external implication is exactly the additional content of a Return Theorem.  $\square$

**Theorem 9.128** (Final AK projection-and-return principle). *An external AK theorem is licensed precisely by a complete theorem-bearing chain of the form in Proposition 9.127. When every invoked arrow is proved and the final Core verdict passes, the declared external conclusion follows. If any required Interface, finite-certificate, reduction, audit, or Return arrow is absent or undefined, no external conclusion is licensed by the AK chain.*

*Proof.* The positive statement is Theorem 9.115 combined with the Chapter 8 pass semantics. The negative statement follows from the typed conjunction and no-outcome-collapse rule: a missing required arrow leaves the corresponding clause undefined and prevents a theorem-bearing composition.  $\square$

**Remark 9.129** (Final AK projection principle).

Projection reveals a structured shadow.  
 Finite certificates may compress its proof.  
 Higher Ext records a natural obstruction only through realization.  
 Defect calculus separates terminal failure from transient residue.  
 Atlases coordinate local evidence, and reduction alone crosses the finite–infinite boundary.  
 Only Return carries the certified statement home.

The asymmetry is deliberate. AK does not identify a shadow with its source; it makes the exact theorem needed for that identification visible.

**Remark 9.130** (Closure of the main body). With this chapter, the AK-HPDST v21.0 main-body architecture is closed. The metric nucleus is sharp where proved; finite evidence is complete only where certified; higher

obstruction is natural only where realized; terminal and transient defects remain separated; local evidence becomes global only through coverage; finite data become infinite only through reduction; and Core truth returns to an external domain only through a proved Return Theorem.

The theory therefore closes not by claiming that every projection is faithful, but by identifying, typing, and auditing every place where faithfulness must be proved. Future versions enlarge this closure only by supplying new proofs, new natural realizations, stronger reduction theorems, verified external Returns, and conforming immutable artifacts.

### 9.32 Provenance and status

This chapter consumes the final-audited Chapters 1–8 and Appendix A. It restates their mathematical results only at their registered strengths, proves the internal eventually-constant reduction theorem, and fixes the final Interface/Return boundary. The v21 Claim Register must register every item at the same or weaker strength and compatible role.

**Release-level and closure definitions.** Definitions [9.61](#), [9.62](#), [9.63](#), [9.64](#), [9.66](#), [9.69](#), [9.70](#), [9.71](#), [9.74](#), [9.78](#), [9.80](#), and [9.83](#).

**Interface, realization, and Return definitions.** Definitions [9.85](#), [9.86](#), [9.87](#), [9.90](#), [9.91](#), [9.94](#), [9.95](#), [9.96](#), [9.97](#), [9.98](#), [9.99](#), [9.102](#), [9.103](#), [9.104](#), [9.105](#), [9.106](#), [9.107](#), [9.109](#), [9.110](#), [9.111](#), [9.112](#), [9.113](#), [9.117](#), [9.120](#), [9.122](#), and [9.124](#).

**Mathematical theorems and closure consequences.** Propositions [9.65](#), [9.67](#), [9.72](#), [9.79](#), [9.81](#), [9.82](#), [9.88](#), [9.92](#), [9.100](#), [9.108](#), [9.114](#), [9.118](#), [9.121](#), [9.123](#), [9.125](#), and [9.127](#); Theorems [9.75](#), [9.115](#), and [9.128](#); Corollary [9.76](#); and Example [9.77](#).

**Boundary and final-closure remarks.** Remarks [9.58](#), [9.59](#), [9.60](#), [9.68](#), [9.73](#), [9.84](#), [9.89](#), [9.93](#), [9.101](#), [9.116](#), [9.119](#), [9.126](#), [9.129](#), and [9.130](#).

**Explicit non-claims.** Section [9.30](#).

**Main-body closure.** Proposition [9.81](#), Proposition [9.82](#), Proposition [9.127](#), and Theorem [9.128](#).

### 9.33 v21 Chapter 9 completion and non-promotion boundary

**Theorem 9.131** (v21 Unified Bridge and Return package). *Chapter 9 supplies:*

- (i) *the universal datum, route classifier, component contract, and assumption/strength composition rule;*
- (ii) *the generic component library and composable Bridge DAG; closure, non-composition, partial Return, regularized Return, and ceiling theorems;*
- (iii) *the complete filtered-chain route of Theorem [9.54](#) and the migrated v20 closure/Interface compatibility layer.*

*No named domain-specific theorem is promoted into the Main body.*

*Proof.* The new assertions are proved above; compatibility migration preserves the old labels at no stronger strength. Non-promotion is Definition [9.1](#). □

## 10 Chapter 10: Validated Returns, Demonstrations, and Successor Problems

**Chapter 10 contract.** This chapter is the v21 validation portfolio. It does not enlarge  $\text{Core}_P$ ,  $\text{Core}_{DG}$ , or the Chapter 8 verifier, and it does not promote a benchmark, replay, numerical calculation, or successful example to a stronger theorem. Its role is to instantiate the final-audited Chapters 1–9 on explicit finite objects, exact matrices, registered v20 recovery tracks, finite-state towers, and a complete external predicate *Return*.

Every validation case records:

- (i) the exact theorem and source strength consumed;
- (ii) the concrete input, expected output, and independently checkable calculation;
- (iii) the proof-DAG boundary and the strongest licensed conclusion;
- (iv) the failure witness obtained when a necessary hypothesis is removed;
- (v) a distinction among mathematical proof, semantic replay, artifact identity, and release governance.

The mandatory v21 completion portfolio consists of a complete declared v20 regression slice, enriched finite-diagram reconstruction, finite object-level descent, and one external *Return* consuming both  $\text{Core}_P$  and  $\text{Core}_{DG}$ . Finite persistence reconstruction, finite-state *Return*, certificate complexity, and negative validation are executed as additional Main validation gates. Passing this portfolio is a main-body validation result. It is not, by itself, the complete v21 release headline.

### 10.1 Scope, validation taxonomy, and evidence strength

**Definition 10.1** (Validation taxonomy). Every portfolio item declares exactly one principal validation class:

- (i) *regression validation*: a v20 theorem, claim, counterexample, or governance boundary is re-executed at exactly its inherited strength;
- (ii) *internal Core validation*: a Chapter 2–5 mathematical construction is instantiated on explicit finite data;
- (iii) *finite-globalization validation*: a Chapter 6–7 atlas or descent theorem constructs an actual global object or morphism;
- (iv) *external Known-Theorem Recovery*: an independently known external theorem is recovered through a quarantined Interface–Core–Reduction–Return chain;
- (v) *new Bridge Return*: a v21 Chapter 9 route closes an external conclusion not merely by citing the v20 recovery registry;
- (vi) *Challenge Track*: a successor-strength route is explored under explicit assumptions and a stopping rule;
- (vii) *negative or no-go validation*: a prohibited implication is refuted by an explicit witness.

A case may carry secondary tags, but its headline class is unique.

**Definition 10.2** (Validation case record). A validation case is a tuple

$$\mathfrak{B} = (\text{ID}, \text{class}, \text{input}, \text{route}, \text{expected}, \text{observed}, \text{strength}, \text{artifacts}, \text{status}, \text{boundary}).$$

Its status is one of the Chapter 8 atomic statuses. A case passes only when its expected and observed outputs agree, every necessary proof-DAG predecessor passes, and the conclusion is consumed at or below the route-admissible strength. A replay script or hash check may support an artifact clause, but it does not replace the theorem proof.

**Definition 10.3** (Portfolio status). Let  $\mathcal{V}_{\text{req}}$  be the required validation cases and  $\mathcal{V}_{\text{opt}}$  the optional or challenge cases. The required portfolio has status:

- (i) **reject** if one required case rejects;
- (ii) **undefined** if no required case rejects and one required case is undefined;
- (iii) **pass** if every required case passes.

Optional cases do not compensate for a nonpassing required case.

**Proposition 10.4** (Validation is strength preserving). *A passing validation case confirms only the theorem statement, source class, equivalence relation, assumption context, and Return target recorded by that case. It does not enlarge the theorem's domain, discharge another route, or promote a partial Return to object reconstruction.*

*Proof.* The validation record is evaluated by the Chapter 8 noncompensating typed proof-DAG and the target-relative strength calculus of Chapter 9. No field in the record changes the registered theorem statement.  $\square$

**Definition 10.5** (Replay hierarchy used by this chapter). The Chapter 10 artifacts distinguish:

- (i) matrix and arithmetic replay;
- (ii) artifact and source-binding replay;
- (iii) semantic replay of the declared route;
- (iv) mathematical verification of each theorem invocation.

The supplied deterministic replay program establishes items (i)–(iii) for the encoded cases. Item (iv) remains the content of Chapters 1–9 and the proofs below.

## 10.2 v20 regression portfolio

**Definition 10.6** (v20 regression suite). The v20 regression suite  $\mathcal{R}_{20}$  contains four required tracks:

- R1 Iwasawa characteristic-growth recovery at the exact relative strength registered as CR-G16-IW;
- R2 fixed-exponent strict-subcritical Serrin recovery at the exact one-way strength registered as CR-G16-NS;
- R3 detector and Type IV boundary witnesses;
- R4 Claim Register, source binding, manifest, and non-claim consistency.

The suite is backward compatibility testing. None of its passing tracks is a new v21 arithmetic or analytic theorem.

**Definition 10.7** (Regression evidence axes). Each inherited recovery track records four independent axes:

- (i) *source identity*: the source statement and registry hashes agree with the frozen v20 recovery manifest;

- (ii) *registered mathematical strength*: the final v20 Claim Register identifies the exact theorem statement, hypotheses, and prohibited stronger readings;
- (iii) *quarantine conformance*: conclusion-side benchmark information is not consumed as precursor, realization, or Core evidence;
- (iv) *replay scope*: Chapter 10 rechecks identity and semantic conformance but does not re-prove the frozen external source theorem.

A pass on these axes is a regression pass for the declared v20 registry slice, not a new proof of Iwasawa's theorem or the Serrin criterion.

**Remark 10.8** (Regression is not theorem re-proving). The external source theorems remain imported theorem nodes with their v20 source bindings. The replay program checks content addressing, registry language, quarantine fields, and the finite boundary witnesses encoded in this chapter. Mathematical validity of the imported source theorem is not inferred from a hash or from successful replay.

**Proposition 10.9** (Regression R1: Iwasawa characteristic-growth recovery). *The v20 Iwasawa regression passes at the following exact strength:*

*relative recovery of the characteristic-growth clause of Iwasawa 1959, Theorem 4, for the registered strictly  $\Gamma$ -finite discrete module class and registered precursor package.*

*The replay verifies the source-statement binding hash*

a5f0a65a73c802492167676aba024b01651cd855bbdc00bc3d3c48d46cd6f0d7

*and source-registry hash*

7a3c2eb157eba3ccbe1613c4ade739991a26f8639083724f1213406a04ebff6c.

*The returned strength excludes the Iwasawa main conjecture, a class-number theorem without a separate arithmetic Interface, full module-structure recovery, and terminal information-preservation claims.*

*Proof.* The content-addressed source binding and registry agree with the v20 recovery manifest. The final v20 Claim Register records CR-G16–IW as a proved Interface theorem and limits its reading to the registered characteristic-growth clause. The benchmark conclusion remains quarantined from realization and precursor construction.  $\square$

**Proposition 10.10** (Regression R2: fixed-exponent Serrin recovery). *The v20 Serrin regression passes at the exact one-way statement*

$$u \in L^6(0, T; L^6(\mathbb{R}^3)) \implies u \text{ is interior regular on } \mathbb{R}^3 \times (0, T],$$

*for the registered unforced three-dimensional Leray–Hopf source regime. The source binding and registry hashes are respectively*

608216fda60bd3fdc055243c0169bdb31b53c337bb99d9b7c9cb43a215996423

*and*

445dcd8863d73cdfd68099ee0539fe3a4e96881f48788d9e817076e61666de18.

*The regression introduces no new analytic estimate and does not assert unconditional three-dimensional regularity.*

*Proof.* The v20 recovery manifest records the strict-subcritical exponent check  $2/6 + 3/6 = 5/6 < 1$ , exact source realization of the scalar Serrin norm bound, and return through the frozen source theorem. The final Claim Register records CR-G16-NS at criterion-level strength and explicitly denies a new analytic proof.  $\square$

**Proposition 10.11** (Regression R3: detector and Type IV separation). *At the repaired defect-profile stage on  $W = [0, 10)$ , with the displayed bars retained in the monitored defect scope, the morphism*

$$0 \longrightarrow k_{[2,3)}$$

*has terminal pair  $(\mu, \nu) = (0, 0)$  but is not an isomorphism, while*

$$0 \longrightarrow k_{[0,10)}$$

*has terminal pair  $(0, 1)$ . Hence the v20 terminal/interior separation and short-residual-defect boundary are reproduced exactly. This regression does not silently choose a threshold that would delete either witness.*

*Proof.* The first cokernel has one noncofinal bar and therefore zero terminal-generic dimension. The second cokernel has one bar cofinal to the right endpoint and therefore terminal-generic dimension one. In the first case the full cokernel is nonzero, proving that terminal zero alone is not full isomorphism.  $\square$

**Proposition 10.12** (Regression R4: governance and G16 closure). *The final v20 Claim Register contains the three mutually compatible entries CR-G16-IW, CR-G16-NS, and CR-G16-PASS. The gate passes because at least one exact-strength complete recovery is registered; two tracks are present. The same register retains the global non-claim excluding the Iwasawa main conjecture, unconditional Navier–Stokes regularity, and other major conjectural strengthenings.*

*Proof.* The replay reads the final Claim Register artifact with SHA-256

56be7b1e0bbde1ef2765961361a1eea1f11cc1d3e462aaade021a72f41e6ab5b

and checks the registered identifiers and exact-strength language. Hash agreement establishes artifact identity; the mathematical and governance meaning is supplied by the registered statements and their source artifacts.  $\square$

**Theorem 10.13** (Complete declared v20 regression slice). *All four tracks of  $\mathcal{R}_{20}$  pass at their exact inherited v20 strengths and at the evidence axes of Definition 10.7. Therefore the v21 Main body is backward compatible with the registered v20 Known-Theorem Recovery, detector/Type IV, and claim-governance boundaries exercised by this suite. The conclusion is limited to this declared regression slice and is not a re-execution of every artifact or theorem in the complete v20 release.*

*Proof.* Combine Propositions 10.9– 10.12 and preserve the four evidence axes. The conclusion is registry, source-binding, quarantine, and boundary compatibility, not new source-domain mathematics.  $\square$

### 10.3 Stable Case S1: finite persistence reconstruction

Let  $A_{[3]}$  be the chain  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ . Define a representation  $V$  by

$$V_0 = k^2, \quad V_1 = k^3, \quad V_2 = k^3, \quad V_3 = k,$$

with structure matrices

$$f_{01} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad f_{23} = (1 \quad 0 \quad 0).$$

**Proposition 10.14** (S1 rank table). *The complete rank table  $r(i, j) = \text{rank}(V_i \rightarrow V_j)$  is*

| $r(i, j)$ | $j = 0$ | $j = 1$ | $j = 2$ | $j = 3$ |
|-----------|---------|---------|---------|---------|
| $i = 0$   | 2       | 2       | 1       | 1       |
| $i = 1$   |         | 3       | 2       | 1       |
| $i = 2$   |         |         | 3       | 1       |
| $i = 3$   |         |         |         | 1       |

and the Chapter 4 Moebius multiplicities are

$$\mu(0, 1) = \mu(0, 3) = \mu(1, 2) = \mu(2, 2) = 1,$$

with every other  $\mu(i, j) = 0$ .

*Proof.* Direct matrix multiplication gives the displayed ranks. Apply Theorem 4.25:

$$\mu(i, j) = r(i, j) - r(i - 1, j) - r(i, j + 1) + r(i - 1, j + 1),$$

with the boundary values zero. The four displayed intervals are the only nonzero values.  $\square$

**Theorem 10.15** (S1 exact reconstruction). *The representation  $V$  is reconstructed up to isomorphism as*

$$V \cong k_{[0,3]} \oplus k_{[0,1]} \oplus k_{[1,2]} \oplus k_{[2,2]}.$$

Consequently an endpoint-complete persistence profile whose Chapter 4 atom restriction is  $V$  is returned exactly, including interval multiplicities.

*Proof.* The Chapter 4 finite-grid encoding is an equivalence on the declared constructible atom class, and the rank-to-multiplicity transform is invertible. The multiplicities in Proposition 10.14 produce the displayed direct sum, whose dimensions and maps reproduce the original matrices. Apply Theorems 4.15 and 4.22.  $\square$

**Definition 10.16** (S1 validation verdict). Case V21-S1 has class `internal_core_validation`, expected output the four interval multiplicities above, observed output the same multiplicities, and status `pass`. Its Return strength is finite persistence object reconstruction on the declared endpoint-complete grid; it is not a Return of an arbitrary filtered source object.

## 10.4 Stable Case S2: enriched finite-diagram Return

Let  $P$  have two objects  $p, q$ , identity morphisms, and two parallel arrows  $a, b : p \rightrightarrows q$ . Over a field of characteristic not two, consider representations  $E, E'$  with

$$E(p) = E(q) = E'(p) = E'(q) = k^2,$$

$$E(a) = E(b) = I_2, \quad E'(a) = I_2, \quad E'(b) = J := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

**Proposition 10.17** (Scalar probes are incomplete in S2). *The two representations have the same vertex dimensions and the same rank of every arrow, but they are not isomorphic.*

*Proof.* All four vertex dimensions are two and  $I_2, J$  both have rank two. Since  $a$  is the identity, an isomorphism would conjugate  $E(b)$  to  $E'(b)$ . However

$$\text{rank}(E(b) - I_2) = 0, \quad \text{rank}(E'(b) - I_2) = 1,$$

a similarity invariant contradiction.  $\square$

**Theorem 10.18** (S2 enriched generator reconstruction). *Let  $G_P = G_p \oplus G_q$  and  $\Lambda_P = \text{End}(G_P)^{\text{op}}$ . The enriched module*

$$M_E = \text{Hom}(G_P, E)$$

*recovers  $E(p), E(q)$ , the two independently labelled arrow maps, and every morphism between such diagrams. Applied to  $E'$ , it recovers the Jordan matrix  $J$ , and therefore distinguishes  $E'$  from  $E$ .*

*Proof.* The idempotent summands of  $\Lambda_P$  recover the two vertex spaces, while the actions of the two arrow elements recover  $E(a)$  and  $E(b)$ . The finite Morita equivalence and exact enriched reconstruction are Theorems 7.21 and 7.23.  $\square$

**Definition 10.19** (S2 validation verdict). Case V21–S2 has status pass. It validates enriched finite-diagram object and morphism Return and simultaneously validates the scalar-probe no-go boundary.

### 10.5 Stable Case S3: finite object and derived descent

Let  $P = \{a, b, c\}$  be the V-shaped poset  $a < c, b < c$ , with the upper Alexandrov convention of Chapter 7. Let

$$U_1 = \{a, c\}, \quad U_2 = \{b, c\}, \quad U_{12} = \{c\}.$$

Define local diagrams

$$\begin{aligned} E_1(a) &= k, & E_1(c) &= k^2, & E_1(a < c)(1) &= e_1, \\ E_2(b) &= k, & E_2(c) &= k^2, & E_2(b < c)(1) &= e_2, \end{aligned}$$

and take the overlap isomorphism  $g_{12} = I_{k^2}$ .

**Theorem 10.20** (S3 object-level descent). *The local data descend to the global diagram*

$$E(a) = k, \quad E(b) = k, \quad E(c) = k^2,$$

*with maps  $1 \mapsto e_1$  and  $1 \mapsto e_2$ . Its restrictions to  $U_1, U_2$  are isomorphic to the declared local objects, and it is unique up to unique descent isomorphism.*

*Proof.* The overlap map is an actual isomorphism and the two-set cocycle condition is tautological. The equalizer construction in Theorem 7.31 gives the displayed global object. Since the cover is finite and the descent category is the declared finite Alexandrov sheaf model, effectivity and uniqueness hold.  $\square$

Define local two-term complexes  $K_1^\bullet$  on  $U_1$  and  $K_2^\bullet$  on  $U_2$  as follows. On the overlap object  $c$ , both have

$$K_j^{-1}(c) = k, \quad K_j^0(c) = k^2, \quad d_c(1) = e_1 + e_2.$$

At  $a$  for  $K_1^\bullet$ , respectively at  $b$  for  $K_2^\bullet$ , the degree  $-1$  term is zero, the degree-zero term is  $k$ , the differential is zero, and the degree-zero structure map is the corresponding coordinate inclusion. Equip their common restriction to  $U_{12} = \{c\}$  with the identity chain isomorphism.

**Proposition 10.21** (S3 strict derived descent). *The local complexes  $K_1^\bullet, K_2^\bullet$  and their identity overlap chain isomorphism descend degreewise to a global complex  $K^\bullet$  with*

$$\begin{aligned} K^{-1}(a) &= K^{-1}(b) = 0, & K^{-1}(c) &= k, \\ K^0(a) &= K^0(b) = k, & K^0(c) &= k^2, & d_c(1) &= e_1 + e_2. \end{aligned}$$

At  $c$ ,

$$H^{-1}(K^\bullet)(c) = 0, \quad \dim_k H^0(K^\bullet)(c) = 1.$$



*Proof.* The local differentials are natural because their degree  $-1$  terms at  $a, b$  vanish, and the identity overlap chain map commutes with  $d_c$ . Theorem 7.34 constructs the displayed global complex from the local datum. The map  $k \rightarrow k^2, 1 \mapsto e_1 + e_2$ , is injective with one-dimensional cokernel.  $\square$

**Definition 10.22** (S3 validation verdict). Case V21-S3 has class `finite_globalization` and status `pass`. It validates actual object construction and strict bounded-complex descent, not arbitrary homotopy-coherent or unbounded descent.

## 10.6 Stable Case S4: finite-state infinite Return

Let  $V_n = k^2$  for every  $n \geq 0$ , and define the inverse transition  $V_{n+1} \rightarrow V_n$  by

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

**Theorem 10.23** (S4 finite-state inverse-limit Return). *The stable image is*

$$W_{\text{st}} = \text{im}(T^2) = ke_1,$$

*the inverse limit is canonically*

$$\varprojlim_n V_n \cong ke_1,$$

*and*

$$\varprojlim_n^1 V_n = 0.$$

*If the apex is  $A = ke_1$  with the evident compatible maps into every  $V_n$ , the apex comparison is an isomorphism.*

*Proof.* Since  $T^2 = T$ , the image chain stabilizes after one step and the system is Mittag-Leffler. A compatible sequence  $x_n = a_n e_1 + b_n e_2$  satisfies  $x_n = T x_{n+1}$ , hence  $b_n = 0$  and  $a_n = a_{n+1}$ . Thus compatible sequences are exactly constant  $ke_1$ -sequences. The derived-limit vanishing and apex criterion are the finite-state theorems 5.84 and 5.85.  $\square$

**Definition 10.24** (S4 validation verdict). Case V21-S4 has status `pass`. The Return is the inverse-limit object, derived-limit clearance, and apex agreement. It does not identify an external arithmetic tower without a separate source realization and Return theorem.

## 10.7 Stable Case S5: unified external Return through both Cores

Let  $C$  be the filtered chain complex over  $k$  with

$$C_0 = kx \oplus kz, \quad C_1 = ky \oplus kw, \quad \partial y = x, \quad \partial w = z,$$

and filtration births

$$\beta(x) = 0, \quad \beta(z) = 1, \quad \beta(y) = 2, \quad \beta(w) = \frac{5}{2}.$$

Take the source breakpoint set

$$K_{\text{src}} = \{0, 1, 2, \frac{5}{2}\},$$

work on  $W = [0, 3)$ , and set  $\tau = 2$ . The excluded endpoint 3 is a window boundary, not a sampled source breakpoint. For the HC component take  $T = C$  with the identity zero-shift filtered contraction.

**Proposition 10.25** (S5 source persistence). *The degree-zero persistent homology barcode is*

$$\mathcal{B}_0(C) = \{[0, 2), [1, \frac{5}{2})\},$$

*and degree-one persistent homology is zero.*

*Proof.* The class of  $x$  is born at zero and becomes a boundary when  $y$  enters at time two. The class of  $z$  is born at one and becomes a boundary when  $w$  enters at time  $5/2$ . The degree-one generators map injectively to the two independent degree-zero generators, so no degree-one homology class appears.  $\square$

**Definition 10.26** (S5 external Return predicate). Let  $\mathcal{L}_{0,W,2}(C)$  be the source-domain statement

$$\text{im}(H_0(F_s C) \rightarrow H_0(F_t C)) = 0 \quad \text{for every } s \leq t \text{ in } W \text{ with } t - s > 2.$$

This is a predicate about the filtered chain complex and its actual homology maps, not merely a barcode label.

**Theorem 10.27** (S5 complete Corep + CoreDG Return). *For the filtered chain complex above, the complete route*

$$C \xrightarrow{\text{HC}_{\text{id}}} C \longrightarrow \text{Corep} \longrightarrow \text{CoreDG} \longrightarrow \text{enriched generator probe} \longrightarrow \mathcal{L}_{0,W,2}(C)$$

*passes. More explicitly,*

$$\begin{aligned} \mathcal{L}_{0,W,2}(C) &\iff T_2^{\text{bar}}(W_W P_0(C)) = 0 \\ &\iff M = 0 \\ &\iff E_{\Gamma_W} = 0 \\ &\iff \mathbf{R} \text{Hom}(G_{A_{[\ell]}}, E_{\Gamma_W}) \simeq 0. \end{aligned}$$

*Thus the external long-span homology predicate is returned at exact predicate strength.*

*Proof.* Both source bars have length at most two, so hard threshold deletion at  $\tau = 2$  removes both. The strict span condition  $t - s > 2$  is exactly the source interpretation of absence of bars longer than two. The source breakpoint set  $K_{\text{src}} = \{0, 1, 2, 5/2\}$ , together with the right-open window boundary policy for  $W = [0, 3)$ , produces the endpoint-complete atom grid of Chapter 9 without evaluating the excluded endpoint 3. The HC predecessor is discharged by the identity filtered contraction  $C \rightleftharpoons C$ . The Chapter 4 restriction equivalence transports zero to the finite linear diagram, and the Chapter 7 projective generator reflects zero. Finally apply Theorem 9.54 and its certified Return corollary.  $\square$

**Proposition 10.28** (S5 Return ceiling). *The S5 route returns the predicate  $\mathcal{L}_{0,W,2}(C)$ , the repaired persistence zero object, and the corresponding zero finite diagram. It does not reconstruct the source filtered chain complex.*

*Proof.* Let  $B$  be the filtered complex with one degree-one generator born at time zero, no degree-zero generator, and zero differential. Then  $P_0(C \oplus B) \cong P_0(C)$ , so the repaired degree-zero persistence object, finite diagram, generator probe, and S5 Return predicate are unchanged. However  $P_1(C) = 0$  while  $P_1(C \oplus B) \neq 0$ . Thus the two sources are not filtered quasi-isomorphic, and a fortiori are not isomorphic as strict filtered chain complexes. The degree-zero predicate route therefore cannot be strengthened to full source-object or full filtered-derived reconstruction.  $\square$

**Definition 10.29** (S5 validation verdict). Case V21-S5 has class `new_bridge_return`, status `pass`, and maximum licensed external strength `predicate_return`. It is the required v21 example consuming both `Corep` and `CoreDG`.

## 10.8 Certificate-complexity validation

Fix  $m \geq 0$  and let

$$N_m = \#\{(i, j) : 0 \leq i \leq j \leq m\} = \frac{(m+1)(m+2)}{2}.$$

Let  $\mathcal{F}_m$  be the family of finite  $A_{[m]}$ -representations with arbitrary nonnegative interval multiplicities. In the declared deterministic, possibly adaptive *coordinate rank-query model*, a query selects one coordinate and returns one value  $r(i, j) = \text{rank}(V_i \rightarrow V_j)$ . Query choice may depend on earlier answers, but no object-dependent advice, arbitrary linear combination, or encoded scalar is permitted.

**Theorem 10.30** (Exact query complexity of rank-table reconstruction). *In the coordinate rank-query model, exact reconstruction of every object in  $\mathcal{F}_m$  requires and is achieved by exactly  $N_m$  rank queries. For  $m = 3$ , the S1 certificate uses ten queries and is query-minimal.*

*Proof.* There are  $N_m$  interval multiplicity coordinates. The complete rank table contains  $N_m$  entries, and the Chapter 4 Moebius formula reconstructs every multiplicity, proving the upper bound.

For the lower bound, write the rank responses as

$$r = A\mu,$$

where  $\mu \in \mathbb{Z}_{\geq 0}^{N_m}$  is the multiplicity vector and  $A_{(i,j),(a,b)} = 1$  exactly when  $a \leq i \leq j \leq b$ . Consider any deterministic algorithm that uses fewer than  $N_m$  possibly adaptive coordinate queries. For every proper coordinate subset  $Q$ , choose after clearing denominators one nonzero integer vector  $h_Q \in \ker A_Q$ . There are finitely many such subsets, so

$$H := \max_Q \|h_Q\|_\infty < \infty.$$

Choose  $M > H$ , run the algorithm on  $M\mathbf{1}$ , and let  $Q$  be the coordinates queried along that transcript. Then both

$$M\mathbf{1}, \quad M\mathbf{1} + h_Q$$

are distinct nonnegative multiplicity vectors and give identical answers on every queried coordinate. Inductively, the algorithm makes the same adaptive query choices on both inputs and produces the same output. It cannot be exact on both. Thus fewer than  $N_m$  queries cannot reconstruct the entire unbounded family. For the full table, the Moebius inverse shows that  $A$  is unimodular; in the  $m = 3$  replay its determinant is one.  $\square$

**Corollary 10.31** (Model-relative certificate and verifier cost). *If every interval multiplicity is at most  $B$ , each rank response is at most  $N_m B$ . A complete rank-table certificate therefore has*

$$q = N_m, \quad b = O(N_m \log(N_m B + 1)), \quad t_{\text{arith}} = O(N_m), \quad a = 0$$

*in the declared Chapter 3 model. The verifier performs at most four integer additions or subtractions per Moebius coordinate, plus nonnegativity and reconstruction checks. Bit complexity is this arithmetic count multiplied by the cost of operations on  $O(\log(N_m B + 1))$ -bit integers.*

**Remark 10.32** (Minimality is model relative). The lower bound does not apply to an oracle returning the full barcode in one answer, an unbounded real encoding all multiplicities, or object-dependent advice. Those are different information models and cannot be used to claim a smaller MinCert in the coordinate rank-query model.

## 10.9 Challenge and Stable tracks

**Definition 10.33** (Two-track validation cycle). A successor validation cycle contains:

- (i) a *Challenge route* targeting a new source class, stronger strength, or unresolved Bridge;
- (ii) a *Stable route* using at least one of the same generic components to close a previously known or lower-strength Return.

The Challenge route and Stable route have separate target identifiers and proof-DAG roots. A Stable pass is not reported as a Challenge pass.

**Definition 10.34** (Challenge stopping rule). Before a cycle begins, its manifest fixes:

- (i) a maximum number of proof-DAG expansion rounds  $N_{\max}$ ;
- (ii) a finite set of admissible new assumptions and source imports;
- (iii) a monotone progress measure  $(|\Gamma_n|, |\mathcal{F}_n|, d_n)$ , consisting of unresolved assumptions, open failure leaves, and target-DAG distance;
- (iv) a forbidden-input and benchmark-quarantine policy;
- (v) the Stable fallback target and components shared with the Challenge route.

The Challenge route stops without theorem promotion if the target remains outside the route-admissible strength set at round  $N_{\max}$ , if the progress measure does not strictly improve for the declared patience window, or if continuing would require prohibited conclusion-side input.

**Proposition 10.35** (Stable fallback is informative but noncompensating). *A Stable fallback pass validates the shared component and preserves a mathematical Return for the cycle, but it neither discharges the Challenge failure leaf nor changes its status.*

*Proof.* The two routes have different target nodes. Chapter 8 noncompensation and immutable revision prevent a pass on one target from modifying another.  $\square$

**Definition 10.36** (Initial v22 Challenge portfolio). The initial successor portfolio contains:

- (i) Iwasawa tower information preservation and a source-certified determinant or characteristic-ideal Return;
- (ii) a variable-exponent or endpoint Serrin route with genuinely new analytic input;
- (iii) the finite-syzygy Tachikawa challenge under an independently meaningful realization;
- (iv) a machine-checkable homotopy-coherent finite descent model.

Each Challenge item is `specification_target` until a new revision closes its proof-DAG.

## 10.10 Negative validation portfolio

**Example 10.37** (N1: incomplete generator). Let  $P$  be the discrete category on  $p, q$ , and probe only with  $G_p$ . The nonzero object supported at  $q$  has  $\text{Hom}(G_p, E) = 0$ . Hence a proper subset of the representable generator family need not reflect zero.

**Example 10.38** (N2: pairwise isomorphism without cocycle). On a triple overlap, over a field of characteristic not two, take one-dimensional local objects and

$$g_{12} = \text{id}, \quad g_{23} = \text{id}, \quad g_{13} = -\text{id}.$$

Every pairwise map is an isomorphism, but  $g_{23}g_{12} \neq g_{13}$ , so the datum is not effective descent data.

**Example 10.39** (N3: finite-prefix fallacy). For any  $N$ , one direct system remains zero forever and another agrees with it through stage  $N$ , then becomes  $k$  with identity maps. Their finite prefixes agree and their colimits differ.

**Example 10.40** (N4: local zero with global nonzero). Let the global barcode consist of one bar  $[0, 2\tau)$ , and cover it by the Chapter 6 windows  $[0, \tau)$ ,  $[\tau/2, 3\tau/2)$ ,  $[\tau, 2\tau)$ . Every local restricted bar has length at most  $\tau$  and is deleted, while the global bar survives. Retained-bar coverage is the missing hypothesis.

**Example 10.41** (N5: synthetic Bridge). Given a detector vector space  $D(P)$ , form the synthetic complex  $D(P)[-n]$ . Then

$$\text{Ext}^n(D(P)[-n], k) \cong D(P)^*.$$

This is a correct computational duality but does not identify an independently meaningful external obstruction, and therefore is not a principal natural Bridge.

**Example 10.42** (N6: false Return strengthening). The S5 route observes only degree-zero persistent homology. Add the filtered complex  $B$  consisting of one degree-one cycle born at time zero and no other generator. Then  $P_0(C \oplus B) \cong P_0(C)$ , so the S5 predicate is unchanged, while  $P_1(C \oplus B) \neq P_1(C)$ . Hence the sources are not even filtered quasi-isomorphic. Predicate Return cannot be strengthened to full source-object or filtered-derived reconstruction.

**Theorem 10.43** (Negative validation closure). *Examples 10.37– 10.42 validate six independent non-composition boundaries: generator completeness, higher coherence, finite-to-infinite reduction, retained-feature coverage, natural realization, and Return ceiling. N1–N4 and N6 are mathematical counterexamples to the corresponding stronger implication. N5 is a typed non-entailment witness: the synthetic duality is true, but the source-semantic realization required by a principal Bridge is absent, so the stronger Bridge conclusion is not licensed rather than mathematically refuted.*

*Proof.* The five counterexamples satisfy their weaker premises while falsifying the stronger conclusion. In N5 the weaker computational statement inhabits a different interface type and supplies no source-realization edge. Thus in all six cases the omitted obligation is logically necessary for the corresponding general route, with reject and undefined/non-licensed outcomes kept distinct.  $\square$

## 10.11 Validation and release gate matrix

**Definition 10.44** (v21 Main validation gates). The Chapter 10 required Main validation gates are:

V1 complete v20 regression at exact registered strength;

- V2 finite persistence reconstruction;
- V3 enriched finite-diagram reconstruction;
- V4 finite object-level and strict derived descent;
- V5 finite-state inverse-limit and apex Return;
- V6 a complete external Return consuming both Cores;
- V7 model-relative certificate upper and lower bounds;
- V8 negative validation of all six mandatory no-go boundaries;
- V9 artifact, replay, proof-DAG, and manifest consistency for this portfolio.

The four mandatory completion categories are witnessed by V1, V3, V4, and V6. This chapter executes the stronger V1–V9 portfolio; V2 supplies the additional finite-persistence reconstruction validation used elsewhere in the portfolio.

**Proposition 10.45** (v21 Main validation gate results). *The gate results are:*

| <i>Gate</i> | <i>status</i> | <i>principal witness</i>                                      |
|-------------|---------------|---------------------------------------------------------------|
| V1          | pass          | <a href="#">Theorem 10.13</a>                                 |
| V2          | pass          | <a href="#">Theorem 10.15</a>                                 |
| V3          | pass          | <a href="#">Theorem 10.18</a>                                 |
| V4          | pass          | <a href="#">Theorem 10.20</a>                                 |
| V5          | pass          | <a href="#">Theorem 10.23</a>                                 |
| V6          | pass          | <a href="#">Theorem 10.27</a>                                 |
| V7          | pass          | <a href="#">Theorem 10.30</a>                                 |
| V8          | pass          | <a href="#">Theorem 10.43</a>                                 |
| V9          | pass          | <i>Chapter 8 closure plus the companion re-audit manifest</i> |

**Theorem 10.46** (Required v21 validation portfolio closure). *The four mandatory portfolio items*

- (1) *v20 regression,*
- (2) *finite diagram reconstruction,*
- (3) *finite object-level descent,*
- (4) *one external Return using  $\text{Core}_P + \text{Core}_{DG}$ ,*

*all pass. In fact, every gate V1–V9 passes. Therefore the v21 Main body has a closed, explicit, and replayable validation portfolio at the strengths stated in this chapter.*

*Proof.* The required items are Theorems [10.13](#), [10.18](#), [10.20](#), and [10.27](#). The remaining gates follow from Proposition [10.45](#). Chapter 8 noncompensation is satisfied because every required gate independently passes. □

**Remark 10.47** (Main validation pass versus release headline). Theorem [10.46](#) is not the complete v21 release headline. A complete release additionally requires the final v21 Claim Register, appendix migration and synchronization, release manifest, source-binding bundle, and every mandatory release-level artifact. Until those release records are constructed and audited, the release headline is undefined even though the Main validation portfolio passes.

## 10.12 Successor problem registry

**Definition 10.48** (Successor classification). Every successor item is assigned exactly one disposition:

`core_promotion_candidate,`      `technical_extension_candidate,`  
`challenge_continue,`      `rejected_no_go.`

The disposition is governance metadata, not theorem status.

**Proposition 10.49** (v22 Core promotion candidates). *The strongest conservative Core promotion candidates are:*

- (i) *a certified compiler from finite-state transition records to the Chapter 5 stable-image and apex certificates;*
- (ii) *a fixed dg or simplicial model implementing finite homotopy-coherent descent with machine-checkable cells and effectivity;*
- (iii) *generator detection and finite Morita over a declared Noetherian base for perfect finite-presentation objects, with all handedness and derived-completion hypotheses explicit;*
- (iv) *a uniform target-relative calculus for compatible partial Returns.*

**Proposition 10.50** (Technical Extension candidates). *The principal Technical Extension candidates are derived-completion comparison, determinant and Fitting-ideal Return, spectral-sequence extension recovery, non-field coefficient UCT/Bockstein packages, and source-specific rigidity lemmas.*

**Proposition 10.51** (Challenge-continuation registry). *The Iwasawa information-preservation route,  $p$ -adic  $L$ -function Return, variable-exponent Serrin route, and finite-syzygy Tachikawa route remain `challenge_continue`. Their current stable fallbacks are respectively the exact characteristic-growth recovery, fixed-exponent Serrin recovery, finite-state tower Return, and finite-diagram generator reconstruction.*

**Proposition 10.52** (Permanent no-go registry). *The following remain `rejected_no_go` without a change of definitions or hypotheses:*

- (i) *a general positive-threshold exact Serre reflector for hard barcode deletion;*
- (ii) *global non-expansiveness of hard positive-threshold deletion;*
- (iii) *pairwise overlap implying higher coherence;*
- (iv) *finite-prefix observation implying a limit theorem;*
- (v) *scalar ranks reconstructing arbitrary finite diagrams;*
- (vi) *terminal Type IV zero implying full isomorphism;*
- (vii) *Core pass implying an external theorem without Return;*
- (viii) *a benchmark conclusion used as precursor evidence for its own recovery.*

### 10.13 Final validation theorem and closure of the v21 Main body

**Theorem 10.53** (Validated v21 Main-body portfolio architecture). *The final-audited v21 Main body simultaneously validates the following typed routes and boundaries:*

- S1 : *finite rank data*  $\longleftrightarrow$  *finite persistence object*,
- S2 : *enriched generator module*  $\longleftrightarrow$  *finite diagram object and morphisms*,
- S3 : *effective finite descent datum*  $\longrightarrow$  *global object and strict bounded complex*,
- S4 : *certified finite-state inverse tower*  $\longrightarrow$  *inverse limit,  $\varprojlim^1$ -clearance, and apex*,
- S5 : *filtered-chain source*  $\longrightarrow$   $\text{Corep} \longrightarrow \text{CoreDG} \longrightarrow$  *external degree-zero predicate Return*.

*These rows form a validation portfolio, not one serial theorem: no output of S3 or S4 is claimed to be a predecessor of S5 unless a separate Chapter 8 proof-DAG and Chapter 9 Bridge record explicitly composes them. The v20 registered recovery slice remains valid at inherited strength, the mandatory v21 portfolio passes, and every prohibited shortcut listed in the negative portfolio has an explicit witness.*

*Proof.* S1–S5 prove the five displayed routes at their declared targets. Their logical independence is preserved by Chapter 8 noncompensation and by the route-specific predecessor graphs of Chapter 9. Theorem 10.13 establishes the declared backward-compatibility slice, while Theorem 10.43 validates the necessary boundaries.  $\square$

**Remark 10.54** (What v21 has actually achieved). The v21 advance is not an unconditional solution of a major external problem. It is a materially stronger mathematical and architectural closure than v20: finite persistence data can be reconstructed as enriched finite diagrams; finite diagrams can be globally descended in a proved finite model; selected infinite tails can be returned from finite-state certificates; and one full external-to-external predicate route has been executed through both Cores. The exact strength and the exact ceiling are both explicit.

**Remark 10.55** (Final AK validation principle).

A validation is credible only when the success case closes at its declared strength, the failure case breaks at the missing hypothesis, the artifact can be replayed, and the Return ceiling is stated as carefully as the Return.

### 10.14 Explicit non-claims

This chapter asserts none of the following.

- (i) A complete v21 release pass before the final appendix, Claim Register, and release-manifest audit.
- (ii) New analytic content in the Serrin regression or new arithmetic structure theory in the Iwasawa regression.
- (iii) Reconstruction of the S5 filtered chain complex from its returned persistence predicate.
- (iv) Universality of the S1 query lower bound outside the declared deterministic coordinate rank-query model, or the bounded-family bit estimate without its  $\log(N_m B + 1)$  response-size factor.
- (v) Arbitrary derived or homotopy-coherent descent from the strict finite S3 example.
- (vi) An external arithmetic interpretation of the S4 inverse tower.
- (vii) Promotion of a Challenge target because its Stable fallback passes.



- (viii) Mathematical truth from hash agreement, successful compilation, or replay alone.
- (ix) A new Bridge theorem from a benchmark, synthetic encoding, or nonfaithful scalar probe.
- (x) Any strengthening beyond the route-admissible Return target and assumption context recorded by the case.

### 10.15 Provenance, replay artifacts, and status

The v20 regression consumes the final v20 Claim Register, the Iwasawa and Serrin source-binding registries and recovery manifests, and the inherited counterexample semantics. It verifies the declared registry slice at the four evidence axes of Definition 10.7; it does not reconstruct every v20 release artifact or re-prove the frozen source theorems. Stable Cases S1–S5 consume only the final-audited Chapters 1–9 and the explicit data displayed in this chapter. The replay artifact `AK_HPDEST_v21_0_0_Chapter_10_Validation_Replay_REAUDITED.py` regenerates the machine-readable portfolio manifest `AK_HPDEST_v21_0_0_Chapter_10_Validation_Portfolio_REAUDITED.json`.

**Core validation theorems.** Theorems 10.13, 10.15, 10.18, 10.20, 10.23, 10.27, 10.30, 10.43, 10.46, and 10.53.

**Validation records and governance.** Definitions 10.1, 10.2, 10.3, 10.5, 10.6, 10.16, 10.19, 10.22, 10.24, 10.26, 10.29, 10.33, 10.34, 10.36, 10.44, and 10.48.

**Boundary results and counterexamples.** Proposition 10.4, Corollary 10.31, Remark 10.32, Proposition 10.28, Examples 10.37– 10.42, and Subsection 10.14.

**Final Main-body status.** The Chapter 10 required validation portfolio is pass. The full v21 release headline remains a separate Chapter 8 governance computation and is not set by this chapter alone.